

AECOM TISHMAN CONSTRUCTION

Safety, Health, and Environmental Program

Updated August 2025

AECOM TISHMAN CONSTRUCTION projects shall incorporate the policies and procedures within this document which apply to the scope of work for which they have committed. The comprehensive expectations enumerated in this document are to be incorporated as performance expectations in the contracts for all subcontractors.

NOTICE OF INTENT

The content of this Safety, Health, and Environmental Program does not contain, address or offer legal interpretations on all aspects of safety compliance with regulations, rules and ordinances applicable to construction nor to all management principles and/or best practices. The content also does not cover all items found in the Tier 1, 2, and 3 policies of AECOM. Many items included in this document are identical to the Tier policies but are not all inclusive. In the event that an item is not covered, please refer to the [AECOM Tier safety programs](#) located on Ecosystem. These programs are the guiding documents for all operations. What it does purport to present is a detailed written safety, health, and safety management plan that will increase the likelihood of maintaining a work environment that is free of recognized hazards and thereby maintain compliance with the myriad of federal, state, and local OSHA Standards, Regulations, Letters of Interpretation and applicable Review Commission Decisions and Case Law. In all cases, AECOM TISHMAN CONSTRUCTION projects shall follow and meet federal, state, and local safety standards and requirements, and this program as a minimum. The most stringent shall apply for the applicable area.

In this respect, it is to be considered a guide for AECOM TISHMAN CONSTRUCTION and our subcontractors working on our projects. Every AECOM TISHMAN CONSTRUCTION employee shall be provided access to this program. If there is any question as to this program's content, interpretations thereof or site application regarding its use on the project, contact your Regional Safety Director.

Each jobsite is expected to develop a site-specific plan that addresses the risks and hazards pertaining to that job's particular scope of work in conjunction with this program. Contact your Regional Safety Director should you need assistance in developing your site-specific plan.

Life-Preserving Principles

AECOM has adopted nine "Life-Preserving Principles" which guide our global Safety, Health & Environment (SH&E) program, Safety for Life. These Principles advance our commitment to achieve our goal of zero work-related injuries to AECOM employees and stakeholders, incident prevention, and protection of the environment while executing our work.



Demonstrated Management Commitment

Our executive, senior and project managers lead the Safety, Health and Environment improvement process by continuously demonstrating their support, commitment and accountability.



Pre-Planning

Effective hazard control will be employed through Pre-Project and Pre-Task planning to design, plan and build the safety and health protection of our employees, stakeholders and the environment into every project.



Safety Orientation and Training

Appropriate safety training will be provided to all employees and partners to set the expectations of our Safety for Life program and to provide skills enabling proactive hazard identification and control.



Employee Participation

Our employees are empowered to engage in our Culture of Caring and Safety for Life program to actively contribute to the physical, psychological, and social well-being of all affected by our work.



Contractor Management

Our project staff will coordinate with our clients, sub-consultants, subcontractors, contractors and Joint Venture Partners to collectively plan for and maintain a safe and healthy workplace.



Incident Investigation

Our managers and safety staff will investigate all recordable and high potential incidents to identify contributing factors and root causes. Lessons learned will be developed, shared and implemented to prevent reoccurrence.



Budgeting and Staffing for Safety

Our safety staff will be appropriately trained, experienced and qualified to provide technical support to our internal and external clients. A budget to support safety and health needs will be included in project proposals.



Recognition and Rewards

Our employees and project participants will be recognized for their commitment to working safely and their positive contribution to our safety efforts.



Fit for Duty

All employees are responsible for reporting fit for the tasks at hand and for remaining so throughout the course of the work period so as not to compromise their health and safety or that of others.

Safety, Health & Environment

Purpose

This policy establishes the framework to safeguard AECOM employees and stakeholders through effective management of risk and commitment to a Culture of Caring.

Commitment

In recognition of the right to a safe and healthful working environment, AECOM is committed to maintaining the physical, psychological, and social well-being of our employees and stakeholders through appropriate risk management strategies.

To advance our Safety, Health & Environment (SH&E) program, AECOM is committed to:

- The ultimate objective of incident prevention for AECOM employees, stakeholders, and the environment.
- Implementation of our Safety for Life program, a comprehensive SH&E management system guided by our Life-Preserving Principles, to empower employees and drive continuous review and improvement.
- Effective management of SH&E risk throughout the project lifecycle, through identification of hazards, development of suitable actions using an established hierarchy of controls, and appropriate resourcing.
- Meet client requirements and comply with all applicable local, state, provincial and national SH&E legal requirements.
- Foster a safety culture where communication, collaboration, and consultation enable ownership for the well-being of self and others.
- Set appropriate SH&E performance goals and Core Value Metrics; work with employees and business partners to meet targets and promote continuous improvement opportunities.
- Establish AECOM as the global provider of choice through safe execution of professional services throughout the project lifecycle.

Participation

Individual engagement with the AECOM Safety for Life program is required through the participation of all parties in our Culture of Caring.

AECOM expects its leaders, managers, supervisors, employees, and subcontractors to:

- Support their own well-being as well as the well-being of all other stakeholders both on and off the job.
- Demonstrate individual commitment through compliance with applicable policies and processes, and properly identify, manage, and eliminate hazards to reduce risk in the workplace.
- Complete applicable planning and training to enable competency and proper and appropriately maintained equipment, materials, and personal protective equipment required to work safely and respond as necessary to emergencies.
- Take a moment to assess SH&E risk and take action to stop work if the work cannot be executed safely or if conditions or behaviors are unsafe or unhealthy.
- Immediately report SH&E incidents, near-misses, unsafe conditions, and at-risk behaviors; participate in investigation processes to enable learning and implementation of corrective and preventive actions.

Accountability

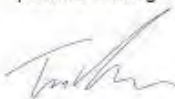
AECOM expects ownership of our Safety for Life program, where everyone participates, and leaders demonstrate and drive accountability to achieve SH&E excellence.

AECOM management, employees, and subcontractors will be held accountable to their commitment and participation through:

- Recognition and reward of those who positively contribute to excellent SH&E culture and performance.
- Inspections, investigations and reporting to assess SH&E management system application and effectiveness; elevation of high potential findings to leadership to enable appropriate action.
- Suitable action such as coaching or disciplinary measures when expectations are not met.

Review and Communication

This Policy and associated SH&E management system will be reviewed annually and will be made available to all persons working for or on behalf of AECOM.



Troy Rudd
Chief Executive Officer

August 29, 2025

Date

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SECTION 1 - AECOM TISHMAN CONSTRUCTION GENERAL WORKPLACE POLICIES

1. PURPOSE and SCOPE

1.1 AECOM TISHMAN CONSTRUCTION is responsible for providing and maintaining, within its scope of work, a safe and healthful workplace for its employees and general public. Subcontractors are subject to audits from AECOM TISHMAN CONSTRUCTION to ensure compliance with all agencies having jurisdiction for their identified scope of works.

1.2 All individuals on all projects must be aware of our policies and procedures. Subcontractors are solely responsible for the safety of their employees at all times. All subcontractors are required to follow all applicable federal, state, local, AECOM, and AECOM TISHMAN CONSTRUCTION laws, standards, policies, and procedures.

1.3 This program applies in its entirety to all AECOM TISHMAN CONSTRUCTION projects, unless a variance is granted by the Vice President of SH&E.

2. TERMS and DEFINITIONS

None Applicable.

3. RESPONSIBILITIES

3.1 The Project Leadership has overall responsibility for establishing and ensuring compliance with these procedures.

3.2 The Safety Manager is responsible for implementing and/or monitoring activities associated with these procedures.

3.3 The Project Leadership shall utilize this program when reviewing Safety, Health, and Environmental operations.

3.4 It is the responsibility of all managers and supervisory personnel to enforce these procedures and of each individual to follow them.

3.5 No work is to be completed unless there is onsite supervision from AECOM or a designated JV partner.

3.6 Employees and subcontractors have the right and responsibility to report an unsafe condition and/ or stop work if conditions or activities are unsafe without the fear or threat of reprisal.

3.7 Further information on requirements and responsibilities can be found in the [AECOM Code of Conduct](#) and [AECOM Employee Handbook](#).

4. GENERAL REQUIREMENTS

4.1 Smoking

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- AECOM TISHMAN CONSTRUCTION has implemented a no-smoking policy inside all of its facilities. For the purpose of this policy, facilities include buildings, company leased or provided vehicles, project sites, and company provided/supported temporary housing. All occupants in such facilities are required to comply with this policy.
- Anyone who wishes to smoke can do so outside company facilities in designated areas. It is the individual's responsibility to limit time spent away from the workplace to smoke and to ensure that break time is not abused.

4.2 **Drug-Free Workplace**

- **Key provisions**

Consistent with this commitment and in keeping with the federal Drug-Free Workplace Act of 1988, it is the policy of AECOM TISHMAN CONSTRUCTION to maintain a drug-free workplace. AECOM TISHMAN CONSTRUCTION policy prohibits individuals from being under the influence of alcohol or drugs or improperly using medication in any way that could diminish, or raise questions concerning, an individual's ability to perform at his or her best while performing services for or on behalf of AECOM TISHMAN CONSTRUCTION. Individuals who are under the influence of alcohol or any controlled substance have the potential for interfering with their own and their coworkers' safe and efficient job performance. Compliance with this policy is considered a condition of employment.

AECOM TISHMAN CONSTRUCTION also prohibits the sale, possession, manufacturing and/or distribution of illegal drugs, and/or other controlled substances in the workplace or while on company business off premises. Violations of this policy will be considered to be gross and willful misconduct and will result in disciplinary action, up to and including termination. Any illegal substances discovered in the workplace will be turned over to the appropriate law enforcement agency and may result in criminal prosecution.

- **Company - Sponsored events**

It is understood that individuals may at times attend company-sponsored events where alcohol is served. While consuming alcohol on such occasions is not prohibited, individuals are expected to exercise proper judgment and must observe professional, legal and company guidelines at all times.

Prior approval from management and Human Resources is required for any onsite AECOM TISHMAN CONSTRUCTION - sponsored event where alcohol is served. In addition, such events must comply with the following conditions:

- Individuals should not be expected to return to work after attending the event.
- Transportation arrangements will be provided for individuals who need such assistance.

- **Testing for drugs and alcohol**

Drug and/or alcohol screening may be required:

1. Of any applicant to whom a job offer has been made.
2. Of any individual where there is reason to believe that he or she may be using illegal or non-prescribed drugs or may be under the influence of drugs and alcohol. "Reason to believe" includes an injury or incident at work where there is reason to believe that individual impairment may have been a factor. "Reason to believe" may be based on objective symptoms such as the individual's appearance, behavior, or speech.

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3. As part of occasional follow-up testing, if the individual is found to have breached these policies but has been permitted to remain employed.
4. As required by client contract, project, or if an individual is employed in a safety-sensitive position. Under these limited circumstances, individuals may also be subjected to pre-employment and random drug screening.

An individual's cooperation with such drug or alcohol screening tests is required as a condition of employment. The individual's refusal to cooperate with such a request and to provide a specimen may result in termination where there is reason to believe that the individual has violated this policy and the individual's refusal to cooperate has prevented a medical determination of his or her condition. Any violation of this policy may result in immediate termination and/or removal from the AECOM TISHMAN CONSTRUCTION project.

AECOM TISHMAN CONSTRUCTION employees that take the initiative to advise their supervisor or Human Resources in advance they have a medical problem with regard to alcohol or drug use and demonstrate a commitment to take the necessary remedial action may be eligible for a medical leave of absence for such purpose and may not be subjected to disciplinary action.

As part of the disciplinary process, and at its sole discretion, AECOM TISHMAN CONSTRUCTION may require our employees who violate this policy to successfully complete a drug abuse assistance or rehabilitation program as a condition of continued employment.

Persons whose positions with AECOM TISHMAN CONSTRUCTION require driving as part of their work may be removed from such positions and/or subject to termination if found to have been driving under the influence of alcohol or controlled substances whether on or off-duty.

For additional information, please refer to AECOM's [Substance Abuse Prevention Policy](#).

- **Drug conviction**

An individual who has been convicted of a felony under a criminal drug statute for a violation occurring on company property or during the individual's working hours must notify AECOM TISHMAN CONSTRUCTION Human Resources no later than five (5) calendar days after the felony conviction becomes final under the law.

4.3 **Harassment-Free Workplace**

AECOM TISHMAN CONSTRUCTION is committed to providing and maintaining a productive work environment that is characterized by mutual respect and is free from any type of harassment or discrimination, including but not limited to that of a sexual nature or otherwise intimidating behavior, threat or assault.

AECOM TISHMAN CONSTRUCTION maintains a strict policy prohibiting harassment in the workplace, including sexual harassment, by any individual, and by third parties such as customers, clients, vendors or visitors. Harassment may include unwelcome or unsolicited speech or conduct based upon age, ancestry, color, creed, or religion, HIV status, marital status, medical condition, national origin, citizenship status, genetic information, physical or mental disability, race, sex, sexual orientation, veteran status, family care leave status or any other protected status under federal, state or local law.

Prohibited harassment includes the following examples:

1. Verbal conduct, such as threats, epithets, derogatory comments or slurs.
2. Visual conduct, such as derogatory posters, photographs, cartoons, drawings or gestures.

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3. Written communications, statements and images which may be offensive to individuals in a particular protected group, such as gender, religion or race.
4. Physical conduct such as assault, unwanted touching or blocking normal movement.

Further, any form of retaliation against an individual who has made, or cooperated with investigation of, a harassment or discrimination complaint is unlawful and will not be tolerated.

- **Responsibility**

All AECOM TISHMAN CONSTRUCTION employees and subcontractors share the responsibility of understanding and preventing harassment and discrimination. Individuals who believe they have been subjected to discrimination, harassment or retaliation must immediately inform the appropriate company representative identified in the Complaint Procedures section below.

Management will take action, as appropriate, to address any violations of the law or company policies, regardless of the manner in which AECOM TISHMAN CONSTRUCTION becomes aware of the unlawful or improper conduct.

- **Sexual harassment defined**

Federal law defines sexual harassment as unwanted sexual advances, requests for sexual favors, or visual, verbal, or physical conduct of a sexual nature including when:

1. Submission to such conduct is made a term or condition of employment.
2. Submission or rejection of such conduct is used as a basis for employment decisions affecting the individual.
3. Such conduct has the purpose or effect of unreasonably interfering with an employee's work performance or creating an intimidating, hostile, or offensive work environment.

This policy prohibits not only conduct and language that constitute unlawful sexual harassment as defined by the law, but also all inappropriate behavior with sexual connotations. The following is a partial list of conduct that is prohibited by AECOM TISHMAN CONSTRUCTION's policy against sexual harassment:

1. Unwanted sexual advances.
2. Offering employment benefits in exchange for sexual favors.
3. Making or threatening reprisals after a negative response to sexual advances.
4. Visual conduct such as leering, sexual gestures and displaying sexually suggestive objects, pictures, cartoons, calendars or posters.
5. Verbal conduct such as making or using derogatory comments, epithets, slurs, sexually explicit jokes or comments about an employee's body or dress in a sexual context or with sexual overtones.
6. Verbal sexual advances or propositions.
7. Verbal abuse of a sexual nature, graphic and/or excessive verbal commentaries about an individual's sexual experiences or activities.
8. Written communication of a sexual nature such as suggestive or obscene letters, e-mails, and notes.

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9. Accessing sexually explicit, pornographic and/or socially offensive websites, chat rooms or other material on the internet or via AECOM TISHMAN CONSTRUCTION company resources.
10. Physical conduct such as unwanted or suggestive touching, assault or impeding or blocking movement.

- **Complaint Procedures**

AECOM TISHMAN CONSTRUCTION strongly encourages the prompt reporting of all incidents of discrimination or harassment. If you believe you are being harassed or have witnessed harassment, or have been discriminated against in any way, you should inform your supervisor, Human Resources and/or in-house Counsel immediately to report any violation of company policy or incident of discrimination, including harassment, sexual harassment or non-compliance with any aspect of AECOM TISHMAN CONSTRUCTION's Equal Employment Opportunity policy. AECOM TISHMAN CONSTRUCTION employees may also use the Hotline 1-888-299-9602 or 1-770- 613-6332 (outside of the US) to report incidents of discriminatory, harassing or retaliatory conduct in the workplace.

Any manager/supervisor who knows of, or thinks he or she knows of, or witnesses a potential violation of this policy, has an added responsibility to act and must notify Human Resources immediately. To the extent possible, all notifications will be maintained with strict confidentiality.

AECOM TISHMAN CONSTRUCTION will perform, or cause to be performed, a prompt, thorough, and objective investigation of all facts necessary to resolve a complaint.

AECOM TISHMAN CONSTRUCTION prohibits all forms of retaliation against any individual filing a complaint under this policy or for assisting in the investigation of a complaint. AECOM TISHMAN CONSTRUCTION will take appropriate disciplinary action, up to and including termination of employment, and/or removal from our projects if: (a) an individual is determined to have violated this policy; (b) an individual files a false complaint or provides false information regarding a complaint; or (c) an individual fails to fully cooperate in the investigation of a complaint of alleged discrimination, harassment, retaliation or other non-compliant behavior.

4.4 Workplace Violence

AECOM TISHMAN CONSTRUCTION is dedicated to providing a workplace that is free from acts or threats of violence. In keeping with this commitment, a zero tolerance policy has been established that prohibits any individual from threatening or committing an act of violence against a coworker, visitor or any other person who is on the premises, or has contact with the individual in the course of his or her duties. This policy applies in the workplace, at client/project sites, while on duty, while on company related business or while operating any vehicle or equipment owned or leased by the company.

Individuals are required to comply with this policy and immediately report any incident involving a threat or act of violence to their supervisor or local manager or if preferred, to Human Resources or in-house Counsel.

4.5 Subcontractors

All subcontractors on all AECOM TISHMAN CONSTRUCTION projects shall meet and/or exceed the policies found in this program.

5. ATTACHMENTS

None Applicable.

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SECTION 2 - CONFINED SPACES

1. TERMS AND DEFINITIONS

1.1 Alternate Entry Confined Space. As permitted by the applicable jurisdiction, a confined space, initially classified as a permit-required confined space that only has an atmospheric hazard that can be completely controlled through ventilation.

1.2 Asphyxiant. An airborne substance that can cause suffocation. Simple asphyxiates (e.g., carbon dioxide, nitrogen, argon, etc.) physically displace oxygen from the atmosphere; chemical asphyxiates (e.g., carbon monoxide, hydrogen cyanide, etc.) prevent the body from utilizing oxygen in the atmosphere.

1.3 Attendant. An individual who is stationed outside of a permit-required confined space to monitor authorized entrants and to initiate emergency response if necessary. As per jurisdictional requirements, an attendant may be required to be in the vicinity of a confined space irrespective of a permit requirement.

1.4 Authorized Entrant. Individual, authorized by the employer, who enters into the confined space to perform the task(s) defined in the entry permit and mitigation/control procedures.

1.5 Blanking or Blinding. The absolute closure of a pipe, line, or duct by the fastening of a solid plate (such as a spectacle blind or a skillet blind) that completely covers the bore and that is capable of withstanding the maximum pressure of the pipe, line, or duct with no leakage beyond the plate.

1.6 Competent Person. The designated individual who evaluates the hazards in the space and confirms the controls and procedures outlined in the plans and permits.

1.7 Confined Space. A space that:

- ☐ Is large enough and so configured that an employee can physically enter.
- ☐ Has limited or restricted means for entry or exit.
- ☐ Is not designed for continuous human occupancy.

1.8 Subcontractor Controlling Contractor. The employer that has overall responsibility for construction at the worksite. The subcontractor controlling contractor may have been contractually issued the responsibility by the property owner or manager, or the subcontractor controlling contractor may be the property owner or manager.

1.9 Double Block and Bleed. The closure of a line, duct, or pipe by closing and locking or tagging two in-line valves and by opening and locking or tagging a drain or vent valve in the line between the two closed valves.

1.10 Emergency. Any failure of hazard control or monitoring equipment, or other event(s) inside or outside a confined space that could endanger entrants within the space.

1.11 Engulfment. The surrounding and effective capture of a person by a fluid (i.e., liquid or finely-divided particulate) substance that can be aspirated and cause death by filling or plugging the respiratory system or that can exert enough force on the body to cause death by strangulation, constriction or crushing.

1.12 Entry. The action by which a person passes through an opening into a confined space. Entry includes ensuing work activities in that space and is considered to have occurred as soon as any part of the body breaks the plane of an opening of a confined space.

1.13 Entry Employer. Any employer who decides that an employee it directs will enter a permit space.

1.14 Entry Permit. An employer authorized written or printed document containing the required information as specified in this procedure that controls entry into a permit-required confined space.

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1.15 Entry Supervisor. An employee responsible for determining if acceptable entry conditions are present, for authorizing entry into a permit-required confined space, for overseeing entry operations, and for terminating entry. After the initial entry authorization, the duties of an entry supervisor may be passed from one supervisor to another supervisor during the course of an entry operation. Entry supervisors may serve as attendants, competent persons, or as authorized entrants, so long as they are properly trained and equipped for each role the entry supervisor fills.

1.16 Flammable Atmosphere. Any atmosphere that contains a concentration of flammable or combustible material in excess of the percentage of the lower explosive limit (LEL) or lower flammable limit (LFL) and below the percentage of the upper explosive limit (UEL) or upper flammable limit (UFL) as specified by the applicable jurisdiction.

1.17 Hazardous Atmosphere. One or more of the following atmospheres that may expose employees to the risk of death, incapacitation, impairment of ability to self-rescue, injury, or acute illness:

- ☐ Oxygen concentrations less than the minimum or in excess of the maximum established by the applicable jurisdiction.
- ☐ Flammable gas, vapor, or mist in excess of the percentage of the lower explosive limit (LEL) or lower flammable limit (LFL) as specified by the applicable jurisdiction.
- ☐ Toxic environments.

1.18 Host employer. The employer that owns or manages the property where the construction work is taking place. If the owner of the property on which the construction activity occurs has contracted with an entity for the general management of that property and has transferred to that entity the information specified in OSHA 29 CFR 1926.1203(h)(1), OSHA will treat the contracted management entity as the host employer for as long as that entity manages the property.

1.19 Hot Work. Any task that may produce a spark or source of ignition (e.g. riveting, welding, cutting, heating, etc.) and requires the employer's written authorization (hot work permit) to perform the operations.

1.20 Immediately Dangerous to Life or Health (IDLH). Any condition that:

- ☐ Poses an immediate or delayed threat to life; or
- ☐ Could cause irreversible adverse health effects; or
- ☐ Potentially interfere with an individual's ability to escape unaided from a confined space.

1.21 Inerting. The displacement of the atmosphere in a permit space by a noncombustible gas (such as nitrogen) to such an extent that the resulting atmosphere is noncombustible. This procedure produces an IDLH oxygen-deficient atmosphere.

1.22 Isolation. The process by which a permit space is removed from service and completely protected against the release of energy and material into the space by such means as: blanking or blinding; misaligning or removing sections of lines, pipes, or ducts; a double block and bleed system; lockout or tagout of all sources of energy; or blocking or disconnecting all mechanical linkages.

1.23 Limited or Restricted Means of Entry or Exit. A condition that has the potential to impede an employee's movement into or out of a confined space such as trip hazards, poor illumination, slippery floors, steep inclining floors and ladders.

1.24 Line Breaking. The intentional opening of a pipe, line, or duct that is or has been carrying flammable, corrosive, or toxic material, an inert gas, or any fluid at a volume, pressure, or temperature capable of causing injury.

1.25 Lockout Tagout. The use of a locking device to confirm that an energy-isolating device and the equipment or system it controls cannot be operated or energized until the lockout device is removed. A disconnect switch,

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circuit breaker, valve or other energy-isolating mechanism is put in the safe or off position, a lock applied and a written warning is attached to it. Tagout is always to be used in conjunction with lockout.

1.26 Lower Explosive Limit (LEL). Also referred to as Lower Flammable Limit (LFL). The lowest concentration of a flammable gas / vapor in air which will ignite.

1.27 Non-Permit Required Confined Spaces (NPRCS). As permitted by the applicable jurisdiction, spaces that do not contain existing or potential hazards, nor will the work performed or natural environment therein create a hazardous condition. Alternate terms may be used to refer to these spaces, depending upon the jurisdiction (e.g. Restricted Space, Confined Space – Low Hazard Atmosphere, etc.). These spaces do not require a confined space entry permit.

1.28 Oxygen-deficient. An atmosphere with an oxygen concentration less than the minimum established by the applicable jurisdiction that should not be entered without wearing an approved Self-Contained Breathing Apparatus (SCBA), or approved supplied air, full face respirator.

1.29 Oxygen-enriched. An atmosphere with an oxygen concentration in excess of the maximum established by the applicable jurisdiction. Oxygen enriched atmospheres will cause flammable materials to burn violently when ignited.

1.30 Permit-Required Confined Space (PRCS). Also called Permit Space. A confined space that exhibits one or more of the following properties:

- ☐ Contains or has a potential to contain a hazardous atmosphere.
- ☐ Contains a material that has the potential for engulfing an entrant.
- ☐ Has an internal configuration that could potentially cause an entrant could be trapped or asphyxiated by inwardly converging walls or by a floor that slopes downward and tapers to a smaller cross-section.
- ☐ Contains any other recognized safety or health hazard.

1.31 Physical Hazard. A nonchemical hazard that may cause cuts, abrasions, suffocation, crushing, trauma, hearing loss, burns, or radiant energy effects (e.g., welding).

1.32 Prohibited Condition. Any condition in a PRCS that is not allowed by the permit during the period in which entry is authorized. A hazardous atmosphere is a prohibited condition unless it is demonstrated that personal protective equipment (PPE) will provide effective protection for each entrant in the permit space and provides the appropriate PPE to each entrant.

1.33 Rescue Service. The individual and/or organization that is properly trained and equipped to carry out a confined space rescue operation.

1.34 Retrieval System. Includes equipment such as retrieval lines, harnesses, wristlets (if appropriate), lifting devices and anchors used for non-entry rescue of persons from permit spaces. Note that a retrieval line differs from a lifeline, which is a type of fall-arrest system.

1.35 Self-Rescue. The act of escaping unaided from a hazardous atmosphere or IDLH situation in a permit space.

1.36 Upper Explosive Limit (UEL). The highest concentration of a flammable gas/ vapor in air that will ignite.

2. GENERAL REQUIREMENTS

2.1 Before it begins work at a worksite, each employer must ensure that a competent person identifies all confined spaces in which one or more of the employees it directs may work, and identifies each space that is a permit space, through consideration and evaluation of the elements of that space, including testing as necessary. All confined spaces shall be considered confined space unless deemed otherwise.

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2.2 If the workplace contains permit spaces, the employer who identifies, or receives notice of, a permit space shall:

- a) Inform exposed individuals by posting danger signs or by any other equally effective means, of the existence and location of, and the danger posed by, each permit spaces; and a sign reading "DANGER – PERMIT-REQUIRED CONFINED SPACE, DO NOT ENTER" or using other similar language would satisfy the requirement for a sign.
- b) In a timely manner inform either verbally or in writing, individuals' authorized representatives and the controlling contractor of the existence and location of, and danger posed by, each permit space.

2.3 Each employer who identifies, or receives notice of, a permit space and has not authorized employees it directs to work in that space must take effective measures to prevent employees from entering that permit space.

2.4 Permit Space Entry Communication and Coordination:

- a) Before entry operations begin, the host employer must provide the following information, if it has it, to AECOM TISHMAN CONSTRUCTION:
 - i. The location of each known permit space;
 - ii. The hazards or potential hazards in each space or the reason it is a permit space; and
 - iii. Any precautions that the host employer or any previous controlling contractor or entry employer implemented for the protection of employees in the permit space.
- b) Before entry operations begin, the controlling contractor must:
 - i. Obtain the host employer's information about the permit space hazards and previous entry operations; and
 - ii. Provide the following information to each entity entering a permit space and any other entity at the worksite whose activities could foreseeably result in a hazard in the permit space:
 - The information received from the host employer;
 - The precautions that the host employer, controlling contractor, or other entry employers implemented for the protection of employees in the permit spaces.
- c) Before entry operations begin, each entry employer must:
 - i. Obtain all of the controlling contractor's information regarding permit space hazards and entry operations; and
 - ii. Inform the controlling contractor of the permit space program that the entry employer will follow including any hazards likely to be confronted or created in each permit space.
- d) The controlling contractor and entry employer(s) must coordinate entry operations when:
 - i. More than one entity performs permit space entry at the same time; or
 - ii. Permit space entry is performed at the same time that any activities that could foreseeably result in a hazard in the permit space are performed.

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2.5 If non-permit spaces are modified or experience any change that causes an increased hazard to entrants, the Entry Employer shall have a competent person reevaluate that space and, if necessary, reclassify it as a permit required confined space.

2.6 If any individual is to enter a permit space prior to the entry their employer shall:

- (a) Inform the employee that the workplace contains permit spaces and that entry must comply with OSHA 29 CFR 1926.1203.
- b) Inform the employee of the elements, including known hazards and experiences from any previous entries into the space, that classify the space as a permit space.
- c) Inform the employee of any controls or procedures implemented to protect employees near the work area.
- d) Coordinate entry operations among client, contractor, and subcontractor personnel when necessary.
- e) Coordinate for rescue services to be available.

2.7 The employer shall debrief the employee at the conclusion of entry operations regarding the permit space entry program and any hazards or problems encountered during the entry operations.

3. IMPLEMENTATION

3.1 Non-Permit Confined Spaces. Entry into non-permit confined spaces will be subject to the following controls:

- a) AECOM TISHMAN CONSTRUCTION shall be informed in advance of the planned entry.
- b) The entry shall be coordinated with any work activities near the non-permit space so that hazardous conditions will not be created in or around the space.
- c) Individuals shall not be allowed to work alone in any confined space.
- d) Entrants will immediately withdraw upon recognition of any hazardous condition.
- e) AECOM TISHMAN CONSTRUCTION will be advised of any unanticipated incidents related to the non-permit space entry.
- f) Non-permit confined spaces shall be continuously monitored for changes that could cause reclassification.

3.2 Permit Required Confined Spaces. Entry into permit required confined spaces will be subject to the following controls:

- a) Unauthorized entries shall be prohibited by the Entry Employer.

(**Note:** The Superintendent and Project Safety Manager shall be notified of any planned entry into a permit space.)

- b) The Entry Employer shall identify and evaluate hazards prior to entry. Hazards to be addressed include, but may not be limited to:

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- i. Chemical exposure (via inhalation, ingestion, or dermal absorption) from the contents or residues of previous contents of the space, from chemicals introduced into the space as part of the entry operation, and from chemicals used near the space.
 - ii. Oxygen deficiency or enrichment.
 - iii. Discharge of steam, high pressure air, water, or oil into the confined space, or against personnel working outside.
 - iv. Structural failure of the space walls, roof, roof support members, swing line cables, or other structural members.
 - v. Tools, debris, or other objects dropping from overhead.
 - vi. Falls through or from the roof or from scaffolds, stairs, or ladders.
 - vii. Tripping over hoses, pipes, tools, or equipment.
 - viii. Slipping on wet, oily surfaces, or colliding with objects in inadequately lighted interiors.
 - ix. Insufficient or faulty personal protective equipment.
 - x. Insufficient or faulty operations, equipment, and tools.
 - xi. Noise in excess of acceptable levels.
 - xii. Temperature extremes that may require additional protection or shorter work periods.
- c) Permit-Required Confined Space Program.
- i. The Entry Employer shall develop and maintain a program to address work activities associated with permit required confined spaces. This program shall meet the requirements outlined in OSHA 29 CFR 1926.1204. In addition, ensure local or state program requirements are included where applicable.
 - ii. Acceptable entry conditions shall be specified (both chemical and physical conditions shall be addressed, and conditions which could arise as a result of operations performed outside the space shall be considered).
 - iii. Procedures to fully isolate the space shall be specified (this may not be feasible in sewers; see Attachment 2).

3.3 Alternate Entry Procedure. A simpler, alternate procedure may be followed for permit required space entry if all of the following conditions are met:

- a) The only hazard posed by the permit space is from an actual or potentially hazardous atmosphere.
- b) Continuous forced air ventilation alone is sufficient to maintain the space safe for entry.
- c) Monitoring and inspection data supporting the above two conditions is collected and documented. Entries conducted to obtain this data must be conducted according to Section 3.2 above.
- d) Supporting data must be made available to each employee who enters the permit space under this alternate procedure.

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- e) The alternate entry procedure shall conform to the following:
- i. Any conditions making it unsafe to remove an entrance cover shall be eliminated before the cover is removed.
 - ii. When entrance covers are removed, the opening shall be promptly guarded by a railing, temporary cover, or other temporary barrier that will prevent an accidental fall through the opening and that will protect each employee working in the space from foreign objects entering the space.
 - iii. Before an employee enters the space, the internal atmosphere shall be tested with a calibrated direct reading instrument for the following conditions, in the order given:
 - Oxygen content.
 - Flammable gases and vapors.
 - Potential toxic air contaminants.
 - iv. There may be no hazardous atmosphere within the space whenever any employee is inside the space.
 - v. Continuous forced air ventilation shall be used, as follows:
 - An employee may not enter the space until the forced air ventilation has eliminated any hazardous atmosphere.
 - The forced air ventilation shall be directed so as to ventilate the immediate areas where an employee is or will be present within the space and shall continue until all employees have left the space.
 - The air supply for the forced air ventilation shall be from a clean source and may not increase the hazards in the space.
 - vi. The atmosphere within the space shall be continuously monitored to ensure that the continuous forced air ventilation is preventing the accumulation of a hazardous atmosphere.
 - vii. If a hazardous atmosphere is detected during entry:
 - Each employee shall leave the space immediately.
 - The space shall be evaluated to determine how the hazardous atmosphere developed.
 - Measures shall be implemented to protect employees from the hazardous atmosphere before any subsequent entry takes place.
 - viii. The employer shall verify that the space is safe for entry and that the measures have been taken through a written certification that contains the date, the location of the space, and the signature of the person providing the certification. The certification shall be made before entry and shall be made available to each employee entering the space.

4. ENTRY PERMIT SYSTEM

4.1 Before authorization to enter is granted, the completion of hazard control shall be documented on an entry permit.

4.2 The entry supervisor identified on the permit shall authorize the entry by signing the completed permit.

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4.3 The signed permit shall be posted or otherwise made available to all authorized entrants so that they may confirm that all pre-entry preparations are in place.

4.4 The duration of the permit may not exceed the time required to complete the assigned purpose of the entry.

4.5 The entry supervisor will terminate the entry and cancel the permit when either of the following occurs:

- a) The operations covered by the permit have been completed.
- b) Any condition not allowed under the entry permit occurs in or near the permit space.

4.6 Review the permit space program using the canceled entry permits. The cancelled permits shall be retained for at least one (1) year. Any problems which occur during an entry shall be noted on the permit.

4.7 Refer to Attachment 2 of this section for Sewer System Entry procedure.

5. ENTRY PERMIT

5.1 The entry permit shall contain the following information:

- a) The permit space to be entered.
- b) The purpose of the entry.
- c) The date and the authorized duration of the entry permit.
- d) The authorized entrants within the permit space, by name or by such other means (for example, through the use of rosters or tracking systems), as will enable the attendant to determine quickly and accurately, for the duration of the permit, which authorized entrants are inside the permit space.

Note: This requirement may be met by inserting a reference on the entry permit as to the means used, such as a roster or tracking system, to keep track of the authorized entrants within the permit space.

- e) Means of detecting an increase in atmospheric hazard levels in the event the ventilation system stops working.
- f) Each person, by name, currently serving as attendants.
- g) The individual, by name, currently serving as entry supervisor, with a space for the signature or initials of each entry supervisor who authorizes entry.
- h) The hazards of the permit space to be entered.
- i) The measures used to isolate the permit space and to eliminate or control permit space hazards before entry.

Note: Those measures can include, but are not limited to, the lockout or tagging of equipment and procedures for purging, inerting, ventilating, and flushing permit spaces.

- j) The acceptable entry conditions.
- k) The results of tests and monitoring accompanied by the names or initials of the testers, and by an indication of when the tests were performed.

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- l) The rescue and emergency services that can be summoned and the means (such as the equipment to use and the numbers to call) for summoning those services.
- m) The communication procedures used by authorized entrants and attendants to maintain contact during the entry.
- n) Equipment, such as personal protective equipment, testing equipment, communications equipment, alarm systems, and rescue equipment, to be provided for compliance with this section.
- o) Any other information whose inclusion is necessary, given the circumstances of the particular confined space, in order to ensure employee safety.
- p) Any additional permits, such as for hot work, that have been issued to authorize work in the permit space.

5.2 All permits shall be made readily available to AECOM TISHMAN CONSTRUCTION.

6. TRAINING

6.1 The Competent Person for the subcontractor entering any confined space shall assure that all employees assigned to tasks under this procedure have been trained, and have the understanding, knowledge, and skills necessary for the safe performance of their duties.

6.2 All training conducted will be documented on appropriate training forms, and copies will be submitted to AECOM TISHMAN CONSTRUCTION when requested.

6.3 Training shall be provided in a language and a vocabulary that the employee understands. Training must be provided to each employee on the following occasions:

- a) Before the employee is first assigned duties under this procedure.
- b) Before there is a change in assigned duties.
- c) Whenever there is a change in permit space operations that presents a hazard about which an individual has not previously been trained.
- d) Whenever the employer has reason to believe either that there are deviations from the permit space entry procedures or that there are inadequacies in the employee's knowledge or use of these procedures.

6.4 The training must establish employee proficiency in the duties required and must introduce new or revised procedures, as necessary.

6.5 Training required by this section shall be documented. The documentation shall contain each individual's name, the name of the trainers, and the dates of training.

6.6 Subcontractors are solely responsible for their employee training.

7. ASSIGNMENT OF DUTIES

7.1 Authorized Entrants. The entry employer shall ensure that all authorized entrants:

- a) Know the hazards that may be faced during entry, including information on the mode, signs, or symptoms, and consequences of the exposure.
- b) Properly use equipment as required.

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- c) Communicate with the attendant as necessary to enable the attendant to monitor entrant status and to enable the attendant to alert entrants of the need to evacuate the space as required by section 7.2 of this procedure.
- d) Alert the attendant whenever:
 - i. The entrant recognizes any warning sign or symptom of exposure to a dangerous situation.
 - ii. The entrant detects a prohibited condition.
- e) Exit from the permit space as quickly as possible whenever:
 - i. An order to evacuate is given by the attendant or the entry supervisor.
 - ii. There is any warning sign or symptom of exposure to a dangerous situation.
 - iii. The entrant detects a prohibited condition.
 - iv. An evacuation alarm is activated.

7.2 Attendant. The entry employer shall ensure that each attendant:

- a) Knows the hazards that may be faced during entry, including information on the mode, signs or symptoms, and consequences of the exposure.
- b) Is aware of possible behavioral effects of hazard exposure in authorized entrants.
- c) Continuously maintains an accurate count of authorized entrants in the permit space.
- d) Remains outside the permit space during entry operations until relieved by another attendant.

***Note:** When the employer's permit entry program allows attendant entry for rescue, attendants may enter a permit space to attempt a rescue if they have been trained and equipped for rescue operations as required by this procedure and if they have been relieved by another attendant.*

- e) Communicates with authorized entrants as necessary to monitor entrant status and to alert entrants of the need to evacuate the space under Section 7.1 of this procedure.
- f) Monitors activities inside and outside the space to determine if it is safe for entrants to remain in the space, and orders the authorized entrants to evacuate the permit space immediately under any of the following conditions:
 - i. If the attendant detects a prohibited condition.
 - ii. If the attendant detects the behavioral effects of hazard exposure in an authorized entrant.
 - iii. If the attendant detects a situation outside the space that could endanger the authorized entrants.
 - iv. If the attendant cannot effectively and safely perform all the duties required under this section.
- g) Summon rescue and other emergency services as soon as the attendant determines that authorized entrants may need assistance to escape from permit space hazards.

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- h) Takes the following actions when unauthorized persons approach or enter a permit space while entry is underway:
 - i. Warn the unauthorized persons that they must stay away from the permit space.
 - ii. Advise the unauthorized persons that they must exit immediately if they have entered the permit space.
 - iii. Inform the authorized entrants and the entry supervisor if unauthorized persons have entered the permit space.
 - iv. Performs non-entry rescues as specified by the employer's rescue procedure.
 - v. Performs no duties that might interfere with the attendant's primary duty to monitor and protect the authorized entrants.
 - vi. Is trained in first aid and CPR.

7.3 Entry supervisor. The entry employer shall ensure that each entry supervisor:

- a) Knows the hazards that may be faced during entry, including information on the mode, signs or symptoms, and consequences of the exposure.
- b) Verifies, by checking that the appropriate entries have been made on the permit, that all tests specified by the permit have been conducted and that all procedures and equipment specified by the permit are in place before endorsing the permit and allowing entry to begin.
- c) Terminates the entry and cancels the permit as required by Section 4.5 of this procedure.
- d) Verifies that rescue services are available and that the means for summoning them are operable.
- e) Removes unauthorized individuals who enter or who attempt to enter the permit space during entry operations.
- f) Determines, whenever responsibility for a permit space entry operation is transferred and at intervals dictated by the hazards and operations performed within the space, that entry operations remain consistent with terms of the entry permit and that acceptable entry conditions are maintained.

8. RESCUE AND EMERGENCY SERVICES

8.1 The following requirements apply to employers who have employees enter permit spaces to perform rescue services.

- a) The employer shall ensure that each member of the rescue service is provided with, and is trained to use properly, the personal protective equipment and rescue equipment necessary for making rescues from permit spaces. Also inform the rescue team of the hazards they may confront when called on to perform rescue.
- b) Each member of the rescue service shall be trained to perform the assigned rescue duties.
- c) Each member of the rescue service shall also receive the training required of authorized entrants under Section 6 of this procedure.
- d) Each member of the rescue service shall practice making permit space rescues at least once every 12 months by means of simulated rescue operations in which they remove dummies, manikins, or actual

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persons from the actual permit spaces or from representative permit spaces. Representative permit spaces shall, with respect to opening size, configuration, and accessibility, simulate the types of permit spaces from which rescue are to be performed.

- e) Each member of the rescue service shall be trained in basic first aid and in cardiopulmonary resuscitation (CPR). At least one member of the rescue service holding current certification in first aid and in CPR shall be available.

8.2 An employer who designates rescue service (outside rescuer) shall ensure that:

- a) The outside rescuer can effectively respond in a timely manner to a rescue summons.
- b) The outside rescuer is equipped, trained and capable of functioning appropriately to perform permit space rescues at the site.
- c) The outside rescuer is aware of the hazards they may confront when called on to perform rescue at the site.
- d) An outside rescuer is selected that:
 - i. Has the capability to reach the victim(s) within a time frame that is appropriate for the permit space hazard(s) identified,
 - ii. Is equipped for, and proficient in, performing the needed rescue services,
 - iii. Agrees to notify the employer immediately in the event that the rescue service becomes unavailable.
- e) The outside rescuer is provided with access to all permit spaces from which rescue may be necessary so that the outside rescuer can develop appropriate rescue plans and practice rescue operations.

8.3 To facilitate non-entry rescue, retrieval systems or methods shall be used whenever an authorized entrant enters a permit space unless the retrieval equipment would increase the overall risk of entry or would not contribute to the rescue of the entrant. Retrieval systems shall meet the following requirements:

- a) Each authorized entrant shall use a chest or full body harness with a retrieval line attached at the center of the entrant's back near shoulder level, above the entrant's head, or other point which the employer can establish will ensure that the entrant will present the smallest possible profile during removal.
- b) The other end of the retrieval line shall be attached to a mechanical device or fixed point outside the permit space in such a manner that rescue can begin as soon as the rescuer becomes aware that rescue is necessary. A mechanical device shall be available to retrieve personnel from vertical type permit spaces more than 5 feet deep.
- c) If an injured entrant is exposed to a substance for which a Safety Data Sheet (SDS) or other similar written information is required to be kept at the worksite, that SDS or written information shall be made available to the medical facility treating the exposed entrant.

9. ATTACHMENTS

9.1 Procedures for Atmospheric Testing

9.2 Sewer System Entry

9.3 Confined Space Entry Permit Example

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The Confined Space Entry Permit is used for documenting activities associated with this procedure. The completed Permits shall be maintained for a period of one (1) year from the date of cancellation, in accordance with section 4 of this policy. All permits used must meet what is found on this attachment, at a minimum.

9.4 Confined Space Decision Flowchart

9.5 Confined Space Sign In/Out Log Example

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Attachment 1: Procedures for Atmospheric Testing

PROCEDURES FOR ATMOSPHERIC TESTING

Atmospheric testing is required for two distinct purposes: evaluation of the hazards of the permit space and verification that acceptable entry conditions for entry into that space exist.

1. Evaluation testing. The atmosphere of a confined space should be analyzed using equipment of sufficient sensitivity and specificity to identify and evaluate any hazardous atmospheres that may exist or arise so that appropriate permit entry procedures can be developed and acceptable entry conditions stipulated for that space. Evaluation and interpretation of these data, and development of the entry procedure should be done by, or reviewed by, a technically qualified professional (e.g., certified industrial hygienist, marine chemist, registered safety engineer, certified safety professional, etc.) based on evaluation of all serious hazards.
2. Verification testing. The atmosphere of a permit space which may contain a hazardous atmosphere should be tested for residues of all contaminants identified by evaluation testing using permit specified equipment to determine that residual concentrations at the time of testing and entry are within the range of acceptable entry conditions. Results of testing (i.e., actual concentration, etc.) should be recorded on the permit in the space provided adjacent to the stipulated acceptable entry condition.
3. Duration of testing. Measurement of values for each atmospheric parameter should be made for at least the minimum response time of the test instrument specified by the manufacturer.
4. Testing stratified atmospheres. When monitoring for entries involving a descent into atmospheres that may be stratified, the atmospheric envelope should be tested a distance of approximately four (4) feet (1.22 m) in the direction of travel and to each side. If a sampling probe is used, the entrant's rate of progress should be slowed to accommodate the sampling speed and detector response.



Attachment 2: Sewer System Entry

SEWER SYSTEM ENTRY

Sewer entry differs in three (3) vital respects from other permit entries. First, there rarely exists any way to completely isolate the space (a section of a continuous system) to be entered; second, because isolation is not complete, the atmosphere may suddenly and unpredictably become lethally hazardous (toxic, flammable or explosive) from causes beyond the control of the entrant or employer; and third, experienced sewer workers are especially knowledgeable in entry and work in their permit spaces because of their frequent entries. Unlike other employments where permit space entry is a rare and exceptional event, sewer workers' usual work environment is a permit space.

1. Adherence to procedure. The employer should designate as entrants only employees who are thoroughly trained in the employer's sewer entry procedures and who demonstrate that they follow these entry procedures exactly as prescribed when performing sewer entries.
2. Atmospheric monitoring. Entrants should be trained in the use of, and be equipped with, atmospheric monitoring equipment which sounds an audible alarm, in addition to its visual readout, whenever one of the following conditions is encountered: oxygen concentration less than 19.5 percent; flammable gas or vapor at 10 percent or more of the lower flammable limit (LFL); or hydrogen sulfide or carbon monoxide at or above their PEL (10 PPM or 50 PPM, respectively); or, if a broad range sensor device is used, at 100 PPM as characterized by its response to toluene. Normally, the oxygen sensor/broad range sensor instrument is best suited for sewer entry. However, substance specific devices should be used whenever actual contaminants have been identified. The instrument should be carried and used by the entrant in sewer line work to monitor the atmosphere in the entrant's environment, and in advance of the entrants' direction of movement, to warn the entrant of any deterioration in atmospheric conditions. Where several entrants are working together in the same immediate location, one instrument, used by the lead entrant, is acceptable.
3. Surge flow and flooding. Sewer crews should develop and maintain liaison, to the extent possible, with the local weather bureau and fire and emergency services in their area so that sewer work may be delayed or interrupted and entrants withdrawn whenever sewer lines might be suddenly flooded by rain or fire suppression activities, or whenever flammable or other hazardous materials are released into sewers during emergencies by industrial or transportation accidents.
4. Special Equipment. Entry into large bore sewers may require the use of special equipment. Such equipment might include such items as atmosphere monitoring devices with automatic audible alarms, escape self-contained breathing apparatus (ESCBAs) with at least ten (10) minute air supply (or other NIOSH approved self-rescuer), and waterproof flashlights, and may also include boats and rafts, radios and rope stand-offs for pulling around bends and corners as needed.

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Attachment 3: Confined Space Entry Permit Example

CONFINED SPACE ENTRY PERMIT

Permit good for one work shift

Date:

Job:

Permit Prepared by:

Subcontractor:

Description of confined space:

Location of confined space:

Work to be performed:

Trained Entrant (s):

Trained Attendant (s):

Trained Supervisor _____ (must remain at scene during confined space work).

Communications procedures ☐ verbal ☐ radio ☐ other

Rescue Procedures ☐ non-entry retrieval ☐ trained stand-by rescue personnel

Site Emergency Phone #: _____ Offsite Emergency Phone #: _____

Gas monitor #:

Daily Test by:

Date monitor calibrated:

Expiration date of calibration gas:

Calibration/Bump Test Levels LEL ____ % OXY ____ % CO ____ PPM H2S ____ PPM

<u>Equipment</u>	<u>N/A</u>	<u>YES</u>	<u>NO</u>	<u>Identify all Potential Hazards</u>
Ventilation (continuous)	()	()	()	____ Traffic__ Vehicle__ Pedestrian
Hoist /Lifelines & Harness	()	()	()	____ Slips, trips and falls
Communications (continuous)	()	()	()	____ Combustible Dust
Ground Fault Circuit Interrupter	()	()	()	____ Engulfment
Fire Extinguisher (s)	()	()	()	____ Temperature extremes (hot/cold)
Personal Protective Equipment	()	()	()	____ Electrical Shock
Hot work Permit	()	()	()	____ Mechanical /Configuration
Proper Lighting	()	()	()	____ Biohazards

Special items needed: _____ Other: _____

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Control of Dangerous Equipment, Energy, Liquids, Gases and Materials

	N/A	YES	NO
Locked or tagged out	()	()	()
Support/securing materials used	()	()	()
Pipes blocked or disconnected	()	()	()

Special precautions when lockout, tagout, blocking or disconnect not possible:

Continuous Air Testing (record initial test every hour)

Safe Exposure level

Time _____ AM or PM

Oxygen _____

19.5 to 22%

LEL (flammable) _____

< 10% LEL

Hydrogen Sulfide _____

< 10 PPM

Carbon monoxide _____

< 35 PPM

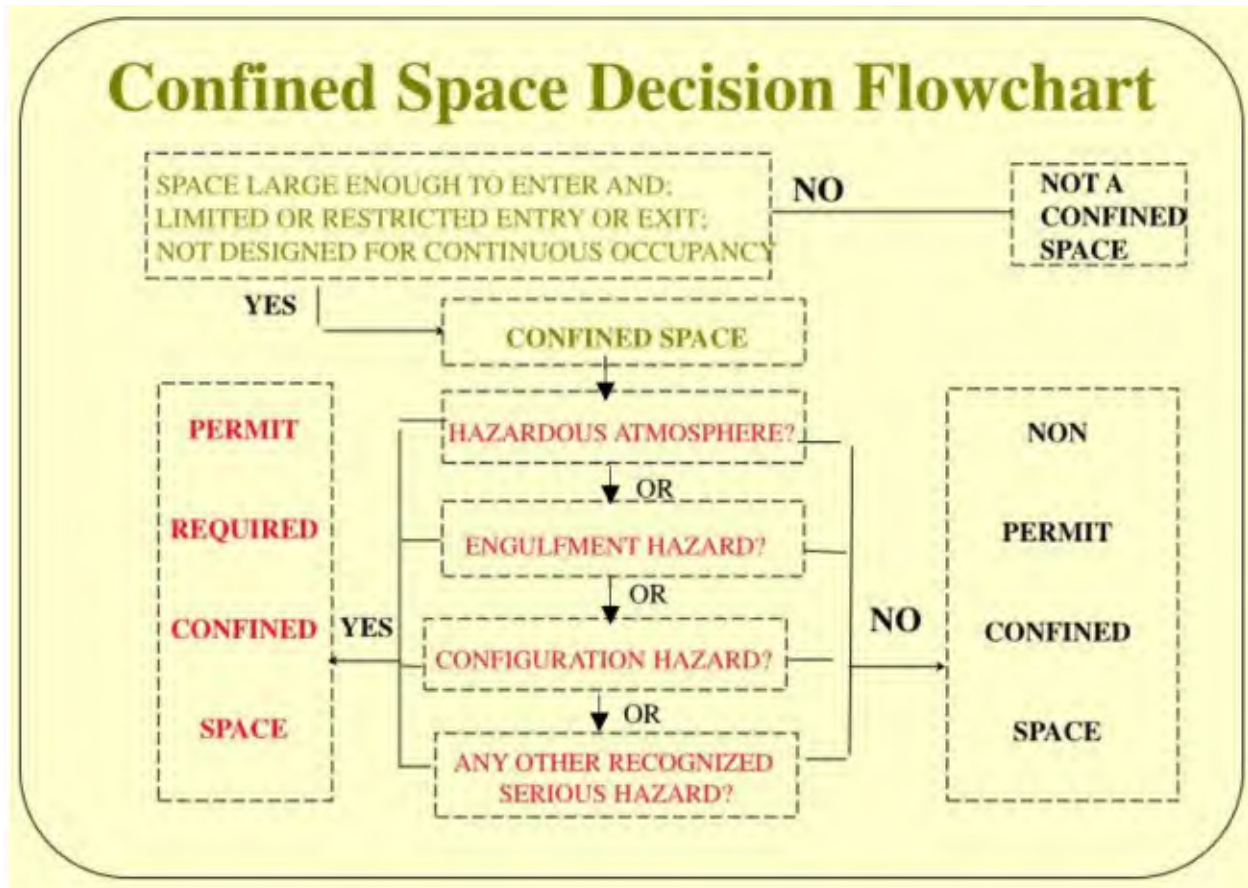
Time In _____

Time Out _____

Trained Air Tester Name: _____

Supervisor's signature/authorization: _____

Attachment 4: Confined Space Decision Flowchart



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Attachment 5: Confined Space Sign In/Out Log Example

CONFINED SPACE PERMIT SIGN-IN/SIGN-OUT SHEET FOR EMERGENCY RESPONSE

(Attach to the Confined Space Permit)

The purpose of this form is to provide a standardized method for maintaining an accurate, real time tracking of entrants in a confined space. Maintaining an accurate log will enable emergency response/rescue personnel to know precisely who and how many entrants are in a confined space at a given time.

Contractor Name			PROJECT:		
SITE LOCATION					
COMPANY	NAME	SIGNATURE	DATE	TIME IN	TIME OUT

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SECTION 3 - CRANE OPERATIONS

1. TERMS and DEFINITIONS

1.1 ASME. American Society of Mechanical Engineers.

1.2 Assembly/Disassembly Director (A/D Director). Individual who meets OSHA requirements for an A/D director, irrespective of the individual's formal job title or whether the person is management or not.

1.3 Competent Person. One who is capable of identifying existing and predictable hazards in the surroundings or working conditions that are unsanitary, hazardous, or dangerous to employees, and who has the authorization to take prompt corrective measures to eliminate hazards.

1.4 Controlling Entity. A prime contractor, general contractor, construction manager, constructor, or any other legal entity that has the overall responsibility for the construction of the project, including planning, quality, operations, safety, and completion.

1.5 Crane. Any power operated equipment designated such that can hoist, lower, and horizontally move a suspended load. This does not include a daily delivery truck boom-lift type of equipment or other equipment that offloads material from the truck to the ground as excepted by 29 CFR 1926.1400(c).

1.6 Crane Hoisted Personnel Platform. Work platform designed and approved to be suspended from a crane while lifting workers. When work must be performed that cannot be accomplished without the use of a Crane Hoisted Personnel Platform, a Crane Hoisted Personnel Platform Permit (Attachment 5) must be completed prior to hoisting any personnel.

1.7 Critical Lift. A hoisting operation in which:

- The load item, if damaged or upset, would result in a release of hazardous material into the environment.
- The load item is unique and, if damaged, would be irreplaceable or not repairable.
- The use of more than any one single piece of equipment is required to lift, stabilize, support, or be in close proximity to the load to facilitate installation.
- The maximum anticipated load is at or above 75% of the crane's rated capacity for mobile cranes and 90% for tower cranes as configured according to the applicable lifting chart, for that particular lift unless otherwise stated in this policy.
- Personnel are being lifted.
- Further site-specific criteria may be developed to supplement those listed above and may include loads which require exceptional care in handling because of size, weight, close-tolerance installation, high susceptibility to damage, or at the discretion of the Person-In-Charge (PIC).

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1.8 Critical Lift Data Sheet. An AECOM TISHMAN CONSTRUCTION document (Attachment 3) that identifies pre-lift information requirements to be completed prior to making the actual critical lift. The documented Critical Lift Data Sheet shall be completed by the subcontractor (or AECOM TISHMAN CONSTRUCTION entity) and submitted to AECOM TISHMAN CONSTRUCTION Project Team for review and retention. The Data Sheet shall also be forwarded to the Regional Safety Director and/or Area Safety Director for review and approval prior to making the lift.

1.9 Critical Item or Load. A part, component, assembly, or piece of equipment ("item") that which dropping, upset, or collision could cause/result in any of the following:

- The load item, if damaged or upset would result in a release hazardous material into the environment.
- The load item is unique and, if damaged, would be irreplaceable or not repairable.
- The use of more than any one single piece of equipment is required to lift, stabilize, support, or be in close proximity to the load to facilitate installation.

1.10 Engineered Lift. Any critical lift requiring load lift sequencing, ground condition improvement, center of gravity calculations along with engineered weights, load lift path restrictions and multiple lifting devices with a review completed by a qualified and registered physical engineer (PE).

1.11 Master Rigger. An individual who has met all of the requirements of Rigger II plus has accumulated 5 or more years of experience rigging loads and is authorized by the employer to take prompt corrective action to eliminate hazards.

1.12 Operator. An individual who is competent and qualified to operate a crane.

1.13 Person-in-Charge (PIC). The key designated supervisor or manager (other than the equipment operator) responsible for the safe handling of a critical lift operation.

1.14 Project Executive (PX)/Project Director (PD). An AECOM TISHMAN CONSTRUCTION employee assigned responsibilities for overall management operations of a single or multiple projects within a region or division.

1.15 Qualified Person. One who by possession of a recognized degree, certificate, or professional standing, or by extensive knowledge, training, and experience has successfully demonstrated his/her ability and given the authority to solve or resolve problems relating to crane processes.

1.16 Qualified Engineer/Qualified Engineering Organization. An engineer or an engineering organization that is accepted by AECOM TISHMAN CONSTRUCTION and, if necessary, the client.

1.17 Qualified Inspector. An individual whose crane inspection competence is recognized by the state in which the critical lift is to be conducted, the U.S. Department of Labor, or another agency authorized to certify inspection personnel.

1.18 Qualified Operator. An operator whose competence to operate equipment associated with a critical lift safely and effectively can be demonstrated to AECOM TISHMAN CONSTRUCTION. This demonstration may be by an operational test or by past first-hand knowledge of the operator's experience on similar lifts, or

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other means that allows AECOM TISHMAN CONSTRUCTION to assess the operator's competence by documented compliance with section 6.4 of this chapter.

1.19 Qualified Rigger. A person with rigging skill and competence demonstrated either by satisfactory and extensive past experience or by successfully completing a demonstration rigging test. See section 5 of this chapter for rigger qualifications.

1.20 Qualified and Authorized Signaler. A person properly trained and possesses the skill and competence to direct the crane operator when moving loads. This person also must have the permission of their employer and the Person - In- Charge to complete signaling tasks.

1.21 Regional Safety Director (RSD). An AECOM TISHMAN CONSTRUCTION employee assigned safety and health oversight responsibilities over a region of the company.

1.22 Rigger I. An individual who works under the supervision of a competent person and on a routine basis performs rigging work engaged in lifting loads, other than rigging for special lifts, and the erection, dismantling, jumping, or reconfiguring of cranes.

1.23 Rigger II. An individual who has accumulated 2 or more years of experience working as a Rigger I and performs rigging work engaged in lifting loads under the direction of a Master Rigger.

1.24 Spotter. A competent individual who is designated and must position themselves to watch for hazards for the operator, effectively communicate with the operator and have proper training per OSHA standards.

1.25 Tandem Lift. Any lift involving two cranes (for purposes of this document).

2. GENERAL REQUIREMENTS

2.1 Documentation.

- a) A thorough annual inspection of the hoisting machinery shall be made by a qualified person or by a government or private agency recognized by the U.S. Department of Labor. A copy of the inspection shall be given to AECOM TISHMAN CONSTRUCTION prior to the crane's use on an AECOM TISHMAN CONSTRUCTION project.
- b) Verification of valid crane operator's license and certification must be completed with copies kept on file.
- c) Crane registration must be completed where required by law.
- d) A preventative maintenance program based on the crane manufacturers or certified agent's recommendation shall be in place for all cranes and be made readily available upon request.
- e) Individual crane packets shall be kept on file for all cranes as outlined in this program.
- f) Subcontractor shall prepare and submit a written crane and hoisting equipment safety program which is to include suitable testing of critical components to AECOM TISHMAN CONSTRUCTION and a written rigging program a minimum of two weeks prior to the mobile crane and a minimum of four

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weeks prior to the tower crane or other hoisting equipment arriving on site. The program shall also include a schedule of daily, weekly, monthly, and other periodic inspections to be performed in accordance with the manufacturer's specifications and ASME B30.5 for mobile cranes and ASME B30.3 for tower cranes. Such equipment shall not be permitted on site unless it is approved by the CM/GC. Failure to submit written programs within the time frame specified may result in a penalty of \$5,000.

2.2 Equipment.

- a) A minimum ten (10) pound ABC rated fire extinguisher must be accessible at the operator's station in the cab of all cranes. The size may be larger per manufacturer's specifications.
- b) Rated load capacities and recommended operating speeds, special hazard warning, or instructions shall be conspicuously posted and visible while the operator is at the control station.
- c) Cranes and derricks with a variable angle boom shall be equipped with a functioning boom angle indicator readily visible to the operator.
- d) An operable boom stop or jib stop is required on any crane where the boom could fall over backwards.
- e) Safe access to the car and cab (by steps and handholds) must be provided as well as proper lockout/tagout procedures followed per the manufacturer.
- f) All platforms and walkways shall have anti-skid surfaces per the manufacturer.
- g) All hooks designed for lifting shall have a working safety latch to prevent the load from becoming unhooked. Grab hooks and/or shake out hooks do not need safety latches unless the manufacturer specifies they do.
- h) All cranes must be equipped with an operable siren or horn warning device controlled by the operator.
- i) All cranes shall be equipped with a positive anti-two-blocking device except for pile driving equipment or when using a wrecking ball, dragline, or clamshell.
- j) All moving parts, such as belts, gears, shafts, pulleys, sprockets, spindles, drums, fly wheels, chains, or other reciprocating, rotating, or other moving parts or equipment, shall be guarded if such parts are exposed to contact by employees (except maintenance or repair employees) or otherwise create a hazard.
- k) All windows in cabs shall be made of safety glass which introduces no visible distortions, including cracks in the glass.
- l) No structural or mechanical crane modifications shall be made without the manufacturer's written approval or per a qualified person as defined by ANSI/ASME B30.5. A copy of the approval shall be kept on file in the AECOM TISHMAN CONSTRUCTION field office, if applicable. A third party inspection is required before returning to service.
- m) Running or hoist rope will be replaced when any of the following conditions are detected:

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- When there are six or more randomly distributed broken wires in any one lay or when there are three or more broken wires in any one strand of any one lay.
 - When one third of an individual outside wire's diameter is worn.
 - When there is kinking, crushing, bird caging, or any other damage or distortion of the wire rope structure.
 - When there is evidence of heat damage.
 - When there are two or more broken wires in any one lay in those sections beyond the end connections or there is one or more broken wire at the end connection.
 - When the rope has been shock loaded.
 - When contact is made with an electrical source.
- n) Non-rotating wire rope is designed to resist the tendency to spin or rotate under a load. Open swivels should not be used unless approved by the rope manufacturer. Non-rotating rope shall not be used for boom hoist reeving except for luffing, boom, or mast attachment systems. Non-rotating rope shall be replaced when any of the following conditions are detected:
- Internal core slippage is detected.
 - Two broken wires in six rope diameters.
 - Four broken wires in thirty diameters.
 - The pitch diameter is less than 18 times the nominal diameter of the rope used.
- o) Wire rope passing through wedge sockets shall not be secured to the live end of the wire rope.

2.3 Operations.

- a) The manufacturer's specifications and limitations applicable to the operation of any and all cranes shall be followed at all times.
- b) A competent person shall inspect all machinery and equipment prior to each use to ensure it is in safe operating condition. All deficiencies shall be repaired prior to use unless otherwise noted in this policy.
- c) A qualified person shall also inspect all machinery and equipment during its use to ensure it is operating in a safe condition.
- d) The operator, in accordance with the manufacturer's operation manual and OSHA requirements, must make daily inspections (visual checks). The date of the inspection and the signature of the operator must be entered into the logbook. The logbook is to be kept in the cab of the crane. Any deficiencies noted must have been reviewed and/or corrected by the appropriate maintenance personnel prior to

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use. The logbooks must be kept with the crane as long as it is on an AECOM TISHMAN CONSTRUCTION project.

- e) Before starting work each shift, the operator will operate each control to ensure proper functioning.
- f) Good housekeeping must be maintained around all cranes. All loose mud, oil, or grease spots must be wiped up to prevent falls. The crane cab shall be kept clean and clear of loose materials and debris.
- g) Personnel shall not mount cranes unless the unit is stopped or the operator indicates that it is safe.
- h) No overhead lifting shall be completed with grab or shake out hooks.
- i) Accessible areas within the counterweight swing radius of the crane shall be barricaded in such a manner as to prevent an employee from being struck or crushed by the crane.
- j) No cranes shall operate near electrical lines except where electrical distribution and transmission lines have been de-energized and visibly grounded, proper safety devices have been put in place, and/or minimum clearances, as outlined by OSHA and/or other governmental authorities, are being followed.
- k) Side loading of booms shall be limited to freely suspended loads. Booms shall not be used for dragging loads sideways unless the boom is specifically designed and constructed to withstand such side loading.
- l) Only qualified and authorized persons shall be permitted to give crane hand signals. All of these persons shall wear orange gloves or have something similar to distinguish them as an authorized signaler. All crane operations shall use qualified and authorized persons as signalers.
- m) Hand signals shall be those prescribed by the applicable ANSI or ASME standard for the crane in use and shall be posted on site by the subcontractor. All verbal communication shall be in English.
- n) Operators shall be physically and mentally capable of operating cranes at all times.
- o) All required lifting equipment tags must be in place and legible.
- p) Six-foot fall protection shall be followed at all times throughout all operations per AECOM TISHMAN CONSTRUCTION's Fall Protection Policy included in this program.
- q) The headache ball, hook or load shall not be used to transport personnel. Only approved Crane Hoisted Personnel Platforms are to be used following proper completion of the Crane Hoisted Personnel Platform Permit.
- r) All crane operations shall be suspended when wind gusts reach 25 mph or per local regulations. Suspension of operations may occur at less than 25 mph if and when the operator or the AECOM TISHMAN CONSTRUCTION Project Superintendent determines that continued crane operations will create a hazard.
- s) When off-loading precast panels from trucks and maneuvering them for lifting with a single or multiple cranes commonly referred to as "tripping panels," an independent engineering firm chosen by the subcontractor who owns the crane(s) or is responsible for the operation shall review and approve the

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operational setup. In all cases, the AECOM TISHMAN CONSTRUCTION Project Superintendent shall be notified in writing that the review and approval has taken place prior to operations commencing.

- t) The primary or secondary crane lines shall not be tied back unless permitted to do so by the manufacturer's specifications. If a manufacturer permits a line to be tied back, a secondary means of securing the line to prevent it from swinging in the event of a line failure shall be used.
- u) All Operators shall be made aware of each prior shift's activities prior to operating the crane.
- v) All work areas where materials must be lifted over shall be reviewed prior to the lifts with proper precautions put in place.
- w) A third party peer inspection shall be completed for all tower cranes, personnel hoists, material hoists, and complicated crane lifts prior to use and shall be documented. The third party must be completely independent of the owner of the tower crane, personnel hoist, material hoist, or complicated crane lift.

3. SUSPENDED LOADS

3.1 Individuals are not permitted to work or pass under any suspended loads. Individuals may stand beside a suspended load and guide it into place with their hands or tag lines when, in view of the height of the load, actual and potential swing of the load and trip hazards, the individuals are not at risk of being struck by the load if it were to fall.

3.2 Tag lines will be attached to suspended loads and will be used for guiding and landing loads unless the use of the tag lines would create a greater hazard. Concrete buckets are excluded from this provision.

3.3 In the event a suspended load must be swung over an area where individuals are working, a horn or whistle shall be used to warn individuals in the path of the load above. Individuals shall maintain visible contact with the load until it is swung away from the work area.

3.4 All personnel shall be kept from the suspended load up until the point where they must be near the suspended load to complete their task(s).

3.5 All suspended personnel platforms used to lift individuals shall be only used as a last resort. The use of such a platform shall be approved by the Project Director and the Regional Safety Director.

3.6 The hoisting of personnel is to be considered a critical lift.

3.7 The total load of the personnel platform shall not exceed 50% of the rated capacity for the radius and configuration of the equipment, except during proof testing.

3.8 The platform shall be designed for the intended purpose of lifting personnel and shall have dedicated rigging. Tools shall be secured from falling. Individuals shall use proper fall protection devices secured to the platform.

3.9 All materials hoisted shall be secured from displacement via banding, strapping, or other methods.

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4. RIGGING EQUIPMENT

4.1 Inspections.

- a) A qualified rigger (as defined by OSHA and this policy) shall inspect the rigging and attachment points prior to each lift.
- b) If any deficiency is identified, the rigger shall make an immediate determination as to whether the deficiency constitutes a hazard.
- c) If the deficiency is determined to constitute a hazard, the rigging equipment shall be removed from service until the deficiency has been corrected.
- d) Any repairs must be completed by a competent person.
- e) ANSI/ASME Standards shall be referred to for further specific requirements.
- f) Nylon straps shall be replaced if the original length changes +/- 3%.

4.2 Below-the-Hook Lifting Devices.

- a) A qualified engineer or a qualified engineering organization shall design all below-the-hook lifting devices.
- b) A competent person shall inspect all below-hook lifting devices prior to use.
- c) All below-the-hook lifting devices shall be designed, tested, and certified per ANSI/ASME B30.20 with documentation made available upon request.
- d) Spreader bars shall be marked with their load capacity.
- e) Full compliance with load testing shall be ensured.
- f) ANSI/ASME Standards shall be referred to for further specific requirements for specific devices.
- g) All hooks shall have latches.

4.3 Wire Rope Slings.

- a) All hoisting wire rope slings must have been recommended by the manufacturer for that service.
- b) When they are not being used, wire rope slings should be stored in an area where they will not be subjected to mechanical damage, corrosive action, moisture, extreme heat, or kinking.
- c) Wire rope slings should be kept lubricated to avoid rusting.
- d) Wire rope slings shall be tagged to show the name or trademark of the manufacturer, rated load capacity for the type of hitches, and the angle upon which it is based, and the diameter or size.

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- e) Wire rope slings shall be removed from service and destroyed when inspections show kinking, crushing, bird caging, or any other distortion or damage from the original wire rope structure.
- f) Use suitable wood or rubber protectors where slings go around corners.
- g) Use suitable softeners to prevent chokers from slipping during a lift.
- h) Proper sling angles shall be used at all times.

4.4 Synthetic Slings.

- a) Synthetic web slings (nylon/polyester) shall be marked or coded to show the name or trademark of the manufacturer, the rated capacities, and the type of material.
- b) Synthetic slings shall be removed from service and destroyed when inspection shows damage from heat (including acid or caustic burns), broken or worn stitches, distortion of fittings or snags, punctures, tears or cuts exposing colored thread.
- c) Proper sling angles shall be used at all times.

4.5 Shackles.

- a) Use only forged North American made alloy steel shackles made within the last 2 years.
- b) Use only those shackles that have a known load rating marked on them.
- c) All shackles must be properly closed and secured.
- d) Inspect per manufacturer's recommendations.
- e) Remove shackles from service when showing signs of abnormal wear or damage.

4.6 Chains.

- a) Identification tags must be affixed stating size, grade, rated capacity, and manufacturer.
- b) All chains shall be inspected prior to use by a competent person with proper documentation per OSHA standards. Whenever the chain wear exceeds the permissible limits as set by OSHA (See 29 CFR 1926.251), it shall be removed from service and destroyed.
- c) All chain lifting attachment(s) shall have at a minimum the same rated capacity of the chain and be made within the last 2 years.

5. RIGGER QUALIFICATIONS AND TRAINING

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5.1 Riggers must be trained and qualified for the task and must be capable of determining if equipment is damaged or not fit for a purpose. AECOM TISHMAN CONSTRUCTION distinguishes three levels of Riggers (Rigger I, II, and Master Rigger).

5.2 Riggers must be familiar with the different and correct rigging techniques, and with all rigging equipment used.

5.3 Individuals must meet and/or exceed the requirements of any governmental agency having jurisdiction, and AECOM TISHMAN CONSTRUCTION training requirements to be qualified as a Rigger.

5.4 All projects shall designate Qualified Riggers.

5.5 Level I Rigger must have basic knowledge of the following:

- a) Crane Communications (voice/radio).
- b) Standard Crane Hand Signals Including Tower Crane (Must wear orange gloves or other specific identification.)
- c) Rigging hardware; including:
 - Hooks and similar attaching devices.
 - Shackles, clips and clamps.
 - Taglines.
 - Any other approved and used devices.
- d) Synthetic/wire rope slings and chains.
- e) Common load configurations positioning.

5.6 Level I Rigger shall demonstrate specific knowledge of crane operations, including:

- a) Communications (voice/radio/hand).
- b) Personal fall protection methods.
- c) Emergency procedures.
- d) Hazards and restrictions associated with working adjacent to overhead electric lines and equipment.

5.7 Level II Rigger shall meet all of the requirements of a Level I Rigger plus have training in the safe application, use, and limitations of rigging equipment as applicable to the work performed, including:

- a) Come-along /chain hoist operations, rigging hitches and knots, anchor points, synthetic ropes for rigging, wire rope, chains, reeving, spreader bars and equalizing beams, synthetic slings, lifting points, dollies, trolleys, tuggers, and winches (manual and power), bars and levers.

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- b) Fiber rope for rigging jacks, jacking systems and rams, links and rings, plate clamps, softeners, and cable dogs/grips.

5.8 Master Rigger shall meet all of the requirements of a Level II Rigger plus have training in the safe application, use, and limitations of rigging equipment as applicable to the work performed, including:

- a) Blind hoists, traveling with the load, work in close quarters, personnel lifting procedures, load dynamics, load weight estimation or determination, specific criteria from the manufacturer or equipment representative in the safe and appropriate methods of erection, dismantling, jumping, and reconfiguring of cranes, specific criteria for lifting a single load with multiple cranes.
- b) Load indicator devices, capacity or load charts, mechanical advantages, center of gravity, effects of angles or indirect pulling, equipment capacity computations.
- c) Drum/diameter (D/d) ratios, vectors and angles, and boom angles and road radius.

6. OPERATOR QUALIFICATIONS AND TRAINING

6.1 Cranes and hoisting equipment shall be operated only by designated qualified personnel. Proof of qualification shall be provided in writing to AECOM TISHMAN CONSTRUCTION.

6.2 In addition to fully qualified crane and hoisting equipment operators, the following personnel may be designated to operate cranes and hoisting equipment under limited conditions (may not perform critical lifts).

- a) Trainees under the direct supervision of the designated operator of the crane or hoist.
- b) Designated maintenance personnel who have completed all operator qualification requirements. Operation is limited only to those functions necessary to perform maintenance or verify performance of a crane or hoist.
- c) Inspectors who have completed all operator qualification requirements. Operation is limited only to functions necessary to accomplish inspection.

6.3 General Requirements.

- a) Crane Operators shall be able to communicate effectively with the lift supervisor, rigger(s), flagmen and other affected employees on site.
- b) Prior to the start of work, documentation of operator qualifications shall be provided to the AECOM TISHMAN CONSTRUCTION Project Office.
- c) All crane operators shall have knowledge of AECOM TISHMAN CONSTRUCTION crane safety requirements and manufacturer requirements and recommendations provided in the crane operator manual.
- d) Crane operators shall demonstrate their ability to read, write and comprehend in the language of the crane manufacturer's operation and maintenance instruction materials, exhibit arithmetic skills and

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load/capacity chart usage and use written manufacturer procedures applicable to the class/type of equipment for which certification is being sought.

- e) Crane operators shall be familiar with the immediate crane they are intending to use prior to the start of operations.

6.4 Crane Operator Certifications. All Crane Operators shall possess the appropriate local license and in addition shall possess at least one of the following licenses or certifications:

- a) Option 1. A current certification by an accredited (a nationally recognized accrediting agency) crane/derrick operator testing organization. The organization shall:
 - i. Administer written and practical tests that assess operator applicants regarding necessary knowledge and skills.
 - ii. Provide different levels of certification based on equipment capacity and type.
 - iii. Have procedures for operators to re-apply and be retested in event operator applicant fails a test or is decertified.
 - iv. Have testing procedures for recertification.
 - v. Have accreditation reviewed by the nationally recognized accrediting agency at least every 3 years.
- b) Option 2. Qualification by a professional source that qualifies crane operators (e.g. independent testing and qualifying company, a union, or a qualified consultant who can be an in-house resource) as long as the program is an audited employer program. Employer's qualification of its employee shall meet the following:
 - i. Administer written and practical tests that assess operator applicants regarding necessary knowledge and skills. These tests shall be either developed by an accredited crane/derrick operator testing organization (see Option 1 above) OR approved by an examiner in accordance with the following:
 - a. The examiner is certified to evaluate such tests by an accredited crane/derrick operator testing organization (see Option 1 above).
 - b. The approval shall be based on the examiner's determination that the tests meet nationally recognized test development criteria and are valid and reliable in assessing the operator applicant's knowledge and skill needed.
 - c. The employer program shall be audited within three (3) months of the beginning of the program and every three (3) years thereafter.
 - d. The employer program shall have testing procedures for recertification.
 - e. Any significant deficiencies identified by the examiner shall be corrected prior to further qualification of any operators.

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- f. Records of audits shall be retained for 3 years and made available to AECOM TISHMAN CONSTRUCTION upon request.
- g. A qualification under this option is non-portable and is valid for 5 years from date of issuance.
- c) Option 3. Qualification by the U.S. Armed Forces. Operator is considered valid if he has a current operator qualification issued by the U.S. military for operation of the equipment. Qualification meets operator qualification requirements of this section for operation of equipment only within the jurisdiction of the government entity and is valid for the period stipulated but no longer than five (5) years from issuance.
- d) Option 4. Licensing by U.S. Federal Government. An examiner that issues operator licenses for operating equipment is considered a government accredited crane/derrick operator examiner if the following criteria are met:
 - i. The requirements for obtaining the license include an assessment by written and practical tests of the operator applicant regarding knowledge and skills, as applicable to the specific type of equipment the individual will operate.
 - ii. The testing meets industry recognized criteria for written testing materials, practical examinations, test administration, grading, facilities/equipment and personnel. Testing shall include:
 - a. The controls and operational/performance characteristics.
 - b. Responsibilities of operator, rigger, signal persons, and lift supervisor.
 - c. Ability to determine the crane configuration, determine size and shape of loads, and the crane's applicable capacity using the load chart.
 - d. Use and limitations of crane safety devices and operator aids.
 - e. Inspection, testing, and maintenance requirements.
 - f. Suitability of ground and surface to handle expected loads.
 - g. Identification of site hazards and site access conditions.
 - h. Outrigger and matting requirements (as applicable).
 - i. Crane set-up, assembly, dismantling, and demobilization procedures.
 - j. Requirements for clearance from power sources, procedures for preventing contact and responding to contact with said sources.
 - k. Signaling and communication procedures.
 - l. Factors that reduce rated capacity.

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- m. Emergency control skills.
- iii. The AECOM TISHMAN CONSTRUCTION Safety Department has determined that the requirements for Option 4 licensing have been met.
- iv. The examiner has testing procedures for recertification designed to ensure that the operator continues to meet the technical knowledge and skills requirements.
- v. A license issued by an examiner that meets the requirements of this option:
 - a. Meets operator qualification requirements of this section for operation of equipment only within the jurisdiction of the government entity.
 - b. Is valid for the period stipulated but no longer than 5 years from issuance.

6.5 Operator Practical Examination Requirements. Crane operators shall pass a practical operational test that demonstrates the following:

1. Ability to recognize, from visual and audible observation, any deficiencies with the crane.
2. Ability to establish a stable foundation and to the crane:
 - i. Operating skills - Raising, lowering, extending, retracting, and swinging the boom, raising and lowering the load line.
 - ii. Attaching the load, holding the load, and moving the load.
 - iii. Reading and applying load, boom angle, and other indicator devices.
 - iv. Maneuvering skills.
 - v. Applying safe shut-down and securing procedures.

7. MOBILE CRANE SPECIFICS

7.1 Set Up.

- a) All mobile cranes shall not be assembled or used unless ground conditions are firm, drained, and graded to a sufficient extent so that, in conjunction (if necessary) with the use of supporting materials, the equipment manufacture specifications for adequate support and degree of level of the equipment are met. The operator will be informed of ground bearing pressure and hazards beneath the equipment set-up area such as voids, tanks, utilities, etc.
- b) If the crane is equipped with outriggers, they must be extended per manufacturer's specifications and all of the crane wheels must be clear of the ground. The operator must know the ground bearing pressure.

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- c) Adequate blocking must be used under all outrigger floats to spread the load and transmit the load without bending or breaking.
 - 1. Blocking must meet one of the following criteria:
 - i. Equal to three times the size in area of the outrigger float or
 - ii. Crane lift capacity (tons), divided by five equals the square footage of blocking needed under each outrigger float or
 - iii. Comply with manufacturers recommendations or
 - iv. Designed by a professional engineer.
 - 2. Blocking must be rigid, level, and tightly spaced.
 - 3. Blocking must be centered and make full contact with the float.
 - 4. Never block under outrigger beams.
- d) If the A/D director or the operator determines that the ground conditions do not meet the requirements listed above then that individual must bring the concerns to the controlling entity for resolution.
- e) Mobile cranes provided by the subcontractor will be less than 20 years old unless they have been certified by a corporate officer or chief engineer from the manufacturer as being reconditioned to “like new” condition.

7.2 Use.

- a) All mobile crane usage shall meet and/or exceed all standards, guidelines, and rules found in this program and all applicable city, state, and federal agency programs.
- b) In the event of multiple standards, guidelines, and rules, the most stringent, based on the location of the crane, shall apply.
- c) The crane operator is responsible for the safe use and operation of the crane at all times. Whenever there is a concern regarding safety, the operator has the authority to stop the operation and refuse to handle loads until a qualified person has determined that it is safe to resume operations.

7.3 Inspections and Checks.

- a) Before a mobile crane is permitted to operate on an AECOM TISHMAN CONSTRUCTION project, an additional check shall be completed by an AECOM TISHMAN CONSTRUCTION approved qualified third-party crane inspector. This check shall include at a minimum items contained in Attachment 2. This form shall be filled out and placed in the crane packet. Any issues shall be resolved prior to mobile crane operation.

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- b) As mentioned previously, all operators shall visibly inspect (check) and document the crane prior to use on each shift.
- c) Once a month, the crane shall be inspected by a competent person qualified to complete crane inspections.
- d) Once every twelve (12) months, the crane shall be inspected by a designated qualified third-party inspector.
- e) Periodic inspections shall be completed for “heavy use” category cranes.
- f) Applicable standards and OSHA guidelines shall be followed for all inspections.
- g) Equipment not in regular use or idle for 3 or more months shall be inspected prior to use by a qualified person following the monthly inspection guidelines.
- h) The A/D director shall implement the use of a checklist designed by the crane operating company for the specific type of crane for assembling and dismantling of the crane. The A/D director shall fill out and initial each part of the assembly/disassembly after it occurs and before the next step begins. The intent is to ensure all steps in assembly and disassembly occur in proper fashion. The form shall be readily available upon request by AECOM TISHMAN CONSTRUCTION.

7.4 A “Crane Packet” for each mobile crane brought onto an AECOM TISHMAN CONSTRUCTION project shall be readily available in the crane for review by authorized personnel. The packet shall include the following at a minimum:

- a) Completed Attachment 2.
- b) Photocopy of Operator's license and physical certification (As required by locality).
- c) Daily inspections (visual checks) (by crane operator using a 2 part form, or collected and copied on a weekly basis).
- d) Annual Inspection.
- e) Maintenance records that have occurred while the crane was on the project.
- f) Lifting plan including matting procedures (including building re-shore requirements when applicable) developed as a result of looking at the allowable bearing capacity of the surface versus the applied load. This needs to be done for all mobile cranes, including crawlers. (Provide licensed engineer's criteria for base compaction as required.).
- g) Load charts including attachments for crane configurations used on the project.
- h) Copy of completed critical lift plans and checklists when applicable.
- i) FAA height permit when applicable.

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- j) A/D Director name and assembly plan.
- k) All diagrams, lift data sheets, rigging and hardware calculations, applicable permits.
- l) Bucket lifting plans.
- m) Applicable training documents for Riggers and Signalers.

7.5 Delivery truck mounted cranes shall follow and meet all of the manufacturer's specifications for setup prior to off-loading materials. All of the guidelines found in this program shall be followed except for the collection of a crane packet for this type of short-term use crane. The operator shall produce an adequate inspection log as required by AECOM TISHMAN CONSTRUCTION staff and shall be qualified to operate the crane.

8. TOWER CRANE SPECIFICS

8.1 Erection and Dismantling Operations.

- a) All tower cranes shall be erected in strict accordance with the tower crane manual as supplied by the crane manufacturer and guidance from the A/D director.
- b) A proposed tower crane layout shall be prepared and submitted to the AECOM TISHMAN CONSTRUCTION Project Team. All permits shall be obtained by the erector, project team, and/or the responsible subcontractor.
- c) Prior to erecting, dismantling, or "jumping" a crane, a documented safety coordination meeting shall take place at least 24 hours prior to the activity between the A/D director and AECOM TISHMAN CONSTRUCTION designated staff to cover all of the following:
 - 1. Scope of work.
 - 2. Roles and Responsibilities.
 - 3. Rigging.
 - 4. Inspection of the rigging.
 - 5. Review of other equipment such as collars, ties, and bolts.
 - 6. Permits and qualifications/training of personnel.
 - 7. Weather.
 - 8. Softening mechanisms on sling edges (if using nylon slings).
 - 9. Review of impact (if any) of erection crane on support of the excavation system and determination of matting requirements.
 - 10. Notification to the local jurisdiction as applicable.

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11. FAA permits, regulations, and use restrictions as applicable.
 12. Coordination between multiple tower cranes on a project site when applicable, especially when overlapping jibs swings are present.
 13. Security at the base of the tower crane to prevent unauthorized access.
 14. Fueling Requirements.
- d) All foundations for tower cranes shall be designed by a Professional Engineer familiar with the varying loads placed on these types of structures.
 - e) The tower crane foundation shall be inspected for conformance with design during and after construction by a third party inspection agency (including soils, reinforcing steel, and concrete).
 - f) When the foundation placement has taken place and it has obtained required strength as verified by a third party testing agency, the tower may begin to be erected.
 - g) All tower cranes shall be properly grounded per manufacturer's specifications.
 - h) The tower crane's capacity, hoist, and trolley limits shall be set once the erection has been completed. Where possible, limits should be set so that the operator cannot come into contact with power lines, other cranes, buildings or other structures.
 - i) Adequate guardrails, handrails, and ladder access shall be provided for the crane operator and inspector at all heights greater than six feet both on the crane and at its base.
 - j) Once fully erected, the tower crane shall be inspected and certified by the erection company with an additional inspection being completed by a competent third party crane inspector.
 - k) All tower crane configurations shall be prepared by a qualified and licensed professional engineer.
 - l) Tower cranes, including tower sections and booms, provided by Subcontractor will be less than 10 years old unless they have been certified by a corporate officer or chief engineer from the manufacturer as being reconditioned to "like new" condition. In no case shall tower cranes, tower sections, and booms be over 20 years old.
 - m) Tower crane bases (including affiliated transformers and power supply equipment) shall be surrounded by an appropriate height (based on location but at least 8 foot high,) 5/8 – inch plywood enclosure with a locked controlled entrance.

8.2 Use.

- a) All operators shall be qualified to run a tower crane within the local jurisdiction.
- b) All operators shall conduct a documented visible inspection of the crane prior to use on each shift.

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- c) Monthly inspections per the operations maintenance manual shall be completed, kept on file, and submitted to AECOM TISHMAN CONSTRUCTION.
- d) At the base of the tower crane and around electrical distribution for the crane shall be kept free of water and debris at all times.
- e) All tower cranes shall be equipped with a self-rescuing device for the operator rated for the height of the crane.
- f) All operators shall stay within the limits of the crane at all times.
- g) Active, designated communication shall be maintained with the operator and signaler at all times. The tower crane operator shall be made aware of each shift's activities prior to operating the crane.
- h) If multiple tower cranes are on a single project, each crane operator shall be equipped with radios so they can communicate movements. They shall also have a specific number or name designation and have anti-collision devices installed.
- i) A site plan for multiple tower cranes shall be submitted to the AECOM TISHMAN CONSTRUCTION project team for review prior to any crane set up.
- j) Rated load capacities, and recommended operating speeds, special hazard warnings, or instructions shall be available onsite for review.
- k) Platforms and walkways shall have anti-skid surfaces as required by the manufacturer.
- l) Any noticeable defect affecting structural, mechanical, or lifting functions shall cause the crane to be taken out of service until the defect is reviewed and corrected (if necessary) by a qualified and authorized inspector.
- m) A system to secure access to a tower crane shall be in place. At the end of the work day the access point shall be secured/locked to prevent unauthorized access to the crane.

8.3 Inspections and Checks.

- a) Before a tower crane is permitted to operate on an AECOM TISHMAN CONSTRUCTION project, an inspection must be completed by an AECOM TISHMAN CONSTRUCTION approved qualified third party crane inspector. Any issues must be resolved prior to tower crane operation.
- b) As mentioned previously, all operators shall visibly inspect (check) and document the crane prior to use on each shift.
- c) Once a quarter, the crane shall be inspected by a qualified inspector.
- d) Once every twelve (12) months, the crane shall be inspected by a designated third party inspector.
- e) Periodic inspections shall be completed for "heavy use" category cranes.

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- f) ANSI, ASME, and OSHA guidelines shall be followed for all inspections.
- g) The A/D director shall implement the use of a checklist designed by the crane operating company for the specific type of crane for assembling and dismantling of the crane. The A/D director shall fill out and initial each part of the assembly/disassembly after it occurs and before the next step begins. The intent is to ensure all steps in assembly and disassembly occur in proper fashion. The form shall be readily available upon request by AECOM TISHMAN CONSTRUCTION.

8.4 A “Crane Packet” for each tower crane erected on an AECOM TISHMAN CONSTRUCTION project shall be readily available in the AECOM TISHMAN CONSTRUCTION project office for review by authorized personnel. The packet shall include the following at a minimum:

- a) Completed Crane Packet Checklist (Attachment 7)
- b) Completed General Crane Inspection Sheet (Attachment 2)
- c) Swing radius/site plan.
- d) Approved foundation (pad) design with PE stamp.
- e) FAA height permit, if applicable.
- f) When required to be tied to a structure, crane tie-in design(s) with PE stamp.
- g) Third party inspection records of soil compaction, reinforcing steel, and concrete testing for crane foundation.
- h) Tower crane erection permit (as required by locality).
- i) Public space permit (as required by locality for swing).
- j) Erection inspection/certification.
- k) Daily inspections (visual checks by crane operator).
- l) Monthly inspections.
- m) Quarterly Inspections (executed by AECOM TISHMAN CONSTRUCTION Tower Crane Crew in the Mid Atlantic; or by a crane rental company as applicable nationally).
- n) Annual inspections.
- o) Photocopy of Operator's license.
- p) Copy of completed critical lift plans and checklists when applicable.

9. POWER LINE SAFETY

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9.1 All cranes shall stay as far from overhead electrical lines as possible. De-energize and ground all lines where possible. Determine the voltage with the power company prior to operations.

9.2 Maintain at least 20 feet clearance and ensure that no part of the crane or attachments come closer.

9.3 Use a trained spotter and visual markers when using the 20-foot rule.

9.4 Follow the table below:

Voltage (nominal, kV, alternating current)	Minimum Clearance Distance (feet)
up to 50	15
over 50 to 200	20
over 200 to 350	25
over 350 to 500	30
over 500 to 750	40
over 750 to 1,000	50
over 1,000	<i>as established by the utility owner/operator or registered professional engineer who is a qualified person with respect to electrical power transmission and distribution</i>

10. CRITICAL LIFTS

10.1 The AECOM TISHMAN CONSTRUCTION Project Superintendent shall review the classification of each lift with the applicable subcontractor to determine whether a lift is ordinary or critical prior to proceeding with work. If any doubt exists, the lift is automatically considered critical.

10.2 A lift shall be immediately designated critical if any of the following conditions are met:

- The load item, if damaged or upset would result in a release of hazardous material into the environment.
- The load item is unique and, if damaged, would be irreplaceable or not repairable.
- The use of more than any one single piece of equipment is required to lift, stabilize, support, or be in close proximity to the load to facilitate installation.
- The maximum anticipated load is at or above 75% of the crane's rated capacity for mobile cranes and 90% for tower cranes as configured according to the applicable lifting chart, for that particular lift unless otherwise stated in this policy.
- Personnel are being lifted.
- Further site-specific criteria may be developed to supplement those listed above and may include loads which require exceptional care in handling because of size, weight, close-tolerance installation, high susceptibility to damage, or at the discretion of the PIC.

10.3 A Person-In-Charge (PIC), as defined in the definition at of this program, will be appointed for the entire operation. This person shall be "competent" and shall be present at the lift site during the entire lifting operation.

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10.4 The PIC shall ensure that a Critical lift plan is prepared and approved, which defines the operation and includes the following:

- a) Identification of the items to be moved, the weight, dimensions, and center of gravity of the load.
- b) Identification of operating equipment to be used by type and rated capacity.
- c) Rigging sketches that include (as applicable):
 - 1. Identification and rated capacity of slings, lifting bars, rigging accessories, and below-the-hook lifting devices. Calculate and provide the rated capacity of equipment in the configuration in which it will be used.
 - 2. Load-indicating devices.
 - 3. Load vectors.
 - 4. Lifting points.
 - 5. Sling angles.
 - 6. Boom and swing angles.
 - 7. Methods of attachment.
 - 8. Crane orientations.
 - 9. Other factors affecting equipment capacity (e.g. load path sketch, etc.)
- d) Experienced operators who have been trained and qualified to operate the specific equipment to be used shall be assigned to make the lift.
- e) Only designated, qualified signalers wearing orange gloves shall give signals to the operator. However, the operator shall obey a STOP signal at all times, no matter who gives the signal.
- f) The PIC shall consider the crane's lifting history when completing the final lift plan. Additional considerations shall include, but not be limited to (a) the crane's inspection history (b) sling angles and sizes (c) inspection and test certification periods; (d) load configuration (e) the presence of hazardous materials and/or operations (f) lifting attachment (e.g., spreader bar) rated capacity, and (g) requirements for a load indication device.
- g) Hoisting shall be stopped when the load is approximately two inches off the supports and a check made to determine any tendency to swing or sway, and any tendency of the slings to slip or change position. If needed, sling positions shall be adjusted prior to continuing the lift. Procedural approval for changes to the sling position, support, and restraints shall be obtained from the PIC.

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- h) Field changes to the subcontractor's Critical Lift Plan/Procedures or equipment adjustments to the subcontractor's equipment may be made providing authorized AECOM TISHMAN CONSTRUCTION project personnel and the PIC concur with the changes prior to implementation.
- i) Nothing in the procedure should be construed as relieving a subcontractor from responsibility to provide a pre-lift/rigging plan. The subcontractor's plan may include additional information not reviewed in this procedure; however, this procedure provides minimum acceptable guidelines.
- j) The Critical Lift Data Sheet (Attachment 3) and the Tandem Lift Data Sheet (Attachment 4) shall be completed before a critical lift or tandem lift procedure is performed. All signatures are required on the applicable sheet prior to the lift commencing.
- k) A concrete or precast operation which involves rapid concrete placement using a concrete bucket or normal precast lifting by use of a tower crane is exempted from the normal /pre-lift /critical lift requirements found herein as long as proper inspections and documentation are being carried out as defined below:
 - 1. Precast: Prior to these operations commencing and on a weekly basis, one Critical Lift Data Sheet (Attachment 3) for precast lifting may be used. The sheet shall list all types and weights of precast to be lifted for the upcoming week. The PIC shall complete the sheets and add additional pages as necessary with all required signatures and information.
 - 2. Concrete Bucket: Prior to these operations commencing and on a weekly basis, one Concrete Bucket Lift Plan Sheet (Attachment 6) shall be completed by the PIC with the appropriate and required signatures as listed on the form which covers all of the concrete bucket lifting ranges for the week.
 - The information shall be properly communicated to the field level supervision. Each week the data sheet(s) shall also be readily available for review by authorized personnel. This pre-lift paperwork exemption is only for concrete bucket use and precast use on a regular daily basis and all other rules apply. In the event that all of the other rules and/or standards are found not being followed, this exemption shall be removed and each individual lift shall require a Critical Lift Data Sheet to be filled out prior to the operation commencing.

11. SPECIAL CRANE REQUIREMENTS

11.1 Please refer to local or state crane regulations if applicable to ensure compliance. Additional local requirements may be attached in a Rider.

12. ATTACHMENTS

12.1 Crane Lift Worksheet

12.2 General Crane Inspection Sheet

12.3 Critical Lift Data Sheet

12.4 Tandem Pick Critical Lift Data Sheet

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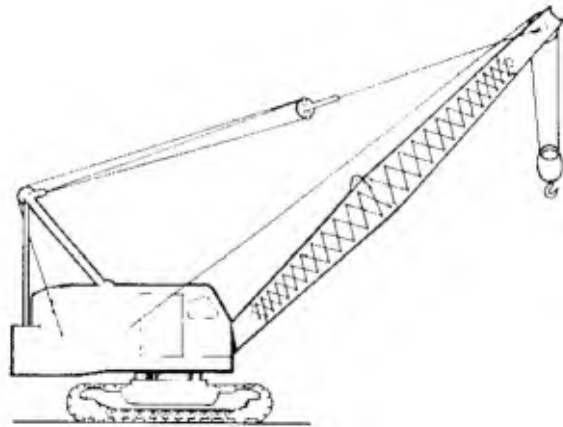
12.5 Crane Hoisted Personnel Platform Permit

12.6 Concrete Bucket Lift Plan

12.7 Crane Packet Checklist

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Attachment 1: CRANE LIFT WORKSHEET



Date: _____

Reviewer: _____

Crane: _____

Description: _____

Weight

Item: _____

Jib: _____

Ball & Hook: _____

Load & Block: _____

Wire Rope: _____

Rigging: _____

Total: _____

Other Specifications

Max Radius to be used: _____ ft.

Chart Capacity @ _____ Radius _____

Minimum Boom Angle: _____

Percent of Capacity: _____ %

Set-up Inspection

Ground Conditions: _____

Wind Conditions: _____

Levelness: _____

Over 75% - Attachment #3 Filled Out Completely

Over 90% - Professional Engineer Review

Any Tandem Lift - Professional Engineer Review

Remarks: _____

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Attachment 2: GENERAL CRANE INSPECTION SHEET

Project Name:		Name of Subcontractor (User):	
Make of Crane:	Model:	Serial Number:	
Rated Capacity:	Owner:	Date:	

Visually Checked: $\checkmark$ = Yes N/A = Not Applicable List Noticed Deficiencies in Comment Section

Crane Packet Information	Floats / Pads	Telescoping Cylinders
Competent Person	Hydraulic Hoses / Tubes / Fittings	Comments:
Operators Certification (Photocopy)	Position Locks	
Daily Inspections	Warning Signs	
Annual Inspection	Cracks / Deformation	
Maintenance Records	Swing Radius Guarded	
Lifting Plan	Comments:	Hoist Lines
Load charts including attachments		Rope Damage / Deformation
Critical Lift Plan		Broken Wires
FAA Height Permit		Lubrication
Comments:		End Connections
	Tires/Crawler Assemble	Wrapping on Drum
	Air Pressure / Wheels	Comments:
	Cuts/Damage	
	Track Pads / Pins	
Operators Cab & Station	Sprockets / Rollers	
Windshield Wiper	Car Body / Side Frames	
Door Restraint	Comments:	Safety Devices / Operational Aids
Cab Access		Boom Angle/Length Indicator
Guard Rails & Hand Holds		Crane Signal Signs
Anti-Slip Walking Surfaces		Radius Indicator
Proper ladder access to cab		Crane Level Indicator
Gauges	Power Plant / Engine	Wind Load Indicator
Load Chart (in Cab)	Exhaust System / Guards	Comments:
Fire Extinguisher (10 lb.)	Belts / Hoses	
Owner or Operators Manual / Instructions	Rotating Parts Guards / Covers	
Seat Belt	Leaks	
Electrocution Sign(s)	Battery Compartment	
Electrical Covers / Guards	Comments:	
Level		
Lights (Internal / External)		Block / Hook
Horn		Capacity & Weight Marking
Load Indicator (LMI)		Sheaves
Controls Marked / Identified	Structure / Boom / Jib	Safety Latches
Swing Break & Lock	Deformation / Cracks / Corrosion	Swivel / Bearing
Comments:	Sheaves	Wedge Socket / End Fitting
	Wear Pads	
	Lattice Members	Comments:
	Cord Members	
	End Connections / Pins	
Outriggers/Stabilizers	Anti-two Block	Other
	Boom Stops	
Beams	Pendants	Load Moment Indicator
Cylinders	Wire Rope Retainer	Two-Block Warning Device

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Any Other Visible Defects

Additional Comments.

I have completed a visual check of the crane above.
The crane meets the structural and operational items
listed on this check sheet.

Circle one

Yes

No

Printed Name and Signature of Person Completing Visual
Check.

Printed Name and Signature of Crane Operator:

Printed Name and Signature of Project Lead Superintendent.

Date: _____

Note: Refer to OSHA 1926 1400 and ANSI/ASME B30 5 for additional standards, as required

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Attachment 3: CRITICAL LIFT DATA SHEET

Subcontractor Name	DATE:
Project Name / Location	
Type of Work:	
Lifted Piece Description	
Person In Charge (PIC)	

CRANE CONFIGURATION:			
Crane Type		Block Size	
Boom Length		Line Parts	
Boom Type		Jib Type	
Tip Type		Jib Length	
Counterweights		Jib Offset	

CRANE FIXED WEIGHTS (lbs.)	
Main Block	
Wire Rope	
Whip Line Ball/Block	
Auxiliary Boom Sheaves (deduct)	
Jib (deduct)	
Rigging Weight	
Spreader	
Slings	
Shackles	
Other	
Total Crane Fixed Weight	

	Check (√)
Load Chart Referenced?	
Pre-Lift Meeting Held?	
Rigging Inspected?	
Crane Setup Reviewed?	
Weather Conditions?	
Tag Lines Installed?	
Proper Communications?	
Electrical Hazards?	
Underground Hazards?	

PIECE WEIGHT (lbs.)	
Piece Weight	
Attachments (lugs, etc.)	
Other	
Total Weight of Piece	

PICK CAPACITY	
Actual Pick Radius	
Chart Radius	
Chart Capacity	
Percent Capacity	

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LIFT WEIGHT SUMMARY	
Total Fixed Weight	
Total Piece Weight	
Total Lift Weight	

SET CAPACITY	
Actual Set Radius	
Chart Radius	
Chart Capacity	
Percent Capacity	

CLEARANCES	
Minimum Clearance from Boom to Obstruction	
Minimum Clearance from Boom to Load	

Diagram of the Lift:



PREPARED BY:	DATE:
PERSON IN CHARGE (PIC):	DATE:
OPERATOR:	DATE:
AECOM TISHMAN CONSTRUCTION LEAD SUPERINTENDENT:	DATE:
AECOM TISHMAN CONSTRUCTION SAFETY:	DATE:

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Attachment 4: TANDEM PICK CRITICAL LIFT DATA SHEET

Project Name _____ Project No. _____ Date _____

Description of lift

Crane Data

Crane 1: Make and Model _____

Crane 2: Make and Model _____

Lattice Boom Model Type: Unit 1 _____ Unit 2 _____

Boom Length: Unit 1 _____ Unit 2 _____

Counterweight: Unit 1 _____ lb. Unit 2 _____ lb.

Chart Maximum Capacity at Radius:

Unit 1 _____ lb. Unit 2 _____ lb.

Load Data

Load Weight _____ lb.

+Rigging Weight: Unit 1 _____ lb. Unit 2 _____ lb.

+Main Block: Unit 1 _____ lb. Unit 2 _____ lb.

+”Effective” Jib Weight: Unit 1 _____ lb. Unit 2 _____ lb.

+Cable Weight: Unit 1 _____ lb. Unit 2 _____ lb.

+Headache Ball Weight: Unit 1 _____ lb. Unit 2 _____ lb.

=Net Load Weight: Unit 1 _____ lb. Unit 2 _____ lb.

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Radius and Angle Data

Maximum Load Radius: Unit 1 _____ ft. Unit 2 _____ ft.

Minimum Load Radius: Unit 1 _____ ft. Unit 2 _____ ft.

Maximum Boom Angle: Unit 1 _____ ° Unit 2 _____ °

Minimum Boom Angle: Unit 1 _____ ° Unit 2 _____ °

Rigging Data

Sling Construction:

Diameter: Unit 1 _____ in. Unit 2 _____ in.

Core Type: Unit 1 _____ Unit 2 _____

Number of Legs: Unit 1 _____ Unit 2 _____

Sling Angle: Unit 1 _____ ° Unit 2 _____ °

Sling Capacity: Unit 1 _____ Unit 2 _____

Means of Fastening Sling or

Hoist Hook to Load: Unit 1 _____ Unit 2 _____

Capacity of Fastener: Unit 1 _____ lb. Unit 2 _____ lb

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Pre-Lift Requirements (ALL requirements must be met)

- ☐ Load chart utilized is for exact crane model, boom type, length, tip and counterweight.
- ☐ Competent Person in charge of lift. Name:
- ☐ Competent Signaler identified. Name:
- ☐ Pre-pick meeting held with crew.
- ☐ Written crane inspection completed within one (1) day of critical pick.
- ☐ Swing path not over personnel.
- ☐ Footing is sound and level (soil conditions/compaction, underground tunnel or utilities).
- ☐ Pre-planning for radio or hand signal communications.
- ☐ Minimum clearances from power lines can and will be maintained.
- ☐ The load radius has been measured with tape measure.
- ☐ Wind speed does not exceed 20 mph. Consider postponing lifts if speeds are more than 10 mph.
- ☐ Load will not touch boom at any time.
- ☐ Four dual crane lift diagrams have been prepared.
- ☐ If on barge, the Regional Safety Director has reviewed stability and potential lift conditions.
- ☐ Barge chart load provided.
- ☐ Tag lines are long enough, tied only to the load, and in good condition – loose end controlled by the designated person.
- ☐ Operating locations are far enough away from shorting, excavations, and trenches to eliminate risk of collapse.
- ☐ Application of hardwood mats has been carefully considered.
- ☐ Outriggers or crawler tacks are fully extended and wheels are clear of ground.
- ☐ Application of blocking under outrigger pads has been carefully considered.
- ☐ Adequate swing clearance (minimum two foot) between the counterweight and any obstacles.
- ☐ Boom composition is correct.
- ☐ Machine is rigged with adequate type of cable and number of parts of hoist line.
- ☐ Regional Safety Director and project superintendent have discussed lift.
- ☐ Load block is of adequate capacity and sheaves are of proper size for hoist cable.
- ☐ All rigging has been inspected for capacity and condition.
- ☐ Underground structures and conditions have been considered.

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Diagram of the Lift:

Approval Signatures

Lead AECOM TISHMAN CONSTRUCTION
Superintendent

Date

Crane Operator, Unit 1

Date

Crane Operator, Unit 2

Date

Qualified P.E.

Date

Regional/Area Safety Mngr.

Date

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Attachment 5: CRANE HOISTED PERSONNEL PLATFORM PERMIT

Date _____

1. This permit is required by AECOM TISHMAN CONSTRUCTION in compliance with OSHA and EM-385 requirements.
2. Prior to use of any personnel platforms suspended from cranes for the hoisting of personnel, the highest level or project management shall determine if the use of conventional means is more hazardous or not possible because of structural design or work site conditions. Reasons shall be stated on line 3.

Approval Signature _____ Title _____ Date _____

3. Justification for use:

Work description/Location (include Elevation):

_____ Crane inspected and approved.

_____ Rigging inspected and approved.

_____ Personnel platform inspected and approved.

_____ Anti-two block device radius and capacity established.

_____ Maximum operation radius and capacity established.

Length of radius: _____

Crane capacity at above radius: _____

(50% of chart at radius given)

_____ Test lift completed.

_____ Proof lift completed.

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_____ Pre-lift safety meeting held with all personnel involved.

_____ Communication methods established.

_____ Signaler designated. Name _____

The persons below acknowledge that they have participated in a pre-lift meeting and acknowledge the above activities were completed satisfactorily.

Person In Charge (PIC w/ title)

Crane Operator

Superintendent in charge of work

Signaler

Regional/Area Safety Manager

Employee(s) on Platform

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Attachment 6: CONCRETE BUCKET LIFT PLAN

Project Name:		Date	
Crane Operator:		Labor Foreman:	
Superintendent:		Person in Charge:	

Crane Type to be used: Tower ☐ Crawler ☐ Truck/Rough Terrain ☐

Crane Model: _____ Boom Length (Other than Tower): _____
Jib Length: _____ Counter Weights (Other than Tower): _____

Has the crane had its quarterly inspection to allow heavy service use? Yes ☐ No ☐

Pour Station Location: _____
 _____ New Pour Station ☐ Existing Pour Station ☐

	Yes	No	N/A
Has the pour station been coordinated with other deliveries and the Superintendent?	☐	☐	☐
Has pedestrian traffic been routed around the station?	☐	☐	☐
If needed, is a certified flagman present to direct traffic and truck staging?	☐	☐	☐
If ramps are used to elevate the rear of the truck, are they in good condition with the area clear?	☐	☐	☐
Are adjacent buildings to remain and passing vehicles protected from concrete splatter?	☐	☐	☐

Bucket Routing

Confirm bucket's path of travel, confirm it does not travel over pedestrians.	☐	☐	☐
Is a whistle, horn, or other warning signal used to alert people on the ground?	☐	☐	☐
Does the load, crane and load path have sufficient clearance from power lines?	☐	☐	☐
Is the load's path of travel within the operator's line of sight at all times?	☐	☐	☐
If not in the line of sight, is a qualified and authorized signaler used to provide adequate communications?	☐	☐	☐

Bucket Rigging and Inspection

Bucket Equipment Identification Numbers: _____			
Has rigging been inspected by a trained and authorized Rigger on a daily basis?	☐	☐	☐
Are connection points to the bucket free from visual defects/cracks?	☐	☐	☐
If a metal bale is present, are the pins free from visual cracks and/or excessive wear?	☐	☐	☐
Are the shackles the proper size and load rating?	☐	☐	☐
Are the shackles free of cracks, excessive deformation, and/or excessive wear (<10% reduction)?	☐	☐	☐
Are the wire rope chokers the proper size and load rating?	☐	☐	☐
Do the wire rope chokers have the proper id tags and/or labeling?	☐	☐	☐
Are wire rope chokers in satisfactory condition?	☐	☐	☐
No more than 10 broken wires in one lay or 5 broken in one strand? No kinks or bird caging.	☐	☐	☐
Are thimbles used in the wire rope chokers in good condition?	☐	☐	☐

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Are the master links the proper size, loading rating and are they free from any deformation?	☐	☐	☐
Is the bucket properly cleaned after each use?	☐	☐	☐
Are there any visible cracks in the welds of the bucket?	☐	☐	☐
Are there any dents or deformed parts of the bucket that would impair its use?	☐	☐	☐
If a side chute or boot is used, is it in good working order with the pin not bent?	☐	☐	☐
Does the chute open and close correctly and stay tightly closed?	☐	☐	☐

Crane Setup (Mobile Cranes Only)

Is the crane set up on level firm ground with proper level dunnage and outriggers fully extended?	☐	☐	☐
Has the soil bearing been checked and the crane independently checked per AECOM TISHMAN CONSTRUCTION policy?	☐	☐	☐

Crane Capacity Evaluation

Crane Chart Capacity @ Radius to Pour Station = _____ lbs. @ _____ Ft Radius

Crane Chart Capacity @ Radius to Point of Placement = _____ lbs. @ _____ Ft Radius

Minimum Crane Capacity = _____ lbs. @ _____ Ft Radius

Weight of Pick = Bucket + Concrete + Rigging = _____ x100 = _____ %

Minimum Crane Capacity

If less than 100% for a tower crane, OK.

If more than 90% for a mobile crane, review and approval by a licensed professional engineer is required

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Attachment 7: CRANE PACKET CHECKLIST

Complete Prior To Placing A Crane In Service

Project Name: _____ **#:** _____ **Subcontractor:** _____ **Date:** _____

	N/A	Rec.	Completed
Crane Assembly and/or Disassembly Director	()	()	()
Critical Lift Plan	()	()	()
Critical Lift Plan – PE Plan required above 90% capacity	()	()	()
Crane Assembly/ Disassembly Plan	()	()	()
Diagram & Calculations of Rigging & Hardware	()	()	()
Current Complete Annual Inspection	()	()	()
Current Monthly Inspection	()	()	()
Cribbing Calculation & Size for Outriggers	()	()	()
Ground Bearing Pressure below Crane Mats	()	()	()
Job/Activity Hazard Analysis (JHA/AHA)	()	()	()
Tandem Lift Data Sheet (when applicable)	()	()	()
Crane Hoisted Personnel Platform Permit (when applicable)	()	()	()
Concrete Bucket Lift Plan (when applicable)	()	()	()
Crane Operator's Certifications and Medical Card	()	()	()
Rigger and Signal Person Training Documentation	()	()	()
FAA and/or Street Permits	()	()	()
Crane Set-up Area Level for Outriggers or Tracks	()	()	()
Above & Below Ground Utility Locations Identified to Operator	()	()	()
Third Party Inspection (check)	()	()	()
Crane Load Chart and Radius Field Verified	()	()	()
Rigging Inspected	()	()	()
Start-up/Daily Crane Check	()	()	()
Owner Certificate of Compliance Signed and Placed in Cab (when applicable)	()	()	()

Competent Person in Charge of Lift: _____ Date: _____

Site Safety Manager Signature: _____ Date: _____

Superintendent Signature: _____ Date: _____

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SECTION 4 - DISCIPLINARY POLICIES

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 All individuals are required to follow company safety, health, and environmental policies and operating procedures. When required, individuals will be provided with additional training and information or retraining to maintain their knowledge.

2.2 Individuals found performing work in an unsafe manner that would endanger the individual or another individual as well as supervisors shall be subject to discipline that may include termination.

2.3 The Project Safety Manager and Project Superintendent will determine the appropriate disciplinary action to be taken based on the circumstances. AECOM TISHMAN CONSTRUCTION Human Resources will be contacted in regards to disciplinary actions pertaining to AECOM TISHMAN CONSTRUCTION employees. The steps to be taken, at a minimum, shall include the following, but not necessarily in this order:

- Verbal Warning. As the first step in correcting an unacceptable behavior or minor infraction, a verbal warning will be issued to the individual. This warning will be documented.
- Written Warning. If the unacceptable performance continues, or the severity of the infraction warrants, the next step will be a written warning. The written warning will clearly state the safety policy that was violated and steps the individual must take to correct it.
- Suspension. If the unacceptable practice continues or the severity of the infraction warrants, the individual will be suspended and given time off without pay. If a subcontractor employee, a letter shall be sent to the home office informing them of the removal from the project action.
- Termination. Individuals may be terminated if they do not improve their safety performance. As above, if a subcontractor employee, a letter shall be sent to the home office informing them of the removal from the project action.
- Immediate Termination. Any individual who commits a serious safety violation may be subject to immediate termination without prior notice in lieu of any verbal and/or written warnings. As above, if a subcontractor employee, a letter shall be sent to the home office informing them of the removal from the project action.

2.4 Fighting, possession or use of illegal drugs or weapons, or flagrant violations or disregard of project safety rules shall be handled in accordance with the company's Human Resources policies.

2.5 Violations of AECOM TISHMAN CONSTRUCTION fall protection policies shall be zero tolerance and up to immediate removal from the project. Refer to Fall Protection Zero Tolerance Policy.

2.6 It is the intent that these disciplinary procedures are educational rather than punitive; however, an individual or subcontractor individual who is terminated/removed from working at any AECOM TISHMAN CONSTRUCTION or AECOM TISHMAN CONSTRUCTION joint venture (JV) project where AECOM TISHMAN CONSTRUCTION is the sponsoring partner may not work on the project again either as an individual of AECOM TISHMAN CONSTRUCTION, a JV partner, or a project subcontractor. Further, individuals who are terminated/removed from a project may not transfer or work on any other AECOM TISHMAN CONSTRUCTION or JV project in any capacity for a minimum period of 180 days. Before being allowed to return to work on an AECOM TISHMAN CONSTRUCTION or a JV project the terminated/removed individual must meet with a Project Safety Director or his/her designee. The purpose of

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this meeting will be to assure that there has been a significant change in the terminated/removed worker's safe work attitude and to ensure appropriate safety training has taken place. The previously terminated/removed individual, if allowed to return to work on AECOM TISHMAN CONSTRUCTION or JV projects, will then be on probation for another 180 days. Any further safety violations by the subject individual during this probationary period will result in their permanent termination/removal from all AECOM TISHMAN CONSTRUCTION and/or JV projects.

2.7 In the event of a disagreement in disciplinary actions, the Regional Safety Director and the Regional EVP of Operations shall make the final determination.

2.8 Any individual on an AECOM TISHMAN CONSTRUCTION project is subject to disciplinary actions up to and including suspension and termination for gross safety infractions.

2.9 AECOM TISHMAN CONSTRUCTION may implement a financial penalty program for noncompliance to safety policies by subcontractors and/or vendors. The plan will be detailed in the site-specific safety program and/ or a safety Rider.

3. ATTACHMENTS

3.1 Example Letter Suspension/Termination to Subcontractors

3.2 Disciplinary Action Form

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Attachment 1: Example Suspension/Termination Letter to Subcontractors

(Insert Company Name & Address)

(Insert Date)

Dear (Insert Subcontractor Name),

This letter is to inform you that *(insert individual's name(s))* has been found to have violated AECOM TISHMAN CONSTRUCTION's Safety Program and has been *(insert suspended or removed)* from our project at *(insert project location)* for *(insert suspension duration)*. The infraction(s) included *(insert infraction(s))*.

At AECOM TISHMAN CONSTRUCTION, safety is a value and we want each worker to leave work the same way they arrive.

If you have any questions, please feel free to contact me.

Sincerely,

(Insert Superintendent signature and contact information)

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Attachment 2: Disciplinary Action Form

Individual Safety Citation

Project:

Name of Offending Party	Job Title/Company	Date of Occurrence
Person Who Observed Unsafe Act/Condition	Job Title/Company	Location of Occurrence
1. ☐ Not Wearing Hard Hat	5. ☐ Horseplay	9. ☐ Housekeeping
2. ☐ Not Wearing Eye, Ear, or Face Protection	6. ☐ Fall Protection	10. ☐ Alcohol/Drugs
3. ☐ Not Wearing Proper Clothing	7. ☐ Fighting/Altercation	11. ☐ Violating OSHA Rules
4. ☐ Throwing Material From Building	8. ☐ Disabling or Unhooking Safety Devices	12. ☐ Other

Description:

Disciplinary Action Taken	☐ Written Reprimand ☐ Suspension ☐ Termination	☐ Subcontractor Office Informed ☐ Subcontractor Site Mgt. Informed
Date and Time	☐ Retraining	☐ Other

Details:

Signature and Title of Person Issuing Citation

NOTE: I have read and acknowledge receipt of this citation and fully understand its contents.

Signature of Employee	Date	Witness Name and Job Title
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SECTION 5 - ELECTRICAL

1. TERMS and DEFINITIONS

1.1 Competent Person. One who is capable of identifying existing and predictable hazards in the surroundings or working conditions that are unsanitary, hazardous, or dangerous to employees, and who has the authorization to take prompt corrective measures to eliminate them.

1.2 Ground Fault Circuit Interrupter (GFCI). Electrical interrupting device that is sensitive to very low levels of current flow to ground.

1.3 Qualified Person. One who by possession of a recognized degree, certificate, or professional standing, or by extensive knowledge, training, and experience has successfully demonstrated his/her ability and given the authority to solve or resolve problems relating to electrical processes.

2. GENERAL REQUIREMENTS

2.1 All electrical site apparatus and equipment shall be of type listed by Underwriters' Laboratories, or Factory Mutual Laboratories for the specific application.

2.2 Flexible cords used with temporary and portable lights shall be designed for hard or extra hard usage as defined by the most current National Electric Code (NEC).

2.3 The National Electric Code, ANSI/NFPA 70 & 70E, in Article 400-4, lists various types of flexible cords, some of which are noted as being designed for hard or extra-hard usage. Examples of these types of flexible cords include hard service cord (types S, ST, SO, STO) and junior hard service cord (types SJ, SJO, SJT, SJTO).

2.4 Lighting shall not be less than:

- 5 foot - candles for access ways and storage areas.
- 5 foot - candles for general construction areas.
- 5 foot - candles for indoor, corridors, halls, exits.
- 10 foot - candles for indoor work areas.
- 30 foot - candles for first aid stations and offices.

All flexible, Temporary Electrical power distribution and Lighting installations shall be a minimum of 80 inches above the walking working surface, or high enough to not pose a position for incidental contact by mechanized equipment. Live cables will not be permitted to be stored, coiled, or banded together and placed on walking working surface at location of termination.

2.5 Perform a light meter review, if a light determination must be made.

2.6 Temporary lights shall not be suspended by their cords and must be equipped with guards to prevent contact with the bulb or must be designed to prevent breakage.

2.7 Light stringer conductors shall have S rated insulation. Open conductor light stringers are prohibited.

2.8 Temporary lighting circuits shall be separate circuits from temporary power distribution circuits except for light stringers designed by the manufacturer to be plug connected.

2.9 Fixtures and lamps used in wet locations shall be designed for such purposes.

2.10 All lamps for general illumination shall be protected from accidental contact or breakage. Metal-case lighting appliances shall be grounded.

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2.11 Competent Persons shall not permit employees to work in such proximity to any part of an electrical power circuit that he/she may contact in the course of his/her work unless the employee is protected against electric shock by de-energizing the circuit and grounding it or by guarding it by effective insulation or other means.

2.12 Before electrical work has begun the Competent Person shall determine whether any part of an electric power circuit, exposed or concealed, is so located that the performance of the work may bring any person, tool or machine into physical or electrical contact therewith. Individuals shall be advised of the hazards, and warning signs shall be posted and maintained.

2.13 All general use extension cords will be of a type listed by Underwriters' Laboratories and will be classified as Type SJ, SJO, SJT, SJTO, S, SO, ST, or STO as described in Table 400.4 of the most current edition of the National Electric Code. Every cord shall be equipped with a grounding conductor plug. Flexible cord shall only be used if durably marked on the surface with the type designation, size, and number of connectors. Electrical cords shall be used only in continuous lengths without splices or tape. Hard service flexible cords No. 12 or larger may be repaired if spliced so that the splice retains the insulation, outer sheath properties, and the usage characteristics of the cord being spliced. To the greatest extent possible, extension cords shall be kept off the floor by the use of hangers, stanchions, or other methods. If they must be on the floor they shall be protected in areas where traffic must run over them. Under no circumstances should extension cords be tied to mobile equipment. Weight-based stanchions shall have means to prevent tipping.

2.14 All installations shall comply with the National Electrical Safety Code, National Electrical Code, or applicable United States Coast Guard regulations.

2.15 Personnel familiar with code requirements and qualified for the class of work to be performed shall perform all work.

2.16 Live parts of wiring or equipment shall be effectively guarded to protect all persons or objects from harmful contact.

2.17 Temporary power lines, switch boxes, receptacle boxes, metal cabinets, and enclosures around equipment shall be plainly marked to indicate the maximum operating voltage. All temporary electrical panels or cabinets will be marked with current contact information for electrical supervisor responsible for identified electrical installation.

2.18 Plugs and receptacles shall not be interchangeable between circuits with different voltage and current ratings.

2.19 All 120 volts single-phase 15 and 20-ampere receptacle outlets which are not a part of the permanent wiring of the structure and which are in use by individuals shall have approved GFCI's. All hand held electrical tools are to be plugged into these GFCI's.

2.20 At the start of each project the Superintendent in conjunction with the Safety Manager shall determine who is responsible for testing and inspection of GCFI protection equipment.

2.21 Only qualified personnel authorized by the responsible electrical subcontractor shall install, modify, disconnect or maintain temporary and permanent electrical installations other than subcontractor owned cord and plug connected devices.

2.22 Subcontractors are required to ensure proper testing of their own equipment in accordance with this section and applicable laws, and shall submit documentation to the project Safety Manager as requested.

2.23 All generators must be GFCI compliant or utilize GFCI adaptors plugged in to its receptacles prior to any power cord.

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2.24 GFCI's do not replace a fuse or circuit breaker. If a receptacle or receptacles are installed as part of the permanent wiring of the building or structure and they are used for temporary electric power, GFCI protection shall still be provided.

2.25 GFCI's should be located as close as possible to the electrical equipment it is protecting.

2.26 GFCI's are to be inspected and tested, prior to first use, before the equipment is returned to use after repair, after any incident involving the equipment, and/or at least monthly or more frequently per specific jobsite requirements.

2.27 All inspections and tests shall be documented to include the identity of each receptacle, cord set, and/or other GFCI piece of equipment. The date and person conducting the inspection and/or test shall also be listed.

2.28 All projects are to be 100% GFCI compliant for all phases of the project. An assured grounding program may only be used in addition to this GFCI program.

2.29 Subcontractors are required to ensure proper testing of their own equipment in accordance with this program and shall submit documentation to the AECOM TISHMAN CONSTRUCTION Safety Manager as requested.

2.30 Overhead transmission and distribution lines shall be carried on towers and poles that provide safe clearances over roadways and structures.

2.31 Clearances shall be adequate for the movement of vehicles and for the operation of construction equipment.

2.32 Work activity adjacent to overhead lines shall not be initiated until a survey has been made to ascertain the safe clearance from energized lines.

2.33 Any overhead wire shall be considered energized unless the person owning such line or operating officials of the electrical utility supplying the line certifies that it is not energized and it has been visibly grounded and tested.

2.34 Operations adjacent to overhead lines are prohibited unless at least one of the following conditions is satisfied:

- Power has been shut off and positive means taken to prevent the lines from being energized.
- Equipment, or any part, does not have the capability of coming within the minimum clearance from energized overhead lines.

2.35 Storage batteries shall be stored in enclosures with outside vents or in well-ventilated rooms and be so arranged as to prevent the escape of fumes, gases, or electrolyte spray into other areas. Provisions shall be made for sufficient diffusion and ventilation of gases from storage batteries to prevent the accumulation of explosive mixtures.

2.36 Facilities for quick drenching of the eyes and body shall be provided for emergency use within 25 ft (7.6 m) of battery handling areas.

2.37 Use only insulated tools in the battery area to prevent accidental shorting across battery connections. For lead acid batteries, bicarbonate of soda to neutralize any acid spillage (1 lb/gal (0.1 kg/L) of water) shall be provided for flushing and neutralizing spilled electrolyte and for fire protection.

2.38 Battery charging installations shall be located in areas designated for that purpose. The charging apparatus shall be protected from mechanical damage.

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2.39 Exit from battery area shall remain unobstructed.

2.40 The following shall be available and used for the safe handling of the battery and protection of personnel:

- Safety glasses with side shields and face shields or goggles.
- Acid-resistant rubber gloves.
- Protective rubber aprons and safety shoes.
- Lifting devices of adequate capacity, when required.

2.41 All terminals and connections in all battery rooms are to be properly protected prior to energization. All entries to these rooms are to be controlled.

2.42 All arc flash exposures in all operations need to be in compliance with NFPA 70 E.

2.43 Open unused knockouts and holes on electrical panels and enclosures shall be protected using appropriately sized manufactured replacement covers.

2.44 All conductors entering the panel, enclosures, and other electrical devices shall have appropriate strain relief.

2.45 All electrical panels, disconnects, breakers, circuits shall be clearly labeled. All panels shall list the name and contact number of the electrical subcontractor responsible for them.

2.46 A shut off device for welders shall be provided on or near the welding machine.

2.47 Temporary underground electrical installations shall be encased on appropriate materials to prevent accidental line strikes. For utility strike avoidance refer to the Excavation Section of this program.

2.48 Follow all of the latest NFPA 70E standards as applicable.

3. ATTACHMENTS

None Applicable.

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SECTION 6 - EMERGENCY RESPONSE PLANNING

1. TERMS and DEFINITIONS

1.1 Emergency. An unplanned situation or event (including natural disasters) resulting in involvement of the public emergency services, police, fire, paramedic, or the environmental regulatory authorities.

1.2 Emergency Response Coordinator. An individual in a worksite or project environment designated to lead and direct the immediate emergency response.

1.3 Local Resilience Coordinator (LRC). A manager designated as the Office or Worksite lead for local level organizational resilience who may or may not be the emergency response coordinator.

1.4 First Aid Provider. A First Aid, CPR, and AED trained volunteer employee who provides emergency first aid or treatment (including performing CPR and applying an AED) to someone who is injured or suddenly ill, before emergency medical services (EMS) arrives. They may use a limited amount of equipment to perform initial assessment and provide immediate life support and care while awaiting arrival of emergency medical services.

2. GENERAL REQUIREMENTS

2.1 Emergency Response Plan (ERP)

- a) An assessment shall be completed by the manager of each location to determine the potential emergency situations and the adequate number of First Aid Providers, first aid supplies and medical requirements, including determining the response time and availability of Emergency Medical Services (EMS).
- b) Managers will establish and implement the location-specific ERP.
- c) The location-specific ERP will include:
 - The location of the muster point, first aid, fire extinguishers, fire exits, AED, and other emergency equipment.
 - Defined roles and responsibilities in the event of an emergency.
 - A contact list that includes, as applicable, fire, police, ambulance, poison control, First Aid Providers on location, fire wardens on location, Site Safety Officer, security, SH&E committee, SH&E Reporting number for reporting all AECOM TISHMAN CONSTRUCTION incidents, and other required emergency contacts.
 - Procedures appropriate to the potential emergency situations.
 - As applicable, maps to appropriate services, such as hospital or medical clinic.
- d) The location-specific Emergency Response Plan (ERP) will comply with all governing regulations.
- e) The location-specific ERP shall be included in the location specific Office Safety, Health & Environment Plan or the project specific SH&E Plan.
- f) If the hazard assessment for the location indicates a need for planned evacuation or rescue, appropriate written procedures will be developed and implemented.

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- Depending upon the various contributing factors to the potential emergencies, the procedures may require coordination with a third party rescue provider, or preparations for mass evacuation away from a site. If applicable, procedures should be developed to assist any personnel with disabilities in the event of an evacuation. Engage Human Resources to ensure the proper measures are in place.
- g) The location-specific emergency plan will be readily available to personnel.
- h) Appropriate methods to account for AECOM TISHMAN CONSTRUCTION employees, subcontractor employees, and visitors shall be established.
- i) Staff will be trained for involvement in an emergency evacuation or rescue; however, all evacuations may require special preparation and arrangements with third party rescue providers in the following circumstances:
 - Work at high elevations
 - Work in confined spaces or where there is a risk of entrapment
 - Work with hazardous substances
 - Underground work
 - Work on or over water
 - Work in remote isolation
 - Workplaces where there are persons who require physical assistance to be moved.
- j) The ERP will address clear paths of travel to and from a working area.
- k) All staff will be advised of the location of first aid services, equipment, and supplies.
- l) The ERP shall be tested for deficiencies through emergency response drills annually or more frequently as required by legislation. Emergency drills such as man-down, earthquake/hurricane/tornado drill, security, active shooter, bomb threat, and first aid are recommended to be conducted and lessons learned documented quarterly. At a minimum, a documented evacuation drill is required twice a year.
- m) The ERP shall be reviewed annually or more frequently as required by legislation.
- n) ERPs shall include the following sections:
 - Emergency Contact List
 - Site Plan with Access Points
 - First Aid Injury
 - Medical Clinic Injury
 - Serious Accidents / 911
 - First Hour Response Checklist
 - Messages for the Media

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- Motor Vehicle Accidents
- Evacuation
- Fire
- Utility Emergency
- Weather
- CPR
- Heat Illness Prevention
- Active Shooter Response
- Bomb Threat
- Other/unique project-specific requirements

2.2 Communications

- a) Supervisors are responsible for confirming that crews have access to communication devices that are in good working order, have reception in the area in which the crews will be working, and meet the needs of the planned check-in and emergency response procedures. Subcontractors shall ensure the proper equipment for their employees.
- b) All individuals are responsible for maintaining the communication devices in good working order before leaving for the field and for ensuring that battery-operated electronic devices have been recharged or have fresh batteries.

2.3 Visitors

- a) In the event of an evacuation, visitors will be escorted by their host to the muster point.
- b) For in-house meetings, safety orientations will be delivered before the meeting begins so all visitors are aware of the evacuation routes and procedures.

3. ATTACHMENTS

3.1 Emergency Response Plan Template Example

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Attachment 1: Emergency Response Plan Template Example

Emergency Response Plan for the _____ Project located at _____

The main project phone number is: _____

Emergency Contact List

(insert names and phone numbers)

- Superintendent:
- Safety Manager:
- Project Manager:
- Project Director/Executive:
- Assistant Superintendent:
- Executive Officer:
- Area/Regional Safety Manager:
- Vice President of Safety:
- Legal Department:
- Insurance Carrier:
- HR/Corporate Communications:

Emergency Services

- EMS:
- Hospital:
- Police:
- Fire Department:
- HAZMAT Response:

Utility Companies

- Power Company:
- Gas Company:
- Phone Company:
- Water Company:
- Traffic Signal Repair:

Governmental Agencies/Offices (List only those applicable)

- OSHA: (Note: only the Safety Department can notify OSHA)
- MSHA: (Note: only the Safety Department can notify MSHA)
- EPA:
- HazMat:
- Coast Guard:

Note: Add other specific numbers relevant to your project.

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First Aid Injury

A. Procedures:

1. 1. Notify:

(insert names and phone numbers)

- Superintendent
 - Project Manager:
 - Safety Manager:
 - On-Site Medic:
 - Others as necessary (e.g. Owner):
2. For non-serious injuries, the injured worker should report to the on-site medic (or personnel trained in first-aid) for evaluation and treatment.
 3. Employer shall conduct post-incident testing in accordance with AECOM TISHMAN CONSTRUCTION's Fitness for Duty policy. (Note: Subcontractors are responsible for testing their own employees. AECOM TISHMAN CONSTRUCTION shall not test subcontractor employees.)
 4. On-site medic and/or Safety Manager shall follow-up with the injured worker as necessary.
 5. All AECOM TISHMAN CONSTRUCTION employee injuries shall be reported to the AECOM Injury Management Hotline (512-419-5016). Non-employee injuries should not be reported to this hotline.
 6. Reference the Incident Reporting, Investigation, and Claims Management section if additional treatment is required.

B. First Aid Trained Personnel:

1. Insert name and phone number of on-site medic here, if applicable.
2. Insert name(s) and phone number(s) of AECOM TISHMAN CONSTRUCTION staff with first aid training here.

C. Reporting:

1. The Safety Manager or member of the project team must submit all first aid incidents into IndustrySafe.

Medical Clinic Injury

A. Procedures:

1. Notify:

(insert names and phone numbers)

- Superintendent:
- Project Manager:
- Safety Manager:
- On-Site Medic:
- Area Safety Manager:
- Regional Safety Director:
- Others as necessary (e.g. Owner):

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2. A Medical Authorization for Treatment Form must be filled out and signed by an AECOM TISHMAN CONSTRUCTION employee, and given to the injured worker to take to the clinic.
3. An AECOM TISHMAN CONSTRUCTION or subcontractor management member shall accompany the injured worker to the clinic for evaluation and treatment.
4. Obtain a copy of the paperwork from the clinic which includes the diagnosis, treatment, work restrictions, and follow-up requirements.
5. Employer shall conduct post-incident testing in accordance with AECOM TISHMAN CONSTRUCTION's Fitness for Duty policy. (Note: Subcontractors are responsible for testing their own employees. AECOM TISHMAN CONSTRUCTION shall not test subcontractor employees.)
6. On-site medic and/or Safety Manager shall follow-up with the injured worker and continue to monitor their condition until any restrictions are removed.
7. Late reporting (>24 hours) of any incident may result in a monetary violation of \$5,000. If a Subcontractor fails to timely submit more than five reports, the monetary violation may increase to \$10,000 per occurrence. The deduction shall be made monthly from the Subcontractor's pay requisition.
8. All AECOM TISHMAN CONSTRUCTION employee injuries shall be reported to the AECOM Injury Management Hotline (512-419-5016). Non-employee injuries should not be reported to this hotline.

B. Medical Clinic Location:

1. Nearest medical clinics:

Insert clinic name, address and link to Google Maps here.

C. Reporting:

1. Safety Manager shall review the paperwork from the clinic to determine if the incident shall be classified as first aid, recordable, recordable with restrictions, or lost time.
2. The Safety Manager or member of the project team must submit all first aid, recordable and lost time incidents within IndustrySafe.

Serious Accidents / 911

A. Procedures:

1. Call 911 if anyone is seriously injured. Administer first aid and/or CPR until medical professionals arrive on the scene. Do not move the injured person(s) if there is a potential for further injury.
2. Secure the site to avoid further damage or injuries.
3. Notify:

(insert names and phone numbers)

- Superintendent:
- Project Manager:
- Safety Manager:
- On-Site Medic:
- Area Safety Manager:

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- Regional Safety Director:
 - Others as necessary (e.g. Owner):
4. Post AECOM TISHMAN CONSTRUCTION employees at access points to receive the ambulance and assist EMS personnel.
 - List each access point and responsible AECOM TISHMAN CONSTRUCTION team members.
 5. Keep someone posted at the job site trailer's telephone.
 - List responsible AECOM TISHMAN CONSTRUCTION team member.
 6. Determine which hospital the injured worker will be transported to. An AECOM TISHMAN CONSTRUCTION or subcontractor management member should accompany the injured worker to the hospital.
 7. Additional Notifications:

(insert names and phone numbers)

- Project Executive:
 - Vice President of Safety:
 - Executive Officer/ Business Unit Leader:
 - Communications VP:
 - Injured Worker's Family
 - AECOM Rapid Response Hotline if applicable: 1-877-232-6607. Contact Project Executive before calling.
8. If media is present, or contacts the project for a statement, notify the Communications Team at media@aecom.com. Only the Communications Team is to answer questions or give a statement.
 9. Take photographs, write down names and addresses of witnesses. Follow the incident investigation and reporting procedures.
 10. Obtain a copy of the paperwork from the hospital which includes the diagnosis, treatment, work restrictions, and follow-up requirements.
 11. Employer shall conduct post-incident testing in accordance with AECOM TISHMAN CONSTRUCTION's Fitness for Duty policy. (Note: Subcontractors are responsible for testing their own employees. AECOM TISHMAN CONSTRUCTION shall not test subcontractor employees.)
 12. On-site medic and/or Safety Manager shall follow-up with the injured worker and continue to monitor their condition until any restrictions are removed.
 13. All AECOM TISHMAN CONSTRUCTION employee injuries shall be reported to the AECOM Injury Management Hotline (512-419-5016). Non-employee injuries should not be reported to this hotline.

B. First Aid/CPR/AED Trained Personnel:

1. Insert name and phone number of on-site medic here, if applicable.
2. Insert name(s) and phone number(s) AECOM TISHMAN CONSTRUCTION staff with first aid/CPR/AED training here.

C. Hospital Locations:

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1. Nearest hospitals with an emergency room for general medical care:

Insert hospital name, address and link to Google Maps here.

2. Nearest hospitals with trauma unit:

Insert hospital name, address and link to Google Maps here.

D. Fire Departments:

1. Responding fire station(s):

Insert fire station name, address and link to Google Maps here.

2. Responding Hazardous Materials (HAZMAT) team:

Insert HAZMAT team name, address and link to Google Maps here.

E. Reporting:

1. Safety Manager shall review the paperwork from the hospital to determine if the incident shall be classified as first aid, recordable, recordable with restrictions, or lost time.
2. The Safety Manager or member of the project team must submit all first aid, recordable and lost time incidents within IndustrySafe.
3. In the event of a critical incident, the most senior AECOM TISHMAN CONSTRUCTION employee available is to call AECOM's Rapid Response Hotline (1-877-232-6607) in order to alert the AECOM TISHMAN CONSTRUCTION crisis team. The Project Executive must give permission for the call to be made.
4. The Regional Safety Director shall notify OSHA as follows:
 - Within 8 hours for fatalities, and
 - Within 24 hours for in-patient hospitalizations of 1 or more employees, amputations, or losses of an eye.

(Note: For subcontractor employees, Regional Safety Directors shall coordinate with the employer to notify OSHA).

First Hour Response Checklist

Superintendent

- Call 911.
- Contact the Safety Manager.
- Initiate site control and make certain that all workers are accounted for.
- If the site will be shut down, tell workers when they are to report back to work and that counseling will be available (if applicable). Also, direct information requests from outside group to project management. Keep select individuals on-site to help with the incident.
- Do not move anything that could be classified as evidence.
- Ensure telephone coverage at the site.
- Post workers to restrict entry to the site.

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- Establish a command center, generally the AECOM TISHMAN CONSTRUCTION office.
- Notify the owner/developer of the project.

Project Manager

- Determine what happened, when/where it happened, and who is involved.
- Notify corporate management (insert name(s)/number(s)).
- Verify the current status of the site (shut down?).
- Determine whether you and/or spokesperson are needed on site and let the Project Director/Executive know.
- Advise the receptionist how to route calls.
- Identify potential spin-off crises.
- Notify human resources.
- Notify the Corporate Safety Department and Risk Management.

Safety Manager

- Follow the incident investigation guidelines.
- Document the entire incident scene.
- Fill out the initial investigation report and submit it.
- Communicate to Regional Safety Director and Risk Management.

Regional Executive Officer

- Maintain close contact with the Superintendent and Project Manager to determine level of involvement.
- Review/approve all statement/communications to the news media.
- Work closely with legal counsel.
- In the event of an injury/fatality, be prepared to designate a person to make the call/visit to the family.
- In the event of a highly visible crisis, be prepared to make the initial statement to the news media with no Q&A.
- Oversee communications with employee base and other outside stakeholders.

Communications Department

- Write, and get clearance for, all statements and releases from the EVP.
- Designate someone to screen your calls from the news media.
- Complete media log sheets.
- Anticipate media questions. If possible, role play a media interview with a colleague before going live.
- Assemble necessary background information and literature.
- Advise reporters of a time and place for future updates.
 - Follow-up on additional media inquiries.

*Leadership must coordinate to determine if the AECOM Rapid Response Hotline (1-877-232-6607) should be called.

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SECTION 7 - ENVIRONMENTAL ISSUES

1. TERMS and DEFINITIONS

1.1 Asbestos. An incombustible, chemical-resistant, fibrous mineral used for fireproofing, electrical insulation, building materials, brake linings, and chemical filters.

1.2 Asbestos containing material (ACM). Any material containing more than 1% asbestos.

- Friable Asbestos. Used for fireproofing, insulation, or sound proofing are considered to be friable, and they readily release airborne fibers if disturbed.
- Non-friable Asbestos. Vinyl-asbestos floor tile or roofing felts are considered non-friable and generally do not emit airborne fibers unless subjected to sanding or sawing operations.

1.3 Chemical Protective Clothing. Apparel that is constructed of relatively impermeable materials with the intent to act as a barrier to physical contact of the worker with potentially hazardous materials in the workplace. Such materials include: Tyvek coveralls (all types), CPF Series coveralls, and polyvinyl chloride (PVC) coveralls and rain suits.

1.4 Class I - Asbestos work. Activities involving the removal of thermal system insulation (TSI) and surfacing asbestos containing material.

1.5 Class II - Asbestos work. Activities involving the removal of ACM that is not TSI or surfacing material. This includes removal of asbestos-containing gaskets, packing, wallboard, floor tile and sheeting, roofing and siding shingles, and construction mastics.

1.6 Class III - Asbestos work. Includes repair and maintenance operations where ACM, including TSI and surfacing material, is likely to be disturbed to the extent that renders ACM friable or generates visible debris. Class III asbestos work is limited to cutting away small amounts of ACM, no greater than the amount which can be contained in one standard sized glove bag or waste bag in order to access a building component. In no event shall the amount of disturbed ACM exceed that which can be contained in one glove bag or waste bag measuring 60 inches in length and width.

1.7 Class IV - Asbestos work. Includes custodial activities during which employees are involved in clean-up activities of waste and debris containing asbestos containing material.

1.8 Competent Person. A designated contractor employee who has the authority to take prompt corrective actions and has received training and certification equivalent to the EPA's Model Accreditation Plan and equivalent training as conducted by the National Asbestos Center, at the manager or supervisor level, and thereby is knowledgeable in:

- Identifying asbestos hazards in the workplace.
- Selecting appropriate control strategies for asbestos exposure.
- The contents of the OSHA asbestos regulations.
- Work practices for safe asbestos removal/clean-up.

1.9 Compliance Map. A document defining and detailing the actions necessary to assure compliance with applicable environmental legal requirements.

1.10 Frostbite. A condition in which part of the body is frozen. Some of the symptoms may include:

- Lost sensation of touch, pressure, and pain, which may occur without awareness of any numbness or other sensation.

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- Just before freezing, the skin becomes bright red.
- At freezing, small patches of white appear on skin.
- Skin becomes elastic.

1.11 Generator of Hazardous Wastes. For example, based on the Resource Conservation and Recovery Act (RCRA) and Environmental Protection Agency (EPA) regulations in the United States, a hazardous waste generator is any person or site whose processes and actions create hazardous waste.

1.12 Hazardous Wastes. According to EPA regulations, a hazardous waste is a waste with properties that make it dangerous or potentially harmful to human health or the environment.

1.13 Heat Stress. A common hazard to individuals working on projects involving exposure to hazardous substances, most particularly when impermeable protective clothing is used. This problem can occur at ambient temperatures below what is normally considered "hot weather." The body normally sheds excess heat primarily through radiation (capillaries in the skin dilate, transferring heat from the body core to the surface), and evaporation of sweat. Thermodynamically, evaporation is much more effective than radiation. Heat stress results when the body's regulating mechanisms are inadequate to dispose internally generated and externally supplied heat.

1.14 Housekeeping. This term refers to the general nature of cleanliness on a project site which is further defined as the absence of clutter and waste building materials, food and beverage containers, pieces of broken machinery or machinery parts or any other materials that could result in any form of trip hazard, visual clutter or unhealthy environment.

1.15 Hypothermia. A condition when a person's body loses heat faster than it can be produced. If body temperature drops to 95°F (98.6°F is normal), uncontrollable shivering occurs. If body temperature continues to cool, the following symptoms may also occur:

- Vague, slow, slurred speech.
- Forgetfulness, memory lapses.
- Inability to use hands.
- Frequent stumbling.
- Drowsiness.
- Impairment of judgment.

1.16 Presumed asbestos containing material (PACM). Thermal insulation and surfacing material found in buildings constructed no later than 1980.

1.17 Storage. Per EPA regulations, storage is defined as the temporary holding of waste before the waste is treated, disposed of or stored somewhere else.

1.18 Subject Matter Expert. A person who is an expert in a particular topic or area based on experience, technical/regulatory knowledge, and/or training.

1.19 Surfacing material. Material that is sprayed troweled-on or otherwise applied to surfaces (such as acoustical plaster on ceilings and fireproofing materials on structural members or other materials on surfaces for acoustical, fireproofing and other purposes).

1.20 Thermal system insulation. ACM applied to pipes, fittings, boilers, breeching, tanks, ducts or other structural components to prevent heat loss or gain.

1.21 Treatment and Disposal. Per EPA regulations, treatment and disposal is any process that changes the physical, chemical or biological characteristics of a waste to minimize its threat to the environment.

2. GENERAL REQUIREMENTS

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2.1 Quality Control.

- a) AECOM TISHMAN CONSTRUCTION will establish and maintain quality control for the environmental items and procedures in this section.
- b) A record will be made on the daily reports of any problems as well as the corrective action taken to maintain environmental control as required by governing regulations and these procedures.

2.2 Protection of Environmental Resources.

- a) AECOM TISHMAN CONSTRUCTION will protect environmental resources inside the project boundaries and those affected outside the limits of work during the entire period of the contract.
- b) Activities will be confined to areas defined by the specifications and drawings.

2.3 Erosion and Sediment Control.

- a) Strict compliance with the approved Erosion and Sediment Control Design Plan shall be maintained. The Plan shall be maintained at the site for use by inspectors during their periodic inspections. Drawings that illustrate silt and water run-off control and details shall be created as necessary.

2.4 Hazardous Spills.

- a) All Subcontractors are required to immediately notify AECOM TISHMAN CONSTRUCTION of any spillage or leakage of hazardous materials, oils, substances, and waste, regardless of the quantity.
- b) Examples of hazardous items include, but not limited to the following: gasoline, fuel, oil, kerosene, acetone, equipment oil (waste or fresh), certain paints and coatings, contaminated soil and water etc. Hazardous items, similar to those mentioned here, are never to be deposited directly on or into the soil or water but must be in a proper container for disposal. Please refer to the Hazard Communication section in this manual.
- c) Containment shall be affected in the following manner:
 - Temporary berms or dikes shall be placed around the spill or leak to contain the hazard until the arrival of the appropriate AECOM TISHMAN CONSTRUCTION or Subcontractor representative to direct the clean-up operations.
 - In case of small spills, the material spilled shall be collected into leak proof containers for salvage or off-site disposal. This shall also apply to soil, water and cleaning materials contaminated with the hazardous spilled material.
 - In the event of a large scale spill or leak, a licensed waste management company may be required to provide the clean-up and disposal service. AECOM TISHMAN CONSTRUCTION shall receive copies of all disposal documents.

2.5 General Construction Waste.

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- a) All construction debris and waste will be placed in a project designated container. It will be removed on a regular basis via a dumpster service and will be disposed of in the proper approved location.
- b) Daily project clean-up will be performed. This includes trash collection, cleaning of on-site roadways and off-site roadways if required. All clean-up activities will be monitored by the AECOM TISHMAN CONSTRUCTION project team and enforced by the AECOM TISHMAN CONSTRUCTION Superintendent (s) on a daily basis.
- c) In addition to the hazardous waste plans presented herein, the waste removal subcontractor shall be required to present the appropriate hazardous and solid waste licenses/permit(s) and landfill/disposal manifest(s), etc., prior to removal of any hazardous and solid waste classified and/or regulated materials. Copies shall be given to AECOM TISHMAN CONSTRUCTION and shall be kept in AECOM TISHMAN CONSTRUCTION permanent project files.
- d) Trash generated by eating, drinking, etc., will be required to be placed in strategically located trash cans or containers. Subcontractors must provide trash cans or containers near employee break areas. Trash will be removed periodically in order to prevent vermin infestation.
- e) Open incineration to dispose of scrap or waste materials at the site shall not be permitted.
- f) Concrete washout stations shall be provided and shall be designed to keep the concrete water from entering the ground.

2.6 Protection of Water Resources.

- a) All natural water resources will be protected from infiltration of waste materials, uncontrolled project water run-off, and deposit of sediment. Existing sewer systems will be protected by covering inlets with filter cloth to prevent infiltration of sediment and spillage from the project, if required, and hay bales will be placed to control mud and sediment. Follow the project sediment/wash out control plan as necessary.
- b) Washing, curing, and waste water directly derived from construction activities will be disposed of properly.

2.7 Protection of Air Resources.

- a) No open burning of any type other than those involving normal construction activities will be allowed on the project unless approved by AECOM TISHMAN CONSTRUCTION management.
- b) Equipment, activities, and work operated or performed shall be in strict accordance with the state and local air pollution statute and Federal emission performance laws and standards.

2.8 Dust/Mist Control.

- a) All dust producing operations shall comply with all federal, state, and local regulations and other procedures of accepted good practice to minimize generation and/or migration of dust from the construction site.
- b) All subcontractors engaged in paint spraying, coating, sealing or other spraying operations shall utilize best practice procedures to prevent material from leaving the jobsite.

2.9 Cleaning of Site Access Roads.

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- a) Wind born dust problems shall be sprinkled with water, covered with suitable restraining barriers, or treated with wetting agent to reduce the generation of dust.
- b) All sweeping operations, whether by hand brooms or power sweepers, shall use a means to prevent dust generation.

2.10 Noise Control.

- a) The hours of construction permitted by the governing jurisdiction shall be strictly followed. During construction, noise shall be kept within acceptable levels as required by all federal, state, and local codes and regulations. If noise rises above these levels, engineering controls and/or administrative controls shall be required to lower the noise level.

2.11 Post-Construction Cleanup and Restoration.

- a) Following completion of the project, all areas in and around the construction project will be cleaned in accordance with contract requirements/language/local laws and regulations.
- b) The construction area will be left in a clean condition and may include off-site disposal of all excess material, debris and rubbish as required.

3. PERMITS

3.1 Air Permits – If a project will result in emissions to the air, the project may require an air permit. It is important to work closely with the client as the permit may need to be applied for by the client.

3.2 Water Permits – For projects that involve discharges to receiving bodies of water (rivers, streams, etc.), storm sewers and sanitary sewers, AECOM TISHMAN CONSTRUCTION needs to work closely with our clients to ensure all water related permits are obtained prior to start-up of the projects.

3.3 Waste Management and Waste Storage Permits – Federal, state, provincial and local environmental agencies have many regulations related to proper waste management (i.e. approval of spill containment plans, wastes placed in drums and containers must be properly labeled, etc.). Ensure appropriate approvals and permits are in place where required may require working closely with the client to obtain required approvals and/or permits. For projects where a client is authorizing AECOM TISHMAN CONSTRUCTION to manage their waste, AECOM TISHMAN CONSTRUCTION employees must never sign a waste manifest indicating AECOM TISHMAN CONSTRUCTION is the generator of the waste.

4. ABRASIVE BLASTING

4.1 Abrasives and the surface coatings on the materials blasted are shattered and pulverized during blasting operations and the dust formed will contain particles of respirable size. The composition and toxicity of the dust from these sources shall be considered in making an evaluation of the potential hazards.

4.2 Aisles and walkways shall be kept clear of steel shot or similar abrasives which may create a slipping hazard.

4.3 Air Compressors

- a) Air compressors must be located in a well-ventilated area.
- b) Due to the high pressure that air compressors create, precautions must be taken to prevent unleashing of strong forces that can cause serious bodily injury.

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- c) Air for abrasive blasting respirators must be free of harmful quantities of dust, mists, or noxious gases and must be inspected daily, prior to use and comply with OSHA 29 CFR 1910.134(i) (Respiratory Protection).
- d) Never adjust the pressure setting on a compressor above the blast equipment maximum working pressure rating.

4.4 Hoses and Connectors

- a) Couplings must have safety wires in place and be secure as required by federal safety regulations.
- b) The operator shall be responsible to ensure that each coupling has safety wires in place.
- c) Whip checks must be installed at bull hose connections.
- d) Operator should hold onto the blast hose until the air pressure from the nozzle drops off to zero.
- e) Tape shall not be used to repair a blown-out hose.
- f) Immediately replace a hose if a blowout or leak occurs.
- g) Hose ends must come into contact with coupling gaskets to prevent leaks and to maintain static electricity conductivity.

4.5 Nozzles and Remote Controls

- a) Blast nozzles shall be bonded and grounded to prevent the buildup of static charges.
- b) Blast cleaning nozzles shall be equipped with an operating valve which must be held open manually.
- c) Never tape, strap, or tie down an air actuated remote control lever or choke electric remote control switch.

4.6 Abrasive Blasting Personal Protective Equipment

- a) A specific work-site Job Hazard Analysis shall be developed where respirators or Type CE blasting hoods/helmets are required to protect the health of the operator. The procedure shall comply with OSHA 29 CFR 1926.103 (Respiratory Protection).
- b) Equipment for the protection of eyes, face and body shall be supplied to the operator when the respirator design does not provide such protection and to any other personnel working in the vicinity of abrasive blasting operations. This equipment shall conform to the requirements of OSHA 29 CFR 1926.102 (Eye and Face Protection).
- c) Protection shall be provided to any other personnel working in the vicinity of abrasive blasting operations.
- d) Secure hoses by tying them to stable secure equipment or material. This material must be easily displaced or moved by pressure from the hose.
- e) Before using any blasting abrasive, check the SDS to find out the chemical composition of the abrasive material.

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- f) Ventilation systems and dust collectors may be necessary in enclosed conditions.
- g) Hearing protection shall be used as necessary per OSHA standards.
- h) Blaster must wear heavy-duty gloves and steel toe boots.
- i) Helmet lenses should be changed as soon as pitting or frosting takes place.

5. ASBESTOS

5.1 General Requirements

- a) A licensed and properly insured asbestos abatement subcontractor must conduct the asbestos removal.
- b) Safety, health, and industrial hygiene oversight must be provided to the subcontractor by an independent environmental/industrial hygiene consultant subcontractor (referred to as environmental subcontractor).

5.2 Activity-Specific Requirements

- a) Wet methods, or wetting agents, to control environmental exposures during asbestos handling, mixing, removal, cutting, application, and cleanup, except where subcontractors demonstrate that the use of wet methods is infeasible due to for example, the creation of electrical hazards, equipment malfunction.
- b) Local exhaust ventilation equipped with HEPA filter dust collection systems.
- c) Enclosure or isolation of processes producing asbestos dust.
- d) Ventilation of the regulated area to move contaminated air away from outside or common areas and toward a filtration or collection device equipped with a HEPA filter.
- e) Use of other work practices and engineering controls to be determined by individual, company or federal, state and/or local requirements.

5.3 Housekeeping

- a) Vacuuming. Where vacuuming methods are selected, HEPA filtered vacuuming equipment must be used. The equipment shall be used and emptied in a manner that minimizes the reentry of asbestos into the environment.
- b) Waste disposal. Asbestos waste, scrap, debris, bags, containers, equipment, and contaminated clothing consigned for disposal shall be collected and disposed of in sealed, labeled, impermeable bags or other closed, labeled, impermeable containers.

5.4 HAZWOPER or Hazardous Waste Activities

- a) Asbestos may be present at HAZWOPER or hazardous waste sites as a soil contaminant or building material. If such contamination is noted, the following requirements will be observed:
 - The site-specific SH&E plan will provide a specific analysis of the asbestos exposure hazard for each task involving the disturbance or handling of ACM/PACM.

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- If any potential is identified for environmental exposures, then specific asbestos exposure control and monitoring procedures will be developed for the work activity and included in the SH&E Plan.

5.5 Asbestos Removal / Abatement Activities

- a) The past widespread use of ACM in buildings is widely recognized as the source of a significant exposure hazard for employees performing building demolition/renovation activities. For the purpose of this procedure, asbestos removal / abatement activities include, but may not be limited to, the following:
 - Conducting Asbestos surveys.
 - Removing and disposing of ACM encountered during renovation and construction activities.
 - Performing visual observations either for the intent of verifying the removal subcontractor is complying with the approved work plan/removal specifications, and/or verifying the asbestos abatement activities conducted by the removal subcontractor have been sufficiently performed.
 - Documenting daily work activities.
 - Collecting air samples that include background, work-in-progress, and final air clearance samples.

5.6 When soil or bedrock is to be disturbed that is suspected as containing asbestos, an assessment shall be conducted to confirm presence and quantity.

- a) Assessment should include consultation of geological surveys.
- b) As per jurisdictional requirements, if the results confirm ACM, appropriate planning and control measures shall be implemented.

5.7 Asbestos Surveys

- a) Prior to commencement of possible Asbestos work activities, a thorough inspection and sampling program will be completed throughout the demolition or renovation area to identify the presence of ACM, PACM and, as applicable, to determine the appropriate classification of the activities.
- b) All individuals conducting asbestos surveys shall have completed and successfully passed the jurisdictionally required training course(s). Any associated refresher training or additional training given the potential hazards shall be completed and successfully passed in accordance with any established timeframes. Subcontractors are solely responsible for this training.
- c) Personal Protective Equipment - Require that appropriate PPE items are available and worn during asbestos surveys.
- d) Respiratory protection shall be worn until such assessments have been conducted, and it is determined that respiratory protection is not warranted.
- e) Where feasible, ACM will be removed prior to commencement of general work activities.

5.8 Asbestos Abatement Plan

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- a) The hazards of the abatement work and associated control measures will be addressed in an Asbestos Abatement Plan, which will be reviewed and approved by the SH&E Manager prior to implementation. It shall include, as applicable:
- Classification and/or project description.
 - Training requirements of workers, supervisor(s) and competent worker(s).
 - Permit and notification requirements.
 - Assessment and monitoring requirements and procedures.
 - Environmental/Exposure controls; including, but not limited to, barriers, signage, air-filtration, required equipment, containment procedures, decontamination facilities, worker hygiene and personal protection equipment.
 - Specific removal, encapsulation or enclosure procedures.
 - Disposal considerations and procedures.
 - A demolition or renovation notification, notice of project, etc. will be filed within the appropriate time frame (if specified) and approved as required by the governing authority (e.g. EPA, inspector, etc.) and applicable local construction regulatory agencies prior to commencing demolition activities.

5.9 Medical Monitoring and Respirator Requirements

- a) Respirator use. All abatement subcontractor employees and others must wear a respirator when entering into an active regulated abatement area. Respirator selection must be in accordance with Table 1 of this section.
- b) All abatement subcontractor employees and others wearing a respirator must satisfactorily be participating in a Respirator Program which meets the minimum requirements found in this program.
- c) Fit-test. Employees must be up-to-date on a respirator fit-test. Fit-tests must be conducted at intervals not exceeding 12 months.
- d) Medical Surveillance. All employees entering into a regulated area and wearing a respirator must be in a medical surveillance program under the direction of a physician. This medical surveillance is frequently referred to as the annual asbestos physical.
- e) Physician(s) written opinion. Prior to donning a respirator and entering a regulated area, each employee must have a current (within the past 12-months) written opinion by a physician indicating that the employee is medically capable of wearing a respirator and is cleared. to work in an asbestos abatement environment.

Table 1:

AIRBORNE CONCENTRATIONS OF ASBESTOS IN FIBERS/CUBIC CENTIMETER (F/CC)	NIOSH/MSHA REQUIRED RESPIRATOR
Not in excess of 1 f/cc [(10) x PEL]	Half-mask air purifying respirator other than a

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	disposable respirator, equipped with HEPA filters.
Not in excess of 5 f/cc [(50) x PEL]	Full face piece air purifying respirator equipped with HEPA filters.
Not in excess of 10 f/cc [(100) x PEL]	Any powered air purifying respirator equipped with HEPA filters or any supplied air respirator operated in continuous flow mode.
Not in excess of 100 f/cc [(1,000) x PEL]	Full face piece supplied air respirator operated in pressure demand mode.
Greater than 100 f/cc [(1,000) x PEL] or unknown concentration	Full face piece supplied air respirator operated in pressure demand mode, equipped with an auxiliary positive pressure SCBA.

5.10 Training

- a) All individuals who work in areas that contain or may contain asbestos and have the potential to disturb the ACM shall complete Asbestos Awareness Training. The training shall be documented and provided to AECOM TISHMAN CONSTRUCTION as requested.

5.11 Regulated Areas

- a) All Class I, II and III asbestos work, or the type or class as determined by the given jurisdiction, will be conducted within regulated areas. All other operations will be conducted within a regulated area if airborne concentrations of asbestos exceed, or there is a reasonable possibility they may exceed the Established Exposure Limit or Excursion Limit. Regulated areas will comply with the requirements below.
- b) Demarcation – The regulated area will be identified and controlled in any manner that minimizes the number of persons within the area and protects persons outside the area from exposure to airborne asbestos.
- c) Post warning signs around perimeter and at all entryways into the work area. Signs shall be at least 20 inches (51 centimeters) by 14 inches (36 centimeters). Signs will be provided and displayed with the following legend:
 - DANGER
 - ASBESTOS
 - CANCER AND LUNG DISEASE HAZARD
 - AUTHORIZED PERSONNEL ONLY
 - RESPIRATORS AND PROTECTIVE CLOTHING ARE REQUIRED IN THIS AREA
- d) Signs and labels shall identify the material which is present, its location, and appropriate work practices which, if followed, will confirm that ACM and/ or PACM will not be disturbed.
- e) Additional information may be required given jurisdictional requirements.
- f) Asbestos spills shall be cleaned up promptly.
- g) The spraying of any substance containing any amount of asbestos in or upon a building or other structure during its construction, alteration, or repair is prohibited.
- h) Where cleaning is performed by vacuuming methods, a high efficiency particulate air filter (HEPA) shall be utilized for removal of asbestos from all air exhausted by the vacuum system

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or device. The system or device shall be checked for filtration efficiency performance on a monthly basis.

- i) All exposed surfaces within the removal area will be final-cleaned by HEPA vacuuming and wet cleaning prior to visual inspection by the competent person to confirm complete removal of visible ACM.
- j) All accessible and exposed surfaces will be visually inspected. Final clearance sampling will be performed as required.
- k) Asbestos waste shall be collected and labeled according to applicable regulations.
- l) Asbestos waste, scrap, debris, bags, containers, equipment or other asbestos-contaminated material consigned for disposal which may produce in any reasonably foreseeable use (handling, storage, processing, disposal, or transportation) employee exposure to airborne asbestos fibers shall be collected and disposed of in double-sealed impermeable 6 mil poly bags or other closed, impermeable containers. Collection shall be completed prior to the end of each work shift.
- m) Caution Labels shall be affixed to all equipment and containers with raw materials, mixtures, scrap, waste, debris, and other materials containing asbestos fibers. Labels shall note the following:
 - DANGER
 - CONTAINS ASBESTOS FIBERS
 - AVOID CREATING DUST
 - CANCER AND LUNG DISEASE HAZARD
- n) Do not mix asbestos waste with other construction debris, industrial waste, or soil.
- o) Individuals shall not be permitted to exit the isolated area until contaminated equipment and clothing have been removed.
 - Removal of asbestos from protective clothing or equipment by blowing, shaking, or any other means which disperses asbestos into the air is prohibited.
 - Individuals shall remove contaminated clothing and place in labeled bags for disposal/cleaning. Employees shall not wear any work clothing or equipment home.
 - Individuals who work in negative-pressure enclosures or in areas where exposures exceed or may exceed the Established Exposure Limits shall shower prior to leaving decontamination area and at the end of the work shift. Respirators shall not be removed until individuals are in the shower.
 - Contaminated protective equipment such as respirators, hoses, tools, etc. shall not be removed from the isolated area until they have been cleaned of visible contamination. Additional cleaning shall be conducted in a decontamination area. Discard used filters (i.e., discarded as asbestos-contaminated).

5.12 Enclosed asbestos wastes shall be placed in a hazardous waste storage area designated by the client. Responsible subcontractor will work with AECOM TISHMAN CONSTRUCTION and the client to confirm that only ACM removed from the designated work site is placed in the storage area. The responsible subcontractor Competent Person shall inspect the storage area to confirm that waste containers are properly labeled, sealed, and stored according to applicable regulations.

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5.13 An approved refuse disposal company will be responsible for transporting the waste to an approved landfill and signing the appropriate shipping manifests. Disposal documentation and identification numbers should be in the client or site owners name for all ACM.

5.14 Records

- a) Training records shall be submitted to and maintained with AECOM TISHMAN CONSTRUCTION project management.
- b) The following documents are required to be maintained in project files for tasks involving asbestos operations for 30 years.
 - Training certificates
 - Medical clearance forms
 - Appropriate and valid licenses
 - Air monitoring results
 - Bulk sampling results
 - Respirator fit test records
 - Air permits, when required by applicable regulations
 - Copies of waste manifest records
 - Inventory of ACM/PACM waste generated and properly disposed of.

5.15 Confirm sensitive data and private personal information (i.e. medical clearance forms) is protected through strict confidentiality measures and restricted access.

6. SILICA

6.1 Silica is the second most common mineral on earth and makes up nearly all of what we call “sand” and “rock.” Silica exists in many forms—one of these, “crystalline” silica (including quartz), is the most abundant and poses the greatest concern for human health. Some common materials that contain silica include:

- a) Rock and sand
- b) Topsoil and fill
- c) Concrete, cement, and mortar
- d) Masonry, brick, and tile
- e) Granite, sandstone, and slate
- f) Asphalt (containing rock and stone)
- g) Fibrous-cement board containing silica

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6.2 Silica is a primary component of many common construction materials, and silica-containing dust can be generated during many construction activities, including:

- Abrasive blasting (e.g., of concrete structures)
- Jackhammering, chipping, or drilling rock or concrete
- Cutting brick or tiles
- Sawing or grinding concrete
- Tuck point grinding
- Road construction
- Loading, hauling, and dumping gravel
- Demolition of structures containing concrete
- Sweeping concrete dust
- Hanging drywall
- Tape and bed of drywall with joint compound and subsequent sanding of installed joints.

6.3 Substitution of less hazardous products for those that contain crystalline silica is required where possible. If substitution cannot occur then engineering controls such as ventilation or wet methods must be used to control silica-containing dusts.

6.4 Due to the significant risk posed by respirable silica, it is critical that all individuals involved in operations that could potentially create silica dust take specific action to ensure that, as much as possible, a hazard is not created. This includes disposing of vacuumed silica containing materials that are placed into dumpsters for removal from the project. Subcontractors shall ensure that a competent person provides oversight for personnel conducting activities that generate silica.

6.5 A job-specific Exposure Control Plan (ECP) for each project should be developed and implemented which outlines in detail the work methods and practices that will be followed on each site. Ensure that the materials (e.g., tools, equipment, personal protective equipment) and other resources (i.e., worker training materials) required to fully implement and maintain this exposure the plan are readily available where and when they are required. A general template is attached to this section for projects to use. Subcontractors will develop a site-specific exposure control plan to cover their project-specific issues (e.g., scope of work, project location and site-specific hazards) and to be kept available at the worksite. Subcontractors shall submit their plan along with a list of their exposed employees with documented silica training to AECOM TISHMAN CONSTRUCTION management prior to the start of work. This plan shall meet all of the requirements found in the overall project plan. Subcontractors must assess when silica dust may be generated and plan ahead to eliminate or control the dust at the source.

6.6 General ECP Elements *(See Attachment 1 for a template)*

- a) Identify all sources of potential silica exposure.
- b) List effective control options to be used to eliminate or reduce the risk to the environment from silica dust exposure. The following hierarchy of control measures shall be followed:

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- Elimination/substitution (e.g., using products with less silica or using work methods that would eliminate the need for surface grinding)
- Engineering controls (e.g., water, local exhaust ventilation, enclosure)
- Administrative controls (e.g., coordination of tasks with subcontractors, signage)
- c) Conduct a periodic review of the ECP effectiveness. This would include a review of the available dust-control technologies to ensure these are selected and used when practical.
- d) Ensure that all required tools and equipment are readily available and used as required by the ECP.
- e) Ensure supervisors and workers are educated and trained to an acceptable level of competency.
- f) Maintain records of inspections (equipment, work methods/practices).
- g) Coordinate the work with the prime contractor and other employers to ensure the safety of the environment.
- h) Plan out all work activities.
- i) The supervisor (foreman and crew lead) is responsible for:
 - Obtaining a copy of the ECP from their employer and making it available at the worksite.
 - Selecting, implementing, and documenting the appropriate site-specific control measures.
 - Providing adequate instruction and training to workers on the hazards of working with silica-containing materials (e.g., concrete) and on the precautions specified in the job-specific plan covering hazards at the location.
 - Directing the work in a manner that ensures the risk to the environment is minimized and adequately controlled.
 - Communicating with the prime contractor and other subcontractors to ensure a safe work environment.
 - Ensuring compliance with federal, state, and/or local regulations, as determined by jurisdiction if silica generating activities are to be conducted.
 - Providing AECOM TISHMAN CONSTRUCTION with documentation as requested.
- j) Each worker is responsible for:
 - Knowing the hazards of silica dust exposure.
 - Setting up the operation in accordance with the site-specific plan.
 - Following established work procedures as directed by the supervisor.
 - Reporting any unsafe conditions or acts to the supervisor.
 - Knowing how and when to report exposure incidents.

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6.7 Local Exhaust Ventilation (LEV) systems and work practices shall include the following:

- a) Vacuum attachment systems to capture and control the dust at its source whenever possible.
- b) Dust control systems (used regularly and well maintained).
- c) Grinding wheels operated at the manufacturers' recommended rpm (operating in excess of this can generate significantly higher airborne dust levels).
- d) Retrofit shrouds or exhaust cowlings for corner grinding; use manufacturer-specified rpm speeds and a well-maintained HEPA vacuum.
- e) Diamond stone grinders, which allow for the use of a more efficient suction casing on the grinder, whenever practicable.
- f) HEPA or good quality, multi-stage vacuum units approved for use with silica dust. [The vacuum units should be capable of creating a target airflow of at least 70 cfm. This should achieve a face velocity at the shroud of about 1.3 m/s (260 fpm)—the higher the face velocity, the more dust captured at source.]
- g) Work planning, so that concrete grinding can be completed when wet (dust release can be significantly reduced).
- h) Good housekeeping work practices (for example, use vacuums with high-efficiency particulate air (HEPA) filters, or use wet sweeping).
- i) Train individuals on how to properly use and maintain the equipment.

6.8 Wet methods for dust control shall include the following:

- a) Pneumatic grinders will be used instead of electric-powered grinders if water is the method of control.
- b) Pressure and flow rate of water will be controlled in accordance with tool manufacturers' specifications (for cutting saws, a minimum of 0.5 liters of water per minute should be used).
- c) When sawing concrete or masonry, use only saws that provide water to the blade.
- d) Wet slurry will be cleaned from work surfaces when the work is completed, using a wet vacuum or wet sweeping.

6.9 Barriers and Enclosures shall include the following:

- a) The subcontractor competent person will determine the type and design of barrier or enclosure (based on the work activity and the work area) and ensure it is constructed in accordance with the work plan. Barriers may be simple hazard-flagging ribbon or more restrictive structures.
- b) Use commercially available negative air units when constructing a full enclosure.

6.10 Where vacuuming methods are selected, HEPA filtered vacuuming equipment shall be used. The equipment shall be used and emptied in a manner that minimizes release of silica into the environment. Disposal of waste silica containing materials from a HEPA vacuum shall remain bagged when disposed. Dry sweeping, shoveling, or other dry clean-up of silica dust and debris is prohibited without proper measures being taken to keep it from becoming airborne.

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6.11 Personal protective equipment

- a) All individuals who wear respirators will do so in adherence with their respirator program.
- b) Respirators must be selected based upon measured exposure levels and the assigned protection factor of respirators.
- c) Only approved respirators will be used.
- d) All individuals who wear respirators will be fit-tested.
- e) Individuals will be properly trained in the use of respirators, and a high standard of supervision, inspection, and maintenance will be followed.
- f) Subcontractors may provide their individuals in a restricted areas with protective clothing that protects other clothing worn by the worker from silica contamination.

6.12 Health monitoring

- a) Exposures to airborne concentrations of Silica must be kept below the permissible exposure limits shown in OSHA 29 CFR 1926.1153 for construction.
- b) Full shift personal samples shall be representative of the individual's regular, daily exposure to silica.
- c) Documentation records must be kept of the following:
 - Individuals who are exposed to respirable silica dust while on the job.
 - Education and training sessions.
 - Respirator fit-testing.
 - Equipment maintenance and repair.
 - Worksite inspections.

6.13 The exposure control plan must be reviewed at least annually and updated as necessary by the employer.

6.14 Education and Training

- a) An individual who will generate silica as part of their work activities is to be informed of the environmental hazards associated with silica released into the environment.
- b) Training is required prior to using silica-containing materials or performing an activity that is known to generate silica working in an environment. Periodic refresher training is also required. Training will contain the following at a minimum:
 - Environmental hazards associated with silica dust.
 - Safe work procedures to be followed (e.g., setup of enclosures, disposal of silica waste).
 - Use of control systems (e.g., LEV and wet methods).

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- How to report an exposure to silica dust.

7. LEAD

7.1 General Requirements

- a) A licensed lead abatement subcontractor must conduct the lead removal.
- b) Safety, health, and industrial hygiene oversight must be provided to the subcontractor by an independent environmental/industrial hygiene consultant subcontractor (referred to as environmental subcontractor).

7.2 Sequencing of Lead Removal

- a) Post appropriate signs to comply with OSHA 29 CFR 1926.62 regulations prior to beginning lead abatement project.
- b) The environmental subcontractor must perform wipe, soil, and water testing (if applicable) to establish background lead levels.
- c) Prepare abatement area as indicated below.
- d) Perform lead paint removal utilizing methods described below.
- e) Pack, test, and label hazardous waste if the Toxic Characteristic Leaching Procedure (TCLP) sample results are in excess of 5.0-ppm lead or construction waste if the results are less than 5.0-ppm.
- f) During the removal process the environmental subcontractor must perform area and personal industrial hygiene air monitoring both inside and outside any containment area.

7.3 Area Security

- a) The removal/work area must be restricted only to authorized, trained, and protected personnel. This may include abatement subcontractor employees, environmental subcontractor employees, AECOM TISHMAN CONSTRUCTION qualified representatives (safety managers), state or federal compliance officers, and any other qualified designated individuals.
- b) A logbook shall be maintained and anyone who enters the work area must record name, unique identification number, time in, and time out for each entry into the abatement area.

7.4 Standard Operating Procedures

- a) The standard operating procedures listed below will be used where demolition activity impact tested positive or presumed lead-based paint during demolition work. These procedures will be in accordance with OSHA 29 CFR 1926.62 (Lead-based paint standard). Following these procedures will ensure maximum protection and safeguard against lead exposure to the abatement subcontractor, AECOM TISHMAN CONSTRUCTION employees, subcontractors, and other potentially exposed people.
 - Wear proper protective clothing and respiratory protection prior to entering an abatement/removal area.
 - There shall be no eating, drinking, smoking, or any other activity in the work area that breaks the seal of the respiratory protection.

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- Packing, labeling, loading, transporting, and disposing of hazardous waste shall be done using sealed DOT approved drums to avoid the release of lead particulate.
- Emergency procedures shall be in place in the event of a medical or safety emergency inside of the work area.
- Appropriate safety procedures shall be in place to minimize the hazards inside the work area (i.e., Lockout/Tagout of existing dangerous equipment such as mechanical systems or electrical systems must be done prior to beginning impacting work operations). The owner as opposed to AECOM TISHMAN CONSTRUCTION may do this and verification may only be necessary.
- The abatement subcontractor shall have a competent person on-site at all times. The competent person is a person who is capable of identifying existing and predictable lead exposure hazards in the work area(s), in the proposed abatement methods, and the typical symptoms employees will have when overexposed to lead, and who has the authority to take prompt corrective measures. The competent person must be trained and licensed as a “Lead Supervisor” with a current certification. The competent person must also be able to perform the following functions:
 - Monitor the set-up of the work area.
 - Control entry and exit into the work area.
 - Ensure that employees are adequately trained in the use of engineering controls, proper work practices, proper personal protective equipment, and decontamination procedures.
 - Ensure that employees use proper engineering controls, proper work practices, proper personal protective equipment, and proper decontamination procedures.
 - Check for rips and tears in work suits and insure that they are repaired immediately or discarded and replaced.
- The abatement subcontractor shall have all workers trained and certified as “Lead Workers.” Certification must be current.

7.5 Respiratory Protection

- a) In all areas tested or assumed to be lead-based paint the subcontractor must provide respiratory protection in accordance with OSHA 29 CFR 1926.62.
- b) Provide all workers, including but not limited to workers, foremen, and superintendents personally issued respiratory protection equipment.
- c) All individuals required to wear respiratory protection must be in a respiratory medical surveillance program for lead-based paint, OSHA respiratory protection guidelines.
- d) All individuals required to wear a respirator must be fit-tested and current on the fit-test certification, OSHA 29 CFR 1926.62, Appendix D.
- e) Respiratory protection equipment must be approved by NIOSH and MSHA.

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- f) When respirators are provided with disposable filters, sufficient filters for replacement must be provided.
- g) For manual demolition, the initial respiratory protection shall be half-face HEPA cartridge respirators.
- h) When using gas, electric or other types of welding and cutting equipment on steel beams containing lead, the initial respiratory protection shall be Powered Air Purifying Respirators (PAPR's).
- i) For activities involving the use of power equipment such as saws and drills, the initial respiratory protection shall be Powered Air Purifying Respirators (PAPR's).
- j) The basic guidelines for respiratory protection are as follows. The following guidelines are based on OSHA requirements relative to industrial hygiene air sampling of the work operations:

Lead Concentration	Type of Respirator
Exposure under 30 ug/m3	None
Exposure 30 to 300 ug/m3	Half-face air purifying respirator with cartridges approved for lead (HEPA).
Exposure 300 to 3,000 ug/m3	Powered Air Purifying Respirator (PAPR) (half or full face piece) with cartridges approved for lead (HEPA).
Exposure over 3,000 ug/m3	Supplied Air Respirators operated in positive pressure mode.

7.6 Protective Clothing

- a) In accordance with OSHA 29 CFR 1926.62, the subcontractor shall:
 - Provide protective disposable clothing to all workers as follows:
 - Full-body coveralls (Tyvek suit with booties and hood).
 - Gloves.
 - 100% eye-protection.
 - Boot covers or booties included in the Tyvek suit.
 - Tape so that all seams can be closed and lead paint cannot get behind the protective clothing.
 - All disposable protective clothing shall be placed in labeled bags in the dirty room of the decontamination chamber. When bags become full, bags shall be double bagged, and placed in drums in the lead storage area and disposed as per federal regulations.

7.7 Medical Surveillance Biological Monitoring. The subcontractor shall conduct employee medical surveillance, urinalysis, and employee blood lead level monitoring in accordance with OSHA 29 CFR 1926.62, Appendix C.

7.8 Decontamination to and From Work Area. All personnel entering and leaving the work area will utilize the following procedures with a 3-stage decontamination chamber:

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- a) Entering from the outside. Change from street clothes into protective clothing at staging area. Wear clean protective clothing and gear. Enter the decontamination chamber (through the clean room, then the shower room, then the equipment/dirty room, and finally into the work area).
- b) Exiting from the work area. Dispose of all protective clothing into appropriately labeled DOT drums or waste bags in the equipment/dirty room. HEPA-vacuum off protective clothing and dispose of contaminated waste. Enter shower from equipment/dirty room and shower completely using soap, including removing and decontaminating the respirators; exit the shower into the clean room, and subsequently out of the clean room into the staging area.
- c) The decontamination chamber. A water source shall be utilized for the shower, and a water pump to retrieve the water from the shower drain and pump it into a 55-gallon drum.
- d) Location of decontamination chamber. Location must be outside the work area so that the opening to the equipment room is attached to an entrance to the work area.
- e) Emergency procedures. Post written procedures in workplace and train all personnel on procedures for evacuation and for the removal of an injured employee. Emergency telephone numbers of all applicable parties shall be promptly posted.
- f) Rescue workers and rescue personnel. Make provisions to minimize exposure to rescue personnel in the event of an evacuation or fire procedure.

7.9 Disposal Activities and TCLP Sampling

- a) A representative sampling of the waste shall be collected. It shall be at least 100 grams and represent the relative proportions of waste components. The results shall be analyzed using Lead Toxic Characteristic Leaching Procedure (TCLP) protocol.
- b) If the TCLP results are greater than or equal to 5.0-PPM lead, then the following disposal protocol must be followed:
 - The subcontractor must package all lead waste in drums approved by the U.S. Department of Transportation (DOT). The subcontractor must comply with DOT CFR 49 and EPA CFR 40 as well as applicable OSHA requirements.
 - Waste drums shall be filled in such a way that at least 3 to 5 inches of space remains at the top of the drum to the lid.
 - The subcontractor must label all generated waste drums as hazardous. TCLP sample results must be submitted to AECOM TISHMAN CONSTRUCTION within seven working days. The TCLP sample results must include the signatures of the sampler, the analyst, and the Certified Industrial Hygienist (CIH).
 - Any hazardous waste water (i.e. from the shower and used in cleaning, and greater than 5.0 ppm lead content as determined by the TCLP analysis) shall be drummed as lead waste.
 - Copies of manifests for each drum must be provided to AECOM TISHMAN CONSTRUCTION Project Manager.

7.10 Air, Wipe, and Soil Sampling

- a) The subcontractor shall employ an independent industrial hygiene consultant and laboratory to perform air, wipe, and soil sampling.

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- b) The laboratory used shall be a current proficient participant in the American Industrial Hygiene Association/National Institute of Occupational Safety and Health (AIHA/NIOSH) PAT Program for the analysis of Lead.
- c) The consultant performing the sampling shall meet all of the following criteria as defined:
 - Certified Industrial Hygienist; or
 - Industrial Hygienist with experience in the monitoring of lead abatement projects.
- d) Air sampling and analysis methods shall be as per NIOSH 7082. Air sampling results must be delivered to AECOM TISHMAN CONSTRUCTION within three working days. The air sampling results shall include the signatures of the sampler, the analyst, and the Certified Industrial Hygienist.
- e) Wipe sampling shall be performed prior to and after demolition of lead-based paint materials.
 - Locations of pre-demolition and post-demolition wipe samples shall be taken as follows:
 1. At least 3 wipe samples from inside the work area;
 2. One exterior work area sample; or
 3. Taken, as contract specifications require.
 - The clearance criteria is as follows:

Substrate	Clearance Level (ug/m3)
Bare and carpeted floors	200
Exterior concrete and other rough surfaces	800

- Wipe sample results shall be submitted to AECOM TISHMAN CONSTRUCTION within seven working days. The wipe sample results must include signatures of the sampler, the analyst, and the CIH.
- f) Soil Sampling. There shall be a baseline conducted prior to demolition activities. After demolition, there shall be one exterior composite soil sample per work area. Soil sampling does not apply if there is no soil. Soil sample results must be submitted to AECOM TISHMAN CONSTRUCTION within seven working days. The clearance criterion is the higher of:
 - 2,000 ug/g; or
 - Baseline lead level in the soil.

8. HOT ENVIRONMENTS

8.1 Hot Environment Precautions

- a) Intake of fluid shall be increased beyond that which satisfies thirst, and it is important to avoid "fluid debt," which will not be made up as long as the individual is sweating.
- b) Two 8 oz glasses of water should be taken prior to beginning work, then up to 32 oz per hour during the work shift; although fluid replacement at frequent intervals is more effective.

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- c) The best fluid to drink is water; liquids like coffee or soda do not provide efficient hydration and may increase loss of water.
- d) If commercial electrolyte drinks (Gatorade, Squincher, etc.) are used, the drink should be diluted 1:1 with water, i.e., 8 oz of water should be taken with each 8 oz of electrolyte beverage.
- e) Additional salt is usually not needed; moreover, salt tablets should never be taken.
- f) Replacement fluids should be cool, but not cold.
- g) Breaks should be taken in a cool, shaded location, and any impermeable clothing should be removed.
- h) Dry clothing and/or towels should be available to help minimize chills when taking breaks.
- i) Manual labor, other than paperwork or similar light tasks, should not be performed during breaks.
- j) Other heat controls that may be used include:
 - Scheduling work at night or during the cooler parts of the day (6 a.m. - 10 a.m. and/or 3p.m. - 7p.m.).
 - Erecting a shade over the work area.
 - Use of cooling garments, although expensive and logistically difficult to implement.
- k) Follow any local or state heat regulations and standards.

8.2 Work Schedules

- a) Work schedules providing periodic rest periods should be implemented when employees are exposed to heat stress. Schedules shall be developed based on instrumental measurements of the environment, by assessing temperature, radiant heat, humidity, and wind speed, with the resulting measurement compared to published guidelines, as outlined by NIOSH REL and ACGIH TLV, for Heat Stress. When using this method, allowance must be made for the use of impermeable protective clothing.
- b) Alternate methods of establishing a work schedule are based on measuring physiological indices, such as heart rate or oral temperature, which correlate to the core body temperature.

NOTE: Oral temperature does not correlate well with increasing core temperature and is subject to effects from drinking replacement fluids. During heat exhaustion, oral temperature may be normal or even depressed due to extreme peripheral vasodilatation.
- c) Measures of heart rate (pulse) appear to correlate well with heat related stress and can be used to estimate the impact of heat stress on individuals with differing levels of fitness. For example, less fit individuals with a pulse of 150 beats/minute experience about the same level of stress as a fit person, even though the work output required for the fit person to reach this rate will be quite higher. The heart rate also incorporates the combined effects of environmental heat, muscle activity, and elevated body temperature, making it a useful and easily measured variable.
- d) The procedure for controlling heat stress via establishment of work schedules based on the measurement of heart rate is presented below.

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8.3 The Project Safety Manager and/or Superintendent shall determine the potential for heat stress based on planned activities and weather forecasts. Subcontractors shall do the same for their employees.

a) If the potential for heat stress exists:

- All site workers should be informed of the potential for heat stress during the daily safety meeting. The Project Safety Manager shall determine if any workers are at particular risk for heat stress due to illness, etc. Also, the Project Safety Manager shall assure that sufficient quantities of portable water and electrolyte drinks are available in the decon area, and that a shaded rest area is available at or immediately outside the decon area. Subcontractors shall do the same for their employees.
- All workers should drink 16 oz of water prior to beginning work, and at least 16 oz during each rest period.

8.4 Training

- a) Those individuals potentially exposed to heat stress shall receive training on heat issues. Training provided as a part of this procedure shall be documented and given to the AECOM TISHMAN CONSTRUCTION Project Management upon request. Topics that may be trained on include:
- Sources of heat stress, the influence of protective clothing, and the importance of acclimatization.
 - How the body handles heat.
 - Heat related illnesses.
 - Preventative/Corrective measures.
 - First Aid procedures for heat stress.

9. COLD ENVIRONMENTS

9.1 General

- a) Cold, wet, windy conditions make prime hypothermia weather. Precautions shall be taken to keep warm.
- b) In cold environments individuals should test for symptoms of hypothermia and frostbite often. Wear loose fitting clothing that will not restrict blood flow to the limbs.
- c) If an individual suspects a co-worker has experienced frostbite, they should seek medical attention immediately.

9.2 Cold Environment Precautions

- a) Workers shall wear warm clothing, such as mittens, heavy socks, etc., when the air temperature is below 40-45 degrees Fahrenheit (°F). Chemical protective clothing, if required by the project, may be used to partially protect the employee from the cold.
- b) When the air temperature is below 30-40°F, depending upon employee comfort, clothing for warmth should be worn in addition to chemical protective clothing as needed. This may include:

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insulated suits; whole-body thermal underwear; wool or polypropylene socks to keep moisture off the feet when engaged in work activity which would cause sweating; insulated gloves when air temperatures are extremely low--less than 5-10°F; gloves with reflective surfaces which reflect body heat back to the hand, boots; and insulated head covering, such as knit (ski) caps.

- c) At air temperature below 35°F, the following work practices should be followed:
- If the clothing of an employee could become wet on the job site, the outer layer of clothing should be impermeable to water.
 - If an individual's underclothing--socks, mittens, etc.--becomes wet in any way, the employee should immediately change into dry clothing. If the clothing becomes wet from sweating, the employee may finish the task causing the sweating before changing into dry clothing.
 - Individuals should be provided a warm area to change from work clothing into street clothing.
 - Individuals should be provided a warm break area. If appropriate, space heaters shall be provided in the work area to warm the hands and feet, etc. Necessary fire and electrical safety practices must be observed when using space heaters. Space heaters should be shut off when the site is not occupied.
 - Hot liquids, such as soups and warm drinks, may be consumed in the break area. The intake of caffeinated beverages should be limited, however, due to adverse diuretic and circulatory effects.
 - The buddy system should be practiced at all times. Any employee observed with severe shivering must leave the cold area immediately.
 - Individuals should layer their clothing. Thinner, lighter-weight clothing should be worn closest to the body, with heavier-weight clothing layered outside the lighter, layer.
 - Individuals should avoid overdressing when going into warm areas or when performing activities which are strenuous. This could lead to a heat stress problem.
 - Auxiliary heated versions of hand wear, footwear, etc., can be used instead of mittens and insulated socks, etc., if extremely cold conditions exist, and if such items do not create hazards in the work area.
 - Individuals handling volatile liquids--gasoline, hexane, alcohol, etc.--must take special precautions to avoid saturating clothing or gloves with these liquids. There is an added danger of injury from overexposure due to evaporative cooling.
 - Work shall be arranged in such a way that sitting still or standing for long periods of time is minimized.

9.3 Training

- a) Those individuals potentially exposed to cold stress shall receive training on cold issues. Training provided as a part of this procedure shall be documented and given to the AECOM TISHMAN CONSTRUCTION Project Management upon request. Topics that may be trained on include:
- Proper first aid treatment.

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- Proper clothing practices.
- Proper eating and drinking habits.
- Recognition of impending adverse health effects.
- Safe work practices.

10. SEVERE WEATHER

10.1 A contingency plan should be in place in the event of excessive rain, floods or high winds. The owner may want to be involved in the decision process when severe weather is forecast. Individual situations will vary but the guidelines below may be helpful:

- a) All winds over 25 mph. All rigging of forms or flat-sided materials should be discontinued until wind subsides.
- b) All crane operations shall be suspended per manufacturer's recommendations, local regulations, when wind gusts reach 25mph or greater, or at the direction of the Project Safety Manager and/or Superintendent.
- c) If heavy rain and winds bring visibility down to 1/8 mile, all hauling traffic should be shut down. This could be done before poor visibility conditions due to the condition of the haul road.

10.2 Flooding

- a) The Superintendent should establish a control station for operations.
- b) Set gang boxes, safety skips, and miscellaneous items on trucks or move to higher elevations.
- c) If possible, have all craft people remove their personal tools.
- d) Tow mobile items such as welders and compressors out of the floodplain.
- e) Remove equipment such as electric welding machines, gas driven pumps, and concrete vibrators.
- f) Remove diesel and gasoline tanks, curing compound, and propane tanks, as well as other containers of petroleum products.
- g) Make sure that life rings, life jackets and a motorized boat are available where appropriate.
- h) Disconnect power to the area.

10.3 Specific Plans

- a) Specific weather plans such as hurricane or tornado plans are available for use. If you have a need please contact your Regional Safety Director.

11. OTHER ENVIRONMENTAL ISSUES

11.1 If a project encounters any other environmental issues please inform your Regional Safety Director immediately and refer to the AECOM Tier III documents found on Ecosystem.

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12. ATTACHMENTS

12.1 Silica Control Program Template

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Attachment 1: Silica Control Program Template

For every project in which AECOM TISHMAN CONSTRUCTION participates, the site construction management personnel- with the assistance of the site and corporate safety team, will develop a SILICA EXPOSURE CONTROL PLAN (SECP). A template of this required plan has been developed and will be used as the guidance document for every project. It is the responsibility of the site construction management and safety teams to enter sufficient information so as to transform the template into a relevant and practical site- specific Silica Exposure Control Plan (SECP).

Steps to implement the AECOM TISHMAN CONSTRUCTION SECP are as follows:

STEP 1 – At the pre-con stage of project development, provide all known information about the project and the activities that will create silica exposures on the Exposure Control Template.

STEP 2 – Plan for and arrange a site safety orientation training class covering silica hazards, methods of exposure control, the role of respiratory protection, and means of silica hazard prevention and protection. This can be integrated into the standard HAZCOM and site safety orientation which is provided to employees on every project.

STEP 3 – Refer to the SECP TABLE 1 attachment and fill in every task that will follow the instructions as specified on Table 1.

STEP 4 – For every task that is not included on TABLE 1, describe the task and explain how silica exposures will be minimized and how employees will be protected from silica exposure.

STEP 5 – Keep accurate and complete records of site silica compliance using the SECP. Add additional information and descriptions as necessary.

ALL CONTRACTORS AND SUBCONTRACTORS MUST MAINTAIN A PROGRAM AT LEAST AS COMPLIANT AS AECOM TISHMAN CONSTRUCTION.

ALL RECORDS OF SILICA COMPLIANCE MUST BE SENT TO THE REGIONAL SAFETY DIRECTOR AT THE CONCLUSION OF THE PROJECT.

AECOM TISHMAN CONSTRUCTION

SILICA EXPOSURE CONTROL PLAN

Date: _____

Person Completing the Plan, Title: _____

Competent Person: _____ Contact Number _____

Description of Project: _____

METHOD OF ASSESSMENT OF SILICA EXPOSURE – choose the method used for the planned tasks. Attach supporting information or reference source of information used to make the exposure determination.

☐ **Performance option.** Assess the 8-hour TWA exposure for each employee on the basis of any combination of air monitoring data or objective data sufficient to accurately characterize employee exposures to respirable crystalline silica.

☐ **Scheduled monitoring option.** Perform initial monitoring to assess the 8-hour TWA exposure for each employee on the basis of one or more personal breathing zone air samples that reflect the exposures of employees on each shift, for each job classification, in each work area. Where several employees perform the same tasks on the same shift and in the same work area, sample a representative fraction of these employees in order to meet this requirement. In representative sampling, sample the employee(s) who are expected to have the highest exposure to respirable crystalline silica.

METHOD OF SILICA EXPOSURE CONTROL

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A. TABLE 1 – indicate # within Table 1 _____

B. ALTERNATIVE METHOD – describe task and control methods including engineering, procedures and personal protective equipment.

TASKS	EXPOSURE CONTROL PLAN & METHOD Engineering Restricted Access Assignment of Competent Person

Add additional sheets as necessary to describe tasks and control methods.

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RESPIRATORY PROTECTION

TYPE
TRAINING ___ YES ___ NO
FIT TEST ___ YES ___ NO
MEDICAL ___ YES ___ NO
DESCRIPTION:

HOUSEKEEPING-DESCRIBE SILICA CONTROL ACITIVITIES

MEDICAL SURVEILLANCE – Will employees wear a respirator for more than 30 days per year on this project?

YES___ NO___

If YES, where are records located?

HAZARD AWARENESS TRAINING

Is this done at orientation training and does it cover these critical topics:

SILICA HAZARDS AND HEALTH EFFECTS	YES___	NO_____
METHODS OF PROTECTION	YES___	NO_____
METHODS OF EXPOSURE CONTROL	YES___	NO_____

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RECORDKEEPING

RECORD	YES	NO	Record Location
EXPOSURE ASSESSMENT			
RESPIRATORY PROTECTION			
SILICA HAZARD AWARENESS TRAINING			
MEDICAL			

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Table 1: Specified Exposure Control Methods When Working With Materials Containing Crystalline Silica

Construction Task or Equipment Operation		Engineering and Work Practice Control Methods	Required Respiratory Protection	
			≤ 4 hours/shift	>4 hours/shift
1	Stationary masonry saws	Use saw equipped with integrated water delivery system that continuously feeds water to the blade. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.	None	None
2a	Handheld power saws (any blade diameter) when used outdoors	Use saw equipped with integrated water delivery system that continuously feeds water to the blade. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.	None	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
2b	Handheld power saws (any blade diameter) when used indoors or in an enclosed area	Use saw equipped with integrated water delivery system that continuously feeds water to the blade. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
3	Handheld power saws for cutting fiber-cement board (with blade diameter of 8 inches or less) for tasks performed outdoors only	Use saw equipped with commercially available dust collection system. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions. Dust collector must provide the air flow recommended by the tool manufacturer, or greater, and have a filter with 99% or greater efficiency.	None	None
4a	Walk-behind saws when used outdoors	Use saw equipped with integrated water delivery system that continuously feeds water to the blade. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.	None	None
4b	Walk-behind saws when used indoors or in an enclosed area	Use saw equipped with integrated water delivery system that continuously feeds water to the blade. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
5	Drivable saws for tasks performed outdoors only	Use saw equipped with integrated water delivery system that continuously feeds water to the blade. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.	None	None
6	Rig-mounted core saws or drills	Use tool equipped with integrated water delivery system that supplies water to cutting surface. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.	None	None
7	Handheld and stand-mounted drills (including impact and rotary hammer drills)	Use drill equipped with commercially available shroud or cowl with dust collection system. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions. Dust collector must provide the air flow recommended by the tool manufacturer, or greater, and have a filter	None	None

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		with 99% or greater efficiency and a filter-cleaning mechanism. Use a HEPA-filtered vacuum when cleaning holes.		
8	Dowel drilling rigs for concrete for tasks performed outdoors only	Use shroud around drill bit with a dust collection system. Dust collector must have a filter with 99% or greater efficiency and a filter cleaning mechanism. Use a HEPA-filtered vacuum when cleaning holes.	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
9a	Vehicle-mounted drilling rigs for rock and concrete	Use dust collection system with close capture hood or shroud around drill bit with a low-flow water spray to wet the dust at the discharge point from the dust collector.	None	None
9b	Vehicle-mounted drilling rigs for rock and concrete	Operate from within an enclosed cab and use water for dust suppression on drill bit.	None	None
10a	Jackhammers and handheld powered chipping tools when used outdoors	Use tool with water delivery system that supplies a continuous stream or spray of water at the point of impact.	None	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
10b	Jackhammers and handheld powered chipping tools when used indoors or in an enclosed area	Use tool with water delivery system that supplies a continuous stream or spray of water at the point of impact.	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
10c	Jackhammers and handheld powered chipping tools when used outdoors	Use tool equipped with commercially available shroud and dust collection system. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions. Dust collector must provide the air flow recommended by the tool manufacturer, or greater, and have a filter with 99% or greater efficiency and a filter-cleaning mechanism.	None	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
10d	Jackhammers and handheld powered chipping tools when used indoors or in an enclosed area	Use tool equipped with commercially available shroud and dust collection system. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions. Dust collector must provide the air flow recommended by the tool manufacturer, or greater, and have a filter with 99% or greater efficiency and a filter-cleaning mechanism.	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
11	Handheld grinders for mortar removal (i.e., tuckpointing)	Use grinder equipped with commercially available shroud and dust collection system. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions. Dust collector must provide 25 cubic feet per minute (cfm) or greater of airflow per inch of wheel diameter and have a filter with 99% or greater efficiency and a cyclonic pre-separator or filter-cleaning mechanism.	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask	Powered Air-Purifying Respirator (PAPR) with P100 Filters
12a	Handheld grinders for uses other than mortar removal for	Use grinder equipped with integrated water delivery system that continuously feeds water to the grinding surface.	None	None

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	tasks performed outdoors only	Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.		
12b	Handheld grinders for uses other than mortar removal when used outdoors	Use grinder equipped with commercially available shroud and dust collection system. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions. Dust collector must provide 25 cubic feet per minute (cfm) or greater of airflow per inch of wheel diameter and have a filter with 99% or greater efficiency and a cyclonic pre-separator or filter-cleaning mechanism.	None	None
12c	Handheld grinders for uses other than mortar removal when used indoors or in an enclosed area	Use grinder equipped with commercially available shroud and dust collection system. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions. Dust collector must provide 25 cubic feet per minute (cfm) or greater of airflow per inch of wheel diameter and have a filter with 99% or greater efficiency and a cyclonic pre-separator or filter-cleaning mechanism.	None	N95 (or Greater Efficiency) Filtering Facepiece or Half Mask
13a	Walk-behind milling machines and floor grinders	Use machine equipped with integrated water delivery system that continuously feeds water to the cutting surface. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions.	None	None
13b	Walk-behind milling machines and floor grinders	Use machine equipped with dust collection system recommended by the manufacturer. Operate and maintain tool in accordance with manufacturer's instructions to minimize dust emissions. Dust collector must provide the air flow recommended by the manufacturer, or greater, and have a filter with 99% or greater efficiency and a filter-cleaning mechanism. When used indoors or in an enclosed area, use a HEPA-filtered vacuum to remove loose dust in between passes.	None	None
14	Small drivable milling machines (less than half-lane)	Use a machine equipped with supplemental water sprays designed to suppress dust. Water must be combined with a surfactant. Operate and maintain machine to minimize dust emissions.	None	None
15a	Large drivable milling machines (half-lane and larger) for cuts of any depth on asphalt only	Use machine equipped with exhaust ventilation on drum enclosure and supplemental water sprays designed to suppress dust. Operate and maintain machine to minimize dust emissions.	None	None
15b	Large drivable milling machines (half-lane and larger) for cuts of four inches in depth or less on any substrate	Use machine equipped with exhaust ventilation on drum enclosure and supplemental water sprays designed to suppress dust. Operate and maintain machine to minimize dust emissions.	None	None
15c	Large drivable milling machines (half-lane and larger) for cuts of four inches in depth or less on any substrate	Use a machine equipped with supplemental water spray designed to suppress dust. Water must be combined with a surfactant. Operate and maintain machine to minimize dust emissions.	None	None
16	Crushing machines	Use equipment designed to deliver water spray or mist for dust suppression at crusher and other points where dust is generated (e.g., hoppers, conveyers, sieves/sizing or vibrating components, and discharge points).	None	None

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		Operate and maintain machine in accordance with manufacturer's instructions to minimize dust emissions. Use a ventilated booth that provides fresh, climate-controlled air to the operator, or a remote control station.		
17a	Heavy equipment and utility vehicles used to abrade or fracture silica-containing materials (e.g., hoe-ramming, rock ripping) or used during demolition activities involving silica-containing materials	Operate equipment from within an enclosed cab.	None	None
17b	Heavy equipment and utility vehicles used to abrade or fracture silica-containing materials (e.g., hoe-ramming, rock ripping) or used during demolition activities involving silica-containing materials	When employees outside of the cab are engaged in the task, apply water and/or dust suppressants as necessary to minimize dust emissions.	None	None
18a	Heavy equipment and utility vehicles for tasks such as grading and excavating but not including demolishing, abrading, or fracturing silica-containing materials	Apply water and/or dust suppressants as necessary to minimize dust emissions.	None	None
18b	Heavy equipment and utility vehicles for tasks such as grading and excavating but not including demolishing, abrading, or fracturing silica-containing materials	When the equipment operator is the only employee engaged in the task, operate equipment from within an enclosed cab.	None	None

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SECTION 8 - EXCAVATION AND TRENCHING

1. TERMS and DEFINITIONS

1.1 Benching (Benching system). One or a series of horizontal levels or steps, usually with vertical or near-vertical surfaces between levels to protect employee from cave-ins.

1.2 Cave-in (collapse). The separation of a mass of soil or rock material from the side of an excavation or the loss of soil from under a trench shield or support system and its sudden movement into the excavation, either by falling or sliding, in sufficient quantity so that it could entrap, bury, or otherwise injure, and immobilize a person.

1.3 Competent person. A person who, by way of training, knowledge, and/or experience, is capable of classifying soils and is also capable of identifying existing and predictable hazards in excavation / trenching work area and who has the authority to take prompt corrective measures to eliminate them. The person shall also be familiar with the requirements in the regulation.

1.4 Excavation. A manmade cut, cavity, trench, or depression in an earth surface formed by earth removal. Examples include trenches, tunnels, shafts, caissons and open cut holes.

1.5 Faces (or sides). The vertical or inclined earth surfaces formed as a result of excavation work.

1.6 Failure. A structural member's integrity and supportive capabilities is compromised, causing a breakage, displacement, or permanent deformation.

1.7 Hazardous Atmosphere. An atmosphere that by reason of being explosive, flammable, poisonous, corrosive, oxidizing, irritating, oxygen-deficient, toxic, or otherwise harmful may cause death, illness, or injury.

1.8 Protective Systems. Devices or methods in protecting employees in an excavation from cave-ins, a collapse, or falling material. Protective systems include support systems, sloping, and benching systems, shield systems, and other systems that provide the necessary protection.

1.9 Ramp. An inclined walking or working surface that is used to gain access to one point from another and is constructed from earth or from structural materials such as steel or wood.

1.10 Registered Professional Engineer. An engineer who can authorize any state of work by his professional designation and is registered in the state, province, or territory is deemed to be a "Registered Professional Engineer" within the meaning of this standard when approving designs for "manufactured protective systems" or "tabulated data" to be used in interstate commerce.

1.11 Shield (Shield system). A structure that is able to withstand the forces imposed on it by a cave-in and thereby protects employees within the structure. Shields can be permanent structures or can be designed to be portable and moved along as work progresses. Additionally, shields can be either pre-manufactured or job-built. Shields used in trenches are usually referred to as "trench boxes" or "trench shields."

1.12 Shoring (Shoring system). A structure such as a metal hydraulic, mechanical, or timber shoring system that supports the sides of an excavation and that is designed to prevent cave-ins.

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1.13 Sloping (Sloping system). An alternative to shoring is trench sloping. This means that the trench walls are cut back to decrease the possibility of cave-ins. The angle of incline required to prevent a cave-in varies with such factors as soil type, environmental conditions of exposure, and application of surcharge loads.

1.14 Stable rock. A natural solid mineral material that can be excavated with vertical side wall; unstable rock is considered to be stable when the rock material on the side or sides of the excavation is secured against cave-in or movement by rock bolts or by another protective system that has been designed by a Registered Professional Engineer.

1.15 Support system. A structure such as underpinning, bracing, or shoring that provides support to an adjacent structure, underground installation, or the sides of an excavation.

1.16 Trench. An open narrow excavation made below the surface of the ground. In general, the depth is greater than the width, but the width (measured at the bottom) is often not greater than 15 feet (4.57 meters). If forms or other structures are installed or constructed in an excavation so as to reduce the dimension measured from the forms or structure to the side of the excavation to 15 feet (4.57 meters) or less (measured at the bottom of the excavation), the excavation is also considered a trench.

1.17 Trench Box. A trench box is a unit of shoring that is an engineered shoring system capable of protecting workers in case of cave-in of trench walls. The space between the trench wall and the trench box shall be backfilled.

2. GENERAL REQUIREMENTS

2.1 Restrictions

- a) Because of their inherent dangers, entry into trenches and excavations shall not be performed if there are means other than entry to perform the work. Where entry into trenches and excavations is necessary, strict adherence to the procedures specified below and OSHA standards is extremely important. Whenever there are questions regarding the safety of trench or excavation entry, contact shall be made with the Competent Person or the SH&E Manager.
- b) No one shall enter any trench or excavation greater than four (4) feet until proper precautions laid out in this program and OSHA standards are followed.
- c) Excavation work shall be completed and inspected in accordance with the written instructions of a competent professional and in accordance with jurisdictional legislative regulations.

2.2 AECOM TISHMAN CONSTRUCTION Excavation and Trenching Permit

- a) As required by the applicable jurisdiction and this AECOM TISHMAN CONSTRUCTION SH&E program, confirmation of the proposed excavation is provided within the required timeframe to the appropriate agencies or governing bodies prior to commencing excavation (e.g. California – CAL/OSHA Excavation Permit for the construction of trenches or excavations that are 5 feet (1.5 meters) or deeper that will be entered; Manitoba WHS Branch notification in order to obtain registration number, etc.).
- b) An Excavation and Trenching Permit shall be completed prior to all excavation or trenching activities.

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- c) The Excavation and Trenching Permit shall be completed and signed by all applicable parties as indicated on the permit.
- d) Excavation and Trenching Permits may be valid for up to one week; however, the permit shall be reviewed at the beginning and end of each shift. A copy shall be kept on file in the AECOM TISHMAN CONSTRUCTION office.

2.3 Planning and Preparation

- a) Prior to beginning any excavation work at a site, the location of all underground and overhead utilities shall be identified and work locations will be carefully planned to avoid any potential for inadvertent contact with them. This is sometimes referred to as the “one call.” Documentation shall be kept on file that this has been completed.
- b) Clearance, hydro excavation and hand exposure, of underground utilities shall be completed in accordance with most stringent authority having jurisdiction. Locates of all utilities shall be available and reviewed with all employees expected to be involved in the excavation prior to commencing any excavation activities.
- c) Identify any overhead power lines and de-energize or protect by other appropriate means.
- d) All Trenches/Excavations at a depth of 19’-6” or deeper shall be designed/reviewed by a Registered Professional Engineer with submitted stamped drawings by said Registered Professional Engineer to AECOM TISHMAN CONSTRUCTION.
- e) Trenches/Excavations with a depth less than 19’-6” are still subject to Registered Professional Engineer Design and Review based on: location, soil conditions, proximity to utilities, active DOT area, client requirements, local jurisdictions having authority, AECOM TISHMAN CONSTRUCTION requiring additional engineering from contractual obligations.

2.4 Excavation Requirements

- a) A Registered Professional Engineer shall be engaged if specified by the applicable jurisdiction and as appropriate to the soil conditions and proposed excavation considerations (e.g. wall slope, shoring requirements, load calculations, etc.).
- b) All personnel involved in the excavation activities shall be appropriately trained to their respective activities and associated hazards. Subcontractors are solely responsible for training their employees.
- c) All personnel involved in the excavation activities shall wear the required PPE, including reflective clothing if mobile equipment or vehicular traffic is present.
- d) A Task Hazard Analysis (THA), identifying applicable hazards and appropriate control measures shall be completed and clearly communicated to all involved personnel as well as to any concurrent operations potentially affected. The work environment shall be monitored for changing conditions and the THA updated accordingly. The THA shall be readily available for review in the field.
- e) Excavation shall be conducted in a manner that minimizes environmental impact.

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- f) Excavated (spoil) material is recommended to be kept 4 ft but no less than 2 ft from the edge of an excavation. Excavated (spoil) material shall be piled in a manner to prevent sloughing of loose material. Various jurisdictions provide specific spoil pile sloping requirements.
- g) If the walls of an excavation or trench are not sloped or cutback, barriers shall be placed around the perimeter. The barrier shall be at least 3.6 ft (1.10 meters) in height.
- h) If water is controlled or prevented from accumulating by the use of water removal equipment, the water removal equipment and operations shall be monitored regularly to confirm proper operation.
- i) If excavation work interrupts the natural drainage of surface water (such as streams), diversion ditches, dikes, or other suitable means shall be used to prevent surface water from entering the excavation and to provide adequate drainage of the area adjacent to the excavation. Excavations subject to runoff from heavy rains will require regular inspections.
- j) All excavations shall be appropriately secured at the end of the day to prevent unauthorized entry or inadvertent entry into the excavation. This may require a protective covering, barriers, fencing, signage or other measures appropriate to the excavation and associated conditions.
- k) Backfill trenches as soon as reasonably possible after work is complete.
- l) Fall protection for individuals shall be maintained around all excavations that are 6 feet or greater in depth.

2.5 Soils Classifications

- a) Soil classification shall be conducted to confirm appropriate measures are taken to protect workers and to secure excavation walls. Measures may include, but are not limited to:
 - Sloping, shoring or shielding.
 - Relocation of equipment or materials.
 - Scheduling to minimize concurrent operations.
- b) Soil characteristics evaluated when classifying include, but are not limited to:
 - Cohesiveness / compaction / compressive strengths (e.g. fissured, hardpan, fractured rock, etc.).
 - Composition (sand, clay, gravel, layered, etc.).
 - Moisture content / submersion.
 - Compaction.
 - Exposure to vibration (e.g. traffic, pile driving, etc.).
 - Previous disturbances.
- c) The soil must be classified by the competent person as stable rock, Type A, B, or C as defined in Appendix A of Subpart P of the OSHA regulations.
- d) Once the soil is classified, the appropriate sloping, benching, shoring or shielding method can be selected from appendices B through F of Subpart P. Situations that do not conform to those allowed must have a protective system designed by a registered professional engineer with all

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tabulated data submitted to AECOM TISHMAN CONSTRUCTION Safety for review. The Foreman in charge of the excavation shall keep a copy readily available in the field.

2.6 Protective System Requirements

- a) Protective systems shall be used to protect workers entering an excavation when there is a potential for cave-in, and is required when:
 - An excavation is greater than 4 feet (1.22 meters) in depth and is not entirely in stable rock.
 - A worker is required to be closer to a trench wall than the height of the trench wall.
- b) The protective system may include sloping the excavation walls, shoring the excavation walls, and/or installing a shielding system. The protective system(s) chosen shall have the capacity to resist, without failure, all loads to be applied to the system as demonstrated by onsite tabulated.
- c) If an excavation may affect the stability of an adjacent building or structure, precautions shall be taken to prevent damage to the structure. The precautions shall be specified in writing by a Registered Professional Engineer.
- d) All sloping, shielding, or shoring shall be conducted in accordance with applicable federal, state, provincial, territorial, or legislative regulations.
- e) Each individual in an excavation shall be protected from cave-ins and trench collapse by an adequate protective system except when:
 - Excavations are made entirely in stable rock.
 - Excavations are less than 4 feet (1.22 meters) in depth and an examination of the excavation by a Competent Person reveals no indication of a potential cave-in.
- f) The depth of the excavation or trench is to be measured at its greatest vertical dimension. Be aware that crouching or kneeling in a trench that is less than 3 ft (0.91 meter) in depth may still pose significant hazard for the employee involved.
- g) Consult the applicable OSHA standards concerning the standards that protective systems shall meet. This may include design and certification by a Registered Professional Engineer.
- h) A Registered Professional Engineer can properly assess the need for and the type of shoring required for specific applications. Shoring may not be needed in all cases, but failure to recognize the need for shoring can be catastrophic.

2.7 Use of Sloping as a Means of Protection

- a) Sloping the walls of the trench or excavation is the preferred, and typically simplest, means of protecting employees who shall enter trenches or excavations which are greater than 4 feet (1.22 meters) in depth or where there is danger of collapse.
- b) If sloping is used as the means of protection, the trench or excavation walls shall be sloped back so that the ratio of the horizontal distance to the vertical rise (H:V ratio) of the sloped wall or

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degree from horizontal is appropriate to the soil type and in compliance with jurisdictional requirements.

- c) In many cases, determining the maximum allowable slope may allow the use of a steeper slope, which will result in a narrower excavation. However, determination of soil classification is complicated and requires that the Competent Person be familiar with the manual and visual tests. Since incorrect soil classification may result in the use of a steeper, and potentially unsafe, slope, it is recommended that an angle of 34 degrees (or less, given specific jurisdictional requirements and unstable soil types) with the horizontal typically be selected.

2.8 Use of Shoring or Shielding as a Means of Protection

- a) Where sloping the walls of the trench or excavation is unfeasible (e.g., when there are dimensional constraints or adjacent structures), the use of shoring or shield systems (e.g., trench boxes) may be necessary.
- b) In all cases listed above, the SH&E Manager shall be contacted before proceeding.
- c) Workers shall be protected whenever shoring is being assembled, installed or removed.
- d) Shoring, stringers or bracing shall be installed from the top of the trench or excavation down. Removal shall be completed from the bottom up unless conditions exist that would make doing so unsafe. A removal method shall be developed that does not require worker entry.
- e) Check hydraulic shoring once per shift at a minimum (leaking hoses and/or cylinders, broken connections, cracked nipples, bent bases, other damaged or defective parts) and more frequently as required.
- f) Hydraulic or pneumatic trench jacks shall have a means of ensuring that they will not collapse in the event of loss of internal pressure.
- g) Shielding and Trench Boxes differ from shoring in that their design is intended primarily to protect workers from cave-ins and similar incidents. They may be used in combination with sloping and benching.
 - The excavated area between the outside of the trench box and the face of the trench should be as small as possible and may be backfilled to prevent lateral movement of the shield.
 - The box shall extend at least 18 inches (0.45m) above the surrounding area.
 - Earth excavation below the shield is permitted only if:
 - The excavation does not exceed a depth of 2ft (0.61m) below the shield.
 - The shield is capable of withstanding the forces calculated for the full depth of the trench, and there are no indications while the trench is open of possible loss of soil from behind or below the bottom of the support system.
 - A shield shall not be subjected to loads exceeding those for which it was designed to withstand.

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- h) Bell-bottom pier holes that are to be entered by workers shall be designed and supported according to written instructions of a Registered Professional Engineer.
 - Employees entering bell-bottom pier holes, or other similar deep and confined footing excavations, should wear a harness with a retrieval line securely attached to it.
 - Retrieval lines shall not be used to handle materials.
 - Rescue equipment shall be individually attended at all times while the employee wearing the retrieval line is in the excavation.

2.9 Work Around the Trench/Excavation

- a) Structural ramps used for excavation access or egress of equipment shall be constructed in accordance with jurisdictional requirements and the instructions or designs of an individual competent and qualified in structural design.
- b) If the ramp has an open side, the ramp shall be constructed in accordance with local jurisdiction having authority.
- c) Confirm equipment placement does not compromise the integrity of the excavation wall and optimizes visibility of work zone and any contact hazards (spoil placement should also take these into consideration). Use wheel chocking or barricades as necessary to prevent encroachment of edge.
- d) If the appropriate setback of equipment is not possible confirm appropriate blocking or matting is used to disperse weight. These requirements may need to be determined by a Registered Professional Engineer.
- e) While workers are in a trench, an aboveground observer or spotter shall be present to warn of earth movements and to advise equipment operators of the presence and location of those in the trench so as to avoid vibrating equipment near trenches or excavations.
- f) If there is a danger of a worker or equipment falling into an excavation, or whenever the edge is not clearly visible, identify the trench or excavation perimeter with visual markers (e.g., barricade tape, wooden railings, stop logs, etc.). If the trench or excavation is 4 feet (1.22 meters) or greater in depth, the visual barrier shall be a minimum of 15 feet from the edge or a fence barrier at 6 feet from edge on all sides, supported every 10 feet.
- g) Personnel shall notify workers of the excavation through flagging, marking, safeguards, or other appropriate and effective means.
- h) If walkways are permitted over excavations or trenches (e.g. trench over 6 feet [1.8 meters] in depth and wider than 30 inches [76 centimeters]), the installation shall be in such a manner as to not compromise the stability of the excavation. Means of Personal Fall Protection must be installed when employees are exposed to a fall of 6 feet or greater around trench or excavation high walls as mentioned earlier. See Fall Protection Section for more information.
- i) Walkways shall be equipped with guardrails and constructed in accordance with OSHA requirements.

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- j) If vehicle crossings over excavations are required, they shall be designed by and installed under the direction of a Registered Professional Engineer.
- k) Precautions shall be taken to isolate or remove loose rocks, trees, or other materials that may slide, roll, or fall into the trench and onto workers prior to entry by workers into an excavation.
- l) While operating heavy equipment in the work area, the equipment operator shall maintain communication with a designated signaler through either direct voice contact or approved standard hand signals.
- m) When mobile equipment is operated adjacent to an excavation or when such equipment is required to approach the edge of an excavation and the operator does not have a clear and direct view of the edge of the excavation, a warning system such as barricades, hand or mechanical signals, or stop logs shall be used. If possible, the grade should be away from the excavation.
- n) All site personnel should maintain a safe distance and remain clear of the swing of operating excavation equipment.
- o) Employees shall be required to stand away from any vehicle being loaded or unloaded to avoid being struck by any spillage or falling materials. Operators may remain in the cabs of vehicles being loaded or unloaded when the vehicles are equipped to provide adequate protection for the operator during loading and unloading operations.
- p) All materials such as pipe, rebar, etc., shall be kept out of traffic lanes and access ways. Materials and equipment shall be stored in a designated area so as not to endanger personnel at any time.
- q) A flagman with roadwork, signs, cones, and high-level warning signs shall be provided when it is necessary to control normal vehicular traffic due to vehicles, such as end-dumps, entering, or leaving the site.

2.10 Work Within the Trench/Excavation

- a) The presence or buildup of water inside any excavation/trench changes the soil condition from whatever classification had been previously determined to Type C soil, with no exceptions.
- b) Personnel shall not be permitted on the faces of sloped or benched excavations at levels above other workers unless those workers at lower levels are adequately protected from the hazard of falling, rolling, or sliding material or equipment.
- c) Employees shall not work in excavations in which there is accumulated water or in excavations in which water is accumulating, unless adequate precautions have been taken to protect employees against the hazards posed by water accumulation. The precautions necessary to protect employees adequately vary with each situation, but could include special support or shield systems to protect from cave-ins, water removal to control the level of accumulating water, or use of a safety harness and retrieval line.
- d) A stairway, ladder, ramp, or other safe means of egress shall be located in excavations or trenches that are 4 feet (1.22 meters) or more in depth so as to minimize lateral travel for employees.

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- e) Structural ramps that are used solely by employees as a means of access or egress from excavations shall be designed by a competent person. Structural ramps used for access or egress of equipment shall be designed by a competent person qualified in structural design and shall be constructed in accordance with the design.
- f) Ramps and runways constructed of two or more structural members shall have the structural members connected together to prevent displacement. Structural members used for ramps and runways shall be of uniform thickness. Cleats or other appropriate means used to connect runway structural members shall be attached to the bottom of the runway or shall be attached in a manner to prevent tripping. Structural ramps used in lieu of steps shall be provided with cleats or other surface treatments on the top surface to prevent slipping.
- g) All utilities must be identified, protected, supported, and regularly inspected once exposed as mentioned earlier. If the utilities are supported or suspended in the air, all supporting devices must be installed and coordinated with the utility provider's recommendations and reviewed. If needed, the supports may need to be approved by a Registered Professional Engineer.
- h) If red tape or concrete is encountered (electrical circuits) while creating an excavation, stop immediately and notify AECOM TISHMAN CONSTRUCTION Site Management.

2.11 Confined Spaces and Hazardous Atmospheres

- a) An excavation may contain hazardous gases, vapors, dusts, fumes or an oxygen deficient or enriched atmosphere.
 - To prevent exposure to harmful levels of atmospheric contaminants, the hazard assessment shall evaluate atmospheric hazards when workers are required to enter trenches and excavations:
 - Greater than 4 feet (1.22 meters) in depth in which a hazardous atmosphere exists, or could reasonably be expected to exist, such as in excavations in landfill areas, where equipment is exhausting nearby, or where hazardous substances are stored nearby.
 - Less than 4 feet (1.22 meters) in depth if workers could be exposed to a hazardous atmosphere (e.g. crouching).
 - Appropriate atmospheric testing is necessary to accurately identify these hazards. Ongoing atmospheric monitoring, use of attendants and rescue equipment may be necessary to address the identified hazards.
- b) Confined spaces may exist in excavations where there is limited access or egress and in which a hazardous gas, vapor, dust, or fume or an oxygen-deficient atmosphere may occur. Confined space entry shall be performed in accordance with the requirements specified in the Confined Space section. Consult the applicable jurisdictional requirements as the excavation may or may not be subject to confined space requirements.
- c) Adequate precautions, such as mechanical ventilation or appropriate respiratory protection, shall be taken prior to entry into trenches and excavations in which hazardous atmospheres exist or could reasonably be expected to exist.

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- d) When controls are used that are intended to reduce the level of atmospheric contaminants to acceptable levels, testing shall be conducted as often as necessary to confirm that the atmosphere remains safe. Atmospheric testing will be conducted in the anticipated breathing zone of the work area to determine oxygen content, combustible gas, and toxic gases and vapors, if applicable.
- e) Appropriate respiratory protection shall be donned prior to entry into any trench or excavation in which airborne levels of toxic substances are present at concentrations in excess of their Threshold Limit Value/Occupational Exposure Limit or Permissible Exposure Limit.
- f) Confirm appropriate emergency response measures are in place as necessary, including but not limited to:
 - Location Specific Emergency Response Plan shall include procedures applicable to the potential emergencies the excavation work may present.
 - Communication methods shall be established.
 - Equipment such as spill kits, breathing apparatus, and retrieval equipment, shall be readily available.
 - Where hazardous atmospheres are present rescue equipment shall be attended when workers have entered the excavation.
- g) Review and follow the confined space section of this program.

2.12 Stability of Adjacent Structures

- a) Where the stability of adjoining buildings, walls, or other structures is endangered by excavation operations, support systems such as shoring, bracing, or underpinning shall be provided to confirm the stability of such structures for the protection of employees.
- b) Excavation below the level of the base or footing of any foundation or retaining wall that could reasonably be expected to pose a hazard to employees shall not be permitted except when:
 - A support system, such as underpinning, is provided to confirm the safety of employees and the stability of the structure; or
 - The excavation is in stable rock; or
 - A Registered Professional Engineer has approved the determination that the structure is sufficiently removed from the excavation so as to be unaffected by the excavation activity.
 - A Registered Professional Engineer has approved the determination that such excavation work will not pose a hazard to employees.
- c) In addition, sidewalks, pavements, and secondary structures shall not be undermined unless a support system or another method of protection is provided to protect employees from the possible collapse of such structures.

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2.13 Inspections

- a) Daily inspections of excavations, the adjacent areas, and protective systems shall be made by a Competent Person for evidence of a situation that could result in possible cave-ins, indications of failure of protective systems, hazardous atmospheres, or other hazardous conditions. Refer to Attachment 3.1: Daily Excavation Checklist.
- b) An inspection shall be conducted by the Competent Person prior to the start of work and as needed throughout the shift. Inspections shall also be made after every rainstorm or other hazard-increasing occurrence.
- c) Where the Competent Person finds evidence of a situation that could result in a possible cave-in, indications of failure of protective systems, hazardous atmospheres, or other hazardous conditions, exposed employees shall be removed from the hazardous area until the necessary precautions have been taken to confirm their safety and the permit reissued or revised.

2.14 Backfilling

- a) Perform any required notifications and inspections within the necessary timeframes prior to backfilling.
- b) Confirm accurate classification of soil types of backfill material and absence of signs of contamination, discoloration, and smell.
- c) Confirm the re-establishment of the original soil integrity using the original material (if suitable) or designated fill material(s). A small cap of material on top of the ditch/hole should be left to allow for sloughing and settling of material.
- d) Backfilling shall be done with care to prevent damage to any exposed utilities or facilities.
- e) A spotter may be necessary to avoid encroachment (e.g. working around other equipment, traveling under overhead lines, working in close conjunction to underground facilities and other workers, compromised line of vision) and to watch for any rocks falling into the excavation, which may damage exposed facilities. If fill contains rocks or hard material, a shield or alternate fill material may be used to protect the facilities
- f) Confirm piping or facilities are properly supported prior to backfilling.
- g) If shoring was used, remove from the bottom - up following the direction of the professional engineer's guidance.
- h) If a trench box has been used it should be placed no more than 24" above the base of the excavation or a sub-trench bed containing the pipe.
- i) Appropriate measures shall be taken to confirm proper backfilling and compaction of the soil below the trench box. Removing and reinserting the trench box multiple times may be necessary to accomplish this.
- j) Dragging of a trench box shall only be permitted if it will not damage facility or disturb the backfill, otherwise it shall be lifted vertically. No worker shall occupy a trench box while it is being moved.

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- k) If compaction is required confirm the appropriate method is employed and compaction testing is conducted in a manner that does not damage any facilities or pipelines in the excavation.
- l) Final grading and cover of the ground disturbance should confirm erosion control. Original state of the area and access shall be considered in completion of backfilling.
- m) Any excess excavation material shall be properly disposed of.

2.15 Line Strike Avoidance Procedure

Utility Strike Prevention Plan (USPP)

The development of the plan should be compiled and verified at the same time as the site-specific safety plan or SWPPP and well in advance of mobilization. The implementation of this plan should in fact mirror the SWPPP plan we are all familiar with managing. This project specific plan should be incorporated into the AECOM TISHMAN CONSTRUCTION internal project kick off meeting and again between AECOM TISHMAN CONSTRUCTION and Subcontractor/s at every sub or project preconstruction meeting.

These procedures apply to any work that could be or is taking place in the vicinity to utilities on or near the jobsite. This includes existing, new or temporary utilities to or from the building or subcontractor temporary jobsite office locations.

1. Discuss potential of line strikes based on scope of work to be performed.
 - a. If there is going to be any ground penetration greater than 6", subcontractor must obtain an excavation permit from Superintendent/Safety before work is to begin.
2. UG/U/SPP map is to be created from plans and line locations to show where all lines are located and what type of line.
 - a. UG/U/SPP map to be located on trailer wall as well as distributed, updated and distributed.
3. Discuss method of procedure (MOP) before any excavation is to take place.
 - a. Plan must be updated to show location of new/existing underground line as well as be color coded to match utility moved/installed (yellow – gas, Red – electrical, Blue – water, Brown – waste water). This must be in a location easily seen by the project team at all times ie: the jobsite trailer.
4. Every time a new, permanent, or temporary utility is put in place the UG/U/SPP must be updated and all previous maps discarded and replaced. Once all plans have been updated, they must be distributed and verified at least once a month.
5. A physical marker must be put in place in the field with every new UG utility [PERM or TEMP, Buildings or trailers etc.] EG, start and end of underground utility must be clearly marked via flagging and signage, color coded wooden stakes or some form of clearly understood identifiable marking.

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- a. These must be updated in plans as well as maintained to completion of job or until work in that area is completed and accepted as substantially complete by the client.

- 6. Consider the use of ground penetrating radar as necessary.

3. ATTACHMENTS

3.1 Daily Excavation Checklist

3.2 Example Excavation/Trenching Permit

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Attachment 1: Daily Excavation Checklist

Daily Excavation Checklist

S3AM-303-FM1

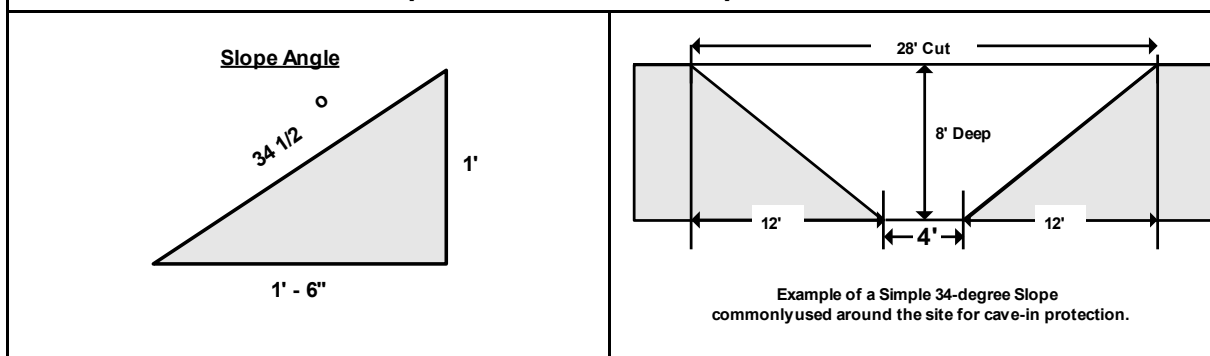
Instructions

1. This form is to be completed and retained in accordance with the Excavation and Trenching Policy.
2. The competent person will use this checklist to inspect the excavation or trench on a daily basis when the potential for employee exposure to the hazards of the trench or excavation exists. Each shift shall inspect all applicable items.
3. If no unsatisfactory conditions are observed, then proceed with the excavation according to the procedure mentioned in Instruction 1. If an unsatisfactory condition (No) is observed, suspend operation of the equipment, and report the condition to the site supervisor immediately.
4. All documentation must be maintained in the appropriate files (project, program, office, etc.).

Daily Excavation Checklist					
Competent Person Name				Date	
Competent Person Signature					
Site Location				Project Number	
Soil Type		Excavation Depth		Excavation Width	
Type of Protective System Used					

Checklist (indicate for each item: Yes, No or N/A for not applicable)	Yes	No	N/A
1. General Information			
a. Is there a potential for a cave-in? If YES, excavation must be sloped, shored, or shielded.	☐	☐	☐
b. Is the excavation deeper than the jurisdictional requirement for the excavation to require sloping, shoring, or shielding?	☐	☐	☐
c. Was a manual method utilized to determine Soil Classification?	☐	☐	☐
d. Was a visual method utilized to determine Soil Classification?	☐	☐	☐
Soil classification comments: 			
e. Is sloping used as your protective system?	☐	☐	☐

Slope Information to Keep in Mind



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Checklist (indicate for each item: Yes, No or N/A for not applicable)		Yes	No	N/A
2. Training				
a.	Have all individuals had Excavation Safety Awareness Training?	☐	☐	☐
b.	Have all individuals reviewed the site-specific Emergency Response Plan?	☐	☐	☐
3. Inspection of Job Site				
a.	Are excavations, adjacent areas, and protective systems inspected by a competent person daily before the start of work?	☐	☐	☐
b.	Does the competent person have the authority to remove all individuals from the excavation immediately?	☐	☐	☐
c.	Are surface encumbrances removed or supported?	☐	☐	☐
d.	Are all individuals protected from loose rock or soil that could pose a hazard by falling or rolling into the excavation?	☐	☐	☐
e.	Are hard hats, safety-toed boots, and safety glasses worn by all individuals?	☐	☐	☐
f.	Is spoil pile set back at least 3.2 feet (1.0 m) from the edge of the excavation and sloped to prevent sloughing and according to jurisdictional requirements?	☐	☐	☐
g.	Are adequate barriers provided at all excavations, wells, pits, shafts, etc.?	☐	☐	☐
h.	Are warning vests or other highly visible clothing provided and worn by all individuals?	☐	☐	☐
i.	Are all individuals required to stand away from vehicles being loaded / unloaded?	☐	☐	☐
j.	Are warning system established and utilized when mobile equipment is operating near the edge of the excavation (e.g., barricade tape, signalpersons, stop logs, etc)?	☐	☐	☐
k.	Are all individuals prohibited from going under suspended loads?	☐	☐	☐
l.	Are materials and equipment set back 3.2 feet (1.0 m) or greater from the edge of the excavation?	☐	☐	☐
4. Utilities				
a.	Are locations of utilities marked?	☐	☐	☐
b.	Prior to the use of equipment, have underground utilities been exposed and located by hand / non-destructive clearance techniques?	☐	☐	☐
c.	Are underground utilities protected, supported, or removed when excavation is open?	☐	☐	☐
d.	Are overhead lines de-energized or protected by other appropriate means?	☐	☐	☐
5. Means of Access and Egress				
a.	Is travel distance to means of egress no greater than the jurisdictional requirement in excavations 4 feet (1.2m) or more in depth?	☐	☐	☐
b.	Do straight ladders used in excavations extend at least 3 feet (0.9m) above the edge of the trench?	☐	☐	☐
c.	Have ramps being used for employee access been designed by the competent person?	☐	☐	☐
d.	Are all individuals are protected from cave-ins when entering / exiting the excavation?	☐	☐	☐
Protection Description:				

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Checklist (indicate for each item: Yes, No or N/A for not applicable)	Yes	No	N/A
6. Wet Conditions			
a. Have precautions been taken to protect workers from the accumulation of water?	☐	☐	☐
b. Is water removal equipment monitored by a competent person?	☐	☐	☐
c. Is surface water or runoff diverted or controlled to prevent accumulation in the excavation?	☐	☐	☐
d. Have inspections been conducted after every rainstorm or other hazard-increasing occurrence (freeze/thaw, local demolition, rerouting of traffic, etc)?	☐	☐	☐
7. Hazardous Atmosphere: The atmosphere within the excavation must be tested where there is a reasonable possibility of an oxygen deficiency or a combustible or other harmful contaminant exposing any individual to a hazard.			
a. Are there exposed sewer or natural gas lines in excavation?	☐	☐	☐
b. Is equipment operating nearby that may produce introduce exhaust into the excavation?	☐	☐	☐
c. Is the excavation near a landfill area, or are hazardous substances being stored close to the excavation?	☐	☐	☐
If you answered YES to A, B, or C then treat the excavation as a confined space.			
d. Has an emergency rescue plan been developed and are rescue services in place?	☐	☐	☐
8. Support Systems			
a. Are materials and/or equipment for support systems selected based on soil analysis, trench depth, and expected loads?	☐	☐	☐
b. Have materials and equipment used for protective systems been inspected and are they in good condition?	☐	☐	☐
c. Have materials and equipment in unsatisfactory condition have been removed from service?	☐	☐	☐
d. Are protective systems installed without exposing all individuals to the hazards of cave-ins, collapses, or the threat of being struck by materials or equipment?	☐	☐	☐
e. Are members of support system securely fastened to prevent failure?	☐	☐	☐
f. Are support systems provided to ensure stability of adjacent structures, buildings, roadways, sidewalks, walls, etc.?	☐	☐	☐
g. Have excavations below the level of the base of a footing been approved by a registered Professional Engineer?	☐	☐	☐
h. Does removal of support systems progress from the bottom, and are members released slowly to allow for detection of any indications of possible failure?	☐	☐	☐
i. Does backfilling progress with the removal of the support system?	☐	☐	☐
j. Is material excavated to a level no greater than 2 feet (0.6m) below the bottom of the support system and only if the system is designed to support the loads calculated for the full depth?	☐	☐	☐
k. Has a shield system been placed to prevent lateral movement?	☐	☐	☐
l. Are all individuals prohibited from remaining in the shield system during movement?	☐	☐	☐

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Attachment 2: AECOM TISHMAN CONSTRUCTION Example Excavation/Trenching Permit

Excavation & Trenching Permit

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Authorization No.: _____

Date & Time Permit Valid: _____

Competent Person: _____

Date & Time Permit Expires: _____

Project Name: _____

Excavation/Trench Location: _____

Description or Job Special Procedures:

ESTIMATED DIMENSIONS:	DEPTH = _____			SOIL TYPE: ☐ Stable Rock ☐ Type A ☐ Type B ☐ Type C ☐ Avg. Compression Strength _____ tsf ☐ Compressed Strength Data _____
	TOP =	W _____	L _____	
	BOTTOM =	W _____	L _____	
SOIL ANALYSIS METHOD(S) USED: ☐ Visual ☐ Manual ☐ Tabulated Data				MANUAL TEST USED: ☐ Plasticity ☐ Dry Strength ☐ Ribbon ☐ Thumb Penetration ☐ Pocket Penetrometer ☐ Dry Testing ☐ Other
SOIL CHARACTERISTICS: ☐ Cemented ☐ Cohesive ☐ Layered ☐ Fissured ☐ Granular ☐ Plastic ☐ Dry ☐ Moist ☐ Saturated ☐ Submerged				
PROTECTIVE SYSTEMS: ☐ Prof. Engineered Protective System Required				
SLOPING/BENCHING: ☐ Vertical (90°) ☐ 3/4 :1 (53°) ☐ 1:1 (45°) ☐ 1 1/2:1 (34°) ☐ 2:1 (26°) ☐ Other				UTILITIES (underground/overhead): ☐ One Call Service Notified ☐ Search Zone Established ☐ Utilities Marked by Public Utilities Locator ☐ Property Owner Contacted ☐ Utility Drawings Reviewed ☐ Utilities Marked by Private Locator
SHORING: ☐ Timber ☐ Trench Shield/Trench Box ☐ Aluminum Hydraulic/Pneumatic				
OTHER: ☐ Means of Egress Required ☐ Confined Space Permit Required ☐ Pre-Entry Air Monitoring Required ☐ Mechanical Ventilation Required ☐ Owner / Designate Required to be Present ☐ Surveys, drawings, Plot Plans, etc. Attached ☐ Landowner Consent Obtained ☐ Other				
LIST OF KNOWN OBSTRUCTIONS: ☐ Electrical ☐ Telephone ☐ Water ☐ Sewer ☐ Steam ☐ Natural Gas ☐ Drain ☐ Process ☐ Alarm ☐ Concrete Encasement ☐ Pilings ☐ Footings ☐ Other				UTILITIES (hand / non-destructive clearance requirements): ☐ Radius Established – ☐ Exposure Interval Established – ☐ Other
SPECIAL INSTRUCTIONS and WORK INSTRUCTIONS				

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All unsafe conditions shall be corrected prior to excavation entry. If any hazardous conditions are observed, the excavation shall be evacuated immediately, and no one is permitted to re-enter until corrective action has been taken.

Signature and Dates
(Manager shall determine which signatures are required)

	Print Name	Signature	Date
Excavation Competent Person (Required)	_____	_____	_____
Supervisor	_____	_____	_____
SH&E Representative	_____	_____	_____
Registered Professional Engineer (if applicable)	_____	_____	_____
Other	_____	_____	_____

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SECTION 9 - FALL PROTECTION

1. TERMS and DEFINITIONS

1.1 Active Fall Protection System. A system that requires workers to take specific actions, including wearing (and otherwise using) personal fall protection equipment and following prescribed procedures. Examples include travel restraint and fall arrest systems.

1.2 Anchor Point. A secure point of attachment for lifelines, lanyards or deceleration devices that can withstand and absorb the arrest force as specified by the appropriate jurisdiction. Anchors may be temporary or permanent, and may be designed and tested or improvised and verified as capable of withstanding the arrest force. Improvised anchors may be a building structural component, crane, or other support.

1.3 Authorized User. An individual who has completed appropriate fall protection training and has been approved to work at heights using Fall Protection / Prevention Systems.

1.4 Base Surface. The area immediately beneath an elevated structure or surface (ground, walking surface, floor, etc.). Used synonymously here with "grade."

1.5 Climbing Protection System. A type of fall protection used on long ladders, poles, and towers that incorporates a permanently installed rail or cable that runs the length of the ladder. The climber's harness is attached to the rail or cable by a sliding device or sleeve that allows climbing freedom but locks the instant a fall is sensed.

1.6 Deceleration Device. A mechanism, such as a rope grab, rip-stitch lanyard, specially woven lanyard, tearing or deforming lanyards, automatic self-retracting lifelines/lanyards, etc., that serves to dissipate a substantial amount of energy during a fall arrest or to otherwise limit the energy imposed on an individual during fall arrest.

1.7 Deceleration Distance. The additional vertical distance a falling individual travels, excluding lifeline elongation and free fall distance, before stopping, from the point at which the deceleration device begins to operate. It is measured as the distance between the location of an individual's body harness attachment point at the moment of activation (at the onset of fall arrest forces) of the deceleration device during a fall, and the location of that attachment point after the individual comes to a full stop.

1.8 Free Fall Distance. The vertical distance from the fall arrest attachment point on the individual's full body harness at onset of the fall to the fall arrest attachment point just before the system begins to apply force to arrest the fall. This distance excludes deceleration distance, and lifeline/lanyard elongation, but includes any deceleration device slide distance or self-retracting lifeline/lanyard extension before they operate and fall arrest forces occur.

1.9 Full Body Harness. A tight-fitting harness that includes shoulder, chest and thigh straps, designed to fit the torso and spread the shock associated with arresting a fall over the entire torso area. Harnesses typically have connecting D-rings at chest level in the front and the back for attaching a lanyard, however connectors may be located at shoulders or waist depending upon the proposed use of the fall protection system (e.g. ladder climbing, work positioning, etc.).

1.10 Lanyard. A line connecting a safety harness to a safety line or structure.

1.11 Lifeline / Safety Line. A flexible line for connection to an anchorage at one end to hang vertically (vertical lifeline), or for connection to anchorages at both ends to stretch horizontally (horizontal lifeline), and which serves as a means for connecting other components of a personal fall arrest system to the anchorage.

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1.12 Passive Fall Protection / Prevention System. Any structure, fence, or barrier that will prevent falls during work at heights. Examples include ladder cages, guardrail systems and scaffolding.

1.13 Personal Fall Arrest System. A system used to arrest an individual in a fall from a working level. It consists of an anchorage, connectors and full-body harness and may include a lanyard, deceleration device, lifeline or suitable combinations of these.

1.14 Rope Access. A means of access by descending or ascending a main line while the worker is protected by a safety line.

1.15 Safety-Monitoring System. An administrative control in which a competent person is responsible for recognizing and warning individuals of fall hazards. AECOM TISHMAN CONSTRUCTION does not permit safety monitoring systems in lieu of positive fall protection.

1.16 Standard Railing. Railing provided to enclose open-sided work platforms and consisting of a top rail, intermediate rail, toe board, self- retracting lifeline, and posts.

1.17 Travel Restraint System. A system designed to prevent individuals from reaching an unprotected edge. Harness and lanyard are connected to a lifeline and suitable anchor, with the system adjusted to limit travel.

1.18 Work at Heights. Any work/job/task to be performed above the normal walking/working surface that necessitates the use of some form of fall protection as determined by the applicable governing rules and regulations. This requirement shall be 6 feet (1.8 meters) or greater in the construction, mining, and demolition industry, and 4 feet (1.2 meters) or greater in other industries (General Industry).

2. GENERAL REQUIREMENTS

2.1 100% positive fall protection is required for all persons exposed to a fall potential of six (6) feet or greater. This requirement includes all operations unless otherwise noted in this program. Positioning devices do not satisfy the positive fall protection requirement and must be used in conjunction with fall arrest system or fall restraint. Fall protection equipment may include but is not limited to personal fall arrest, personal fall restraint, guardrails, catch platforms, nets, signage, etc. All fall protection must be positive in nature, meaning there must be physical means to prevent an individual(s) from falling.

2.2 All orientations, safe-starts, task/job hazard analysis and/or other preplanning activities shall include and cover these requirements.

2.3 All operations shall be overseen by a competent person. Subcontractors shall ensure all procedures in this document are incorporated into their project plans. They are solely responsible for their employees.

2.4 Prior to work commencing:

- a) The appropriate Fall Protection Plan, Fall Rescue Plan, and as applicable, the Alternative Fall Protection Permit, shall be in place and be reviewed (including its associated requirements and fall protection regulations) with all affected individuals.
- b) Individuals shall be trained appropriately for the fall protection system and fall rescue system to be used. Subcontractors are solely responsible for the training of their individuals.
- c) Control or restrict access below or around the individuals working at elevation.
- d) Confirm the use of barricades, caution tape, and signs identify restricted areas.

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2.5 Individuals will not undertake climbing under personal conditions of fatigue, sickness, or injury, nor will they expose themselves to unusual or unacceptable risks.

2.6 Individuals using medications that may interfere with alertness or physical strength will not perform work at elevations.

2.7 Managers will not allow individuals known to be experiencing personal or psychological stress to undertake work at elevations.

2.8 Any equipment chosen to support a person at height should be such that it cannot be accidentally removed, dislodged or become unfastened from the rope, lanyard, or anchor point while a person is suspended from it.

2.9 Fall arrest equipment shall meet applicable jurisdictional requirements and bear the required standards (e.g., NFPA, ASTM, CSA, ANSI, CE, etc.) label, and be tagged as formally inspected and certified annually according to the manufacturer's specifications.

2.10 Visually inspect positioning, arrest, and restraint components before each use for wear, damage, or deterioration. Defective components are removed from use and tagged out to prevent others from using them.

2.11 Any fall arrest components which are damaged or have been involved in a fall shall be tagged out and removed from service. If possible and permitted, the components shall be repaired, and recertified prior to return to service.

2.12 Individuals shall use the buddy system to continually check each other's harness and D-ring to confirm that the harness is not too loose and/or the D-ring is appropriately positioned (e.g., has not slipped down the back).

2.13 Fall protection equipment shall be properly assembled, inspected, used, maintained, stored, and disassembled.

2.14 Use Personal Protection Equipment in accordance with manufacturers' specifications.

2.15 Do not attempt to repair or modify equipment unless properly trained and certified to do so.

2.16 If there is a question of unacceptable risk in working at any elevated work location, the work shall not be done until such questions are dealt with. Alternatives to climbing (elevators, aerial lifts, etc.) shall be used when practical.

2.17 All incidents, including falls and near misses, shall be reported immediately.

2.18 A task hazard analysis (THA) is required prior to initiating a task, including work at height tasks.

2.19 Each recognized work at height hazard shall be mitigated to comply with applicable OSHA regulations. Mitigation control methods shall adhere to the hierarchy of controls described below:

- a) The preferred form of work at height hazard mitigation is elimination or substitution such as moving planned work to ground level or substitution of a process, sequence, or procedure so that individuals are no longer exposed to a fall hazard.
- b) Passive fall protection such as isolating or separating individuals from the fall hazard through the use of guardrails or covered floor openings is the next preferable form of work at height hazard mitigation.

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- c) If the previous two methods are not feasible, active fall protection systems may be employed:
 - i. Travel restraint, which includes securing an individual to an anchor using a lanyard that is short enough to prevent the individual's center of mass from reaching the actual fall hazard. Travel restraint systems are only to be used on walking or working surfaces with a slope of less than 18.4°.
 - ii. Fall arrest, which includes systems to stop an individual's fall after it has begun (e.g. personal fall arrest systems, safety nets, etc.).
 - iii. A safety net.
- d) Administrative controls may be employed; including a control zone and/or boundary markings (i.e., tape, cable, signs and/or barricades) located 6 feet (2 meters) or more from an exposed edge, and maintaining three-points of contact.

2.20 No individual shall perform work at heights alone.

2.21 All fall protection equipment shall be stored appropriately to prevent any damage.

2.22 100% fall protection is required in mobile elevated work platforms (MEWPs).

2.23 Self-retracting lanyards should be used in lieu of any bungee style lanyards.

2.24 100% fall protection shall be used when climbing any formwork or rebar.

2.25 Tool tethers shall be used wherever work must be completed at height and there is potential for a tool to fall to a lower level. Unless it creates a great risk or is infeasible, materials being installed at the perimeter of the building should be tethered until secured in place.

3. GUARDRAIL SYSTEMS

Guardrail systems shall meet all applicable OSHA standards, and the following at a minimum:

3.1 Top rails of guardrail systems must be 42 inches above the walking/working surface plus or minus 3 inches. Guardrail systems must be able to withstand a force of 200 pounds in any outward or downward direction at any point along the top edge without deflecting to a height less than 39 inches. If wire rope is used for top rails, it must be flagged at not more than 6 foot intervals with high visibility material.

3.2 Midrails must be used between the top rail and the walking/working surface when there is no wall or parapet wall at least 21 inches high. Midrails must be able to withstand downward or outward pressure of 150 pounds.

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3.3 When guardrail systems are used around holes that are used as points of access (such as ladder ways), they must be provided with a gate or be so offset that a person cannot walk directly into the hole.



3.4 A toe board shall be installed as part of each guardrail system. The toe board is required to be a minimum of 3½ inches high (4 inches in California) with a maximum of ¼ inch of clearance off the floor. The toe board members shall be installed to withstand a downward or outward force of 50 pounds. The distance between toe board members shall not be greater than 1 inch. When netting is used as part of a guardrail system, it must be secured to the floor.

3.5 Wooden guardrails shall be designed as follows:



3.6 Wire rope guardrails shall meet all applicable OSHA standards, and:

- a) Use a minimum 3/8 inch diameter cable.
- b) Top rail (cable) will be installed 39 to 45 inches (42 inches in California) above the top of the finished slab elevation. The intermediate rail will be installed at the mid-point between the top of the finished slab and railing. High visibility material such as construction tape will be installed on the top rail at 6 foot intervals.
- c) Cables are to be installed so that cable tension and cable supports (columns and intermediate posts) prevent cables from deflecting below 39" above the walking/working surface inches when a 200 pounds downward force is applied.
- d) Intermediate stanchions at approximately 10 feet on center will be required unless they are engineered at a different spacing to comply with OSHA standards.

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- e) Cables must pass through holes in the column webs or be secured at the interior side of each structural column with substantial supports that can support 200 pounds without bending or breaking (no welding washers onto the columns instead of using nuts, etc.).
- f) Eye splices must be utilized for cable splicing with a minimum of three wire rope clamps. No bypass or Molly Hogan splices are permitted.
- g) Cables must be terminated with eye splices around each end support; no dead end terminations are permitted.
- h) 18 inch turnbuckles are to be installed every third bay or 100 feet, whichever is less.
- i) A minimum of three wire rope clamps shall be used at each tie back.
- j) Single cables running past each other with one or two clamps are not acceptable.
- k) A third rail of cable guardrail shall be installed wherever a leading edge exceeds 20 feet to the level below. The third rail shall be installed above the OSHA minimum top rail at 60 inches plus or minus 3 inches. Safety debris netting will also be required for any leading edge areas over 20 feet tall, including all catwalks and stair tower landings. Safety debris netting in a ¼ inch mesh pattern is required. A request for a variance must be approved by Regional SH&E.

3.7 Hanging tarps from a guardrail top rail or mid rail is prohibited. If hanging tarps, the tarps must be hung from a separate independent cable system or the bottom of the posts.

3.8 When individuals are using stilts, the top height of the top rail shall be increased an amount equal to the height of the stilts, and they shall not be allowed within 10 feet of an edge or guardrail system. Stilts are prohibited in the state of California.

4. SAFETY NET SYSTEMS

Safety net systems shall meet all applicable OSHA standards, and the following at a minimum:

- 4.1 Safety nets must be installed as close as practicable under the walking/working surface but in no case more than 30 feet below.
- 4.2 They must extend outward from 8 to 13 feet depending on the vertical distance to the net.
- 4.3 Unless it is unreasonable to do so, they must be drop-tested after installation, relocation, major repair, and at 6-month intervals if left in one place.
- 4.4 They must be inspected at least once a week for wear, damage, and other deterioration.
- 4.5 Materials, scrap pieces, equipment, or tools that fall into the net must be removed as soon as possible and at least before the next shift.

5. PERSONAL FALL ARREST SYSTEMS

Personal fall arrest systems shall meet all applicable OSHA standards, and the following at a minimum:

- 5.1 Shall consist of a full-body harness, a self-retracting lanyard, and a legal anchor point. Where self-retractable lifelines are not feasible, double lanyards shall be utilized.
- 5.2 Shall have anchor points that support at least 5,000 pounds per person attached.

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5.3 Shall limit the shock / arresting force on the individual to 1,350 pounds (6kN).

5.4 Personal fall arrest systems shall be installed and used following manufacturer requirements. All systems shall be inspected for damage prior to use.

5.5 The attached point (D-ring) of the body harness must be located in the center of the wearer's back near shoulder level.

5.6 Personal fall arrest systems may not be attached to guardrail systems unless they are designed for that purpose by a competent person and capable of supporting 5,000 pounds for each person attached.

5.7 Positioning devices must be rigged so the individual cannot free-fall more than 2 feet. They must be secured to an anchorage capable of supporting 3,000 pounds.

5.8 Personal fall arrest systems used during any hot work activity (e.g. welding, torch cutting, grinding, brazing, or soldering) shall comply with the following requirements:

- a) Lanyards, lifelines and harnesses shall be rated for hot work (heat resistant and flame retardant).
- b) Personal fall arrest systems shall be monitored so that they are not allowed to come into contact with heated materials, flame, or welding slag.
- c) Personal fall arrest systems shall be inspected prior to, and during use, for burns and other signs of heat damage. Personal fall arrest systems showing signs of heat damaged shall be removed from service immediately.
- d) Sheathing or other protective devices used to protect personal fall arrest systems shall be designed for this purpose and shall be inspected regularly. Hose, wraps, and other non-flame retardant or non-fire rated material shall not be used to protect personal fall arrest systems from fire or heat damage.



6. WARNING LINE SYSTEMS

Warning line systems shall meet all applicable OSHA standards, and the following at a minimum:

6.1 May be used on low-sloped roofs, defined by OSHA as slope less than or equal to 4 in 12 (vertical to horizontal).

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6.2 The warning line must be erected around all sides of the roof work area. When motor or human propelled wheeled equipment used for roofing work, except wheelbarrows and mop carts, is not being used, it must be erected not less than 6 feet from the roof edge for roofers. When motor or human propelled wheeled equipment used for roofing work, except wheelbarrows and mop carts is being used, the warning line must be not less than 6 feet from the parallel edge and not less than 10 feet from the perpendicular edge. This only applies to workers engaged in roofing work activities. All other trades must utilize a warning line system setup at least 15 feet back from the exposed edges.

6.3 Warning lines must consist of ropes, wires or chains, warning signs, and supporting stanchions erected as follows:

- a) The rope, wire or chain must be flagged at not more than 6 foot intervals with high visibility material.
- b) Its lowest point, including sag, may be no less than 34 inches and the highest point not more than 39 inches.
- c) Stanchions (after erected) must be able to withstand a force of 16 pounds applied horizontally against the stanchion 30 inches above the walking/working surface perpendicular to the warning line in the direction of the edge.
- d) The rope, wire or chain must have a minimum tensile strength of 500 pounds.
- e) The line must be attached to each stanchion such that pulling on the line in one section will not result in taking up slack in the adjacent section before the stanchion tips over.

6.4 Warning line systems for all steel decking, precast deck operations, cast in place concrete operations, and floor sheathing operations shall be set back a minimum of 6 feet from the edge and anyone passing the 6 feet shall use proper fall protection. All other operations shall refer back to OSHA standards.

7. FALL PROTECTION TRAINING

7.1 Only properly trained individuals shall be permitted to perform work at height tasks.

7.2 At a minimum, all training shall be documented and, at a minimum, shall include:

- a) Identifying the working at heights hazards and methods of elimination and control;
- b) Selection of fall protection systems and the associated equipment according to potential hazards and equipment compatibility (e.g. guardrail, travel restraint, personal fall arrest, safety net, warning line, safety monitoring systems, controlled access zones, etc.);
- c) Proper use, including assembly and disassembly, of fall protection equipment;
- d) Potential effects on the human body of a fall from heights while using fall protection systems;
- e) Calculation of fall distances, and pendulum effect;
- f) Care of fall protection equipment, including pre-use inspections and rejection criteria, maintenance, and storage in accordance with manufacturer's specifications;
- g) Limitations of fall protection systems and the various components;
- h) Signs and symptoms of orthostatic intolerance (suspension trauma), the causes of orthostatic intolerance while suspended; and recommended prevention and rescue measures.

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7.3 Retraining will occur when any of the following conditions occur:

- a) Deficiencies are identified in training.
- b) Workplace changes.
- c) Fall protection systems or equipment changes that render previous training obsolete.
- d) After a fall protection incident.

7.4 No individual shall attempt to perform work that he/she has not been trained to safely perform or that he/she considers unsafe, including specific activities involving work at heights.

7.5 Proof of training shall be presented to AECOM TISHMAN CONSTRUCTION management upon request.

8. FALL PROTECTION PLANS

8.1 Where Personal Fall Arrest Systems are required, a task-specific Fall Protection Plan shall be developed prior to work commencing to confirm that individuals exposed to work at heights are protected from falls for 100 percent of the duration of their task, including the time it takes them to locate to their task (e.g. climbing a ladder, transferring materials, descending a pit or vault, and working within a cage arrangement).

8.2 Fall Protection Plans shall be approved by a qualified and/or competent person and the SH&E AECOM TISHMAN CONSTRUCTION superintendent before work commences.

8.3 The Fall Protection Plan shall specify:

- a) The fall hazards at the work site. Consult the applicable Task Hazard Analysis (THA).
- b) The fall protection system to be used (e.g. guardrails, travel restraint, fall arrest) and as applicable specify:
 - i. Adequate anchor points available at each location where Active Fall Protection Systems are used (Confirm any engineered anchor point documentation and inspection data are available).
 - ii. Clearance distances and applicable system components.
 - iii. Assembly/disassembly, maintenance, inspection and use of the applicable system components.
 - iv. Site-specific procedures for the Personal Fall Arrest System(s) to be used.
 - v. Limitations of the systems to be used.
- c) A rescue plan including the procedures to be used if a worker falls, is suspended by a personal fall arrest system or safety net and needs to be rescued, or requires rescuing from the working height.

8.4 The Fall Protection Plan shall be available at the work site and reviewed with all individuals, including affected workers, before beginning any work where there is a risk of falling. This includes any training or review of equipment usage.

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8.5 Identify fall hazards on an ongoing basis and review fall protection equipment needs. If conditions change or it is determined the system in use does not provide adequate protection, the work from heights shall be suspended. The Fall Protection Plan shall be appropriately revised and reissued prior to work recommencing.

9. FALL RESCUE PLANS

9.1 A fall rescue plan shall be developed and included in the Fall Protection Plan for each specific project in their emergency action plan where individuals will be working at height. The fall rescue plan shall incorporate procedures that will confirm prompt rescue in the event that any individual falls.

9.2 At a minimum, the rescue plan should include:

- a) Effective rescue procedures shall be established in advance of an event requiring the use of personal fall arrest systems. Rescue measures shall be capable of being initiated within 5 minutes of a fall and the worker shall be able to be relieved of strap pressure within 15 minutes.
- b) Specific equipment and procedures to facilitate self-rescue, if applicable and appropriate.
- c) As applicable, procedures for utilizing outside rescue services (e.g., client facility emergency services, local fire department rescue services, etc.).

9.3 The appropriate rescue option shall be selected from the following:

- a) Self-Rescue - A reliable method for self-rescue shall be developed and deployed which shall include the use of the “buddy system” when personnel are working in conditions where fall protection systems (other than standard guard rail systems) are employed. The “buddy” or assigned safety person shall remain within visual/verbal range to initiate rescue of the fallen worker if required and shall have a reliable method of communication (cell phone, radio, etc.) to summon rescue.
- b) Assisted Rescue - Provided either by a professional rescue agency, the fire department or by qualified company personnel.
- c) Qualified personnel may be used to provide prompt rescue of fallen workers. If outside professional rescue agencies cannot be relied upon to promptly rescue fallen personnel, then company personnel shall be identified and trained to serve as rescue personnel. Training shall be conducted by a qualified training agency or company and shall include specific training on the methods and equipment which may be required during site-specific rescue scenarios. It shall include hands on training and certification to demonstrate proficiency in rescue and shall be of the duration recommended by the training agency/company. Equipment procured for use as well as practice requirements (drills) and recurring formal training requirements for the company rescue personnel shall conform to the recommendations and requirements set forth by the training agency/company and OSHA requirements.

10. HOLES AND OPENINGS

10.1 All individuals shall be protected from falling through holes and/or openings by way of personal fall arrest systems, covers, and/or guardrails.

10.2 All individuals shall be protected from tripping into the holes and/or openings.

10.3 All walking or working surfaces shall have all holes and/or openings protected.

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10.4 All skylights shall also be protected along with wall openings (including those with chutes attached) where the outside bottom edge of the wall opening is six (6) feet or more above lower levels and the inside bottom edge of the wall opening is less than 39 inches above the walking/working surface.

10.5 Standard railings around holes and/or openings shall consist of a top rail where the top edge shall be 42 inches plus or minus 3 inches above the walking/working level, mid-rails halfway between the top rail and the walking/working level (half the overall distance), and toe boards. See section 3 of this program section.

10.6 Standard hole/opening covers made from appropriately sized plywood may be used on any hole and/or opening equal to or less than 2 feet by 2 feet in size.

10.7 If a plywood hole/opening cover is to be used on a hole or opening on a non-concrete floor greater than 2 feet by 2 feet, a secondary means of protection should be provided (except for steel erection activities), which includes but is not limited to:

- a) A tether strap attached to the plywood cover at each end, independently anchored into the surrounding concrete floor.
- b) A standard guardrail system within 12 inches around the hole or opening edges. In this case, a plywood cover would not be necessary.

10.8 Embedded safety netting, specifically designed for this purpose, should be installed in all holes in cast in place concrete that are a minimum of 1x2. The safety netting shall have a minimum breaking strength of 1,300 lbs. and be constructed at a minimum of #375 Nylon Mesh with holes that are less than 3" X 3".

- a) If an operation must penetrate the hole, such as running a pipe or duct up through the area, the safety netting may be cut, per the manufacturer's recommendations, to allow for the penetration. The amount of safety netting cut should only be the amount required to install the work and upon completion, the plywood cover, if applicable, shall be modified and reinstalled around the penetration.
- b) The embedded netting must be installed per the manufacturer's specifications.
- c) This embedded netting is only to be treated as a secondary means of protection. Each hole/opening shall also have plywood covers and/or guardrails as detailed in this policy.
- d) Each project team shall communicate this policy to their local code enforcement officer prior to each project to ensure they have no issues with it. If they do, please contact the AECOM TISHMAN CONSTRUCTION Safety Department immediately.

10.9 All covers shall support without failure at least twice the weight of the employees, equipment and materials that may be imposed on the cover at any one time.

10.10 All covers shall be secured and/or placed per applicable OSHA standards.

10.11 All covers shall be color coded or marked with the words "hole" or "cover".

10.12 Core drilled holes also fall under these requirements.

10.13 Holes / penetrations / openings in vertical / angle surface walls: Any through wall hole / penetration / opening greater in size than 16 inches in width and greater than 16 inches in vertical dimension, that permits the potential for personnel or debris to fall to a lower surface beneath the surface

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of approach to the hole/penetration must be covered or have guardrail installed. 200 lb psf impact netting will be acceptable as well for wall hole/opening protection.

10.14 Other systems or methods, approved by the AECOM TISHMAN CONSTRUCTION Regional Director of Safety, may be utilized to meet the intention of this section.

11. ATTACHMENTS

11.1 Personal Fall Arrest System Clearance Distances

11.2 Personal Fall Arrest System Inspection Checklist Example

11.3 Fall Protection Training Record Example

11.4 Standard Form for Release of Floor or Stairs

11.5 Fall Rescue Devices

11.6 Zero Tolerance Policy for Fall Protection

11.7 Shaft Work Permit

11.8 Guardrail/Fall Protection Removal Permit

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Attachment 1: Personal Fall Arrest System Clearance Distances

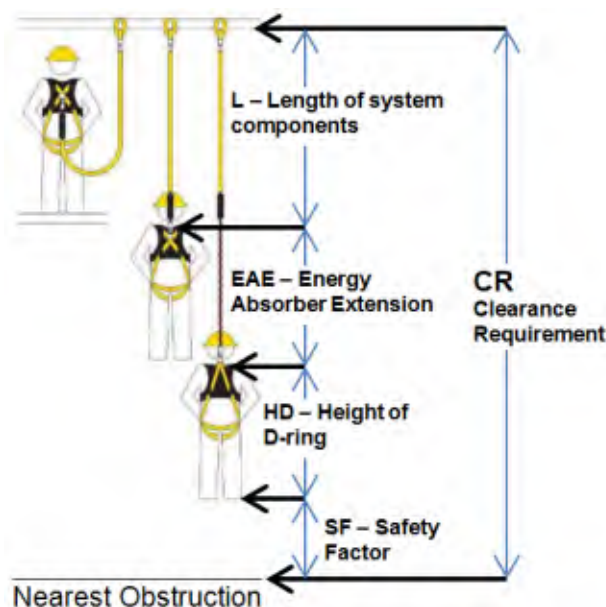
Personal Fall Arrest System Clearance Distances

Before using any Personal Fall Arrest System, the individual shall confirm they have selected a system that allows for ample clearance.

Consideration shall be given to the stretch in the harness, flipping and sliding of the D ring, stretch and elongation of the energy absorbing lanyard, elongation of the vertical lifeline (if present), and elongation of the horizontal lifeline (if present).

Calculations shall be completed given the components of the fall protection system to be used, the distance to the next level or obstruction below and a safety factor, typically 2 feet (0.6m), added.

Calculations of clearance requirements can be completed using the following formulas:



L – Initial total length of system components (lanyard, energy absorber, & connecting hardware).

HD – Height of D-ring from user's feet.

DA – Distance between anchor point and unguarded edge.

EAE – Energy absorber extension.

DS – D-ring slippage (typically 1 foot [0.3m]).

SF – Safety factor (typically 2 feet [0.6m]).

Measured from rigid anchor point at unguarded edge:

$$CR = L + EAE + HD + DS + SF$$

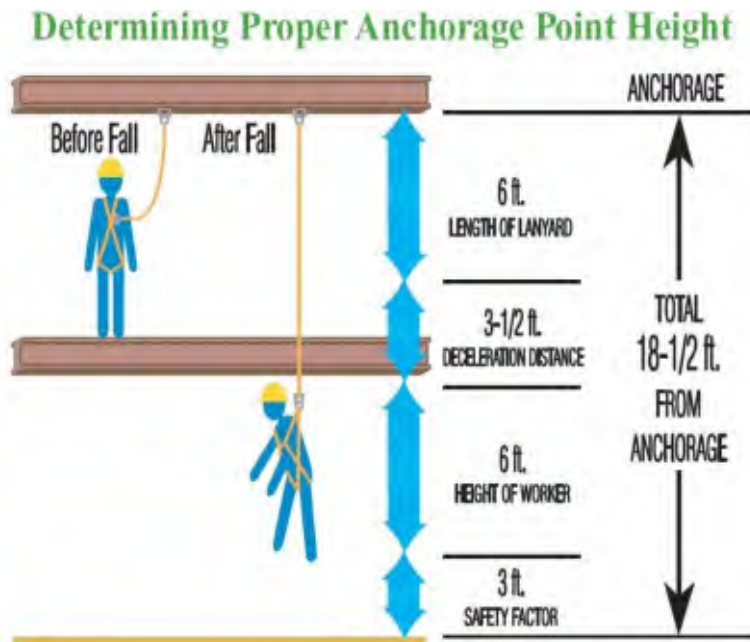
Measured from rigid anchor point away from unguarded edge:

$$CR = L + EAE + HD + DS + SF - DA$$

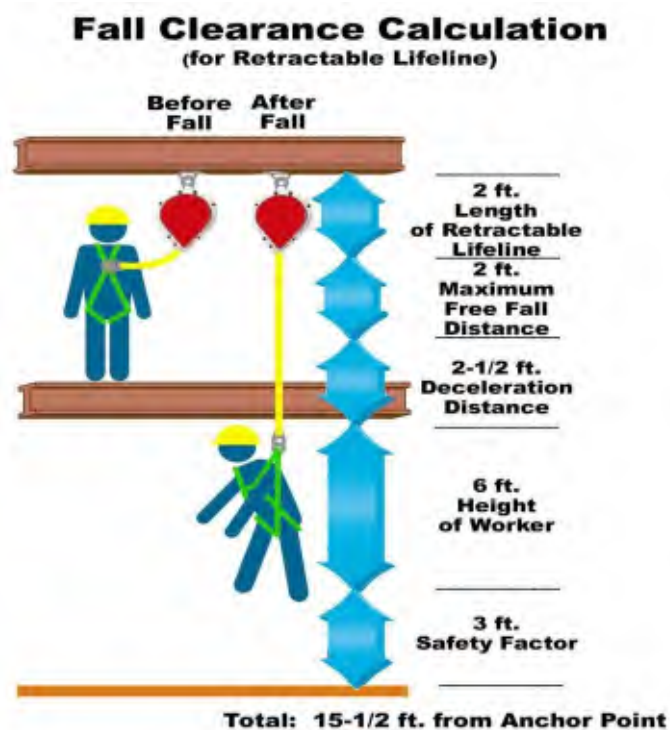
Remember to confirm equipment manufacturer's specifications are consulted when calculating clearance requirements.

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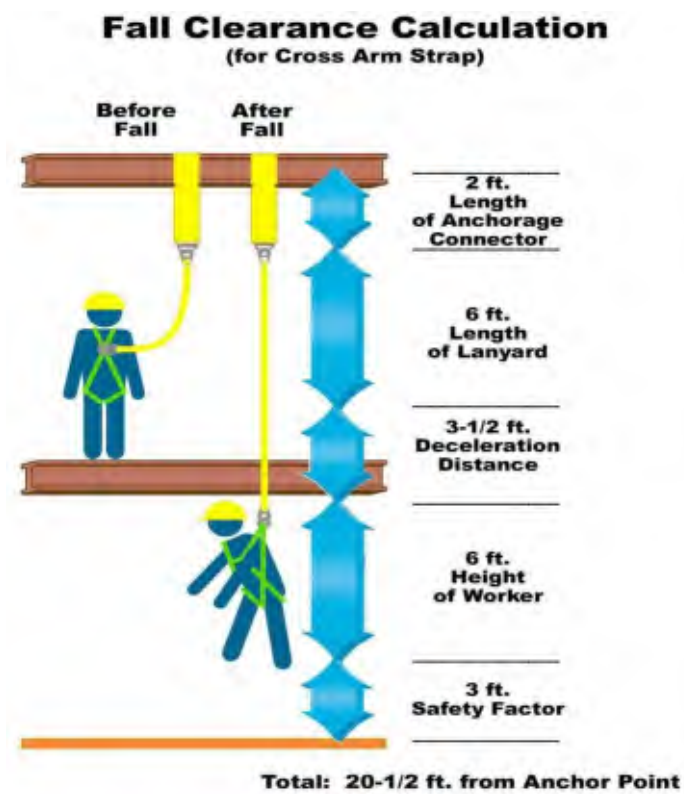
Anchorage Clearance with 6 feet long lanyard.



Anchorage Clearance with Retractable Lifeline.



Anchorage Clearance with a Cross Arm Strap.



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Attachment 2: Personal Fall Arrest System Inspection Checklist Example

Harness Inspection Checklist

- ☐ Inspect the label:
 - Labels shall be attached to equipment; any equipment without a label shall be removed from service immediately.
 - Label shall include appropriate Standards identification.
 - Label shall include inspection verification per manufactures instructions.
- ☐ Inspect the webbing:
 - Inspect the entire surface of webbing for damage. Beginning at one end, bend the webbing in an inverted "U."
 - Look for frayed edges, broken fibers, pulled or broken stitches, cuts, or chemical damage, abrasion, excessive wear, altered or missing straps, burns, and heat or chemical exposure of the webbing.
 - Broken webbing strands generally appear as tufts on the webbing surface.
 - Inspect under any removable padding.
 - Inspect load indicators per manufactures instructions.
 - Replace according to manufacturers' guidelines.
- ☐ Inspect buckles and hardware:
 - Several types of buckles are used on fall protection equipment. Common ones are mating buckles, tongue & grommet buckles and quick connect (seat belt) buckles.
 - Inspect the buckles and hardware (including D-ring and snaphooks for cracks, burrs, distortion, sharp edges, pitted surfaces, discoloration from bending, rusting, corrosion, chemical damage, burns or any other significant defects. Components shall be straight and compare to others for consistent appearance.
 - Carefully check corners and attachment points of the center bar. They should overlap the buckle frame and move freely back and forth in their sockets. The roller should turn freely on the frame.
 - Inspect the belt for loose, distorted, or broken grommets.
 - Check that rivets are tight and cannot be moved. The body side of the rivet base and outside rivet burr should be flat against the material. Make sure the rivets are not bent.
 - Inspect for pitted or cracked rivets that show signs of chemical corrosion.
 - Replace the belt if the "D" ring is not at a 90 degree angle and does not move vertically independent of the body pad or "D" saddle.
 - Check bag rings and knife snaps to see that they are secure and working properly. Check tool loop rivets. Check for thread separation or rotting, both inside and outside the body pad belt.
 - Snaphook keeper (latch) should be seated into the snap nose without binding and should not be distorted or obstructed. The keeper spring should exert sufficient force to close the keeper firmly.

Lanyard Inspection Checklist

- ☐ Inspect shock-absorbing lanyard per the style, manufactures instructions and webbing inspection above.
- ☐ Inspect snap hooks and connection points for signs of damage or wear, free movement, self-closing/self-locking gates, excessive gate play per the manufactures instructions.

Lifeline Checklist

- ☐ Inspect for any signs of abrasion, excessive wear, burns, and UV, heat or chemical exposure/contamination.
- ☐ Inspect hardware or snap hooks as described in Harness or Lanyard inspection above.
- ☐ Rotate the rope and inspect from end to end for fuzzy, worn, broken, cut fibers or hockling

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(kinking and/or untwisting due to constant turning in the same direction or shock loading). Weakened areas have noticeable changes in the original rope diameter.

- ☐ Inspect wire rope for bird nesting, kinks, broken strands, corrosion, or stretched eyelets.
- ☐ Replace when rope diameter is not uniform throughout or does not meet any of the criteria above.

Rope Grab Inspection Checklist

- ☐ Confirm smooth operation of moving parts, including hinge, cam lever and thumb screws.
- ☐ Apply rope grab to a suitable rope to confirm locking mechanism fully engages. Confirm rope grab has appropriate labelling (e.g. indicating direction of rope).

Cable Sling Inspection Checklist

- ☐ Inspect for cut fibers or damaged stitches inch by inch by flexing the strap in an inverted "U." Note cuts, frayed areas, or corrosion damage.
- ☐ Check friction buckles for slippage and sharp buckle edges.
- ☐ Check connection points or D rings for signs of damage or misuse.
- ☐ Remove from service any strap that does not pass inspection criteria.

Retractable Lifelines Inspection Checklist

- ☐ Refer to Manufacturer's instructions on annual inspection criteria.
- ☐ Retractable lifelines contain a cable wound around a drum with a spring that removes slack from the line and an inertial latching device that stops a sudden decent. Slowly pull entire cable out and inspect for damage. Pull rapidly to check inertia latch.
- ☐ Inspect load indicator.
- ☐ Remove from service any retractable lifeline that does not pass inspection criteria.

Inspect Hardware and Anchorage Points Inspection Checklist

- ☐ Inspect rope grabs, carabiners and other hardware for signs of damage, and free movement of pivot points and self-closing/self-locking components.
- ☐ Carabiners and similar connectors shall be equipped with screw-gates or self-locking methods of closure to prevent inadvertent unfastening or release.
- ☐ Confirm rope grab is properly sized for application and vertical lifeline.
- ☐ The anchorage point is the position on an independent structure to which the lanyard, lifeline or anchor adaptor is attached. It should be capable of supporting a minimum static load per the local regulatory requirements.
- ☐ Confirm any engineered anchorage has the appropriate documentation.
- ☐ If there are any concerns about hardware or anchorage points refer the matter to the site Supervisor or Competent Person for further review.

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Attachment 3: Fall Protection Training Record Example

Fall Protection Training Record

Individual Name:

Date:

Last 4 Digits of Social Security Number:

Badge Number:

INITIAL TRAINING

- ☐ A copy of the project required Fall Protection Program is given to the individual.
- ☐ The program is explained either orally or through a video presentation.
- ☐ The equipment in use on this project at this time is demonstrated.

Brands and equipment covered:

- ☐ The individual has been instructed in how to perform the daily inspection.
- ☐ The individual has also been instructed that if he/she suspects that anything is wrong or amiss with the equipment or system that they are asked to use, they immediately report to their supervisor and do not use that component until deemed safe to do so, replaced or repaired.
- ☐ If the individual ever has any questions or concerns about their safety, they shall report the concerns to their supervisor, the project Safety Manager or to management.

Signature of Trainer

Date

Signature of Individual

Date

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Attachment 4: Standard Form for Release of Floor or Stairs

STANDARD FORM FOR RELEASE OF FLOOR OR STAIRS

Job Name / Number: _____

Date: _____

Steel Erector: _____

Per OSHA requirements, AECOM TISHMAN CONSTRUCTION will only accept control and responsibility of perimeter safety cables that meet all requirements listed in OSHA 29 CFR 1926.502 and this program.

Note: There are currently 22 State OSHA programs. States must set job safety and health standards "at least as effective as" comparable federal standards (most states adopt standards identical to federal ones).

DESCRIBE AREA REQUIRING ACCEPTANCE (floor levels and/or gridlines)

Yes	No	N/A	
			Guardrail top rails are at least 42 inches (+/- 3 inches) above walking/working level.
			Guardrail mid-rails are ½ way between the top rail and walking/working level.
			Cable guardrails are flagged every 6 feet for visibility.
			Cable guardrails meet all other requirements of OSHA 29 CFR 1926.502.
			All decking panels are secured (welded) in place.
			All floor openings > 2 inches are covered or protected by guardrails.
			Other (specify):

This form is to be completed jointly by the Steel Erector and AECOM TISHMAN CONSTRUCTION authorized representatives.

Signatures verify that the above listed items are deemed to comply with regulatory requirements, and the deck is deemed acceptable for use by all trades, and the Controlling Contractor accepts control and responsibility from this date forward. **If any item is deemed out of compliance, then acceptance by AECOM TISHMAN CONSTRUCTION is declined and the deck shall return to the Steel Erector for correction of the deficiencies.**

Signature of AECOM TISHMAN Authorized Representative

Printed Name

Signature of Steel Erector Authorized Representative

Printed Name

Attachment 5: Fall Rescue Devices

Fall Rescue Devices

A variety of fall rescue devices can be found at any of the following websites or by calling the manufacturer directly:

3M Fall Protection
<http://3m.com/fallprotection/>
(800) 328-1667

Miller Fall Protection/Honeywell
<http://www.millerfallprotection.com/>
(800) 873-5242

Guardian Fall Protection
<http://www.guardianfall.com/>
(800) 466-6385



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Attachment 6: Zero Tolerance Policy for Fall Protection

Zero Tolerance Policy for Fall Protection

If an individual violates AECOM TISHMAN CONSTRUCTION's Fall Protection Policy, the following actions will be taken:

Safety Staff and/or Supervision will:

- Immediately correct the situation.
- Notify immediate supervisor and the AECOM TISHMAN CONSTRUCTION Superintendent.
- Recommend disciplinary action to Superintendent, in compliance with the following guidelines:
 - 1st Offense – up to removal or 3 days suspension without pay depending upon the circumstances. Return to project conditional upon Fall Protection training.
 - 2nd Offense – Removal from AECOM TISHMAN CONSTRUCTION projects.

Superintendent will:

- Notify the subcontractor if the violation is by a subcontractor's employee.
- Implement the above outlined disciplinary actions.

All disciplinary actions must be documented and kept on file.

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Attachment 7: AECOM TISHMAN CONSTRUCTION Shaft Work Permit

SHAFT WORK PERMIT

Contractors must submit a shaft work permit request 48 hours prior to and receive approval from AECOM TISHMAN CONSTRUCTION Superintendent before commencing any shaft work. Request forms can be submitted to the AECOM TISHMAN CONSTRUCTION field office.

No shaft work is permitted without a permit.

Subcontractor: _____ Date(s) work will occur: _____

Shaft Location & No.: _____ Floor: _____

Competent Person/Foreman & Phone Number: _____

Reason/type of work being performed: _____

Start Time: _____ Completion Time: _____

I, hereby request to commence with work in the elevator/mechanical shaft at _____ construction project. I will adhere to all safety rules and regulations as provided in AECOM TISHMAN CONSTRUCTION safety programs while I execute my work in the shaft/s.

- ☐ Any other permits required i.e. Hot Work Permits, Confined Space. (Check One) ☐ Yes ☐ No
Notifications: All trades to be made aware of this shaft work and the Shaft Permit must be posted.
- ☐ Pre-entry Safety meeting (Pre-Task Safety Briefing) and THA review with signatures.
- ☐ Barricades/Entry Protection: lighting, proper egress, tool/equipment tethered and overhead protection in place prior to the start of work.
- ☐ Fall protection (lifelines attachments) installed and inspected by the competent person prior to use.
- ☐ Proper access: Scaffolding Tie-off over 6'-0" – Rigging Plan and inspection.
- ☐ Elevator platform safety block installed underneath prior to any work.

Work platform built by design specifications and inspected by competent person/foreman prior to use (Check one) ☐ Yes ☐ No

AECOM TISHMAN Safety Rep/ Superintendent Signature

Date

Requestor's Signature and Printed Name

Date

NOTE:

Work in the shaft/s without permit will result in an automatic removal of the worker and site access badge for the work and a 1-week suspension for the foreman. A second offense under the same foreman will result in a permanent removal from the project. Full inspection of the area assessing the safety implementation as well as any other exposure prior to work commencing will be required.

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Attachment 8: AECOM TISHMAN CONSTRUCTION Guardrail/Fall Protection Removal Permit

GUARDRAIL/FALL PROTECTION REMOVAL WORK PERMIT

Contractors must submit for a Guardrail/Fall Protection removal permit request 48 hours prior to and receive approval from AECOM TISHMAN CONSTRUCTION Superintendent before commencing any removal work. Request forms can be submitted to the AECOM TISHMAN CONSTRUCTION field office. No removal of a guardrail or other fall protection means is permitted without a permit.

Subcontractor: _____ Date(s) work will occur: _____

Removal Location: _____ Floor: _____

Competent Person/Foreman & Phone Number: _____

Reason/type for removal: _____

Start Time: _____ Completion Time: _____

I/we, hereby request to commence with work where we have to remove a guardrail or other fall protection at _____ construction project. I will adhere to all safety rules and regulations as provided in AECOM TISHMAN CONSTRUCTION safety programs while I execute my work.

- ☐ Any other permits required i.e. Hot Work Permits, Confined Space. (Check One) ☐ Yes ☐ No
Notifications: All trades to be made aware of this guardrail/fall protection removal and the Guardrail/Fall Protection Removal Permit must be posted.
- ☐ Pre-removal Safety meeting (Pre-Task Safety Briefing) and THA review with signatures.
- ☐ Alternate means of fall protection provided: _____
- ☐ Competent person in charge during removal: _____
- ☐ Protection provided for other tradespersons during removal.
- ☐ Plan to put guardrail/fall protection immediately back when complete.

AECOM TISHMAN Safety Rep/ Superintendent Signature

Date

Requestor's Signature and Printed Name

Date

NOTE:

Removal of a guardrail/fall protection without a permit will result in an automatic removal of the worker(s) and site access badge for the work and a minimum 1 week suspension for the foreman. A second offense under the same foreman will result in a permanent removal from the project.

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SECTION 10 - FIRE PROTECTION

1. TERMS and DEFINITIONS

1.1 Combustible Liquid. A liquid that has a flashpoint at or above a temperature of 100 degrees Fahrenheit (°F) or 37.8 degrees Celsius (°C).

1.2 Combustible Material. Any material that may ignite when introduced to an ignition source (e.g., wood, paper, cardboard and plastic).

1.3 Emergency Response Coordinator. An individual in a worksite or project environment designated to lead and direct the immediate emergency response.

1.4 Fire Watch. Worker(s) designated to monitoring hot work and the surrounding area for incipient fires and changing conditions.

1.5 Hot Work. A work activity that by the nature of the operation (e.g., grinding, burning, thermo cutting/welding, brazing, etc.) creates an open source of ignition (primary source of ignition or secondary source of ignition) or that could produce temperatures high enough to cause the ignition of flammable gases and combustible materials.

1.6 Hot Work Permit. Document issued prior to the start of hot work, which is used to verify the presence of appropriate fire prevention and protection measures.

1.7 Flammable liquid. Any liquid having a flashpoint at or below 199.4 °F (93 °C).

1.8 Flashpoint. The lowest temperature at which a liquid gives off enough vapor to be ignited at the surface of the liquid.

1.9 Sources of ignition. In locations where flammable vapors may be present, precautions shall be taken to prevent ignition by eliminating or controlling ignition sources. These may include open flames, lightning, smoking, cutting and welding, hot surfaces, frictional heat, sparks (static, electrical, and mechanical), spontaneous ignition, chemical and physical-chemical reactions, and radiant heat.

2. GENERAL REQUIREMENTS

2.1 All projects shall develop and maintain of an effective fire protection and prevention program throughout all phases of construction. All individuals shall be trained in the proper use of fire extinguishers prior to their start of work and at least annually thereafter. Subcontractors are solely responsible to train their workforce.

2.2 The necessary fire protection equipment shall be provided with adequate access.

2.3 All individuals shall be constantly alert for conditions that might contribute to a fire and shall remove or report the hazard.

2.4 Portable fire extinguishing equipment, rated not less than 10-20lb ABC and suitable for the fire hazard involved, will be placed at convenient, conspicuously accessible locations throughout each project. Subcontractors are solely responsible to provide fire extinguishers for their operations, as applicable under OSHA law, not less than 10lb ABC.

2.5 Fire protection equipment shall be conspicuously located.

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2.6 Fire protection equipment shall be inspected regularly and maintained in good operating condition.

2.7 Travel distance from any point in the building to the nearest fire extinguisher shall not exceed one hundred feet (100').

2.8 Smoking is not permitted on any project unless in a designated area.

2.9 Indoor storage shall not obstruct or adversely affect means of exit.

2.10 Clearance shall be maintained around any source of heat to prevent ignition of combustible materials.

2.11 Portable heaters shall be equipped with an approved automatic shut-off device. Such heaters having inputs above 50,000 BTU/hour shall have either a pilot, which must be lighted and proven to be lighted prior to main burner ignition, or an electrical ignition system. (Does not apply to heaters under 750 BTU/hour when used with 1 ¼ -pound containers.)

2.12 Containers of one pound LP-Gas capacity or greater when connected for use must stand on a firm, substantial, and level surface and secured in an upright position.

2.13 Heaters used for temporary heating shall be at least six feet (6') from any LP-Gas container. (This does not apply to units designed for this purpose.)

2.14 When two or more heater /container units are located in an un-partitioned area on the same floor, the containers of each unit shall be separated by at least twenty feet (20').

2.15 Before any hot work is performed, the immediate area shall be inspected by the supervisor responsible for authorizing hot work.

2.16 A hot work permitting system shall be implemented on all projects. This system shall include the use of fire watches.

2.17 Storage areas will be kept free from accumulations of unnecessary combustible materials and shall not obstruct means of egress.

2.18 Open fires of any type are strictly prohibited on all projects.

2.19 All doors to the outside of buildings shall be marked with exit signs.

3. ATTACHMENTS

3.1 Fire Extinguisher Information Guide

3.2 Hot Work Permit Example #1

3.3 Hot Work Permit Example #2

Attachment 1: Fire Extinguisher Information Guide

Fire Triangle:

To understand how fire extinguishers work, you need to understand a little about fire. Fire is a very rapid chemical reaction between oxygen and a combustible material, which results in the release of heat, light, flames, and smoke.



For fire to exist, the following four elements must be present at the same time:

- Enough oxygen to sustain combustion.
- Enough heat to raise the material to its ignition temperature.
- Some sort of fuel or combustible material.
- The chemical reaction that is fire.

How a Fire Extinguisher Works:

Portable fire extinguishers apply an extinguishing agent that will either cool burning fuel, displace or remove oxygen, or stop the chemical reaction so a fire cannot continue to burn. When the handle of an extinguisher is compressed, the agent is expelled out of the nozzle.



All portable fire extinguishers must be approved by a nationally recognized testing laboratory to verify compliance with applicable standards. [OSHA 29 CFR 1910.157(c)(2)] Equipment that passes the laboratory's tests are labeled and given an alpha-numeric classification based on the type and size of fire it will extinguish.

NOTE: AECOM TISHMAN CONSTRUCTION projects require ABC type fire extinguishers unless there is a specific hazard requiring a different type.

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Let's take a look at the label pictured. The classification is:

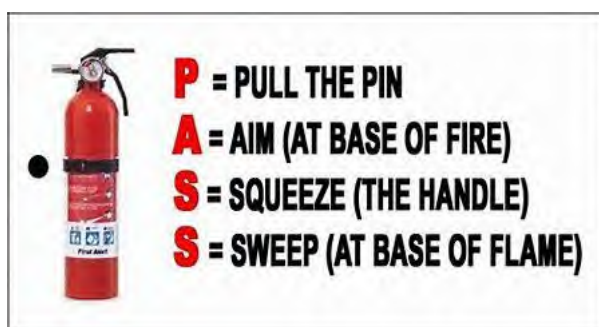
1-A: 10-BC

The letters (A, B, and C) represent the type(s) of fire for which the extinguisher has been approved.

The number in front of the A rating indicates how much water the extinguisher is equal to and represents 1.25 gallons of water for every unit of one. For example, a 4-A rated extinguisher would be equal to five (4 x 1.25) gallons of water.

The number in front of the B rating represents the area in square feet of a class B fire that a non-expert user should be able to extinguish. Using the above example, a non-expert user should be able to put out a flammable liquid fire that is as large as 10 square feet.

Using a Fire Extinguisher:



Types of Fire Extinguishers

Different types of fire extinguishers are designed to fight different types of fire. The three most common types of fire extinguishers are: air pressurized water, CO₂ (carbon dioxide), and dry chemical. The following table provides information regarding the type of fire and which fire extinguisher should be used.



Water

Ordinary Combustibles

Fires in paper, cloth, wood, rubber, and many plastics require a water type extinguisher labeled A.



CO₂

Flammable Liquids

Fires in oils, gasoline, some paints, lacquers, grease, solvents, and other flammable liquids require an extinguisher labeled B.



OR



**Dry
Chemical**

Electrical Equipment

Fires in wiring, fuse boxes, energized electrical equipment, computers, and other electrical sources require an extinguisher labeled C.



**Multi-
Purpose**

Ordinary Combustibles, Flammable Liquids, or Electrical Equipment

Multi-purpose dry chemical is suitable for use on class A, B, and C.



Class D

Metals

Fires involving powders, flakes or shavings of combustible metals such as magnesium, titanium, potassium, and sodium require special extinguishers labeled D.

Class K

Kitchen Fires

Fires involving combustible cooking fluids such as oils and fats.



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Attachment 2: Hot Work Permit Example #1

Specific task to be performed: _____

Location: _____

CONDITION	YES	NO	CONDITION	YES	NO
Welding			Gas powered equipment		
Cutting			Electrical equipment, portable		
Use of power tools			Portable heaters		

PRE-WORK CHECKLIST All items must be completed.	YES	NO	N/A
Has the work area been inspected by Site Safety Representative prior to beginning the Hot Work?			
Has a fire watch been established? Name?			
Have holes, vents or other floor penetrations been identified and protected?			
Has a "No Access Hot Work Zone" been established on the floor/floors below Hot Work activities?			
Is the fire extinguisher appropriate for the media? (CO2, ABC, PKP)			
Is it readily accessible?			
Is the work area clear of all trash and combustible debris?			
Is the equipment properly grounded?			
Is the oxygen level above 20.0% and below 23.5%?			
Will combustible gas indicator(s) (CGI) be used constantly during Hot Work?			
If no, why?			
List personal protective equipment (PPE) to be worn: Hard Hat __, Safety Glasses __, Face Shield __, Leathers __, Gloves __, Fall Protection __, Tripod & Harness __, Other _____			
Is welding or cutting on closed systems prohibited?			
Have closed system cutting procedures been established?			

Special Requirements: _____

CERTIFICATION BY AECOM TISHMAN CONSTRUCTION REPRESENTATIVE THAT HOT WORK MAY COMMENCE:

_____ Date __/__/__ Time ____:____ Permit Expiration Time ____:____

(Signature)

(As necessary) Owner acknowledgment of work to be performed

_____/_____
(Owner Operations Signature) (Date)

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Attachment 3: Hot Work Permit Example #2

Hot Work Permit

S3AM-332-FM1

Name of Contractor: _____	Location: _____	Project #: _____
Date: _____ Time: _____	Weather: _____	
Person Preparing Permit: _____	Title: _____	THIS PERMIT BECOMES VOID: 1. At the end of the shift, or 2. Whenever conditions change significantly, or 3. On any emergency signal
Person Conducting Hot Work: _____	Title: _____	

TYPE OF HOT WORK
☐ PSI CLASS I Location (Complete full permit) ☐ PSI NON-CLASS I Location (Complete section A, D, and E) ☐ SSI CLASS I Location (Complete section B, and E)
Description of Hot Work to be Conducted:

A. WORK AREA PREPARATION FOR HOT WORK			
	YES	NO	N/A
1. Is the hot work equipment in proper working order?	☐	☐	☐
2. Are the proper safety data sheets (SDS) available for review?	☐	☐	☐
3. Have combustible materials been relocated more than 50 feet (15.24 meters) from hot work or been properly shielded?	☐	☐	☐
4. Is firefighting equipment readily available and in proper working order (e.g. fire extinguishers, sand, shovels, etc.)?	☐	☐	☐
5. Have ducts, drains, and sewers been adequately covered to prevent sparks from entering?	☐	☐	☐
6. If hot work is to be performed indoors, have combustible floors been wet down or properly shielded?	☐	☐	☐
7. If arc welding equipment will be used, have measures been taken to protect personnel from shock?	☐	☐	☐
8. If required, is a 20-pound (9.07-kilogram) fire extinguisher available and appropriate for the type of fire expected?	☐	☐	☐
9. If hot work is to be performed in or on containers, vessels, tanks, or similar equipment, have they been cleaned, purged, or ventilated, or have other precautions been taken to verify that they are safe for hot work?	☐	☐	☐
10. If required, have lockout tagout procedures been completed in accordance with <i>Lockout Tagout Policy</i> ?	☐	☐	☐

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11. Have others required to work in the hot work area been notified of the work involved and safety precautions, and do they have the required PPE?	☐	☐	☐
---	--------------------------	--------------------------	--------------------------

LIST ANY PERSONAL PROTECTIVE EQUIPMENT THAT IS REQUIRED (OTHER THAN THE MINIMUM REQUIRED):

B. AIR MONITORING (Initial and continuous monitoring are required for all Class I Hot Work [PSI and SSI])

DOCUMENT WHERE AIR MONITORING WILL OCCUR, WHAT POTENTIAL GAS OR VAPOR SOURCES EXIST WITHIN THE HOT WORK AREA, AND THE WIND DIRECTION:

INSTRUMENT 1 SERIAL NUMBERS: _____	DATE AND RESULTS OF LAST CALIBRATION: _____	PRE-USE CALIBRATION CHECK PERFORMED: _____	YES circle	NO circle	NA circle
INSTRUMENT 2 SERIAL NUMBERS: _____	DATE AND RESULTS OF LAST CALIBRATION: _____	PRE-USE CALIBRATION CHECK PERFORMED: _____	YES circle	NO circle	NA circle

Initial Reading %LEL (if >0% fill out section C)

If O ₂ level below 10% do not proceed without alternate testing	Time	%O₂	%LEL	Time	%Oxygen	%LEL	Time	%Oxygen	%LEL

Continuous Reading %LEL

Time	%LEL	Time	%LEL	Time	%LEL	Time	%LEL	Time	%LEL

C. LOWER EXPLOSIVE LIMIT (LEL) MITIGATION PLAN (Required if initial or continuous monitoring LEL is greater than 0%)
--

% LEL DETECTED:		% LEL AFTER CONTROLS IMPLEMENTED		NOT APPLICABLE		<i>(If not applicable, continue monitoring source)</i>
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DOCUMENT THE LOCATION OF CONFIRMED SOURCE(S) OF GAS / VAPOR AND WHAT CONTROLS (IF ANY) HAVE BEEN IMPLEMENTED
(Note: Hot work is prohibited if the LEL reading is greater than or equal to 10% in the hot work area.)

D. FIRE WATCH			
	YES	NO	
1. Will PSI Class I work be performed?			<i>Note: If the answer to any of the fire watch conditions to the left is YES, a dedicated fire watch must be assigned. Note: The fire watch is authorized to perform air monitoring.</i> Fire Watch 1: Fire Watch 2:
2. Will PSI Non-Class I work be performed within 50 feet (15.24 meters) of combustible materials?			
3. Will PSI hot work be performed where wall or floor openings within 50 feet (15.24 meters) would compose combustibles in adjacent areas?			
4. Is PSI hot work to be performed where combustibles are adjacent to the opposite side of partitions, ceilings, or roofs being worked on?			
5. Will fire alarms or suppression systems be disabled for hot work?			

E. AUTHORIZING SIGNATURES							
AUTHORIZED AIR MONITOR: <i>I have performed initial air monitoring with a survey of the hot work area. I agree to perform continuous air monitoring and record results on this form while hot work is being performed in a Class I area and to stop hot work if the LEL reading is greater than or equal to 10%.</i>							
Name		Signature		Name		Signature	
FIRE WATCH: <i>I have been trained in the use of the applicable fire fighting equipment and agree to maintain a line of sight with the hot work operation at all times, to have immediate access to a fire extinguisher and to remain at the hot work area for 30 minutes after the hot work is completed to verify the area if fire safe.</i>							
Name		Signature		Name		Signature	
HOT WORK OPERATORS: <i>I have reviewed and understand the permit conditions specific to the scope of work. I agree to perform hot work operations within these conditions, to stop any work that I deem to be unsafe, and to notify the performing authority upon completion or interruption of this permitted work.</i>							
Name		Signature		Name		Signature	
Name		Signature		Name		Signature	
SITE SUPERVISOR: <i>The location where this work is to be done has been examined, necessary precautions have been taken, and permission is granted for this work.</i> Permit Expires:							
Name		Signature		Time Hot Work Started		Time Hot Work Completed	

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SECTION 11 - FIRST AID

1. TERMS and DEFINITIONS

1.1 Automated External Defibrillator (AED). A portable electronic device that automatically diagnoses the potentially life-threatening cardiac arrhythmias of ventricular fibrillation and ventricular tachycardia in a patient, and is able to treat them through defibrillation, the application of electrical therapy which stops the arrhythmia, allowing the heart to re-establish an effective rhythm; are used in the resuscitation of a patient in full cardiac arrest.

1.2 Cardiopulmonary Resuscitation (CPR). An emergency procedure in which the heart and lungs are made to work by:

- Manually compressing the chest overlying the heart, or
- Both manually compressing the chest and performing rescue breaths that force air into the lungs.

CPR is applied to a victim in respiratory distress and/or to maintain circulation when the heart stops pumping (cardiac arrest), which may be due to heart tissue damage (heart attack), disease, electrical shock, drug overdose, drowning, suffocation, stroke or trauma.

1.3 First Aid Provider. A First Aid, CPR, and AED trained volunteer employee who provides emergency first aid or treatment (including performing CPR and applying an AED) to someone who is injured or suddenly ill, before emergency medical services (EMS) arrives. They may use a limited amount of equipment to perform initial assessment and provide immediate life support and care while awaiting arrival of emergency medical services.

2. GENERAL REQUIREMENTS

2.1 All injuries requiring medical treatment, including first aid must be reported in accordance with the Incident Reporting, Investigation, and Claims Management section of this Program. Failure to report an injury immediately may be grounds for disciplinary action.

2.2 A project specific emergency response plan must be developed and communicated to all affected personnel.

2.3 The AECOM TISHMAN CONSTRUCTION Project Manager shall ensure adequate first aid supplies are available and an adequate number of trained First Aid Providers/CPR (but not less than one) are available during hours of normal operation. Subcontractors shall provide the same for their workforce.

2.4 For certain long-term, heavily staffed, high hazard projects, and/or projects of \$250 mm or more, AECOM TISHMAN CONSTRUCTION may opt to establish a first aid medical station onsite.

2.5 First aid kits must be inspected weekly to ensure contents meet jurisdictional requirements given the number of staff, work location, and potential hazards prior to being placed in the determined location or sent to site and, as a minimum, monthly thereafter.

2.6 An inventory (listing required and approved items) and weekly inspection form shall be posted on or inside each first aid kit. Any items not listed on the inventory (listed as required or approved for the kit) will be removed during the weekly inspection unless specifically approved by a health care professional for inclusion and added to the inventory.

2.7 Each item in first aid kits shall be individually sealed to protect the contents from contamination. The first aid equipment and supplies shall be maintained in a clean, dry and serviceable condition, contained in a

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material that protects the contents from the environment, and clearly identified as first aid equipment and supplies.

2.8 Any employee who recognizes a medical emergency must immediately initiate an emergency response in accordance with the project Emergency Response Plan. For more specific information regarding emergency response, please refer to the Emergency Response Planning section of this program.

2.9 First Aid Providers should assess the emergency scene to determine and initiate the appropriate course of action based on their observations, the victim's condition, their training, and according to the location-specific Emergency Response Plan.

2.10 As is applicable to the victim's condition, First Aid Providers transfer the victim's care to the EMS agency for appropriate advanced medical treatment and provides a report including:

- The initial time of the event.
- Any care given prior to the First Aid Provider's arrival.
- Victim's condition upon the First Aid Provider's arrival.
- Treatment rendered to the victim by the First Aid Provider.
- Available medical information about the victim.

2.11 If an AED was used, leave the defibrillator attached to the victim until instructed to remove it by EMS personnel or higher medical authority.

2.12 AEDs shall only be used by individuals with current and proper training.

2.13 If corrosive, irritating or otherwise hazardous materials are used, review applicable safety data sheets to assist in determining whether eyewash and body flush (shower) facilities must be provided. Subcontractors shall be required to provide appropriate facilities for their operations.

2.14 These facilities should be highly visible, clearly identified and, if possible, within 10 seconds of the hazard. The water source / flushing fluid must be tepid, pressure controlled, and maintained to prevent freezing and contamination of the fluid per applicable OSHA requirements.

2.15 First Aid Provider(s) shall possess a valid certificate in First Aid, CPR, and AED training from an approved provider for the applicable jurisdiction that can be verified by documentary evidence.

2.16 All first aid given shall be properly documented.

2.17 Local emergency numbers and medical providers shall be posted on every project.

2.18 Universal precautions should be taken when giving first aid to prevent the transfer of Bloodborne Pathogens. These precautions include:

- Gloves.
- Eye protection.
- CPR masks.
- Other, as necessary.

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2.19 Dispose of all contaminated materials in the correct fashion, based on the type of material.

2.20 When blood or other potentially infectious material has come into contact with a work surface or machinery, use an approved disinfectant to decontaminate the affected areas. A one to ten dilution of ordinary household bleach and water is recommended - approximately 1-½ cups of bleach added to 1 gallon of water.

3. ATTACHMENTS

3.1 List of Project First Aid Providers

3.2 Example Emergency Number Posting Form

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Attachment 1: List of Project First Aid Providers

Project First Aid Providers

AECOM TISHMAN Employees

Subcontractor Employees

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Attachment 2: Example Emergency Number Posting Form

Emergency Contacts *(insert name and phone numbers)*

- Superintendent:
- Safety Manager:
- Project Manager:
- Project Executive:
- Assistant Superintendent:
- Area Safety Manager:
- Regional Safety Director:
- Vice President of Safety:
- Risk Management Department:
- Legal Department:
- Insurance Carrier:
- HR/Corporate Communications:

Emergency Services

- EMS:
- Hospital:
- Police:
- Fire Department:
- HAZMAT Response:

Utility Companies

- Power Company:
- Gas Company:
- Phone Company:
- Water Company:
- Traffic Signal Repair:

Governmental Agencies/Offices *(List only those applicable)*

- OSHA: (Note: only the Safety Department can notify OSHA)
- MSHA: (Note: only the Safety Department can notify MSHA)
- EPA:
- HazMat:
- Coast Guard:

Note: Add other specific numbers relevant to your project.

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SECTION 12 - HAND AND POWER TOOLS

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 All hand and power tools shall be inspected prior to use and maintained in a safe condition. Any noticed defect shall cause the tool to be tagged out and removed from service.

2.2 When power operated tools are designed for guards, they shall be so equipped when in use.

2.3 All handheld power tools shall be equipped with a constant pressure switch that will shut off power when pressure is released.

2.4 AECOM TISHMAN CONSTRUCTION will not issue or permit the use of unsafe hand or power tools.

2.5 Electric power operated tools shall either be approved double insulated or grounded per NEC standards.

2.6 Electric cords shall not be used to hoist or lower tools.

2.7 Safety clips or retainers shall be securely installed and maintained on pneumatic impact tools to prevent attachments from being accidentally expelled.

2.8 Compressed air shall not be used for cleaning purposes unless the pressure is less than 30 psi and then only with the use of effective chip guarding and a wand with a continuous pressure handle.

2.9 Only trained employees shall be allowed to operate a powder-actuated tool. Proper training certifications shall be provided upon request.

2.10 Powder-actuated tools shall be tested each day before loading to see that all safety devices are functional. (Method of testing shall be in accordance with manufacturer's recommendations.) The tool shall not have powder charges left in tool when unattended. All used or discarded shots shall be properly disposed of per the manufacturer.

2.11 Any tool not in working order or in which a defect occurs shall be immediately removed from service and tagged as 'Out of Service'. The user shall collect all unspent rounds and remove them from the site for proper disposal by their employer.

2.12 All tools and/or other handheld equipment shall be equipped with tethers back to the worker when working at heights.

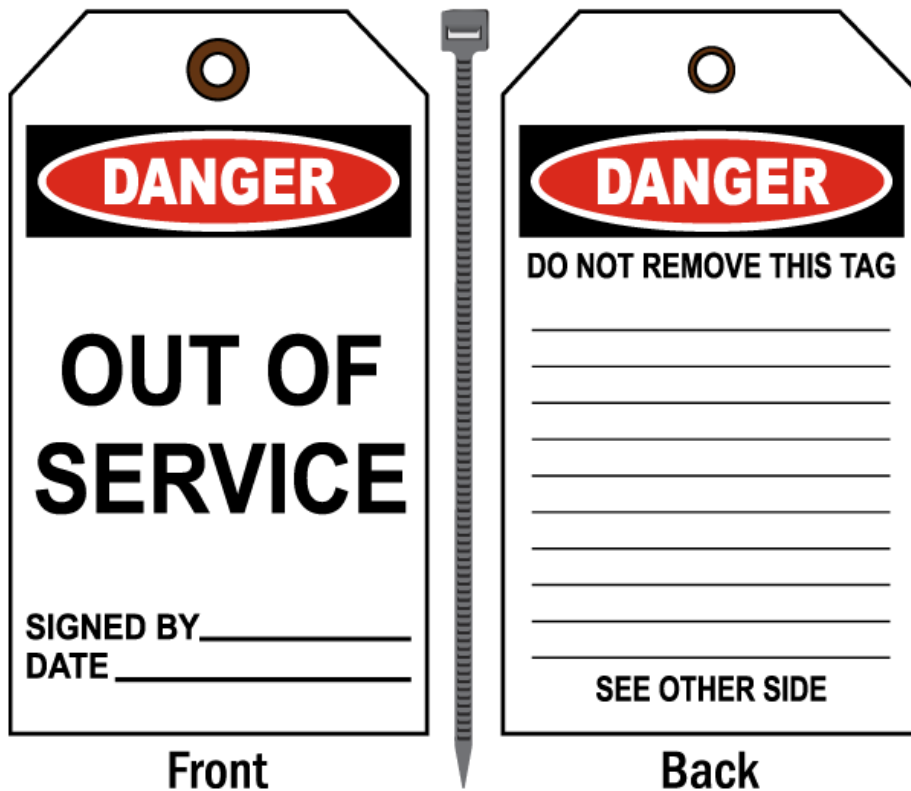
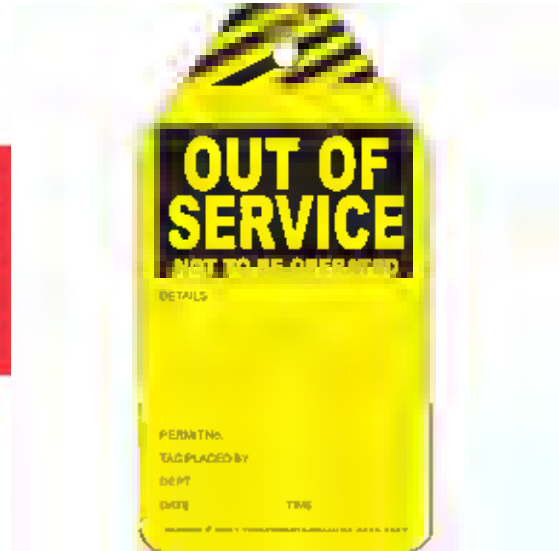
2.13 Battery powered tools shall be used in lieu of corded tools wherever possible.

3. ATTACHMENTS

3.1 Examples of "Out of Service" Tags

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Attachment 1: Examples Out of Service Tags



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SECTION 13 - HAZARD COMMUNICATION

1. TERMS and DEFINITIONS

1.1 Acute Effect. An adverse effect on the human body with immediate onset of symptoms.

1.2 Article. A manufactured item: (1) which is formed to a specific shape or design during manufacture; (2) which has end use function(s) dependent in whole or in part upon its shape or design during end use; and, (3) which does not release or otherwise result in exposure to, a hazardous chemical, under normal conditions of use.

1.3 Carcinogen. Those chemicals appearing in any of the following reference sources are established as carcinogens for hazard communication purposes:

- National Toxicology Program (NTP) Annual Report on Carcinogens.
- International Agency for Research on Cancer (IARC) Monographs, Volumes 1-34.

1.4 Chemical Name. The scientific designation of a substance in accordance with the nomenclature system developed by the International Union of Pure and Applied Chemistry or the system developed by the Chemical Abstracts Service.

1.5 Chronic Effect. An adverse effect on the human body with symptoms which develop slowly over a long period of time or which frequently recur.

1.6 Combustible Liquid. Any liquid having a flash point at or above 100°F (37.8°C) but below 200°F (93.3°C), except any mixture having components with flash points of 200°F (93.3°C), or higher, the total volume of which makes up 99% or more of the total volume of the mixture.

1.7 Common Name. Any designation or identification such as code name, code number, trade name or brand name used to identify a substance other than by its chemical name.

1.8 Container. Any bag, barrel, bottle, box, can, cylinder, drum, reaction vessel, storage tank or the like that contains a hazardous chemical. For purposes of this procedure, pipes or piping systems, and engines, fuel tanks, or other operating systems in a vehicle are not considered to be containers.

1.9 Location. Any separate and distinct AECOM office, laboratory, project or other company facility.

1.10 Exposure. Any situation arising from work operations where an individual may ingest, inhale, absorb through the skin or eyes or otherwise come into contact with a hazardous substance.

1.11 Flammable. A substance that falls into one of the following categories:

- Flammable Aerosol. An aerosol that when tested by the method described in 16 CFR 1500.45, yields a flame projection exceeding 18 inches at full valve opening or flashback (a flame extending back to the valve) at any degree of valve opening.
- Flammable Gas. A gas that at ambient temperature and pressure:
 - Forms a flammable mixture with air at a concentration of 13% of volume or less; or
 - Forms a range of flammable mixtures with air wider than 12% by volume, regardless of the lower limit.

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- Flammable Liquid. Any liquid having a flash point below 100°F (37.8°C), except any mixture having components with flash points of 100°F (37.8°C) or higher, the total of which make up 99% or more of the total volume of the mixture.
- Flammable Solid. A solid, including a powdered, granular or pasty mixture of a substance that is liable to cause fire through friction, absorption of moisture, spontaneous chemical change or retained heat from manufacturing or processing or which can be ignited readily and when ignited burns so vigorously and persistently as to create a serious hazard. Flammable Solids do not include blasting agents or explosives as defined in 8 CCR 5237(a).

1.12 Flash Point. Minimum temperature of a liquid at which it gives off sufficient vapors to form an ignitable mixture with the air near the surface of the liquid or within the container used.

1.13 GHS. The Globally Harmonized System of Classification and Labelling of Chemicals developed by the United Nations with the goal of an international system to define and classify the hazards of chemical products and communicate health and safety information on labels and safety data sheets.

1.14 Hazardous Chemical. Those chemicals appearing in any of the following reference sources are established as hazardous chemicals for hazard communication purposes.

- OSHA 29 CFR Part 1910, Subpart Z, Toxic and Hazardous Substances, OSHA.
- Hazardous Products Act, R.S.C. 1985, c. H-3, Part II, Canada.
- For operations within the state of California, the list of hazardous substances prepared by the California Director of Industrial Relations pursuant to Labor Code.
- Section 6382. The concentrations and footnotes, which are applicable to the list, shall be understood to modify the same substance on all other source lists or hazard determinations set forth in § 8 CCR 5194(d)(3)(B) and (d)(5)(D).

1.15 Hazardous Substance. A hazardous chemical or carcinogen, or a product or mixture containing a hazardous chemical or carcinogen provided that:

- The hazardous chemical is 1% or more of the mixture or product or 2% if the hazardous chemical exists as an impurity in the mixture; or
- The carcinogen is 0.1% or more of the mixture or product;
- Manufacturers, importers and distributors will be relied upon to perform the appropriate hazard determination for the substances they produce or sell.
- The following materials are not covered by the Hazard Communication Standard:
 - Any hazardous waste as defined by the Solid Waste Disposal Act, as amended by the Resource Conservation and Recovery Act of 1976, as amended (42 USC 6901 et seq.) when subject to regulations issued under that act by the Environmental Protection Agency.
 - Tobacco or tobacco products;
 - Wood or wood products. Note: Wood dust is not exempt since the hazards of wood dust are not “self-evident” as are the hazards of wood or wood products;

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- Consumer products (including pens, pencils, adhesive tape) used in the work place under typical consumer usage;
- Articles (i.e. plastic chairs);
- Foods, drugs, or cosmetics intended for personal consumption by individuals while in the work place;
- Foods, drugs, cosmetics in retail store packaged for retail sale; and
- Any drug in solid form used for direct administration to the patient (i.e., tablets or pills).

1.16 Hazardous Substance Inventory (HSI) Log. A listing of all chemicals stored or used at an office or project site. Note that the list may be imbedded in a project Health and Safety Plan.

1.17 Immediate Use. Means that the hazardous chemical will be under the control of and used only by the person who transfers it from a labeled container and only within the work shift in which it is transferred.

1.18 National Fire Protection Association (NFPA). The NFPA is a trade association that issues standards and codes concerning risks associated with fire. A system of categories has been established by NFPA standard 704; colors and numbers, to provide basic hazard information concerning hazardous materials. It enables firefighters and other emergency personnel to easily decide whether or not to evacuate an area or proceed with emergency control operations. The three principal categories of identification are Health, Flammability and Instability. A numerical range of "0 to 4" indicates the severity of the hazard. A "4" indicates the most severe and a "0" indicates a minimal hazard.

1.19 Mixture. Any solution or intimate admixture of two or more substances which do not react chemically with each other.

1.20 Reactivity. A measure of the tendency of a substance to undergo chemical reaction with the release of energy.

1.21 SDS. A Safety Data Sheet prepared pursuant to state and federal regulations, OSHA Form 174.

1.22 SDS Administrator. The individual or group designated by the Office Manager (Operations) or Project Manager to maintain the location-specific inventory list or log and the SDS binder required if that location uses or stores hazardous substances.

1.23 Solubility. The ability of substance to blend and mix uniformly with another.

1.24 Specific Gravity (density). Ratio of the weight of a substance to the weight of the same volume of another substance. As used in this directive, specific gravity or density refers to the weight of substance as compared to the weight of an equal volume of water.

1.25 Vapor Density. The weight of a vapor-air mixture resulting from the vaporization of a volatile liquid at equilibrium temperature and pressure conditions, as compared with the weight of an equal volume of air under the same conditions.

2. GENERAL REQUIREMENTS

2.1 Hazardous substances in the workplace in some forms, concentrations, and job activities, pose potential acute and/or chronic health hazards to individuals who are exposed to these substances.

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2.2 All individuals have a right and a need to know the properties and potential hazards of substances to which they may be exposed, as such knowledge is essential to reducing the incidence and cost of occupational disease and industrial accidents. Individuals that do not read and speak English to tasks with chemical exposures, communications will be provided in the language understood by that individual.

2.3 SDS's shall be requested from all suppliers. All new SDSs will be forwarded to the project safety and health staff and/or designated SDS administrator.

2.4 Pursuant to the requirements of OSHA regulations to make chemical safety and health information available to all individuals, this directive prescribes the use of safety data sheets, by regional management, to provide the required information to inform individuals.

2.5 Subcontractors are solely responsible to comply with all provisions of this program and all federal, state, and local laws with and for their employees.

3. IMPLEMENTATION

3.1 Supervisory Responsibility.

- a) Before any individual is permitted to work with a hazardous substance or hazardous mixture, the potential fire, toxic, or reactivity hazards, which are likely to be encountered in the handling, application, or utilization of such a material, shall be determined.
- b) In order to ascertain the hazards, supervisors shall have available a SDS or an essentially similar form, provided by the supplier, for each hazardous substance used.
- c) Supervisors shall notify individuals of hazardous substances used in their work area for which SDSs are available, making such SDSs available on a timely and reasonable basis, upon request of an individual or his authorized representative. Such notification may be:
 - Posted lists of SDSs where they are readily available to workers.
 - Prominently displayed binders containing SDSs used in the work area.
- d) AECOM TISHMAN CONSTRUCTION Supervisors shall notify subcontractor employers with individuals working for AECOM TISHMAN CONSTRUCTION of any hazardous substances to which their individuals may be exposed and provide suggestions for appropriate protective measures. This shall be done in coordination with the field safety and health staff.

3.2 Training and Information Program.

- a) Supervisors, in coordination with the field safety and health staff shall furnish individuals who may be exposed to a hazardous substance with an appropriate explanation of information on the contents of the SDS. Such information must be comprehensible to the workers. Information should also relate to the entire class of hazardous substances to the extent appropriate and relevant to the job, i.e., acids, caustics, chlorinated hydrocarbons, etc.
- b) For hazardous substances to which the individual may be exposed, they shall be informed of their right of access to an SDS, their right to request an SDS from their employer for material they suspect is hazardous, and the right of their representative or physician to obtain such SDS.
- c) All individuals who may be exposed to a hazardous substance shall be given training and information:
 - Upon a timely and reasonable basis.

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- Prior to assignment of an individual to an area containing a hazardous substance for which previous training has not been received.
- d) Subject training and information shall be accomplished by either formal classroom instruction or by jobsite safety meetings and shall be documented accordingly.
- e) A completed SDS, when available, or equivalent information for each hazardous substance or hazardous mixture used or encountered by individuals, shall be maintained in the workplace, and made accessible to individuals in their work area. This shall include, but is not limited to:
 - Cleaning agents, such as solvents, detergents, acid inhibitors, etc., used for industrial cleaning.
 - Hazardous substances or hazardous mixtures contained in process vessels.
 - Janitorial and housekeeping substances used by individuals at AECOM TISHMAN CONSTRUCTION facilities. Not included are consumer products used in accordance with manufacturer's instructions, and comparable to typical consumer use.

3.3 Information Required on a SDS.

- a) Section 1, Identification includes product identifier; manufacturer or distributor name, address, phone number; emergency phone number; recommended use; restrictions on use.
- b) Section 2, Hazard(s) identification includes all hazards regarding the chemical; required label elements.
- c) Section 3, Composition/information on ingredients includes information on chemical ingredients; trade secret claims.
- d) Section 4, First-aid measures includes important symptoms/ effects, acute, delayed; required treatment.
- e) Section 5, Fire-fighting measures lists suitable extinguishing techniques, equipment; chemical hazards from fire.
- f) Section 6, Accidental release measures lists emergency procedures; protective equipment; proper methods of containment and cleanup.
- g) Section 7, Handling and storage lists precautions for safe handling and storage, including incompatibilities.
- h) Section 8, Exposure controls/personal protection lists OSHA's Permissible Exposure Limits (PELs); Threshold Limit Values (TLVs); appropriate engineering controls; personal protective equipment (PPE).
- i) Section 9, Physical and chemical properties lists the chemical's characteristics.
- j) Section 10, Stability and reactivity lists chemical stability and possibility of hazardous reactions.
- k) Section 11, Toxicological information includes routes of exposure; related symptoms, acute and chronic effects; numerical measures of toxicity.
- l) Section 12, Ecological information.
- m) Section 13, Disposal considerations.
- n) Section 14, Transport information.

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- o) Section 15, Regulatory information.
- p) Section 16, Other information, includes the date of preparation or last revision.

3.4 Obtaining Information.

- a) SDS Binders.
 - A master file of SDSs will be maintained by the SDS Administrator.
 - SDS binders shall be maintained locally by area management for materials used at that location.
 - Copies of SDSs may be obtained upon request.

3.5 AECOM TISHMAN CONSTRUCTION and Subcontractor Employee Information and Training.

- a) Individuals shall be provided with information and training on hazardous substances in their work area at the time of their initial assignment, as well as whenever a new hazard is introduced into their work area by their employer. This orientation and training shall address the following:
 - Regarding onsite investigations or site remediation of unknown materials, every effort shall be made to categorize the hazard(s) before starting work. Safety meetings will be used to inform individuals of the hazards.
 - Individuals shall be informed about what an SDS is, and of the contents of the SDS for hazardous substances to which individuals may be exposed, or equivalent information shall be provided, either in written form or through training programs.
 - When training individuals on the contents of an SDS, supervisors shall explain: any health hazards associated with use of the substance or mixture, proper precautions for handling, and necessary personnel protective equipment or other safety precautions necessary to prevent or minimize exposure. Emergency procedures for spills, fire, disposal, and first aid are also to be explained.
 - When the Project Safety Manager receives a new or revised SDS, such information shall be provided to individuals in a timely manner. This is especially the case if such new information indicates significantly increased risks to, or necessary protective measures for, individual health, as compared to those stated on a SDS previously provided.
- b) Individuals shall be informed of their right:
 - To personally receive information regarding hazardous substances to which the individual may be exposed.
 - For their physician or collective bargaining agent to receive information regarding hazardous substances to which the individual may be exposed.
- c) Individuals shall be informed of:
 - Any operations in their work area where hazardous substances are present.
 - Location and availability of this written hazard communications program/procedure.
- d) Individuals shall be trained in the following:

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- Methods and observations that may be used to detect the presence or release of hazardous substance in the work area, such as personal monitoring, visual appearance, or odor of hazardous substances being released, etc. Also, the physical and health hazards of substances in the work area, and measures and procedures AECOM TISHMAN CONSTRUCTION has implemented to protect individuals.
- Details of this Hazard Communication Program/Procedure and how appropriate hazard information can be obtained and used.

3.6 Trade Secrets.

- a) Some hazardous substance suppliers may claim the information requested on SDS's is proprietary and, therefore, may not provide the information to AECOM TISHMAN CONSTRUCTION.
- b) When suppliers claim trade secrets, the name of the supplier and the hazardous substance in question shall be supplied to the field safety and health staff for further action.
- c) The field safety and health staff will either obtain the necessary information to make a hazard assessment or reject the material for use by AECOM TISHMAN CONSTRUCTION.

3.7 Labeling.

- a) All hazardous substances received from outside suppliers shall conform to legal requirements, and shall have on each container, as a minimum, the following:
 - A pictogram.
 - A signal word.
 - Hazard and precautionary statements.
 - The product identifier.
 - Supplier identification.

Note: Any failure to have proper labeling on the container at the time of receipt will be cause to refuse delivery of the product.

- b) Stationary process containers or pipes may have signs, placards, process sheets, batch tickets, operating procedures, or other written material as a substitute for fixed labels on the containers, as long as the alternative method conveys appropriate hazard information. The written materials shall be readily accessible to the individuals in the work area.
- c) Portable containers of hazardous substances need not be labeled when the substance is transferred from labeled containers, and which are intended for immediate use by the individual who performs the transfer by law however AECOM TISHMAN CONSTRUCTION requires all containers to be labeled.
- d) Containers of hazardous substances transferred from labeled containers and not intended for the immediate use of the individual performing the transfer shall be labeled in accordance with the hazardous materials identification system (HMIS), as outlined in Attachment 1 of this procedure, or an equivalent commercial system.
- e) Labels on incoming containers shall not be removed or defaced.

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- f) Labels or other forms of warning shall be legible, written in English, prominently displayed on the containers, and/or readily available throughout each work shift.

4. ATTACHMENTS

4.1 Hazardous Materials Identification System (HMIS)

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Attachment 1: Hazardous Materials Identification System (HMIS)

This HMIS is based on a comprehensive system for hazardous materials identification. This system conveys chemical identity by chemical, common names, code numbers, or other descriptive terms, which clearly identify the material for hazardous information purposes. The acute health, flammability, and reactivity hazards are communicated by numerical ratings imprinted on the label itself. An alphabetical designation is used to denote a combination of proper personal protective equipment. The designation is also imprinted on the label. The field safety and health staff will provide guidance on what hazard index and what personal protection index should be used.

Hazard Index (HI)

4	Severe Hazard
3	Serious Hazard
2	Moderate Hazard
1	Slight Hazard
0	Minimal Hazard

These designations will be imprinted (written) with indelible ink next to each hazard category on the label, i.e., Health Flammability, Reactivity, etc. Such information is obtainable from the SDS, among other sources.

Definitions of Hazard Categories

Health Hazard

4	Danger: May be fatal on short exposure. Specialized protective equipment required.
3	Warning: Corrosive or toxic - avoid skin contact or inhalation.
2	Warning: May be harmful if inhaled or absorbed.
1	Caution: May cause irritation.
0	No unusual hazard.

Fire Hazard

4	Danger: Flammable gas or extremely flammable liquid.
3	Warning: Flammable liquid, flash point below 100°F.
2	Caution: Combustible liquid, flash point of 100°F to 200°F.
1	Combustible if heated.
0	Not combustible.

Reactivity Hazard

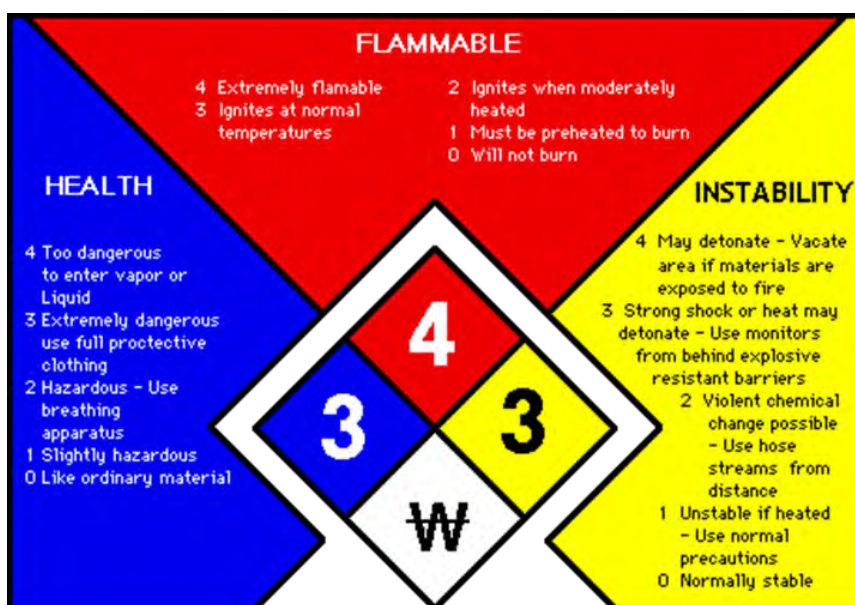
4	Danger: Explosive material at room temperature.
3	Danger: May be explosive if shocked, heated under confinement, or mixed with water.
2	Warning: Unstable, or may react if mixed with water.
1	Caution: May react if heated or mixed with water.
0	Stable: Not reactive when mixed with water.

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Hazardous Materials Labeling System (HMLS) labels

Chemical Name	
CAS #	
HEALTH	☐
FLAMMABILITY	☐
REACTIVITY	☐
SPECIFIC	☐

NFPA Diamond



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Personal Protection Index (PPI)

Alphabetic designations A through K are used to indicate personal protection required. The letter X indicates that the individual should ask the supervisor for guidance.

The alphabetical designations are:

A	Safety glasses
B	Safety glasses, gloves
C	Safety glasses, gloves, apron
D	Face shield, safety glasses, gloves
E	Safety glasses, gloves, vapor respirator
F	Safety glasses, gloves, apron, dust respirator
G	Safety glasses, gloves, vapor respirator
H	Safety glasses, gloves, apron, vapor respirator
I	Safety glasses, gloves, dust and vapor respirator
J	Splash goggles, gloves
K	Airline hose or mask, gloves, full suit, boots, dust and vapor respirator
X	Ask your supervisor for guidance

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SECTION 14 - HOT WORK

1. TERMS and DEFINITIONS

1.1 Authorized Air Monitor. Worker(s) who have been designated to monitor the presence of flammable gas in the area where hot work is planned using appropriate air-monitoring equipment.

1.2 Class I Location – Gases or Vapors. Hazardous area in which flammable gases or vapors may be present in sufficient quantities to be explosive or ignitable. Local regulations should be reviewed to ensure all conditions are met prior to entering work zone.

1.3 Class II Location – Combustible Dust. Hazardous area in which combustible dust under normal operating conditions is present at concentrations sufficient to produce explosive or ignitable mixtures. Local regulations should be reviewed to ensure all conditions are met prior to entering work zone.

1.4 Class III Location – Fibers. Hazardous area in which fibers or materials capable of producing combustible fibers are present in concentrations sufficient to produce ignitable mixtures. Local regulations should be reviewed to ensure all conditions are met prior to entering work zone.

1.5 Combustible Material. Any material that may ignite when introduced to an ignition source (e.g., wood, paper, cardboard and plastic).

1.6 Designated Area. A specific area designed or approved for hot work, such as a maintenance shop, a detached outside location that is of non-combustible or fire- resistive construction, or an outdoor location (many greenfield construction sites) essentially free of combustible and flammable contents, and suitably segregated from adjacent areas. A Hot Work Permit is not required in a Designated Area.

1.7 Fire Watch. Worker(s) designated to monitoring hot work and the surrounding area for incipient fires and changing conditions.

1.8 Hot Work. A work activity that by the nature of the operation (e.g., grinding, burning, thermo cutting/welding, brazing, etc.) creates an open source of ignition (primary source of ignition or secondary source of ignition) or that could produce temperatures high enough to cause the ignition of flammable gases and combustible materials.

1.9 Hot Work Control Areas. Fire-hazardous areas such as cable-spreading rooms, cable trays, conveyor galleries, rubber-lined piping equipment and structures, potentially explosive atmospheres, and similar hazardous hot work areas identified by project safety personnel.

1.10 Hot Work Operator. Worker(s) who will handle the hot work equipment and conduct the hot work processes.

1.11 Hot Work Permit. Document issued prior to the start of hot work, which is used to verify the presence of appropriate fire prevention and protection measures.

1.12 Intrinsically Safe. A form of protection based on the restriction of electrical energy within the system, equipment or tool to a level below that which may cause ignition by either sparking or heating effects. Most electronic communication devices, flashlights, cameras, and power tools are not rated as intrinsically safe.

1.13 Lower Explosive Limit (LEL). The minimum concentration of a particular flammable or combustible gas or vapor necessary to support its combustion in air. The safety factor for hot work is 10 percent of the LEL as measured using a calibrated combustible gas detector. Note: At 10 percent of the LEL, the atmosphere may be considered immediately dangerous to life and health (IDLH). In some locations, the safety factor for hot work is 5 percent of the LEL.

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1.14 Primary Source Ignition (PSI) Hot Work. Any work with equipment and tools that use high - energy sources (open flames, electric arcs or incandescent spark) that may ignite flammable or combustible atmospheres, solid materials or liquids when used in a normal manner. PSI hot work is often referred to as 'naked flame' hot work. Examples of PSI hot work include, but are not limited to, the following:

- Welding and burning.
- Grinding.
- Torch cutting and soldering.

1.15 Secondary Source Ignition (SSI) Hot Work. Any work with equipment and tools that may create low-energy sparks and ignite a flammable or combustible atmosphere when used in a normal manner or due to errors or malfunction. SSI hot work is also referred to as 'spark potential' hot work. Examples of SSI hot work include, but are not limited to, the following:

- Sandblasting.
- Using electrical or electronic equipment that is not intrinsically safe or explosion proof (e.g., most electronic communication devices, flashlight, etc.).
- Using spark-ignition engines (including vehicles) in a Class I area (e.g., vehicle entry into a tank dike).
- Using a rotating steel brush.
- Electrical isolation testing.
- Producing a friction spark, typically from a rusty surface.

1.16 Sources of ignition. In locations where flammable vapors may be present, precautions shall be taken to prevent ignition by eliminating or controlling ignition sources. These may include open flames, lightning, smoking, cutting and welding, hot surfaces, frictional heat, sparks (static, electrical, and mechanical), spontaneous ignition, chemical and physical-chemical reactions, and radiant heat.

2. GENERAL REQUIREMENTS

2.1 As applicable, alternative methods to hot work shall be employed when fire-hazardous areas (e.g., cable-spreading rooms, cable trays, conveyor galleries, rubber-lined piping equipment and structures, etc.), potentially explosive atmospheres, and similar hazardous areas may be encountered.

2.2 At a minimum, a SH&E Plan and a task hazard analysis (THA) shall be prepared prior to any hot work activities.

- a) The SH&E Plan will identify the hot work activities and address any required environmental monitoring including gas monitoring, dust, noise, metals, radiation or other monitoring as may be appropriate for site conditions.
- b) The SH&E Plan will provide the measures required to eliminate or otherwise control the hazards associated with the hot work.
- c) The Hot Work Permit Applicability decision flow chart contained in the procedure shall be used to identify air monitor, fire watch and hot work permit requirements.
- d) All SH&E Plan requirements will be followed by all personnel.

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- e) The location specific emergency response plan shall be in place, contain procedures applicable to the potential emergencies presented by the operations, and be reviewed with all personnel potentially affected. Additional fire watch may be required depending on site conditions.

2.3 Before any hot work is performed, and as part of developing the SH&E Plan, the area shall be evaluated for flammables or combustibles by the Supervisor responsible for authorizing hot work. Examples may include, but are not limited to:

- a) Identifying flooring materials.
- b) Determining if any materials such as paper, straw or wood shavings are within a radius of 50 feet (15.24 meters) of the proposed hot work.
- c) Verifying the contents of any containers or tanks within 50 feet (15.24 meters) of the proposed hot work.
- d) Testing to determine if flammable atmospheres exist in the proposed hot work area.
- e) Testing the atmosphere in vessels that may be welded or cut to verify adequate cleaning or purging.
- f) Assessing the dryness of grass or trees in the hot work area and vicinity.

2.4 Where hot work is used, all hot work activities shall be controlled and isolated from flammables and combustibles.

2.5 Individuals, who have the technical and procedural competencies as defined by their employers, as well as their roles within the hot work activity, shall provide input to any hot work permit as necessary to address all hazards and permit conditions related to the hot work.

2.6 Engineering controls shall be implemented to control hot work hazards to the extent feasible. Specific controls to be considered include:

- a) Non-intrinsically safe equipment is prohibited from all locations that may contain a flammable or combustible atmosphere.
- b) Positive air shut-off valves on internal combustion engines may be required to prevent ignition of flammable or combustible atmospheres.
- c) When possible, objects to be welded, cut, or heated shall be moved to a designated safe location. If this is not possible, all movable combustibles in the workspace shall be taken away to a safe place.
- d) If the object to be welded, cut, or heated cannot be moved and all combustibles cannot be removed (e.g., equipment, walls, floors, etc.), positive means shall be taken to confine the heat, sparks, and slag to protect the immovable combustibles.
- e) No welding, cutting, or heating shall be done where application of flammable paint, presence of other flammable compounds, or heavy dust concentrations create a possible hazard.
- f) Wherever there are openings or cracks in the flooring that cannot be closed, precautions shall be taken so no sparks or slag will drop through the floor. The same precautions shall be taken in the presence of cracks or holes in walls, open doorways, and open or broken windows.
- g) Provide metal buckets for disposal of electrode stubs.
- h) When feasible, ventilation shall be in place to control atmospheric hazards.

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2.7 Identify the type of metal to be worked on and protective coatings that have been applied. Where coatings are flammable or toxic, the coating shall be stripped a minimum of 4 inches (10 centimeters) from the areas of heat application.

- a) Before welding, cutting, or heating is commenced on any surface covered by a preservative coating whose flammability is not known, a test shall be made by a competent person to determine its flammability.

2.8 Position welding screens or shields to protect workers and passers-by from welding arc rays.

2.9 All hot work equipment shall be used only for operations for which it is approved and as recommended by the manufacturer and maintained in good repair in accordance with manufacturer's specifications.

2.10 All personnel involved in a hot work operation have the authority and responsibility to stop any work that they consider to be unsafe.

2.11 Individuals assigned to operate or maintain oxygen/fuel-gas supply equipment and resistance welding equipment will be thoroughly instructed in the safe use and maintenance of such equipment by a qualified person.

2.12 If other individuals are required to work in the vicinity of hot work operations, they shall be protected, given the required PPE, and be notified of the work and all safety procedures prior to starting the hot work (e.g., screens, signs identifying workpieces that are hot, barricades and restricted entry to hot work area, etc.).

2.13 Exposure to welding operations can result in eye damage, burns, or respiratory illness. If an incident occurs, seek medical attention and report it to the manager or supervisor and to AECOM TISHMAN CONSTRUCTION.

2.14 Training shall be provided as to the use of Hot Work Permits to all associated individuals.

- a) Individuals shall be trained and, as applicable, certified in accordance with anticipated tasks (e.g. welding, air monitor, fire watch, etc.), associated hazards, and jurisdictional requirements.
- b) Training shall be provided as to the use of Hot Work Permits. Subcontractors are solely responsible for any and all training for their employees.

2.15 Personal Protective Equipment

- a) Selection and use of personal protective equipment shall comply with AECOM TISHMAN CONSTRUCTION standards. See Personal Protective Equipment section.
- b) Contact lenses should not be worn if there is a potential exposure to chemicals that can irritate the eye.

2.16 Hot Work Permit

- a) A Hot Work Permit is required for and shall be prepared before any hot work.
- b) A Hot Work Permit is valid for no more than one work shift.
- c) If hot work is suspended during a shift, the permit shall be revalidated before further hot work can continue.
- d) Revalidation involves inspecting the hot work area for any change to previous conditions and conducting air monitoring.

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2.17 LEL Mitigation Plan

- a) Section C of Hot Work Permit documents the location of confirmed sources of flammable gas and the controls needed to eliminate, reduce or maintain the LEL reading to less than 10 percent (or 5% in certain jurisdictions) of the LEL.
- b) An LEL Mitigation Plan shall be developed and implemented if the presence of flammable gas is detected (LEL greater than 0 percent) at any time during the hot work.
- c) The following information shall be documented in Section C of Hot Work Permit or equivalent form:
 - The percent of the LEL that was measured.
 - The identified source(s) of the flammable gas within the hot work area.
 - The controls, if any, that shall be implemented to effectively eliminate, reduce or maintain the flammability level to less than 10 percent (or 5% in certain jurisdictions) of the LEL within the hot work area.
 - The percent LEL measured after controls, if any, have been implemented.

***Note:** If at any time the LEL reading is greater than or equal to the safety factor of 10 percent (or 5% in certain jurisdictions) of the LEL, the hot work shall stop immediately and shall not resume until controls are implemented to reduce the LEL level to less than the safety factor (as per jurisdiction, 5% or 10%) of the LEL.*

2.18 Fire Watch Requirements

- a) A Fire Watch is required whenever the hot work meets any of the following criteria:
 - The hot work consists of Class 1 hot work.
 - The hot work will be performed:
 - Within 50 feet (15.24 meters) of shielded combustible material or at the direction of AECOM.
 - Within a 50-foot (15.24-meter) radius of wall or floor openings that expose combustible materials.
 - Adjacent to metal partitions, walls, ceilings or roofs that are in contact with combustible materials on the other side and are likely to be ignited by conduction or radiation.
 - Where fire alarms or suppression systems have to be disabled.
- b) In some cases more than one fire watch may be necessary (e.g. various levels in a building). This shall be identified in the SH&E Plan and the THA.
- c) The Fire Watch shall be in the ready position at all times while hot work is being performed. The ready position consists of the following:
 - Being attentive to the hot work being performed, while performing no other tasks.
 - Properly positioning the fire extinguisher prior to the start of work.

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- Always maintaining a line of sight to the hot work being performed.
- d) The Fire Watch shall stop the work if he or she deems that:
 - Unsafe conditions have developed.
 - The work is exceeding the scope described in the Hot Work Permit.
- e) A Fire Watch shall be maintained for at least 30 minutes, or as dictated by the JHA, after completion of hot work operations so that possible smoldering fire can be detected and extinguished.
- f) They shall be familiar with facilities and procedures in the event of a fire.

2.19 Welding and Cutting General Requirements

- a) Avoid looking directly at the welding arc.
- b) Avoid prolonged exposure to welding/paint fumes.
- c) Avoid touching recently welded joints.
- d) Welding/cutting on containers.
 - No welding, cutting, or other hot work shall be performed on empty drums, barrels, tanks, or other containers until they have been thoroughly cleaned and it is absolutely certain that there are no flammable materials present or any substances such as greases, tars, acids, or other materials which, when subjected to heat, might produce flammable or toxic vapors. Any connection to the drum or vessel shall be disconnected or blanked off.
 - All hollow spaces, vacancies, or containers shall be ventilated to remove gases before preheating, cutting, or welding. Purging with inert gas is recommended.
 - Welding/cutting in confined spaces such as a tank, boiler, pressure vessel, or small compartment shall require the following precautionary measures:
 - Local exhaust ventilation shall be provided, unless workers wear supplied-air respirators.
 - Gas cylinders and/or welding machines shall be placed outside the confined space.

2.20 Gas Welding and Cutting Safety Requirements

- a) Fuel-gas hoses and oxygen hoses shall be easily distinguishable from each other. The contrast shall be made by different colors or by surface characteristics readily distinguishable by touch. Oxygen and fuel-gas hoses shall not be interchangeable. A single hose having more than one gas passage shall not be used.
- b) When parallel sections of oxygen and fuel-gas hose are taped together, not more than 4 inches (10 centimeters) out of 12 inches (30.5 centimeters) shall be covered by tape.
- c) All hoses in use shall be inspected at the beginning of each work shift. Defective hose shall be immediately removed from service.
- d) Hoses, cables, and other equipment shall be kept clear of walkways, ladders, and stairs.

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- e) Clogged torch tip openings shall be cleaned with approved cleaning wires, drills, or other devices designed for this purpose.
- f) Torches to be used shall be inspected at the beginning of each work shift for leaking shutoff valves, damaged hose couplings, and clogged tip connection. Defective torches shall not be used.
- g) Torches shall be ignited by friction lighters or other approved devices only. Matches and flame lighters shall not be used to ignite a torch.
- h) Oxygen and fuel-gas pressure regulators, including related gauges, shall be in proper working order and equipped with “Flashback” arrestors attached to the gauges. NOTE: Flashback arresters are in addition to “Backflow” devices.
- i) All oxygen cylinders and fittings shall be kept away from oil or grease.
 - Cylinders, cylinder caps and valves, couplings, regulators, hose, and apparatus shall be kept free from oil or greasy substances and shall not be handled with oily hands or gloves.
 - Oxygen shall not be directed at oily surfaces or greasy clothes, or used within a fuel oil or other storage tank or vessel.
- j) Torches and hoses shall be completely depressurized (bled) of pressurized gas prior to storage or at the end of each shift.
- k) Torches and hoses shall not be stored in enclosed areas (e.g., gang boxes, lockers) while connected to cylinders and gauges shall be removed at the end of shift.
- l) Oxygen connections shall include a means to prevent backflow.
- m) Fuel-gas cylinders shall be provided flashback protection.

2.21 Arc Welding and Cutting Safety Requirements

- a) Electrode holders that are designed for arc welding/cutting and are capable of safely handling the maximum rate current shall be used.
- b) Any current-carrying parts passing through the holder which the arc welder or cutter grips in his/her hand, or the outer surfaces of the jaws of the holder, shall be fully insulated against the maximum voltage encountered and properly grounded.
- c) Grounding is required of the welding machine and the welding circuit.
 - Grounding the welding machine is not required if the machine is not double insulated, as indicated by the box-within-box symbol on the rating plate.
 - The ground clamp or ground lead (separate from the workpiece connection or workpiece lead) shall connect the workpiece, or the metal table the workpiece rests upon, to a suitable ground, such as a metal building frame.
- d) All arc welding/cutting cables shall be completely insulated and flexible, capable of handling the maximum current requirements of the work.

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- e) Only cables free from repair or splices for a minimum distance of 10 feet (3 meters) from the electrode holder shall be used. Cables with standard insulated connectors or splices with insulating quality that is equal to that of the cable are permitted.
- f) If it is necessary to splice lengths of cable, insulated connectors equivalent to that of the cable shall be used. If connections are made by cable lugs, they shall be securely fastened together and provide good electrical contact. Exposed metal parts of the lugs shall be completely insulated.
- g) If electrode holders are left unattended, the electrodes shall be removed and the holder placed so that they cannot make electrical contact with individuals or conducting objects.
- h) To avoid the possibility of electric shock, particularly in humid or damp environments, appropriate and dry PPE shall be worn. Electrode holders shall not be dipped in water.
- i) When the arc welder or cutter leaves work, stops work for any length of time, or when the arc welding cutting machine is to be moved, the power supply to the equipment shall be turned off.
- j) Any faulty or defective equipment shall be reported to the Supervisor and tagged out of service until repaired.
- k) All arc welding/cutting operations shall be shielded by non-combustible or flameproof screens to protect individuals and other persons working in the vicinity from the direct ray of the arc.

2.22 Storage and Handling of Compressed Gas Cylinders

- a) Compressed gas cylinders shall be legibly marked with either the chemical or trade name of the gas in accordance with Globally Harmonized System (GHS) or Workplace Hazardous Materials Information System (WHMIS) standardized labelling. Such markings will be stenciled, stamped, or labelled and shall not be easily removable.
- b) The marking shall be located on the shoulder of the cylinder.
- c) Compressed gas cylinders shall be equipped with approved connections.
- d) Acetylene cylinders shall be stored and used valve end up.
- e) Cylinders shall not be stored near highly combustible/flammable materials, especially oil or grease.
- f) All cylinders shall be stored in an upright and secured position (with chains) with caps installed and separated from fuel-gas cylinders or combustible materials (especially oil or grease), by a minimum distance of 20 feet (6 meters), or by a non-combustible barrier at least 5 feet (1.5 meters) high and having a fire-resistance rating of at least one half hour. Cart storage shall take place per the manufacturer specifications and OSHA regulations.
- g) Cylinders shall be not dropped, struck by objects, or permitted to strike each other violently.
- h) Cylinder valves shall be closed, gauges removed and caps installed before moving.
- i) Cylinder valves shall be closed and gauges removed at the end of the shift or when work is finished. OSHA's 24-hour Tank storage rule does not apply.
- j) Valves of empty cylinders shall be closed.
- k) Cylinders shall be kept far enough away from the actual welding/cutting operation so that sparks, hot slag, or flames shall not reach them.

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- l) Cylinder valves shall always be opened slowly.
- m) An acetylene cylinder valve shall not be opened more than one and one-half turns of the valve stem and preferably no more than three-fourths of a turn.
- n) Do not use acetylene at a pressure in excess of 15 pounds per square inch (psi) gauge pressure, or 30 psi absolute.
- o) Where a special wrench is required to operate a cylinder valve, it shall be left in position on the stem of the valve while the cylinder is in use. In the case of manifolded or coupled cylinders, at least one such wrench shall be available for immediate use.
- p) Regulators shall be removed, valve caps in place, and valves closed when cylinders are transported by vehicles. All vehicles used to transport cylinders shall have a proper support rack installed.
- q) A suitable cylinder truck, chain, or other steadying device shall be used to prevent cylinders from being knocked over while in use.
- r) Cylinders shall not be placed where they may become part of an electric circuit. Tapping of an electrode against a cylinder to strike an arc shall be prohibited.
- s) Only use warm, not boiling, water to thaw cylinders and valves.
- t) Pressure Reducing Regulators:
 - Pressure regulators, including the gauges, shall be in proper working order while in use. If not, immediately remove from service.
 - Regulators shall be of an approved type for the type of gas to be utilized.
 - When a pressure-reducing regulator is attached to a compressed gas cylinder, the cylinder valve shall be opened just slightly at first, so that the regulator can take on pressure slowly. After which, the valve may be turned open to its normal position. If the regulator takes on pressure too quickly, it can damage the regulator and pressure gauges.
 - The operator shall stand to the side of glass covered gauges and not in front of them.

2.23 Mechanical Ventilation

- a) Mechanical ventilation shall consist of either general dilution systems or local exhaust systems. Local exhaust systems are preferred. The THA shall identify ventilation requirements suitable to the work environment and anticipated hazards.
- b) Mechanical ventilation shall be of sufficient capacity and so arranged as to produce the number of air changes necessary to maintain welding fume and smoke within safe limits.
- c) General dilution ventilation may not be used as the only means of control when toxic metals are involved in the operation.
- d) Local exhaust ventilation shall consist of freely movable hoods intended to be placed by the welder or burner as close as practicable to the work. This system shall be of sufficient capacity and so arranged as to remove fumes and smoke at the source and keep the concentration of them in the breathing zone within safe limits

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- e) Contaminated air exhausted from a working space shall be discharged into the open air or otherwise clear of the source of intake air. Environmental regulations may require filtering or other cleaning of exhausted air.
- f) All makeup air shall be clean and suitable for breathing.
- g) Oxygen shall not be used for ventilation purposes, comfort cooling, blowing dust from clothing, or for cleaning the work area.
- h) The subcontractor shall provide appropriate methods and controls in the case of specific requirements (including welding rods and fluxes, paints and coatings) for materials containing zinc, lead, mercury, beryllium, cadmium, and stainless steel to be cut, heated, and/or welded.

3. ATTACHEMENTS

3.1 Example Hot Work Permit

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Attachment 1: Example Hot Work Permit

Hot Work Permit

S3AM-332-FM1

Name of Contractor: _____ Date: _____ Time: _____ Person Preparing Permit: _____ Person Conducting Hot Work: _____	Location: _____ Weather: _____ Title: _____ Title: _____	Project #: _____ THIS PERMIT BECOMES VOID: 1. At the end of the shift, or 2. Whenever conditions change significantly, or 3. On any emergency signal
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TYPE OF HOT WORK		
☐ PSI CLASS I Location (Complete full permit)	☐ PSI NON-CLASS I Location (Complete section A, D, and E)	☐ SSI CLASS I Location (Complete section B, and E)

Description of Hot Work to be Conducted:

A. WORK AREA PREPARATION FOR HOT WORK			
	YES	NO	N/A
1. Is the hot work equipment in proper working order?	☐	☐	☐
2. Are the proper safety data sheets (SDS) available for review?	☐	☐	☐
3. Have combustible materials been relocated more than 50 feet (15.24 meters) from hot work or been properly shielded?	☐	☐	☐
4. Is firefighting equipment readily available and in proper working order (e.g. fire extinguishers, sand, shovels, etc.)?	☐	☐	☐
5. Have ducts, drains, and sewers been adequately covered to prevent sparks from entering?	☐	☐	☐
6. If hot work is to be performed indoors, have combustible floors been wet down or properly shielded?	☐	☐	☐
7. If arc welding equipment will be used, have measures been taken to protect personnel from shock?	☐	☐	☐
8. If required, is a 20-pound (9.07-kilogram) fire extinguisher available and appropriate for the type of fire expected?	☐	☐	☐
9. If hot work is to be performed in or on containers, vessels, tanks, or similar equipment, have they been cleaned, purged, or vented, or have other precautions been taken to verify that they are safe for hot work?	☐	☐	☐
10. If required, have lockout tagout procedures been completed in accordance with <i>S3AM-325-PR1 Lockout Tagout</i> ?	☐	☐	☐

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11. Have others required to work in the hot work area been notified of the work involved and safety precautions, and do they have the required PPE?	☐	☐	☐
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LIST ANY PERSONAL PROTECTIVE EQUIPMENT THAT IS REQUIRED (OTHER THAN THE MINIMUM REQUIRED):

B. AIR MONITORING (Initial and continuous monitoring are required for all Class I Hot Work [PSI and SSI])
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DOCUMENT WHERE AIR MONITORING WILL OCCUR, WHAT POTENTIAL GAS OR VAPOR SOURCES EXIST WITHIN THE HOT WORK AREA, AND THE WIND DIRECTION:

INSTRUMENT 1 SERIAL NUMBERS: _____	DATE AND RESULTS OF LAST CALIBRATION: _____	PRE-USE CALIBRATION CHECK PERFORMED: _____	YES NO NA circle
INSTRUMENT 2 SERIAL NUMBERS: _____	DATE AND RESULTS OF LAST CALIBRATION: _____	PRE-USE CALIBRATION CHECK PERFORMED: _____	YES NO NA circle

Initial Reading %LEL (if >0% fill out section C)

If O ₂ level below 10% do not proceed without alternate testing	Time	%O ₂	%LEL	Time	%Oxygen	%LEL	Time	%Oxygen	%LEL

Continuous Reading %LEL

Time	%LEL	Time	%LEL	Time	%LEL	Time	%LEL	Time	%LEL

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C. LOWER EXPLOSIVE LIMIT (LEL) MITIGATION PLAN

(Required if initial or continuous monitoring LEL is greater than 0%)

% LEL DETECTED:		% LEL AFTER CONTROLS IMPLEMENTED		NOT APPLICABLE		<i>(If not applicable, continue monitoring source)</i>
DOCUMENT THE LOCATION OF CONFIRMED SOURCE(S) OF GAS / VAPOR AND WHAT CONTROLS (IF ANY) HAVE BEEN IMPLEMENTED <i>(Note: Hot work is prohibited if the LEL reading is greater than or equal to 10% in the hot work area.)</i>						

D. FIRE WATCH

	YES	NO	
1. Will PSI Class I work be performed?			<i>Note: If the answer to any of the fire watch conditions to the left is YES, a dedicated fire watch must be assigned. Note: The fire watch is authorized to perform air monitoring.</i>
2. Will PSI Non-Class I work be performed within 50 feet (15.24 meters) of combustible materials?			
3. Will PSI hot work be performed where wall or floor openings within 50 feet (15.24 meters) would compose combustibles in adjacent areas?			
4. Is PSI hot work to be performed where combustibles are adjacent to the opposite side of partitions, ceilings, or roofs being worked on?			Fire Watch 1: Fire Watch 2:
5. Will fire alarms or suppression systems be disabled for hot work?			

E. AUTHORIZING SIGNATURES

AUTHORIZED AIR MONITOR: *I have performed initial air monitoring with a survey of the hot work area. I agree to perform continuous air monitoring and record results on this form while hot work is being performed in a Class I area and to stop hot work if the LEL reading is greater than or equal to 10%.*

Name		Signature		Name		Signature	
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FIRE WATCH: *I have been trained in the use of the applicable fire fighting equipment and agree to maintain a line of sight with the hot work operation at all times, to have immediate access to a fire extinguisher and to remain at the hot work area for 30 minutes after the hot work is completed to verify the area if fire safe.*

Name		Signature		Name		Signature	
------	--	-----------	--	------	--	-----------	--

HOTWORK OPERATORS: *I have reviewed and understand the permit conditions specific to the scope of work. I agree to perform hot work operations within these conditions, to stop any work that I deem to be unsafe, and to notify the performing authority upon completion or interruption of this permitted work.*

Name		Signature		Name		Signature	
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Name		Signature		Name		Signature	
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SITE SUPERVISOR: *The location where this work is to be done has been examined, necessary precautions have been taken, and permission is granted for this work.*
Permit Expires:

Name		Signature		Time Hot Work Started		Time Hot Work Completed	
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SECTION 15 - HOUSEKEEPING & SANITATION

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 AECOM TISHMAN CONSTRUCTION, through use of its own personnel and those of its subcontractors, will require daily clean-up of the project worksite. All construction debris shall be gathered and placed in a dumpster or suitable collection point. AECOM TISHMAN CONSTRUCTION's Project Superintendent will establish the locations.

2.2 All hazardous debris shall be collected and disposed of in compliance with all appropriate regulations.

2.3 Subcontractors shall be responsible for daily clean-up of construction debris and waste related to their scope of work.

2.4 If a subcontractor fails to maintain a satisfactory clean-up program, AECOM TISHMAN CONSTRUCTION will give 24 hours' notice, perform the clean-up, and back-charge the responsible subcontractor for the costs.

2.5 AECOM TISHMAN CONSTRUCTION will have the option to perform all clean-up on the project. If this option is exercised, the amount included will be deducted from the subcontractor's bid and the general contractor will be responsible for all clean-up including identifiable debris.

2.6 All tools, parts, and other materials entering the work area shall be free of dirt, grease, oil, and visible loose particles.

2.7 Clothing worn by personnel shall be clean and in such condition so as not to contribute to the contamination of the work area or component being serviced.

2.8 Labeled containers shall be used for combustible and non combustible trash and shall not be placed under cable trays or near material or equipment vulnerable to fire damage.

2.9 If poor housekeeping practices are observed by anyone, the condition observed shall be discussed with the first line supervisor and corrective action promptly provided. Cluttered work areas shall not be tolerated. All supervisors shall follow through and make sure that these conditions are not allowed to recur.

2.10 All personnel are responsible for good housekeeping practices and shall be instructed to the requirements of this procedure in orientations.

2.11 Work areas may be divided by whatever means necessary to ensure the proper housekeeping requirements. All designated work and/or cutting stations shall have appropriate trash receptacles next to them.

2.12 Projects may assign groups or teams to certain areas to ensure housekeeping is completed.

2.13 Materials, trash, or other objects must not be thrown from buildings or structures unless approved by AECOM TISHMAN CONSTRUCTION project management with appropriate precautions being taken.

2.14 Where large accumulations of materials occur on a non routine basis, such as the stripping of concrete forms, the material shall be promptly removed or stored neatly away from heavily traveled areas.

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2.15 Materials and equipment delivered to the work area shall be placed so that they are accessible but do not hinder construction progress. Material and equipment shall be so positioned that it will not be damaged by construction activity and shall be placed on proper dunnage and/or wheels for movement.

2.16 The use, location, and deployment of construction tools, supplies, and equipment shall be regulated to keep access and work areas clear.

2.17 All flammable and hazardous materials are to be stored in centrally located areas designated by AECOM TISHMAN CONSTRUCTION's Project Superintendent. Locations of designated areas are to be communicated to all subcontractors onsite.

- All storage areas are to be marked with adequate "Flammable Storage" signage legible from a distance of at least 50 feet. "No smoking" perimeters shall be established and designated with appropriate signage.
- Fire extinguishers of the appropriate type shall be provided in the immediate areas of flammable storage.

2.18 The Project Safety Manager in conjunction with the Project Superintendent will develop, implement and monitor a hazardous waste accumulation and disposal plan.

- This plan will include the identification of hazardous or potentially hazardous materials, storage, appropriate disposal of any containers or excess materials.
- The Project Superintendent for accumulation of all such materials shall establish designated storage areas. These areas shall be appropriately marked and shall afford necessary protection against "spills" or other events. These areas shall include storage of all materials so identified for both contractor and subcontractor.

2.19 Toilets shall be provided as follows (at a minimum):

<u># of Individuals</u>	<u>Minimum Facilities per Gender</u>
20 or less	One
21 to 199	One toilet seat and one urinal for every 40 workers
200 or more	One toilet seat and one urinal for every 50 workers

2.20 All toilet facilities shall be approved for use by the state and local governments. In the event a state or locality requires differing amounts of toilet facilities, the policy requiring the most shall apply.

2.21 All toilet facilities shall be constructed with four (4) walls, a roof (for outdoor units), lighting as required, a toilet seat, a urinal, an adequate vent to the outside, and a self- closing door with a latch.

2.22 All toilet facilities shall be serviced and maintained on a regular basis (at least twice a week or more, as needed).

2.23 All toilet facilities shall be supplied with adequate amounts of toilet paper and a means for sanitizing the user's hands.

2.24 All toilet facilities shall be made readily available to workers. As a general guide, they should be placed within a ten (10) minute or less walking distance from a work area. Separate secured facilities will be provided for opposite genders.

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2.25 All toilet areas shall be inspected on a weekly basis for graffiti. If found, the project team shall take steps to have it properly removed as soon as possible. Each project shall create a plan for cleaning and graffiti removal prior to work commencing.

2.26 Drinking water shall be obtained from a potable water source or sources. Water shall be provided in covered containers meeting current applicable regulations. Individual drinking cups shall be provided. Each water dispenser location will be provided with a disposal unit for spent cups. Subcontractors are responsible to provide adequate drinking water and single-use cups for their employees. Subcontractors are also responsible to supply potable water containers that must be sanitized and sealed before each use.

3. ATTACHMENTS

None applicable

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SECTION 16 - INCIDENT REPORTING, INVESTIGATION, & CLAIMS MANAGEMENT

1. TERMS and DEFINITIONS

1.1 Fatality. Loss of life of any AECOM TISHMAN CONSTRUCTION individual, AECOM TISHMAN CONSTRUCTION subcontractor personnel, client personnel or member of the general public.

1.2 Lost Time Injury/Illness. A work-related injury or illness that has caused an individual to be absent from work following the day that the injury or illness occurred.

1.3 Recordable Injury/Illness. A work-related injury or illness that results in any of the following:

- a) Fatality;
- b) Medical treatment beyond first aid;
- c) Days away from work;
- d) Restricted work or transfer to another job;
- e) Loss of consciousness; and/or
- f) A significant injury or illness as diagnosed by a physician or licensed health care professional.

1.4 Restricted Work (also called "Modified Work"). Occurs when an individual, after the initial day of injury, because of an individual's work-related injury or illness:

- a) Is assigned to another job on a temporary basis; also known as "job transfer;"
- b) Works at their permanent job less than full time;
- c) Works at their permanent job but cannot perform all duties normally performed at least once per week; also known as "routine job functions;"
- d) Performs light duty work at the direction of a Supervisor.

1.5 SH&E Incidents. The following events or situations as applied to AECOM TISHMAN CONSTRUCTION individuals and/or AECOM TISHMAN CONSTRUCTION-controlled operations are considered SH&E Incidents:

- a) Any injury or illness to an AECOM TISHMAN CONSTRUCTION individual or subcontractor, even if it does not require medical attention. This would include work-related injuries/illnesses that have become significantly aggravated by the work environment.
- b) An injury to a member of the public, or clients, occurring on an AECOM TISHMAN CONSTRUCTION-controlled work site.
- c) Re-occurring conditions such as back pain or cumulative trauma disorders (e.g., carpal tunnel syndrome).
- d) Fire, explosion, or flash that is not an intended result of a planned event (e.g., remediation process, laboratory procedure).

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- e) Any incident involving company-owned, rented, or leased vehicles (including personal vehicles used for company business).
- f) Any failure to comply with the requirements of a regulatory permit issued to AECOM TISHMAN CONSTRUCTION.
- g) Any notice of violation or notice of non-compliance received from a regulatory authority.
- h) Property damage resulting from any AECOM TISHMAN CONSTRUCTION or subcontractor activity.
- i) Unexpected release or imminent release of a hazardous material.
- j) Confirmed or suspected chemical or biological exposure.
- k) A safety-related complaint from the public regarding AECOM TISHMAN CONSTRUCTION activities.
- l) SH&E-related incidents that could result in adverse public media interest concerning AECOM TISHMAN CONSTRUCTION or an AECOM TISHMAN CONSTRUCTION project.
- m) Any inspection by a safety, health, & environmental enforcement agency.

1.6 Significant SH&E Incident. Any SH&E incident that meets/involves the following criteria:

- a) Fatality;
- b) Amputation;
- c) Hospitalization for treatment for more than 24 hours (admission);
- d) Any single event resulting in more than one individual requiring medical treatment or more than one individual being away from work more than three days;
- e) Any SH&E-related Consent Agreement/Order/Lawsuit or enforcement action seeking more than \$10,000 or alleging criminal activity;
- f) Any spill or release of a hazardous material that is reportable to a regulatory agency;
- g) Any Notices of Violation resulting from not operating within a regulatory agency permit/license or consent;
- h) Any incident resulting in property damage expected to exceed \$10,000 US dollars;
- i) Any security-related incident that could have caused significant harm to an AECOM TISHMAN CONSTRUCTION individual or subcontractor; and/or
- j) Any Near Miss that may have resulted in any of the above but because of “luck” did not happen.

1.7 Near Miss. This is defined as an unplanned event involving interaction or contact with a hazard but resulting in no negative outcome - that is it had the potential to cause injury, compromise health, environmental impairment, business interruption, or property damage – but did not. Near Misses should be addressed by reporting and, as applicable, investigation, and implementation of corrective action to minimize recurrence.

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1.8 Security Incident. Any security related incident that could cause harm to or is associated with an AECOM TISHMAN CONSTRUCTION individual or subcontractor in the course of duty. Refer to Global Resilience Group resources for additional information.

1.9 SH&E Incident Report (IR) Form. Form used to document incidents which shall be completed when there is no internet access to the on-line incident management system (e.g., IndustrySafe). IR's shall be completed within 4 hours for a Significant Incident and 24 hours for all other SH&E incidents. IR's shall be submitted to the supervisor and manager.

2. GENERAL REQUIREMENTS

2.1. In an emergency/life-threatening situation, use the appropriate local emergency phone numbers and seek immediate medical care. Follow the project ERP (Emergency Response Plan).

2.2 For work-related non-emergency medical support, workers are to contact their supervisor and the designated AECOM TISHMAN CONSTRUCTION SH&E personnel, regardless of time of day/day of week.

2.3 In the event of any work-related injury to an AECOM TISHMAN CONSTRUCTION employee, the AECOM Injury Management Hotline is to be called: 512-419-5016 immediately. If it is an emergency situation, the Emergency Response Plan is to be followed first. If the employee is unable to make the call, the call must be made on their behalf. The AECOM Injury Management Hotline Poster should be posted on all jobsites.

2.4 Do not discuss the incident with members of the media. Direct any inquiries to media@aecom.com.

2.5 Immediately notify AECOM TISHMAN CONSTRUCTION management by telephone (or other direct means such as our Rapid Response Program) of incidents involving an AECOM TISHMAN CONSTRUCTION individual or subcontractor:

- a) A bomb, terrorist threat, or serious fire;
- b) The arrival of media on site for any reason;
- c) Severe damage to the site as a result of severe weather or other serious property damage (>\$5,000);
- d) Injury to any member of the public as a result of our work at a project;
- e) Significant environmental release/ exposure (i.e. fuel, chemical, gas, asbestos, lead) which could cause immediate or future injury, media attention, or compliance fines;
- f) A single incident resulting in three or more injuries where medical attention is required;
- g) A single incident which may potentially lead to a fatality but is unknown at the time;
- h) Any incident resulting in a fatality;
- i) A company vehicular incident occurring offsite involving the hospitalization of an employee or member of the public;
- j) Trade issues on or near the project which may lead to or has caused injury, property damage, and/or media attention;
- k) Any serious situation otherwise not listed which can significantly affect AECOM TISHMAN CONSTRUCTION in any type of negative way;

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Note: only the highest-ranking and informed AECOM employee available should place the call to the Rapid Response Hotline. In the event the Rapid Response Hotline is called, the crisis team (leadership, safety, legal, communications) will be alerted so appropriate action is taken. The Rapid Response number is to be distributed to leadership via the yellow Rapid Response cards and not posted publicly.

2.6 For critical injuries and illnesses, transportation shall be provided by emergency services (e.g., ambulance, paramedics). Individuals should not drive themselves to obtain treatment for critical injuries/illnesses.

2.7 For non-critical injuries and illnesses that require off site medical treatment, transportation with an AECOM TISHMAN CONSTRUCTION project team member and/or subcontractor management team member shall be provided by company owned or rented vehicle as applicable.

2.8 Drug and alcohol testing shall be performed in accordance with the AECOM TISHMAN CONSTRUCTION alcohol (substance abuse) testing program in the [Substance Abuse Prevention Policy](#).

2.9 An initial incident form on IndustrySafe, AECOM TISHMAN CONSTRUCTION's online incident reporting software, shall be completed within 24 hours of an incident's occurrence to communicate basic incident information. Incident type determination is to be made at the best of the project's ability at the time of entering the incident report and updated as applicable. Determining OSHA recordability of an injury or illness shall follow OSHA recordkeeping standards and can be referenced here: <https://www.osha.gov/recordkeeping/entry-faq>.

As additional information becomes available, applicable documentation shall be attached to the incident report such as:

- a) Police report;
- b) Photographs of incident scene;
- c) Witness statements;
- d) Federal/State/Province specific forms;
- e) Timeline;
- f) Root cause analysis;
- g) Medical paperwork if applicable;

2.10 Near misses shall be reported in IndustrySafe within 24 hours of the event occurring.

2.11 Review the incident report or the near miss report for completeness and accuracy. Ensure the reports are distributed properly. Assist with incident evaluation and investigation as directed by management. The preliminary investigation shall be completed within 24 hours of an incident. It can be modified moving forward.

2.12 Complete external reporting requirements if required. The Safety Manager shall notify regulatory agencies of the incident (e.g., reporting to OSHA of severe injuries and/or fatalities using the toll free number, or the reporting application located on OSHA's public website) after discussing it with the Corporate Safety Department. All work-related fatalities must be reported orally to OSHA within eight hours of learning of the incident. Inpatient hospitalization, amputations, and the loss of an eye must be orally reported to OSHA within 24 hours of learning of the incident.

2.13 Enter corrective actions as a result of incident investigation into IndustrySafe and monitor to completion.

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2.14 Perform root cause investigations on the following types of incidents:

- a) A recordable injury or illness of an AECOM TISHMAN CONSTRUCTION or subcontractor individual;
- b) A vehicle incident involving an AECOM TISHMAN CONSTRUCTION individual (while working) where either the individual or a member of the public is hospitalized;
- c) Incidents that result in significant adverse public media interest in AECOM TISHMAN CONSTRUCTION or an AECOM TISHMAN CONSTRUCTION project;
- d) Any near miss or incident occurring on projects undertaken for specific clients, where AECOM TISHMAN CONSTRUCTION has contractually agreed to participate in safety systems that dictate that all near misses and incidents undergo a root-cause analysis;
- e) Damage to Company or Government/Client-owned equipment that exceeds \$25,000, unless otherwise required by a client;
- f) A release to the environment of a substance in excess of a mandated reportable quantity, and requiring the response of external response organizations;
- g) Any near miss that an SH&E Manager determines—if taken to its logical conclusion—would have resulted in a fatality, multi-day lost-workday case, and/or inpatient hospitalization;
- h) Any incident that the Vice President of SH&E or SH&E Manager deems appropriate for an investigation.

2.15 The following actions will be taken to investigate an incident:

- a) The Project Manager or Superintendent will notify the responsible SH&E Manager that a recordable incident has occurred;
- b) The investigation may require that an investigation team be assembled. The team will consist of personnel from operations and SH&E;
- c) The responsible SH&E Manager may include other appropriate members of management in an investigation team, and may solicit input from AECOM TISHMAN CONSTRUCTION Legal regarding the investigation;
- d) The person leading the investigation will be trained in investigation techniques and root-cause analysis;
- e) If deemed appropriate by the Vice President of SH&E, the responsible SH&E Manager will complete the investigation under the direction of AECOM TISHMAN CONSTRUCTION Legal Counsel and the Vice President of SH&E (or designee);
- f) Following an incident, immediately determine the sources of evidence. Evidence may include a listing of people, equipment, and materials involved; a recording of environmental factors such as weather, illumination, temperature, noise, ventilation; and physical factors such as fatigue and medical conditions;
- g) The investigation process will include interviews with those involved or those directly witnessing the event;

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- h) The investigation team lead should ensure that investigation tools are available (e.g., cameras, protective equipment, tape measures, marking devices, etc.);
- i) Store evidence (e.g., witness statements, photographs, documentation, etc.) that is collected during the investigation in a secured and locked location;
- j) If photos are taken, a photo log should be created as the photos are taken during the investigation;
- k) Prepare an (EIR) Executive Incident Review summary for any investigation within 7 working days of the incident, as directed by the Regional/Area Safety Manager;
- l) Investigation reports will identify the critical factors involved in the incident. Develop direct and contributing causes to identify the root cause(s) of an incident. Investigators will evaluate causes associated with human activities, physical causes, and systems causes. The report will identify corrective actions, and assign a responsible party and due dates;
- m) The responsible Manager (or designee) shall complete a Lessons Learned within 7 working days of the incident in conjunction with the EIR. This Lessons Learned shall be forwarded to the SH&E Manager. This may then be used for communication to AECOM TISHMAN CONSTRUCTION staff in order to facilitate the communication of lessons learned. The Lessons Learned will contain only basic facts; will be without reference to a site, location, or individual; and will be developed solely for the purpose of conveying lessons learned to prevent a similar incident, illness, or injury. Lessons Learned information may be posted on the Company intranet for other individuals to access. In addition, the involved subcontractor (s) designated company safety representative shall provide an action plan/lesson learned, and Task Hazard Analysis to prevent reoccurrence. The injured, his/her work crew and supervisor must be retrained on the task, hazard recognition and protection. Failure to comply may result in a \$5,000 violation to the subcontractor. All tiered subcontractors shall comply with AECOM TISHMAN CONSTRUCTION in all facets of this process.

2.16 Release of the investigation report outside of AECOM TISHMAN CONSTRUCTION may only be authorized by the business group Vice President of SH&E (or designee).

2.17 Following the completion of the incident investigation and development of corrective actions, it is critical that the requirements for preventing a reoccurrence of the event be implemented prior to re- starting the activity.

2.18 All active AECOM TISHMAN CONSTRUCTION projects are to submit the latest version of the Monthly Safety Reporting Template each month by the 5th to report information for the month prior. This report includes project work hours and incidents, as well as observations and inspections completed. This template will be provided by the safety department.

2.19 All subcontractors shall submit a monthly report to AECOM TISHMAN CONSTRUCTION project management that includes at a minimum: a summary of work hours, incidents, inspections, and any other material the project requires.

3. CLAIMS MANAGEMENT

3.1 All claims information shall be held in strict confidence.

3.2 Project insurance programs vary by project. Specific project claim notification process shall be identified and relayed to relevant parties. Subcontractors shall follow their own company policies unless covered by a project program or is otherwise mentioned in this document.

3.3 AECOM TISHMAN CONSTRUCTION shall provide transitional work (light duty) for AECOM TISHMAN CONSTRUCTION employees to enable an injured worker to return to work. Subcontractors shall

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provide transitional work (light duty) for their employees to enable an injured worker to return to work. The return-to-work hierarchy includes the following:

- a) Return to own job;
- b) Return to own job with accommodations/modifications;
- c) Return to another job at AECOM TISHMAN CONSTRUCTION with or without accommodations/modifications;
- d) Placement in alternate jobs through telecommuting or other job assignments determined on a case-by-case basis;
- e) Risk manager or claim coordinator is to maintain regular contact with employees who are out of work or have restrictions (contact at least weekly by phone or email).
- f) Subcontractor shall provide applicable medical paperwork as well as a return- to- work/light duty plan for their employees. The plan shall provide alternate job assignment/modified duty, where practical, until the worker can return to work full duty. Failure to provide the return-to-work plan may result in a \$1,500 violation. In addition, the Subcontractor may be assessed a penalty of \$1,500 weekly until such time as the employee is returned to work. The fine will be issued via Deduct Change Order or any other method deemed appropriate by the Owner to any Subcontractor who fails to return an injured employee to work in accordance with the requirements outlined above.

3.4 If a physician/medical provider has indicated that a follow-up appointment is needed or the individual will need to schedule future follow-up appointments, the individual is required to attend the appointment regardless if the individual is feeling better or having no symptoms. The visit is required to ensure proper recovery. The individual is required to obtain an updated work status form at every appointment with the treating physician, and the individual is required to provide the updated work status form to their supervisor.

3.5 After every medical provider visit (whether the employee is off work/restricted work or is just having a follow-up) an AECOM TISHMAN CONSTRUCTION employee is required to contact the AECOM TISHMAN CONSTRUCTION Claim Coordinator/Risk Manager. The AECOM TISHMAN CONSTRUCTION employee shall provide an update as to the medical provider visit, plan of care, to ensure that the employee is getting the appropriate care in a timely manner, and to ensure that the physician is being reimbursed accordingly. The employee must also contact their supervisor and/or SH&E representative to inform him/her of their work status.

3.6 It is the individual's responsibility to provide a medical clearance (e.g., return to work slip) to their Supervisor and AECOM TISHMAN CONSTRUCTION project management, from the treating physician regardless of the extent of treatment. Under no circumstances will individuals be permitted to return to full or modified duty status without appropriate documented medical clearance.

3.7 Subcontractors shall comply to all project insurance policies and procedures. Subcontractors shall communicate the status of any injured employee upon request by AECOM TISHMAN CONSTRUCTION safety or claims staff.

4. ATTACHMENTS

4.1 AECOM Employee Injury Management Hotline Poster

4.2 If the Media Calls Poster

4.3 IndustrySafe Incident Definitions



AECOM Injury Management Hotline

If an AECOM employee experiences a work-related injury or illness, they shall immediately notify their supervisor and safety representative.

After making the required notifications, contact the AECOM Injury Management Hotline to speak to a Registered Nurse (RN) free of charge. **If you are experiencing a medical emergency, call 911 or your local emergency number first.**

Please note: The Injury Management Hotline is for work-related injuries and/ or illnesses only. It is not intended to replace your personal physician for non-work related concerns and will not provide in-person medical examinations or prescribe medication.

1-512-419-5016



The Nurses

- Are available 24 hours a day, 7 days a week, 365 days a year
- Are licensed Registered Nurses (RN)
- Provide guidance and support for work-related injuries and/or illnesses
- Monitor status of AECOM employee's injury and/ or illness
- Ensure AECOM employees receive the right level of care at the right time

The Injury Management Hotline is for AECOM employees only.

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Attachment 2: If the Media Calls Poster



WHAT TO DO IF THE MEDIA CONTACTS YOU

AECOM

Now more than ever, news outlets are interested in our company and industry.

If a reporter, editor, producer or someone you do not know approaches you for a statement, project details or any information about AECOM, always refer them to the company's Communications Team and follow the guidance below.

DO	DON'T
✓ DO inform the journalist that they can contact AECOM's Communications Team directly at media@aecom.com ✓ DO take down the journalist's name, contact information, a brief summary of their request and deadline. Advise them that a Communications Team representative will return their call. ✓ DO forward the inquiry details to the Communications Team at media@aecom.com.	✗ DON'T answer any of the journalist's questions or confirm any information about our company, projects or employees. ✗ DON'T transfer the journalist to an AECOM project manager, leader or executive. ✗ DON'T provide the journalist with any contact information for an AECOM project manager, business leader or executive.

Attachment 3: IndustrySafe Incident Definitions

Incident Type Definitions

Incident Type	Definition
Coronavirus	Used to track work-related Coronavirus cases and exposures.
Environmental	An incident with a negative environmental impact i.e. oil spill
First Aid	A minor occupational injury (work related), where treatment by a healthcare professional is not required and after one time first aid treatment allows the worker to return to normal duties on the same day or at the start of the next scheduled shift.
Near Miss	An incident that has the potential to cause harm but resulted in no injury, loss or damage.
Non-work (Not related to work)	Non-work-related injuries or illnesses are those that are deemed to have occurred outside of the course of employment.
Property Damage	Equipment or property is damaged, but no injury occurs
Recordable Injury (medical treatment beyond first aid)	Any occupational (work related) injury requiring medical intervention by a doctor (or equivalent) on or off site that allows the worker to return to normal duties on the same day or at the start of the next scheduled shift.
Recordable Injury (Lost Time Case)	Any occupational injury which results in at least one full day absent from work beyond the day or shift it occurred (does not include time travelling off site for injury diagnosis).
Report Only	Injury / illness with potential future compensation claim
Security	Any security related incident that could cause harm to or is associated with an AECOM employee in the course of duty.
Vehicle	Incidents involving vehicles, not including commuting to and from your usual place of work.
Regulatory Agency Visit	Any non-routine announced or unannounced visit or inspection of an AECOM office or project site by a regulatory agency involved with Occupational Health and Safety and Environmental laws and regulations

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SECTION 17 - INSPECTIONS

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 AECOM TISHMAN CONSTRUCTION personnel will conduct safety inspections on the project.

2.2 Inspections and review of site safety is the responsibility of every individual on the project.

2.3 Each AECOM TISHMAN CONSTRUCTION staff member will be expected to review the site on an ongoing basis and report any unsafe condition to his/her supervisor for immediate correction.

2.4 On at least a once per day basis, a project safety inspection will be conducted. This inspection may be formal or informal.

2.5 All deficiencies noted on inspections shall be immediately corrected or have a plan of corrective action but in place. If an unsafe condition or act could result in an injury, the work should be stopped until the situation is resolved. All individuals exposed to injury should be removed from the exposure and only those required to correct the problem allowed to remain with appropriate protection for the circumstances.

2.6 Follow up actions shall take place to ensure deficiencies have been corrected.

2.7 Subcontractors are solely responsible to correct any found deficiency. Subcontractors are responsible to follow up with their designated AECOM TISHMAN CONSTRUCTION representative once all noted deficiencies have been corrected. In addition, subcontractors shall conduct their own inspections for their work areas. If corrective action is not taken, a telephone call will be made and/or a letter sent to the subcontractor's chief executive officer requesting immediate attention. The letter will outline remedial action to be taken by AECOM TISHMAN CONSTRUCTION if corrections are not made.

2.8 A weekly documented safety inspection is required by all AECOM TISHMAN CONSTRUCTION Superintendents.

2.9 A quarterly Project Safety Evaluation (PSE) is required by a senior member of the safety department for every active project. This evaluation shall include a jobsite inspection and adhere to all requirements included in Attachment 4: Project Safety Evaluation Guidelines.

2.10 AECOM TISHMAN CONSTRUCTION staff shall log inspections into IndustrySafe and Lifeguard per AECOM policies.

3. REGULATORY AGENCY VISITS

3.1 Visits or inspections by regulatory agencies include:

- Any non-routine announced or unannounced visit or inspection of an AECOM office or project site by a regulatory agency involved with Occupational Health and Safety and Environmental laws and regulations.
- Periodic visits from service or local agencies to project sites to approve certain activities such as utility hook-up, utility clearance, etc. are not considered regulatory visits or inspections.

3.2 In the event of a regulatory agency visit, the following procedure shall be followed:

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1. Confirm a senior AECOM TISHMAN CONSTRUCTION employee meets with the agency representative in order to obtain copies of credentials, determine the purpose of the visit or inspection and to request applicable documentation.
2. Confirm the visit or inspection is reported in IndustrySafe as a “regulatory agency visit” incident within one business day.
3. Confirm notification of the relevant Regional SH&E Director and SH&E Manager of the visit or inspection.
4. Notify the client a regulatory visit or inspection is occurring on their project site so the client may have representation during the visit or inspection, at their discretion
5. Confirm an appropriate AECOM TISHMAN CONSTRUCTION escort is identified and accompanies the agency representative at all times to address questions, observe activities and monitor evidence gathering (this may include obtaining duplication of evidence, such as dual photos and split samples).
6. Confirm copies of the visit record and other documents or postings that the agency representative may issue are obtained.
7. Request a debrief session.

3.3 In the event a citation, violation, or penalty is issued:

1. Confirm direction is given to the local employees to stop the performance of the cited activity until the appropriate corrective actions are implemented.
2. Confirm any required posting is completed in accordance with the directions outlined by the agency.
3. Confirm a formal response is submitted to the issuing agency and facilitate payment of any associated monetary fine after approval from VP-SH&E and Counsel.
4. As applicable, serve or designate an appropriate individual to serve as AECOM TISHMAN CONSTRUCTION’s informal and formal agency negotiation meeting representative to discuss the citation, violation, or penalty.

4. ATTACHMENTS

4.1 Example Jobsite Safety Checklist

4.2 Example Jobsite Inspection Report

4.3 Project Safety Evaluation (PSE)

4.4 Project Safety Evaluation Guidelines

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Attachment 1: EXAMPLE JOBSITE SAFETY CHECKLIST

Project Name and Number: _____ Date: _____

Person Completing the Checklist: _____

Yes	No	Personal Protective Equipment Required
-----	----	--

- | | | |
|-------|-------|--|
| _____ | _____ | 1. Head Protection – Hard Hats. |
| _____ | _____ | 2. Hearing Protection - Ear Plugs. |
| _____ | _____ | 3. Foot Protection – Work Boots. |
| _____ | _____ | 4. Eye and Face Protection- Safety Glasses. |
| _____ | _____ | 5. Respiratory Protection – Masks Dependent on Exposure. |

Yes	No	Job Site Protective Equipment
-----	----	-------------------------------

- | | | |
|-------|-------|---|
| _____ | _____ | 1. First aid kit suitable for crew size (office – dry shack – tool van). |
| _____ | _____ | 2. Fire protection – check extinguisher date & type – must be correct for your situation. |

Yes	No	Employee Facilities
-----	----	---------------------

- | | | |
|-------|-------|--|
| _____ | _____ | 1. Toilet. |
| _____ | _____ | 2. Wash Area. |
| _____ | _____ | 3. Dry Shack – lunchroom. |
| _____ | _____ | 4. Are facilities kept clean and sanitary? |

Yes	No	Safety Management Items
-----	----	-------------------------

- | | | |
|-------|-------|---|
| _____ | _____ | 1. Accident Prevention Program. |
| _____ | _____ | 2. SDS Hazard Communication Program. |
| _____ | _____ | 3. Assured Grounding Program. |
| _____ | _____ | 4. Minutes of Tool Box Meeting. |
| _____ | _____ | 5. First aid training certification cards- Superintendent, Foreman. |
| _____ | _____ | 6. Safety bulletin board and required posting. |
| _____ | _____ | 7. Is weekly management safety inspection done? |
| _____ | _____ | 8. Safety standards for construction work on site? |
| _____ | _____ | 9. Emergency phone number listing. |

Yes	No	Housekeeping
-----	----	--------------

- | | | |
|-------|-------|--|
| _____ | _____ | 1. Is work area clean and orderly? |
| _____ | _____ | 2. Deck free of protruding nails, holes, loose boards? |
| _____ | _____ | 3. Aisles and walkways clear of obstructions? |
| _____ | _____ | 4. Guardrails in place around where there is a drop of more than 6 feet. |
| _____ | _____ | 5. Remove any trip hazards. |

Yes	No	General Job Site Safety Items
-----	----	-------------------------------

- | | | |
|-------|-------|--|
| _____ | _____ | 1. All trailers and tool van step railings. |
| _____ | _____ | 2. Check extension cords and power equipment for shorts. |

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Yes	No	Hand and Portable Power Tools
_____	_____	1. Tools are inspected daily to assure they are in proper working order.
_____	_____	2. Damaged or defective tools are tagged out and returned to the tool room immediately
_____	_____	3. Power saws and grinders have proper guards in place at all times.
_____	_____	4. Power tools are never lowered or hoisted by the end of the cord or hose.
_____	_____	5. Cords and hoses are kept out of walkways, off stairways and ladders.
_____	_____	6. All portable electric tools are grounded.
_____	_____	7. All electric cords and cables are covered or elevated to protect them from damage & to eliminate tripping hazards. Steel wire is not to be used to hang cords leads.
_____	_____	8. An approved safety check valve must be installed at the manifold outlet to each supply line for a handheld pneumatic tool.
_____	_____	9. All pneumatic hose connections must be fastened securely.
_____	_____	10. Only employees who possess a valid credential are permitted to use powder actuated tools.
_____	_____	11. All electric tools are checked for good condition.
_____	_____	12. Shots properly stored and disposed of.

Yes	No	Grinding Wheels
_____	_____	1. Always use double eye protection when using grinders.
_____	_____	2. The tongue (work rest) is 1/8" from the wheel periphery on the bench grinders.
_____	_____	3. Using grinding wheels within limits as recommended by the manual.
_____	_____	4. Grinding wheels are provided with safety guards.
_____	_____	5. Grinders are being used as they are intended.

Yes	No	Scaffolding and Ladders
_____	_____	1. Each scaffold is inspected and approved by a responsible supervising person prior to initial use and after alteration.
_____	_____	2. Toprails, midrails and toeboards are installed on all open sides of scaffolding over 6'0" in height.
_____	_____	3. Proper access ladders are provided on all scaffolding.
_____	_____	4. Scaffold planks are cleated on both ends of all 2x lumber planking.
_____	_____	5. Adequate mudsills are provided capable of withstanding intended loads.
_____	_____	6. Scaffolds 26'0" high and 30'0" long are tied off to building.
_____	_____	7. Scaffolds are not loaded to exceed 1/4 of their rated capacity.
_____	_____	8. Barrels, boxes, kegs are not being used as work platforms.
_____	_____	9. When personnel are passing under a scaffold a 1/2" wire mesh is installed between the toeboards and toprail.
_____	_____	10. Height of rolling scaffolds don't exceed four times the minimum base dimension.
_____	_____	11. Personnel working off floats are wearing safety harnesses and tied off to structure.
_____	_____	12. All straight ladders extend 36" above landing and are tied off.
_____	_____	13. All stepladders are a Type I ladder.
_____	_____	14. Stepladders are not being used as straight ladders.
_____	_____	15. Personnel standing not higher than the third step on stepladder over five feet in height.
_____	_____	16. Personnel are facing ladder when ascending and descending ladder.
_____	_____	17. All straight ladders have safety shoes and are secured at the top and bottom.
_____	_____	18. All straight ladders are set to a 1 to 4 ratio.

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_____ Yes	_____ No	19. Scaffold is set within 14" of building face.
_____ Yes _____ No Cranes, Hoists, Motor Vehicle/Heavy Equipment 1. All cranes, hoists, motor vehicles are inspected prior to use on each shift. 2. Rated load capacity charts are posted in all cranes and hoists. 3. Standard hand signal posters are posted on cranes. 4. Accessible areas within the swing radius of all cranes are barricaded to prevent employees from getting hurt. 5. A 5-lb fire extinguisher is located in cab of each crane. 6. A designated employee is used to assist in all operation where the crane operator's vision is obstructed. 7. A monthly crane inspection is being filled out and is on file. 8. Personnel are not riding in the bed of trucks. 9. Back-up alarms are installed on all vehicles that have obstructed rear view. 10. Operators have valid driver's license and/ or operator's license where required.		
_____ Yes _____ No Welding/Burning Operations 1. All work areas are clear of combustible materials when welding and cutting. 2. Suitable fire extinguishing equipment is available when welding and burning. 3. Personnel are using proper eye protection with suitable filter lenses for welding. 4. All personnel are protected from arc rays by screens or eye protection. 5. All welding machines are grounded. 6. Spark igniters are used to light torches- not lighters or matches. 7. Personnel are wearing appropriate gloves during burning and welding operations. 8. Protective caps are on all cylinders when not in use. 9. Cylinders are properly tied off to structure and are not left unsecured.		
_____ Yes _____ No Flammable/Combustible Liquids 1. Approved metal containers, labeling and storage. 2. Approved metal safety cans are being used to handle flammable liquids in quantities greater than one gallon. 3. Storage areas are free of combustible materials. 4. A 10-20 lb fire extinguisher is located within 25 feet of flammable liquid storage area. 5. No smoking signs are posted at storage area.		
_____ Yes _____ No Chemicals 1. SDS sheets are on file for all chemicals used on the project. 2. All chemicals are being stored in a safe manner in accordance with manufacturer recommendation. 3. All containers are tightly closed when not in use. 4. Proper protective equipment is being used when handling chemicals. 5. Personnel using chemicals have been properly trained in handling the chemicals, including first aid treatment. 6. The guidelines for the storage and disposal of waste chemicals are being followed		

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(ie: proper containers, legibly marked).

___ ___ 7. Fire extinguishers are located in the storage area.

Yes No **Electrical**

___ ___ 1. All electrical tools and equipment are grounded.
 ___ ___ 2. Defective electrical tools are tagged out and taken out of service.
 ___ ___ 3. Temporary electrical cords are to be covered or elevated to prevent damage or tripping hazards.
 ___ ___ 4. Temporary lighting has guards over the bulbs.
 ___ ___ 5. Broken and burned out bulbs are replaced immediately.
 ___ ___ 6. All energized electrical panels have covers in place.
 ___ ___ 7. Hazardous electrical areas are barricaded and warning signs provided.
 ___ ___ 8. Temporary electrical power on projects has "ground fault interrupters" (GFI) or assured grounding system.

Yes No **Confined Entry**

___ ___ 1. Prior to entry into confined space, an air equality test is taken.
 ___ ___ 2. Prior to entering a vessel or tank, all moving machinery is properly locked out.
 ___ ___ 3. A watchman is posted at the entry of the confined space at all times for assistance and emergency.
 ___ ___ 4. Written log of confined entry is maintained.
 ___ ___ 5. "Confined Space" signage is displayed at entry of confined space.

Yes No **Other Safe Practices**

___ ___ 1. No personnel working under suspended loads.
 ___ ___ 2. No excessive speed of vehicles on site.
 ___ ___ 3. No personnel engaged in horseplay.
 ___ ___ 4. No personal vehicles on construction site.

Yes No **Administrative Procedures**

___ ___ 1. Project has proper quantity of first aid kits, stretchers, blankets and pillows.
 ___ ___ 2. Projects have bulletin boards and are updated regularly.
 ___ ___ 3. Emergency telephone numbers are posted at each telephone.
 ___ ___ 4. All supervisors have valid first aid cards.
 ___ ___ 5. OSHA forms are current.
 ___ ___ 6. All required posters are posted.
 ___ ___ 7. Toolbox meetings are being conducted regularly and notes are filed chronologically.
 ___ ___ 8. Site layout for fire extinguishers in file and posted.
 ___ ___ 9. Subcontractor summary current and complete.
 ___ ___ 10. Safety violation notices used and filed.
 ___ ___ 11. Daily inspection logs are being performed and filed chronologically.

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Attachment 2: EXAMPLE JOBSITE INSPECTION REPORT

JOBSITE INSPECTION REPORT

Project: _____ Job No. _____

Date: _____

Location: _____ Time _____ AM / PM

Item Number	Observation Deficiency	Corrective Action
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

Inspector: _____

CC: _____

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Attachment 3: PROJECT SAFETY EVALUATION (PSE) – 2024 LIFEGUARD EDITION





Questions

Specific Audit Related

- 1: Project Name/Location
- 2: OIC
- 3: Head Site Safety Representative
- 4: Head Superintendent
- 5: Number of Subcontractors on Job
- 6: Total Project Headcount
- 7: Evaluator
- 8: Evaluation Date
- 9: Dollar Value of Project
- 10: Contract Type
- 11: Type of Insurance
- 12: Phase of Job
- 13: Client Name
- 14: Is AECOM CM the safety lead?

Using set: CM Quarterly Project Safety Evaluation

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Audit Questions

1: Administration/Project Setup & Staffing

Please Select ▼



2: Subcontractor Prequalification/JHAs/THAs/Safe Starts

Please Select ▼



3: Housekeeping/Site Logistics/Facilities

Please Select ▼



4: Cranes/Hoisting Equipment

Please Select ▼



5: Excavations/Trenches

Please Select ▼



6: Fall Protection/Perimeter Systems/Tool Tether/Floor Openings

Please Select ▼



7: Hot Work/Fire Extinguishers/Fuel Storage

Please Select ▼



8: Ladders/Stairs/Walkways

Please Select ▼



9: Scaffolds

Please Select ▼



10: Motorized Equipment

Please Select ▼



11: MEWPS

Please Select ▼



12: Site Protection/Public Protection/Signage

Please Select ▼



13: PPE

Please Select ▼



14: Steel Erection/CAZs/Floor Permits

Please Select ▼



15: Electrical/LOTO/GFCI/ Temp Lighting

Please Select ▼



16: Training/Standdowns/Orientations

Please Select ▼



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17: Environmental and Health

Please Select



18: Local/ State/Regional/ Client Requirements (Optional)

Please Select



Comments and Feedback

Comment or Follow-Up Actions

Feedback / Discussion

Drop up to 10 files here to upload
or click to select files

Either click above, or drag images onto the highlighted area. Up to 10 JPEG files are allowed.

Disclaimer

The auditor believes the information contained within this report to be correct at the time of audit completion. The auditor does not accept responsibility for any consequences arising from the use of the information herein.

This audit is based on matters which were observed or came to the attention of the auditor during the project assessment and should not be relied upon as an exhaustive record of all possible risks or hazards that may exist or potential improvements that can be made. Safety, health, and environmental deficiencies identified during the audit were discussed with the project team with corrective actions being addressed.

Signature...

Attachment 4: PROJECT SAFETY EVALUATION (PSE) GUIDELINES

**Guidelines for Conducting CM Quarterly Project Safety Evaluations
(2024 Revision)**

Purpose:

To develop an objective system of quarterly project safety evaluation ensuring all ratings are conducted with methods that are consistent throughout AECOM Construction Management in order to achieve reliable and dependable project safety performance data.

These quarterly evaluations are intended to review and rate the overall project. This includes the performance of company employees as well as the performance of subcontractors working onsite.

It should be noted that there will be ratings that, by nature, will be purely subjective. All evaluators are to be fair and as consistent as possible when scoring projects. It is the expectation that every active AECOM CM project be evaluated using the CM Quarterly Project Safety Evaluation by the safety director or a designee approved by CM VP of Safety. Site safety is not to evaluate their project using this method. All deficiencies identified in this evaluation are to be discussed with the project team with corrective actions being addressed.

In addition, our Management Internal Audits will be utilizing similar evaluations on a random basis to ensure proper reporting and compliance. Their report will be sent directly to Senior Management for review. All information included in this evaluation is reflective of the moment in time the evaluation was performed and should not be relied upon as an exhaustive record of all possible risks or hazards that may exist or potential improvements that can be made.

Audit Questions with Scoring Guidance:

Section 1: Jobsite Information

1. **Project Name/Location:** Project name and geographic location.
2. **OIC:** Name of the AECOM Officer in Charge from operations.
3. **Head Site Safety Representative:** The head AECOM safety rep if applicable on the project.
4. **Head Superintendent:** The most senior AECOM superintendent on the project.
5. **Number of Subcontractors on Job:** The number of subcontractor companies working onsite.
6. **Total Project Headcount:** Average total number of workers on the project.
7. **Evaluator:** The person who conducted this evaluation. This is a safety director or approved designee, not site safety.
8. **Evaluation Date:** The date the project was evaluated.
9. **Dollar Value of Project:** Current contract value of project.
10. **Contract Type:** Type of contract. Examples include CM at Risk, Agency, CM, GC, Consultant, etc.
11. **Type of Insurance:** Type of insurance program for the project. Examples include Standard, CCIP, OCIP, DCIP, Mixed, etc.
12. **Phase of Job:** Describe the current construction phase of the project (not a percentage of completion). Examples include Demolition, Excavation, Concrete, Steel Interiors, etc.
13. **Client Name:** Our client for the project.
14. **Is AECOM CM the safety lead?** Answer Yes or No on if we are the lead on safety for the project or if the project is using another safety program.

Rating System for Section 2 of Audit:

- Each item is rated on a scale of 0 to 5, with 0 being the lowest and 5 being the best possible. The evaluator also has a Not Applicable option which doesn't impact scoring. The following suggests

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ratings based on observations, but evaluator discretion applies. Lifeguard will automatically total the total number of items evaluated versus the rating score to calculate the final percentage rating.

- If a particular subject does not apply, select Not Applicable in the rating column. Do not select 0, as this will skew the calculations for the final score.
- The evaluator cannot input half scores (i.e. 3.5, 2.5, etc.) or break a score down into more than a whole number. (2.8, 3.7 etc. is not acceptable). Use whole numbers from 0-5.
- If the reason for the low score was corrected before the evaluation was completed note that in the comments on the audit form.
- Correcting a deficiency does not automatically guarantee that the score will be raised, or that the deficiency will not be noted in the evaluation because it has been fixed.
- Each topic lists items to consider and review. The lists are not all inclusive of all possible risks or hazards that may exist or potential improvements that can be made.
- The comments in the evaluation should detail what is compliant and what needs improvement. Attach additional comments as necessary.
- Be open and honest with the evaluation as this is how our projects will learn and improve.
- This is a living document and will be updated and distributed as necessary.

Section 2: Audit Questions

1. Administration/Project Setup & Staffing

- ❖ Is there adequate safety resources assigned per CM policy? (\$50mm> requirement). See Safety Staffing memo from 3/28/23.
- ❖ Has medical staff needs been reviewed per CM policy (\$400mm> requirement)? See Safety Staffing memo from 3/28/23.
- ❖ Is there a site-specific safety program in place?
- ❖ Are incident reports being submitted within 24 hours of occurrence into IndustrySafe?
- ❖ Does senior project staff participate in investigations, safety meetings, and orientations?
- ❖ Does safety staff have proper safety equipment and office space (meters, laptop, etc.)?
- ❖ Is there a documented project rewards and recognition program for employees and subcontractors in a highly visible way?
- ❖ Are all staff completing observations, SMAs, and audits/inspections as required by CM policy? See CVM report requirements.
- ❖ Do employees and subcontractors understand that they have the responsibility to Stop Work if conditions or activities are not safe? (Red Card)
- ❖ Has proper safety communications been set up to ensure all individuals feel comfortable bringing safety issues to their manager without threat or fear of reprisal?
- ❖ Does the site safety program have documented processes to encourage participation by employees in work planning and safety improvement efforts?
- ❖ Is there a project safety committee in place that includes trade representatives?
- ❖ Are processes in place for work activity coordination with all parties on the project?
- ❖ Is a project emergency response plan (ERP) in place and current for fire, injury, environmental impact or other serious hazards?
- ❖ Is staff aware of requirements and procedures for incident reporting, including time frame, to whom needs notified, proper phone #s and online incident management system (e.g. IndustrySafe, Rapid Response)?

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- ❖ Is there a proactive health program on the project? (i.e. stretch and flex)
- ❖ Are return-to-work programs implemented and supported by management?
- ❖ Is a response process to Safety Department and Insurance Loss Control audits in place?
- ❖ Is an evacuation plan in place? Has it been practiced?
- ❖ Is there a site security plan in action?
- ❖ Other?

Ratings

- 5 - All of the mentioned required items are being completed.
- 4 - Most of the mentioned required items are being completed with one exception.
- 3 - Some of the mentioned required items are being completed with two or three exceptions.
- 2 - Mentioned required items are being completed with more than three exceptions.
- 1 - Proper safety staff on site but many of the mentioned required items need implemented.
- 0 - Many mentioned required items are not being followed and/or there is not adequate safety staff or site specific program in place on the project.

2. Subcontractor Prequalification/JHAs/THAs/Safe Starts

- ❖ Are safe start preplanning meetings being held and documented with all subcontractors prior to them starting work?
- ❖ Are all subcontractors properly prequalified?
- ❖ Are insurance requirements being checked and documented?
- ❖ Does each subcontractor have a documented overall JHA in place for their work?
- ❖ Are daily, pre-shift THAs being completed by ALL subcontractors?
- ❖ Are THAs being collected and properly filed?
- ❖ Are THAs completed in a language all crew members understand and being signed by all crew members?
- ❖ Are company field staff and subcontractor staff conducting documented weekly toolbox talks to review work activities, hazards and mitigations?
- ❖ Do subcontractors submit monthly reports regarding training and lagging indicator performance?
- ❖ Do subcontractors cascade the requirements of the safety program to all subcontractor's personnel and visitors?
- ❖ Are subcontractor's at-risk behaviors and unsafe conditions addressed immediately?
- ❖ Other?

Ratings

- 5 - All of the mentioned required items are being completed.
- 4 - Most of the mentioned required items are being completed with one exception.
- 3 - Some of the mentioned required items are being completed with two or three exceptions.
- 2 - Mentioned required items are being completed with more than three exceptions.
- 1 - Subcontractors are sporadically completing THAs, THAs are not proper, and/or toolbox talks are sporadic.
- 0 - Subcontractors are working without being prequalified, without having a safe start, and/or without submitting a proper JHA prior to starting.

3. Housekeeping/Site Logistics/Facilities

- ❖ Is the project clean and neat?
- ❖ Are aisles and walkways free and clear of debris?
- ❖ Are egress paths properly demarcated with exits marked?
- ❖ Are there enough garbage receptacles throughout the project?
- ❖ Are evacuation plans in place and routes marked?
- ❖ Are evacuation drills being held twice a year and documented?

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- ❖ Is there appropriate relief and wash facilities for all tradespeople (separate facilities for men and women)?
- ❖ Are the facilities clean and free from graffiti?
- ❖ Is the site secure with proper fencing?
- ❖ Is there a delivery truck plan in place?
- ❖ Are spotters being used as part of the logistics plan and per policy?
- ❖ Is a general logistics plan in place?
- ❖ Are hand and power tools in good repair?
- ❖ Are all tool safety features such as handles and guards being used?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

4 - Most of the mentioned required items are being completed with one or two minor exceptions.

3 - Some of the mentioned required items are being completed with three minor exceptions.

2 - Mentioned required items are being completed with more than four exceptions with some being more significant in nature.

1 - Significant housekeeping issues observed, minimal logistics plans in place, many other mentioned required items need addressed.

0 - Housekeeping is poor with egress ways blocked and/or no proper facilities or logistic plan in place.

4. Cranes/Hoisting Equipment

- ❖ Are annual inspections being collected when cranes arrive on the project?
- ❖ Are crane operators providing a copy of his / her certification prior to lifting any load?
- ❖ Are cranes receiving a third-party inspection prior to use?
- ❖ Are cranes being visually checked by the operator prior to use each shift with documentation?
- ❖ Is each crane layout thoroughly planned per company policy, looking for hazards such as overhead power lines or soil depressions?
- ❖ Does each crane have an anti-two-block device with the swing radius demarcated?
- ❖ Are general crane components in good shape? All limits working properly?
- ❖ Is all rigging tagged and in proper working condition?
- ❖ Are cables and spools in good shape without significant wear?
- ❖ Is crane radius protected?
- ❖ Is the equipment free of leaks?
- ❖ Do all parts of hoisting operations have rated proper capacities? Marked?
- ❖ Are hoist /crane cables visibly in good condition?
- ❖ Is rigging equipment in good shape? Stored properly?
- ❖ Are cable lifting devices properly tagged?
- ❖ Are riggers properly qualified/certified? Documentation?
- ❖ Do hoisting operations have proper protection around them if falls exist?
- ❖ Are personal hoists third-party inspected?
- ❖ Are crane packets being used to coordinate crane operations?
- ❖ Is crane dunnage and ground conditions adequate?
- ❖ Are the proper worksheets filled out for the specific crane lifts?
- ❖ Are personal hoists entry and exits properly protected with no gaps for material to fall through?
- ❖ If multiple cranes on site, is there a certified crane director or coordinator?
- ❖ Are hoist operators properly qualified/certified to operate the specific hoist?
- ❖ Are proper tag lines being used on all lifts?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

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- 4 - Most of the mentioned required items are being completed with one minor exception.
- 3 - Some of the mentioned required items are being completed with two or three minor exceptions.
- 2 - Mentioned required items are being completed with more than three minor exceptions.
- 1 - Significant crane issue observed, missing required documentation, many other mentioned required minor items need addressed, significant issues currently being addressed but not to the extent the crane needs shut down.
- 0 - No annual or third-party inspections, no certified operator, set ups have many deficiencies, poor set up, poor component(s), crane or hoist must be shut down.

5. Excavations/Trenches

- ❖ Are all excavations (including trenches) under the direction of a competent person?
- ❖ Do excavations greater than 4 ft. deep have a means of egress within 25 ft. of the individuals working in the excavation?
- ❖ Do excavations greater than 5 ft. deep have sides that are shored, sloped, boxed or benched with documented daily checks?
- ❖ Do excavations greater than 6 ft. deep have fall protection or physical barricades in use at the top edge?
- ❖ Are foundation excavation top edges protected with cable guardrail wrapped with snow fence or similar material?
- ❖ Are all excavation protection 19 feet -6 inches or greater in depth designed by a professional engineer with drawings on file and readily available?
- ❖ Are precautions being taken if there is vehicular traffic around the excavation?
- ❖ Are underground utilities being properly identified (one call/USA dig/etc.)?
- ❖ Was a soil analysis completed? Documented?
- ❖ Are excavations being inspected with documentation on a shift basis?
- ❖ Are all material and equipment stored more than 2 ft. from the edges of excavations?
- ❖ Are proper precautions being taken so equipment does not blow exhaust into excavations?
- ❖ Are hazards in the bottoms of excavations being addressed (water, atmosphere, electric, etc.)?
- ❖ Is a permitting system in use? Documented?
- ❖ If undermining is occurring, is the structure braced from collapse?
- ❖ Is emergency rescue equipment available?
- ❖ Are atmospheres being tested on a regular basis with documentation?
- ❖ Are workers properly trained?
- ❖ Other?

Ratings

- 5 - All of the mentioned required items are being completed.
- 4 - Most of the mentioned required items are being completed with one minor exception.
- 3 - Some of the mentioned required items are being completed with two or three minor exceptions.
- 2 - Mentioned required items are being completed with more than three minor exceptions.
- 1 - One significant excavation/trench issue observed, missing documentation, spoils closer than 2 feet to top edge, egress present but not within 25 feet of workers.
- 0 - No competent person, no daily inspections, excavations greater than 5 feet not protected, area must be shut down until corrected.

6. Fall Protection/Perimeter Systems/Tool Tether/Floor Openings

- ❖ Is zero tolerance 6-foot fall protection being followed on the project?
- ❖ Are guardrails and barriers being maintained in an acceptable condition at all times?
- ❖ Are personal fall arrest systems (harness and retractables) or other fall protection methods being used when there are no guardrails?
- ❖ Are fall protection plans being submitted in writing by subcontractors before implementation?
- ❖ Are controlled access zones (CAZ) being established with physical barriers and proper warning signs?
- ❖ Are fall protection devices being stored properly and inspected (harnesses, cables, guardrails, nets)?

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- ❖ Are cables properly flagged/demarcated?
- ❖ Are floor openings larger than 2 inches properly protected?
- ❖ Are floor hole covers properly secured in place with a warning marking?
- ❖ Are secondary means of protection used as necessary?
- ❖ Are cable guardrails taut with minimal deflection?
- ❖ Do wooden guardrails have the proper posts, top rail, and mid-rails?
- ❖ Does all perimeter protection have proper toe boards and/or means to protect items from falling from a floor?
- ❖ Are debris nets in place when necessary?
- ❖ Are trades utilizing tool tethering where necessary?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

4 - Most of the mentioned required items are being completed with one exception.

3 - Some of the mentioned required items are being completed with two or three exceptions. No significant exposures.

2 - Mentioned required items are being completed with more than three exceptions.

1 - Many mentioned required items are not being followed. Failure to properly cover floor openings and/or to protect penetrations. Perimeter protection is not maintained per policy. Project team has recognized the problem and began addressing it prior to this audit.

0 - Failure to properly cover multiple floor openings and/or to protect penetrations and/or to erect, maintain, or demarcate perimeter protection with no action by the project team. One or more individuals observed without proper fall protection. Tool tethering program not in place. Exposures to falls. Area(s) must be shut down.

7. Hot Work/Fire Extinguishers/Fuel Storage

- ❖ Are fire extinguishers in place and inspected monthly, serviced annually?
- ❖ Are fire extinguishers at least of the 10 lb. ABC dry chemical extinguisher type?
- ❖ Is a fire prevention plan in place and being adhered to?
- ❖ Does mobile equipment have fire extinguishers attached?
- ❖ Are proper protocols with fuel trucks onsite being followed?
- ❖ Is a proper hot work permit procedure in use?
- ❖ Are fire watches in place for all hot work?
- ❖ Do temporary standpipes meet all applicable codes and OSHA standards?
- ❖ Are proper fuel storage processes in place for all gas and liquid fuels?
- ❖ Are plastic containers being prohibited?
- ❖ Is adequate collision protection in place for fuel tanks next to equipment routes?
- ❖ Are workers properly trained?
- ❖ Is smoking prohibited on the project?
- ❖ Are flammable liquids stored in labeled fire-resistant cabinets?
- ❖ Do containers have bonding clamps readily available with proper venting?
- ❖ Is proper PPE being used during hot work?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

4 - Most of the mentioned required items are being completed with one exception.

3 - Some of the mentioned required items are being completed with two or three exceptions.

2 - Mentioned required items are being completed with more than three exceptions.

1 - Significant hot work, fuel storage, and/or fire extinguisher issues observed, missing documentation.

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0 - No hot work plan in place, improper amount of fire extinguishers onsite, multiple situations of fuel not being stored correctly.

8. Ladders/Stairs/Walkways

- ❖ Are ladders being properly used, inspected, and maintained - both commercial and job-built?
- ❖ Are stairs properly constructed and maintained? (This applies to job-built stairs, scaffold stairs, pan stairs, etc.)
- ❖ Are handrails being properly constructed and maintained?
- ❖ Is access and egress to ladders and stairs clear and free of debris?
- ❖ Are stair towers being inspected and tagged on a daily basis?
- ❖ Are metal ladders being prohibited onsite per policy?
- ❖ Are walkway platforms erected correctly with handrails when required?
- ❖ Is overhead protection in place on walkways entering or exiting a building or where else required?
- ❖ Are trades properly trained in ladder use?
- ❖ Do extension ladders extend three feet above the landing with proper walk-through swing gates and/or diversions, tie off, and grab bars?
- ❖ Are portable ladders of commercial grade?
- ❖ Other?

Ratings

- 5 - All of the mentioned required items are being completed.
- 4 - Most of the mentioned required items are being completed with one exception.
- 3 - Some of the mentioned required items are being completed with two or three exceptions.
- 2 - Mentioned required items are being completed with more than three exceptions.
- 1 - Significant ladder, stairway, walkway issues that are evident but will be addressed.
- 0 - Significant ladder, stairway, walkway issues that are obviously a systemic project problem and/or requires areas to be shut down.

9. Scaffolds

- ❖ Are scaffolds being constructed, removed, and/or used under the direction of a competent person?
- ❖ Are all scaffolds being inspected daily by the competent person?
- ❖ Are proper scaffold inspection tags being used (Green, Yellow, Red)?
- ❖ Are scaffolds free of any physical damage?
- ❖ Are cross bracing, planking, access, and guardrails proper and in place?
- ❖ Are footings or anchorage of sound, rigid, and capable of carrying the maximum intended load without settling or displacement?
- ❖ Are guardrails and toe boards installed on all open sides and ends of platforms more than six feet (6') above the ground or floor? Toe boards must be on all working levels.
- ❖ Are the supports spaced no more than eight feet (8') apart?
- ❖ Are egress or work areas under scaffolds properly protected?
- ❖ Is the scaffold plank in use of scaffold grade and installed correctly with overlaps a minimum of 12 inches?
- ❖ Are swing gates, diversions, or similar means installed on fixed ladders on supported scaffolds at each working level to allow for a safe transition from the ladder to the working platform?
- ❖ Are direct connections to roofs and floors, and counterweights used to balance adjustable suspension scaffolds, capable of resisting at least 4 times the tipping moment imposed by the scaffold operating at the rated load of the hoist, or 1.5 (minimum) times the tipping moment imposed by the scaffold operating at the stall load of the hoist, whichever is greater?
- ❖ Are all scaffolds over one hundred twenty-five feet (125') designed by a registered Professional Engineer (PE)?
- ❖ Are working levels free from debris and other hazards?
- ❖ Is a process in place to remove snow, ice, or other accumulations?

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- ❖ Is the amount of material being stored on scaffolds per policy (ie: no more than can be used in one shift/immediate use)?
- ❖ Is fall protection being used during erection and/or dismantling operations?
- ❖ Are scaffolds in proper distance from energized power lines?
- ❖ Are atypical scaffolds designed by a registered Professional Engineer (PE)?
- ❖ Are wheels being locked on baker scaffolds?
- ❖ Are guys, ties, and braces installed according to the scaffold manufacturer's recommendations or at the closest horizontal member to the 4:1 height and be repeated vertically at locations of horizontal members every 20 feet or less thereafter for scaffolds 3 feet wide or less, and every 26 feet or less thereafter for scaffolds greater than 3 feet wide?
- ❖ Are mud sills or other rigid footing, capable of withstanding the maximum intended load, being used?
- ❖ Are scaffold workers properly trained?
- ❖ Other?

Ratings

- 5 - All of the mentioned required items are being completed.
- 4 - Most of the mentioned required items are being completed with one minor exception.
- 3 - Some of the mentioned required items are being completed with two or three minor exceptions.
- 2 - Mentioned required items are being completed with more than three minor exceptions.
- 1 - Significant scaffold issues are evident but will be immediately addressed.
- 0 - Significant scaffold issues are evident. No competent person or inspection taking place. No guardrails or other fall protection means. No engineering where required.

10. Motorized Equipment

- ❖ Is motorized equipment being inspected when it arrives to the project and at least daily (pre-shift) with documentation?
- ❖ Is there manufacturer documentation for any noticed modifications to motorized equipment?
- ❖ Are seat belts being worn?
- ❖ Is there a mounted fire extinguisher on the equipment?
- ❖ Are manufacturer manuals available?
- ❖ Are operators properly trained with documentation? Specific to the equipment?
- ❖ Is motorized equipment that is being observed being used in a proper, safe fashion?
- ❖ Are guards and /or other safety features in place?
- ❖ Are windows free from cracked glass and/or other distortions?
- ❖ Is roll over protection provided where required?
- ❖ Is equipment not in use properly staged and locked out from incidental use?
- ❖ Are all lights, horns, and backup alarms working properly?
- ❖ Are spotters in use when required around the equipment?
- ❖ Other?

Ratings

- 5 - All of the mentioned required items are being completed.
- 4 - Most of the mentioned required items are being completed with one minor exception.
- 3 - Some of the mentioned required items are being completed with two or three minor exceptions.
- 2 - Mentioned required items are being completed with more than three minor exceptions.
- 1 - Significant mobile equipment issues observed, improper training documentation, no daily inspections, many other mentioned required items need addressed.
- 0 - Multiple significant mobile equipment issues observed that may lead to significant incidents.

11. MEWPS

- ❖ Are the workers trained and authorized to be in the equipment? Documentation?

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- ❖ Are the platforms being tagged in and out where required?
- ❖ Are all controls plainly marked as to their function?
- ❖ Are daily, pre-shift inspections occurring?
- ❖ Is proper PPE being used by the operators, including PFAs where required?
- ❖ Are load limits being followed?
- ❖ Are there audible and/or visual warning indicators for movement if required?
- ❖ Is the working area free from hazards such as depressions or floor holes?
- ❖ Are spotters being used as necessary when the equipment is moved?
- ❖ Are all debris and electric cords away from the floor in the area of operation?
- ❖ Are operators standing on the floor platform and not the guardrails?
- ❖ Is the chain or rail guards in place prior to use?
- ❖ Are there emergency stops and lowering devices?
- ❖ Are outriggers being properly used where installed?
- ❖ Is there a rescue procedure in place for removing individuals from the equipment when necessary?
- ❖ Is the operator manual available?
- ❖ Does the daily work plan state how the equipment will be used?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

4 - Most of the mentioned required items are being completed with one minor exception.

3 - Some of the mentioned required items are being completed with two or three minor exceptions.

2 - Mentioned required items are being completed with more than three minor exceptions.

1 - Significant MEWPS issues observed, improper training documentation, no daily inspections, many other mentioned required items need addressed.

0 - Multiple MEWPS issues observed that may lead to significant incidents. No fall protection when required.

12. Site Protection/Public Protection/Signage

- ❖ Are the 9 Life-Preserving Principles being represented on the project?
- ❖ Are 'No Smoking' / 'No Open Flame' signs posted and location restrictions adhered to?
- ❖ Are 'Laser In Use' signs used?
- ❖ Are security guards in use with proper post commands?
- ❖ Are there proper egress signs? Exits marked?
- ❖ Are there enough fire extinguishers posted throughout the project in conspicuous locations?
- ❖ Is there a project fence installed and in good condition?
- ❖ Are sidewalk covers built per code and in good condition with proper lighting?
- ❖ Are gates named and being used properly?
- ❖ Is there a dust control program in place?
- ❖ Are there specific barriers preventing pedestrians from entering the project?
- ❖ Are signs conspicuously posted on the perimeter warning of requirements for entry and hazards?
- ❖ Does the project have proper signage to assist and guide visitors through the project?
- ❖ Are chemicals and liquids properly stored and labeled correctly?
- ❖ Are there specific designated areas for fuel or other hazardous material storage?
- ❖ Are perimeter pedestrian walkways free from hazards?
- ❖ Are general protective measures established to prevent access by unauthorized persons, and to protect persons who may be employees of the owner or customers of the owner when passing near or through the jobsite during the normal course of business?
- ❖ Is a first aid kit readily available with weekly inspections?
- ❖ Other?

Ratings

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- 5 - All of the mentioned required items are being completed.
- 4 - Most of the mentioned required items are being completed with one minor exception.
- 3 - Some of the mentioned required items are being completed with two or three minor exceptions.
- 2 - Mentioned required items are being completed with more than three minor exceptions.
- 1 - Significant site protection/public protection issues observed, little signage, pedestrian hazards exist, many other mentioned required items need addressed.
- 0 - Multiple site protection/public protection issues observed that may lead to significant incidents that need immediate attention. No control over project boundaries or public flow.

13. PPE

- ❖ Is PPE being used by all individuals in the field appropriately (including but not limited to hard hats/helmets, ANSI approved safety glasses, work boots, gloves, reflective shirts and/or vests, work attire)?
- ❖ Do individuals using prescription safety glasses have properly attached side shields?
- ❖ Are face shields being used where and when required?
- ❖ Is hearing protection being utilized (90 dBA>)?
- ❖ Is a hearing conservation program in place on the project?
- ❖ Are individuals using gloves as appropriate for project rules?
- ❖ Are respirators being used with proper programs in place where required?
- ❖ Do workers have shirts with at least a quarter sleeve (>4 inches)?
- ❖ Is proper work attire being worn onsite (ie: no sweatpants or other soft work pants)?
- ❖ Is protective clothing worn suitable for existing hazards?
- ❖ Are respirators being used where respiratory hazards exist?
- ❖ Do hot work employees have the proper protective clothing and face shields?
- ❖ Is proper PPE being utilized for CSE?
- ❖ Are workers protected from chemical burns?
- ❖ Other?

Ratings

- 5 - 100% compliance. All required PPE being worn by all personnel at all times.
- 4 - 95% to 99% of all required PPE being worn by all personnel at all times.
- 3 - 90% to 95% of all required PPE being worn by all personnel at all times. Project team is addressing.
- 2 - 85% to 90% of all required PPE being worn by all personnel at all times. Project team is addressing.
- 1 - 75% to 85% of all required PPE being worn by all personnel at all times. Project team is addressing.
- 0 - Less than 75% of all required PPE being worn by all personnel at all times. Major infractions noted. A systemic problem.

14. Steel Erection/CAZs/Floor Permits

- ❖ Is a floor permitting system in place and being utilized properly per AECOM, OSHA, and local policy?
- ❖ Does the floor permitting system require a sign-off for other trades to work after a floor inspection has been completed by the project team?
- ❖ Are barriers being used for CAZs?
- ❖ Are the CAZs being properly enforced?
- ❖ If caution tape is used as a barrier, is it properly maintained?
- ❖ Are all other trades being kept apart from steel erection activities per policy?
- ❖ Are the proper forms being used for steel erection activities?
- ❖ Are Christmas-treeing rules (<3 pieces) being followed?
- ❖ Is the project notifying the steel erector and fabricator of the proper curing of concrete in footings, piers, etc. for steel columns in writing, prior to the commencement of steel erection?
- ❖ Is the project providing the steel erector a safe site layout?
- ❖ Is a proper approved steel erection plan in place?
- ❖ Is written communication occurring on any damages to anchor bolts?

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- ❖ Is there documentation designating the competent and qualified person(s) for the erector who will conduct rigging inspections prior to each work shift?
- ❖ Are there plans in place to properly land decking?
- ❖ Is six-foot fall protection being followed at all times? Is a site-specific plan from the erector in place?
- ❖ Are deck holes and openings properly protected?
- ❖ Is the erector installing a minimum of four anchor bolts per column?
- ❖ Are there any repairs that have occurred to anchor bolts/rods? If so, is there documentation from the structural engineer on the repair approval?
- ❖ Are all stored materials secured from displacement?
- ❖ Is proper overhead protection being used to protect other tradespeople?
- ❖ Are all perimeter cables a minimum of 3/8" diameter and maintained?
- ❖ Have all tradespeople working on steel erection activities been properly trained?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

4 - Most of the mentioned required items are being completed with one or two minor exceptions.

3 - Some of the mentioned required items are being completed with three minor exceptions.

2 - Mentioned required items are being completed with more than three minor exceptions.

1 - Significant steel erection, CAZ, or permitting issues observed, many other mentioned required items need addressed but it is in process.

0 - Multiple issues observed that need immediate attention. No proper paperwork being utilized. Areas must be shut down.

15. Electrical/LOTO/GFCI/Temp Lighting

- ❖ Is adequate lighting provided in work areas and in egress routes per applicable OSHA/local standards?
- ❖ Is temporary lighting being maintained?
- ❖ Are GFCIs being used on all 120v power supplies?
- ❖ Are portable GFCIs being used after permanent power has been maintained and on generators not equipped with built-in GFCI units?
- ❖ Are GFCIs being inspected and documented?
- ❖ Are extension cords heavy duty, 3-prong grounded with all 3 prongs in place, and all insulation in good condition?
- ❖ Are electric cords kept off the floor or otherwise maintained in a safe fashion?
- ❖ Are electric cords being inspected and removed if they are damaged?
- ❖ Are overhead lines being maintained at a height as defined by OSHA for the voltage?
- ❖ Is temporary power installed and maintained at a minimum of 80 inches from the floor?
- ❖ Are energized electric panels labeled and secured with no missing breakers without blanks?
- ❖ Are electrical rooms kept free from debris, labeled, and secured?
- ❖ Is proper contact information for the electrical subcontractor(s) listed on electrical panels?
- ❖ Are breakers properly labeled?
- ❖ Are outlets properly covered?
- ❖ Is there a proper lockout/tagout program in place? Is it being followed?
- ❖ Are powerline safety distances being practiced with proper signage?
- ❖ Are trades properly trained on electrical and/or lockout/tagout procedures?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

4 - Most of the mentioned required items are being completed with one minor exception such as a missing ground prong on a cord.

3 - Some of the mentioned required items are being completed with two or three minor exceptions.

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2 - Mentioned required items are being completed with more than three minor exceptions.

1 - Significant electrical issues observed, little required electrical signage present, significant amounts of other mentioned required minor items needing addressed but has been identified and being corrected.

0 - Multiple significant electrical issues observed, no GFCI program in place, open and/or unlabeled panels with missing breakers, no LOTO program in place.

16. Training/Standdowns/Orientations

- ❖ Have non-safety supervisors been trained per policy (ie: Speak up/Listen up, OSHA 10/30, DOB, etc.) Documented?
- ❖ Has a new hire orientation been conducted for employees, supervisors, and subcontractor employees prior to beginning work?
- ❖ Does project management participate in project orientations?
- ❖ Is a tracking system in place for all trainings and orientations?
- ❖ Is a project standdown being held on at least a monthly basis?
- ❖ Is the standdown being run by project management?
- ❖ Are other training requirements (specified by law, regulation, company procedures) specified for each position with company employees and subcontractors?
- ❖ Is the general training adequate for the project?
- ❖ Is a Hazard Communication program in place and documented? Training completed? Chemicals labeled?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

4 - Most of the mentioned required items are being completed with one exception.

3 - Some of the mentioned required items are being completed with two or three exceptions.

2 - Mentioned required items are being completed with more than three exceptions.

1 - Training is sporadically completed with many of the mentioned required items needing implementation.

0 - Many mentioned required items are not being followed and/or an orientation process and/or hazard communication program are not in place on the project.

17. Environmental and Health

- ❖ Are environmental issues being addressed by scope of work including silica control, dust control, infection control, storm water runoff, containment and clean-up of spills (fuel, hydraulic fluid, concrete, form oil, or other chemicals) procedures and policies?
- ❖ Is there a bloodborne pathogen cleanup procedure in place on the project?
- ❖ Are all wastes being properly disposed of?
- ❖ Are noise reduction techniques being utilized to reduce noise in the environment?
- ❖ If there are remediation processes ongoing (such as asbestos or lead) do they possess the proper permits? Competent and trained people? Documented daily activities?
- ❖ Are remediation processes properly separated from normal work activities?
- ❖ Are there procedures in place to address heat stress and cold stress?
- ❖ Are there procedures in place to address severe weather?
- ❖ Are procedures in place if animals or pests are creating a problem on the project?
- ❖ Other?

Ratings

5 - All of the mentioned required items are being completed.

4 - Most of the mentioned required items are being completed with one minor exception.

3 - Some of the mentioned required items are being completed with two or three minor exceptions.

2 - Mentioned required items are being completed with more than three minor exceptions.

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1 - Significant deficiencies observed, little procedures in place, significant amounts of other mentioned required minor items needing addressed but are in process of being corrected.

0 - Multiple significant deficiencies observed, disregard of AECOM policies, significant environmental and/or health impacts without control. Immediate remediation needed.

18. Local/ State/Regional/ Client Requirements (Optional)

Many states, clients, and localities have their own regulations which can be added in here. Each region may add in other specific items they wish their project to follow. The grading will follow the similar process of the other questions.

Comments and Feedback

Summarize your review and follow up actions. List positive findings as well as areas for improvement/ concerns. These will help the project recognize strengths and improvement areas for the future. No project is perfect, and all can improve.

In addition, consider the required 9 Life-Preserving principles when creating the summary comments:

Demonstrated Management Commitment

Our executive, senior and project managers will lead the Safety, Health and Environment improvement process and continuously demonstrate support, commitment, and accountability.

Employee Participation

Our employees are empowered to engage in our Culture of Caring and Safety for Life program to actively contribute to the physical, psychological, and social well-being of all affected by our work.

Budgeting and Staffing for Safety

Our safety staff will be appropriately trained, experienced and qualified to provide technical support to our internal and external clients. A budget to support safety and health activities will be included in project proposals.

Pre-Planning

Effective hazard control will be employed through Pre-Project and Pre-Task planning to design, plan and build the safety and health protection of our employees, stakeholders, and the environment into every project.

Contractor Management

Our project staff will coordinate with our sub-consultants, subcontractors, contractors, and Joint Venture Partners to collectively plan for and maintain a safe and healthy workplace.

Recognition and Rewards

Our employees and project participants will be recognized for their commitment to working safely and their positive contribution to our safety efforts.

Safety Orientation and Training

Appropriate safety training will be provided to all employees and partners to set the expectations of our Safety for Life program and to provide skills enabling proactive hazard identification and control.

Incident Investigation

Our managers and safety staff will investigate all recordable and high potential incidents to identify contributing factors and root causes. Lessons learned will be developed, shared, and implemented to prevent reoccurrence.

Fit for Duty

All employees are responsible for reporting fit for the tasks at hand and for remaining so throughout the course of their work period so as not to compromise their health and safety or that of others.

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SECTION 18 - LADDERS

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 All ladders must be inspected by a competent person prior to each use.

2.2 The use of broken or defective ladders is prohibited. When such ladders are discovered, they shall be removed from service immediately and tagged or destroyed.

2.3 Only portable non-conductive Type 1A or Type 1AA ladders shall be used on AECOM TISHMAN CONSTRUCTION projects. No portable metal ladders are to be used on AECOM TISHMAN CONSTRUCTION projects.

2.4 Manufactured wood, fiberglass, and fixed ladders shall be in accordance with accepted standards. (A manufacturer's certification label shall be affixed to the ladder).

2.5 All individuals using ladders shall be trained in their proper use and inspection.

2.6 Portable ladder feet shall be placed on a substantial base and the area around top and bottom of ladder shall be kept clear of debris for clear access.

2.7 Portable ladders shall be used at a pitch of one-foot (1') horizontal distance for every four feet (4') of height to support its intended weight.

2.8 Ladders shall not be placed where they might be displaced by other activities unless protected by barriers.

2.9 Side rails shall extend thirty-six inches (36") above the landing. When this is not practical, grab rails shall be installed. Straight ladders shall be equipped with safety shoes, tied, blocked, or otherwise secured to prevent displacement.

2.10 All ladders shall be secured from movement.

2.11 Job - made ladders are not permitted over heights of four feet.

2.12 Stair towers or other manufactured access means shall be used in lieu of job-made ladders.

2.13 If a straight ladder is to be up and in place for more than two weeks then a stair tower shall be used in lieu of it.

2.14 All stair towers shall be tagged and inspected on a daily basis.

2.15 Walk through grab bars shall be used on extension ladders for access to upper levels.

2.16 All landings shall have a swing gate, offset guardrail, or other diversion to protect individuals once they exit the ladder, as well as individuals walking by the ladder access point.

2.17 All stepladders shall be of the podium and/ or lean-safe type and not be used as straight ladders unless designed for that purpose.

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2.18 No type of work requiring the use of both hands shall be performed on a ladder when the mid-section of the user must break the side rail plane unless a safety harness is worn, and the safety lanyard is properly secured.

2.19 No objects that restrict the use of both hands for climbing shall be carried in the climber's hands. Workers shall maintain three points of contact while ascending or descending ladders at all times.

2.20 A ladder shall not be placed in front of a door opening toward the ladder, unless the door is blocked open, locked, sign posted, guarded, or removed.

2.21 Ladders must be properly commercially rated for the task with a minimum capacity of 300 pounds. Residential rated ladders shall not be permitted on projects.

2.22 A lifting rope and container shall be installed next to any temporary fixed access straight ladder.

2.23 Aerial lifts, scaffolds, or other approved means shall be prioritized for work during preplanning stages to eliminate the need for ladders where possible.

3. ATTACHMENTS

None Applicable.

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SECTION 19 - LOCKOUT TAGOUT

1. TERMS and DEFINITIONS

1.1 Affected Individual. A trained person whose job requires him/her to operate or use a machine or piece of equipment on which servicing or maintenance is being performed under LOTO, or whose job requires him/her to work in an area in which such servicing or maintenance is being performed. The Affected Employee is not performing work under LOTO, but rather is potentially affected by the work being performed by the Authorized Individual(s).

1.2 Authorized Individual. A person who is appropriately trained and handles LOTO of a machine, piece of equipment, or system in order to perform servicing, construction, or maintenance on that machine, system or equipment.

1.3 Cord and Plug-connected Equipment. Equipment where the only energy source is electrical power provided by a plug-in connection.

1.4 Energy Source. Include, but are not limited to, any electrical, mechanical, hydraulic, pneumatic, chemical, radiation, thermal, or compressed gas energy source; fluid systems; flammable systems; energy stored in springs; and potential energy from suspended objects (gravity) that may injure personnel, cause property damage, and/or cause a release of hazardous substance to the environment.

1.5 Energized. Connected to an energy source or containing residual or stored energy.

1.6 Energy-isolating Device. A mechanical device that physically prevents the transmission or release of energy. This includes locks, hairpins, tongs, lockable valves, clamshell devices for valves, blank flanges for piping systems, and restraining devices to prevent movement of parts.

1.7 Isolation. A physical activity using a device that prevents the transmission or release of energy. Examples of devices used to isolate equipment/systems include, but are not limited to restraint blocks, electrical circuit breakers, disconnect switches, fuses, slip gates, slip blinds, or double valves. Control circuit devices, motor controllers, etc., are NOT acceptable isolation devices.

1.8 Locking Device. A device that utilizes a lock, key to hold an energy-isolation device in the safe position for the purpose of protecting personnel.

1.9 Lockout. The use of a locking device to confirm that an energy-isolating device and the equipment it controls cannot be operated until the lockout device is removed.

1.10 Lockout / Tagout- (LOTO) Specific Procedure. A written procedure developed specifically for each piece of machinery or equipment capable of unexpectedly releasing energy. This procedure outlines in detail of how LOTO will be performed.

1.11 Normal Production Operations. The utilization of a machine or piece of equipment to perform its intended production function.

1.12 Qualified Person. A person, familiar with the construction and operation of the equipment and the hazards involved, trained and competent in the applicable LOTO procedures and assigned authority for the coordination and overseeing of Authorized Employee(s) implementation of LOTO protection. A Qualified Person may be referred to by a different term depending upon jurisdiction (e.g. Requesting Authority, Lockout Authority, Person in Charge, etc.).

1.13 Servicing and/or Maintenance. Workplace activities such as constructing, installing, setting up, adjusting, inspecting, modifying, and maintaining and/or servicing machines or equipment. These activities include

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lubrication, cleaning or unjamming of machines or equipment, and making adjustments or tool changes where employees could be exposed to the unexpected energization or start-up of the equipment or a release of hazardous energy.

1.14 Tagout. Use of a warning device, such as a tag and its means of attachment, to establish that an energy-isolating device and the equipment it controls may not be operated until the tagout device is removed.

2. GENERAL REQUIREMENTS

2.1 A Lockout/Tagout Program shall be established for all projects whenever the release of energy during major replacement, repair, renovation, or modification of machines or equipment, or during installation of new machines or equipment may cause injury to personnel, create property damage, or release a harmful substance to the environment.

2.2 Where design permits, a lock shall be applied in support of a "DANGER" tag to ensure the energy source is kept isolated.

2.3 No device shall be operated with tag or lock attached regardless of circumstances.

2.4 In an emergency, or if the person who placed the tag is not available, the Project Manager and/or the Project Superintendent shall have the authority to remove the danger tags and locks ONLY AFTER POSITIVELY DETERMINING THAT THE EQUIPMENT OR SYSTEM IS SAFE FOR OPERATION AND THAT ALL PERSONNEL ARE IN THE CLEAR.

2.5 Personnel deviating from these instructions, or unauthorized persons removing danger tags, shall be subject to disciplinary action that may include their removal from the project.

2.6 Where possible, isolation devices shall provide visible indication of the position of the device, by using push buttons, selector switches, or other types of position indicators of the circuit control device.

2.7 After removal, the tag shall be destroyed. Simply placing the tag in a trash can does not constitute proper tag disposal. Potentially contaminated tags shall be disposed of at the controlled entry point to the site.

2.8 Use of Locks and Tags.

- a) Locks and "DANGER" tags shall be installed by authorized personnel. If locks cannot be used, an alternate method of isolating the system must be implemented. A tag shall always be used in conjunction with the locking device or system.
- b) Affected employees, and contractors if applicable, shall be notified of the placement of locks/tags prior to locking out the equipment for repair or service work.
- c) The authorized individual shall install the lock and tag on the equipment or system requiring the energy source to be controlled. In operations where more than one individual is working on the equipment or system, each authorized individual shall install a lock and tag. The authorized employee shall retain the key to each lock he/she places.
- d) The individual installer shall verify that the equipment or system has been properly isolated. Verification shall be accomplished by operating or testing the equipment or system for energy release. If there is any doubt regarding the isolation of the equipment or system, the field safety and health staff shall be contacted.
- e) The required service/maintenance work shall then be performed. If the required work extends over a single work shift and will be continued by craft workers on multiple shifts, a system that guarantees the safe transfer of energy control between shifts shall be implemented.

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- f) After the service/maintenance work is completed, a physical check shall be conducted of the area to ensure all personnel working on the equipment or system and all affected employees and contractors are accounted for and have been notified of the imminent reenergizing of the equipment before the authorized employee who installed the lock and tag removes them for the area/activity under his/her control.
- g) The installer shall return the lock to the designated area cabinet location, enter into the logbook the date when the lock and/or tag were removed, then destroy the tag.
- h) No combination locks permitted.

2.9 Lock and Tag Cabinets.

- a) Lock and tag cabinets shall be provided at central locations in each area. The area manager or designee shall control the cabinets.
- b) Each cabinet shall contain numbered locks and tags.
- c) The cabinets shall be locked, and the key of one cabinet shall not open the lock of any other cabinet.

2.10 Lock and Tag Requirements. The Project Superintendent, or designee, shall specify the physical appearance of the locks and tags to be used on the project. Tags shall contain a space to identify the person placing the tag. Locks and tags shall be sufficiently strong to resist inadvertent removal or casual vandalism and shall not be significantly affected by the ambient environmental conditions.

- a) The area manager or designee shall number each lock (serial numbers on locks can be used) before placing it in a cabinet to ensure it can be accounted for. The area manager or designee shall keep a list of lock numbers.
- b) The key of one lock shall not fit any other lock. Master keys are not permitted.
- c) Each tag shall be numbered and logged in the logbook by the area manager/designee before being placed in a cabinet to ensure accountability.
- d) The authorized individual removing the tag from a cabinet shall complete the information required on the tag, such as:
 - i. Log book location.
 - ii. Job identification.
 - iii. Name of installer and signature (manager/supervisor/foreman).
 - iv. Component tagged.
 - v. Component position.
 - vi. Any special instructions to employees.

2.11 Log Book Requirements.

- a) A logbook entry is required to account for each tag and lock issued.

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- b) Each time a lock and tag is removed from a cabinet, the following information shall be recorded in the logbook:
 - i. Tag number.
 - ii. Lock number.
 - iii. Work order.
 - iv. Craft number.
 - v. Location of the tag and/or lock.
 - vi. System or component affected.
 - vii. Date and time when lock and/or tag were attached.
 - viii. Name of individual using lock and tag.
 - ix. Date and time lock and/or tag were returned to cabinet.

2.12 Training.

- a) All affected individuals shall be properly trained to complete lockout/tagout.
- b) All affected individuals shall be trained in and utilize the THA process in conjunction with lockout/tagout.
- c) Subcontractors are solely responsible to train their employees on this program and all relevant OSHA standards.
- d) Retraining shall occur as necessary.
- e) All training will be documented and kept on file by the employer and made available upon request by AECOM TISHMAN CONSTRUCTION project staff.

2.13 No work shall be initiated (including LOTO procedures) prior to obtaining any required permits or completed daily THAs.

2.14 Each Authorized Individual shall affix their own individual lock and tag in a manner that will hold the energy-isolating device in a safe or off position.

2.15 Test the machine(s), equipment or system(s) prior to working on it to verify all sources of energy have been isolated prior to beginning work.

2.16 Locks and tags will remain in place until the work has been completed.

2.17 The locks, tags, and equipment shall not be tampered with by any individual.

2.18 The LOTO shall be specific to the type of energy isolation that is trying to be accomplished.

2.19 All work under platforms in elevator shafts shall utilize a safety block installed to prevent the platform hitting individuals if it fails. The block shall be capable of stopping the platform. This shall be part of the elevator LOTO procedures. (See Shaft Work Permit in the Fall Protection Section).

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3. ATTACHMENTS

3.1 Lock & Tag Log

3.2 Lockout Tagout Checklist

3.3 Lockout Tagout Permit Example

3.4 Unattended & Emergency Lock Removal Form

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Attachment 1: Lock & Tag Log

Lock & Tag Log

S3AM-325-FM1

Name of Qualified Person: _____ Job Name: _____ Date: _____

Job Description: _____ Job Location: _____

TAG NUMBER	LOCK NUMBER	AUTHORIZED EMPLOYEE	LOCATION USED	COMPONENT AFFECTED	DATE/TIME ATTACHED	DATE/TIME REMOVED

Name of Qualified Person: _____ Signature: _____ Date: _____
(print)

Name of Supervisor / Manager: _____ Signature: _____ Date: _____
(print)

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Attachment 2: Lockout Tagout Checklist

LOTO Verification Checklist

S3AM-325-FM2

Equipment ID (#)/LOTO Location (S) – Device Type and #: LOTO Reference Number:	Name: Date:
---	----------------

Lockout Tagout Checklist (LOTO)	Yes	No	Initials
Employee Notification Have all Affected Employees been informed that a LOTO is necessary and the reason for the LOTO?			
Energy Source Identification Has the type and magnitude of all energy sources and the respective method of control been identified?			
Equipment Shutdown Has the machine/equipment been shut down by the normal stopping procedure (depressing the stop button, open switch, close valve, etc.)?			
Applying Energy-Isolating Device Have all energy-isolating devices been applied so that the machine/equipment is isolated from all energy sources?			
Lockout Have all involved Authorized Employees placed their assigned individual lock(s) on all appropriate energy-isolating devices?			
Tagout Have all involved Authorized Employees placed their assigned individual tag(s) the associated lock(s) or, as appropriate, on all applicable energy-isolating devices?			
Energy Dissipation Has all stored/residual energy (such as in capacitors, springs, elevated machine members, rotating flywheels, hydraulic systems, air, steam, or water pressure) been dissipated/restrained by methods such as grounding, repositioning, blocking, bleeding down, etc?			
Zero Energy State Verification Has verification been made that the equipment is disconnected from all energy sources by first checking that no personnel are exposed, then verifying the isolation of the equipment by operating the push button or other normal operating controls?			
IF SO, THE EQUIPMENT IS NOW LOCKED OUT			
Restoring Equipment to Service	DATE:		
Job Completion Verification Has the machine/equipment and immediate area been checked to make sure that nonessential items have been removed and the machine/equipment components, including guards, are operationally intact?			
Personnel Verification Have all personnel been notified of the intent to remove LOTO, safely positioned or removed from the area and all controls are in neutral?			

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Lockout Removal And Equipment Startup Have all involved Authorized Employees removed their assigned individual LOTO devices and the machine reenergized?			
Employee Notification Have all Affected Employees been notified that the LOTO is complete and that the machine/equipment is ready for use?			

NOTES:

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Attachment 3: Lockout Tagout Permit Example

Lockout Tagout Permit

S3AM-325-FM4

Equipment/System to be Isolated:

Building:

Floor:

Column:

Other Location:

Purpose of Isolation:

Type of Isolation:

Authorized Employee:

Date:

Special Instructions:

Lockout Tag Out Performed:

Tag No.	Device ID	Bldg/Floor/Col.	Installed By	Removed By

Verification System Is Safe for Specified Work to Start

Authorized Employee:

Date:

Time:

Accountability

Accepts Protection

Qualified Person:

Date:

Time:

Accountability

Release Protection

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Qualified Person:

Date:

Time:

Lockout/Tagout Removal Authorization

Authorized
Employee:

Date:

Time:

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Attachment 4: Unattended & Emergency Lock Removal Form

Unattended & Emergency Lock Removal

S3AM-325-FM3

This form shall be completed any time an unattended lock must be removed or for removal of a lock in an emergency situation. For this form, an emergency is defined as an event that may cause injury, fire, explosion, over exposure, or other hazards to the general public, the environment, or personnel.

☐ Unattended Lock

☐ Emergency Lock Removal

1. NAME of Authorized Employee whose LOTO is to be removed:		
2. METHOD(s) used to contact Authorized Employee whose LOTO is to be removed (include details such as number of attempts, times, by whom, messages left, etc.):		
☐ Contact Efforts Successful Prior to Removal ☐ Contact Efforts Unsuccessful Prior to Removal		
3. LOCATION of LOTO:		
4. REQUIRED CONTACTS: Contact the following AECOM personnel to locate and notify all Affected Employees:		
Contact Name 1:		
Office Phone #	Home Phone #	Cell #
Contact Name 2:		
Office Phone #	Home Phone #	Cell #
Contact Name 3:		
Office Phone #	Home Phone #	Cell #
5. NOTIFICATION: An AECOM representative has been contacted: ☐ Yes ☐ No Notification verified by: _____ (print) _____ (sign) OR, the special conditions for not contacting AECOM are as follows:		
6. WALK DOWN: A walk-down of the equipment/system has been performed to ensure that all personnel are removed from hazardous locations: ☐ Yes ☐ No Walk-down verified by: _____ (print) _____ (sign)		

	Print Name	Signature	Date
Project / Location Manager			
SH&E Manager			
AFTER COMPLETION OF THESE STEPS, THE LOCK AND TAG MAY BE REMOVED. This form must be provided to the AECOM Project / Location Management Team (Manager and			

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SH&E Manager) so the Authorized Employee whose lock was removed (if not previously notified) and any Affected Employees off shift are notified prior to the Employees' next shift of the LOTO removal.

Notification verified by: _____ (print)
_____ (sign)

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SECTION 20 - MACHINERY AND EQUIPMENT

1. TERMS and DEFINITIONS

1.1 Competent Person. One who is capable of identifying existing and predictable hazards and authorized to take prompt corrective measures to eliminate them.

1.2 Qualified Person. One who is recognized by a degree, certificate or professional standing, or possesses extensive knowledge, training and experience.

2. GENERAL REQUIREMENTS

2.1 All equipment received by each project is to be inspected by a competent person prior to being placed in service. Any unsafe conditions shall be corrected before operation.

2.2 Any modifications, alterations, etc. to equipment from the original manufacturer's condition, shall be approved in writing by the manufacturer or by an equivalent entity.

2.3 No equipment will be operated within 10' of energized power lines unless proper preplanning has taken place.

2.4 Equipment requiring an operator shall have the initial receiving inspection in writing and be kept on file. A daily pre-shift inspection shall be conducted for all mechanized equipment that requires an operator.

2.5 All operators of mobile equipment, including earthmoving equipment, elevated work platforms, forklift trucks, and lift cranes must be qualified through training and/or experience to operate the equipment. Employer must provide documentation of employee's equipment training competency upon request.

2.6 The equipment operator's manual shall be available on the equipment and reviewed by the operator.

2.7 All mobile equipment shall have at least a five-pound fire extinguisher attached to it.

2.8 Inspections and tests shall be made at scheduled intervals to assure safe operating conditions.

2.9 Preventative maintenance procedures recommended by the manufacturer shall be followed.

2.10 Any machinery or equipment found to be unsafe shall be tagged out of service and its use prohibited until unsafe conditions have been corrected.

2.11 Inspections of determinations of road conditions and structures shall be made in advance to assure that clearances and load capacities are safe for the passage or placing of any machinery or equipment.

2.12 Riding on equipment or loads is prohibited unless designed for such use. Getting off or on any equipment while it is in motion is prohibited.

2.13 Machinery or equipment requiring an operator shall not be permitted to run unattended.

2.14 Equipment shall be locked or secured when not in use to prevent starting by unauthorized persons.

2.15 Machinery or equipment shall not be operated in a manner that will endanger persons or property nor shall the safe operating speeds or loads be exceeded.

2.16 All machinery and equipment shall be shut down and positive means taken to prevent its operation while repairs or manual lubrications are being done.

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- 2.17 All repairs on machinery or equipment shall be made at a location that will provide protection from traffic for repairmen.
- 2.18 Machinery, equipment or parts thereof which are suspended or held apart by use of slings, hoists or jacks shall also be substantially blocked or cribbed before workers are permitted to work underneath or between them.
- 2.19 Bulldozer scraper blades, loader buckets, forklift forks, or similar attachments shall be lowered to rest when not in use.
- 2.20 No guard, safety appliance or device shall be removed from machinery or equipment or made ineffective except for the purpose of making immediate repairs, lubrications or adjustments, and then only after the power has been shut off. All guards and devices shall be replaced immediately after completion of repairs and adjustment.
- 2.21 Suitable protection against the elements, falling or flying objects, swinging loads and similar hazards shall be provided for operators of all machinery or equipment. If glass is used for windshields or in cabs, when furnished, it shall be "Safety Glass." Broken or cracked glass shall be replaced immediately.
- 2.22 Machinery rollover protection shall comply with applicable federal & state standards.
- 2.23 Highway operated equipment shall be equipped with turn signals, brake lights, tail lights and head lights, and shall be approved for such use.
- 2.24 All machinery or equipment and material hoists operating on rails, tracks or trolleys shall have positive stops or limit devices either on the equipment, rails, tracks or trolleys to prevent over-running safe limits and shall be equipped with over speed devices.
- 2.25 Stationary machinery and equipment shall be placed on a firm foundation and secured before being operated.
- 2.26 All points requiring lubrication during operation shall have fittings so located or guarded to be accessible without hazardous exposure.
- 2.27 All mobile equipment shall have adequate headlights and taillights when operating in hours of darkness.
- 2.28 Service or maintenance equipment that will be parked or moving slower than normal traffic on haul roads at night shall have a yellow flashing light mounted to be visible from all directions.
- 2.29 Mobile - type equipment shall have a braking system or systems capable of stopping and holding the equipment full loaded from operating speed on any grade of operation.
- 2.30 Heavy - duty haulage equipment shall, in addition to the service braking system, have an emergency brake system. The emergency brake system shall automatically stop the equipment upon failure in the service brake system. The system shall also be manually operable from the driver's position.
- 2.31 Electric wheel and rubber-tired haulage equipment not equipped with an effective parking brake shall have the wheels blocked to prevent movement in any direction whenever the equipment is parked.
- 2.32 Operators shall not leave the cab while equipment is being loaded when exposed to danger from suspended or overhead loading equipment or methods.
- 2.33 Mechanized equipment shall be shut down prior to and during fueling operations. Closed systems with automatic shut off, which will prevent spillage if connections are broken, may be used to fuel diesel powered equipment.

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2.34 All towing devices used on any combination of equipment shall be structurally adequate for the weight drawn and securely and properly mounted.

2.35 Persons shall not be permitted to get between a towed and towing piece of equipment until the towing equipment has been stopped.

2.36 All machinery or equipment operating on rails, tracks, or trolleys shall be provided with substantial track scrapers or track cleaners, effective in both directions on each wheel or set of wheels.

2.37 Spotters shall be provided as necessary for equipment operations.

2.38 All self-propelled construction equipment, except light service trucks, panels, pick-up trucks, station wagons, crawler-type cranes, power shovels and drag lines, whether moving alone, or in combination, shall be equipped with a reverse signal alarm. The alarm shall be audible and sufficiently distinct to be heard under prevailing conditions. Alarm shall operate automatically upon commencement of backward motion. Alarm may be continuous or intermittent (not to exceed three-second intervals) and shall operate during the entire backward movement.

2.39 All belts, gears, shafts, pulleys, sprockets, spindles, drums, flywheels, chains or other reciprocating, rotating or moving parts of equipment shall be guarded as such parts are exposed to contact by persons or otherwise create a hazard.

2.40 Aftermarket steering wheel knobs and/or steering assist devices will be prohibited.

3. FORKLIFTS (INDUSTRIAL LIFT TRUCKS)

3.1 Only certified operators who are authorized and trained in the safe operation of the industrial trucks or industrial tow tractors shall be permitted to operate such vehicles.

3.2 Stunt driving and horseplay is prohibited and may result in immediate termination.

3.3 Seat belts shall be worn at all times.

3.4 Individuals shall not ride on the forks of lift trucks.

3.5 Drivers shall document a daily inspection of the vehicle and all attachments prior to use and at least once each shift. If found to be unsafe, have it repaired before continuing operations.

3.6 Attention shall be given to the proper functioning of tires, horn, lights, battery, controls, brakes, steering mechanism, and the lift system of the forklift (forks, chains, cable, and limit switches).

3.7 Vehicles shall not exceed the authorized or safe speed.

3.8 The driver shall slow down and sound the horn at cross aisles and other locations where vision is obstructed.

3.9 Operators should always face their destination.

3.10 When leaving a vehicle unattended, the power shall be shut off, brakes set, and the mast brought to the vertical position with the forks left in the down position. When left on an incline, the wheels shall be blocked. The operator shall not move more than 25 feet from the equipment on the same level at any time.

3.11 Do not attempt to exceed the capacity of the forklift or carry loads that are too heavy or unbalanced.

3.12 A loaded forklift shall not be moved until the load is safe and secure.

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3.13 Where possible, avoid operating near ditches, embankments and holes.

3.14 Reduce speed when turning, crossing slopes, and on rough, slick or muddy surfaces.

3.15 When traveling, watch out for overhead obstruction.

3.16 Rigging over the forks is prohibited without manufacturer approved attachments or without following manufacturer specifications.

4. AERIAL AND SCISSOR LIFTS

4.1 Only trained and authorized operators shall be permitted to operate an aerial lift.

4.2 A malfunctioning lift shall be shut down and tagged out until repaired.

4.3 The controls shall be plainly marked as to their function.

4.4 The controls shall be tested each day prior to use to determine that they are in a safe operating condition.

4.5 All personnel on the platforms of all lift types shall wear an approved full body harness and proper self-retracting lanyard (SRL) attached to the platform's attachment point per the manufacturer's requirements.

4.6 Load limits specified by the manufacturer shall not be exceeded.

4.7 Instruction and other placards must be posted and legible.

4.8 Lifts shall not be used near electrical power lines unless the lines have been de-energized or adequate clearance is maintained.

4.9 Individuals using lifts shall be instructed to recognize and avoid unsafe conditions and hazards.

4.10 Ground controls shall not be operated unless permission has been obtained from the personnel in the platform, except in case of an emergency.

4.11 A competent person shall perform regular inspections of the lifts.

4.12 Personnel shall always stand on the floor of the platform, not on boxes, planks, railing or other devices for a work position unless permitted by OSHA and/or the manufacturer.

4.13 All chain or rail guards shall be in place prior using the lift.

4.14 If equipped, outriggers shall be properly used as recommended by the manufacturer.

4.15 A daily walk-around inspection is required before using any machine. Special lift concerns:

- Inspect platform assembly for loose or missing parts, visible damage, lock pins and bolts in place, foot switch (safety shutoff) in place and working. Clear all debris and materials from the platform prior to use and at end of shift.
- Tire and wheel assembly properly secured, no loose or missing lug nuts, no visible damage (no cuts on tires) and tires are properly inflated.
- Ground and platform control switches are operable, no visible damage, and placards secure and legible.
- Check emergency stop switch for proper operation.

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- Check manual decent control valves for proper operation in the event they may be needed due to total power failure.
- Check to ensure that the operating manual is stored on the equipment and is readily accessible.

5. MATERIAL HOISTS

5.1 All material hoists shall:

- Have a dedicated communication system between the operator and the different floor openings.
- Have an electronic kill switch on hoist.
- Have a third party hoist inspection prior to use.

5.2 Visual confirmation shall be established when the operator moves the hoist.

5.3. Written procedure for operator / spotter to follow before moving the hoist.

6. MATERIAL HANDLING EQUIPMENT (OTHER)

6.1 Only operators qualified through training and/or experience shall operate the equipment.

6.2 Read and understand all safety precautions and warnings before operating the equipment. Perform a daily inspection at the beginning of each shift.

6.3 In an open area, check the operation of all controls while moving them slowly and complete a daily documented inspection.

6.4 Do not start the equipment or move controls if there is a warning tag hanging on the controls.

6.5 Do not jump off equipment. Use grab rails.

6.6 Do not smoke while fueling or when near batteries.

6.7 Riding on loads, fenders, running boards, or tailgates, or allowing legs or arms to dangle over the sides is not permitted.

6.8 Do not ride on equipment not designed for passengers.

6.9 The parking brake must be set whenever the equipment is parked. Equipment parked on an incline must have the wheels chocked. Lower all attachments to the ground before leaving the equipment.

6.10 Do not back up any vehicle or equipment when the view to the rear is obstructed, unless:

- It is equipped with an operating back-up alarm which is audible above the surrounding noise for a distance of 200 feet; or
- An observer/spotter wearing a Class II vest signals that it is safe to back up.

6.11 Know the traffic pattern of the jobsite road and obey flaggers, road signs and signals.

6.12 Keep the equipment, especially the deck and steps, free of foreign material. Secure all loose items.

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6.13 Make sure all safety guards and covers are secured in place.

6.14 When operating digging equipment, know the exact location of buried utility lines (Call appropriate state authority prior to digging) and obtain any necessary permits.

6.15 When loading trucks, never swing over the truck cab and make sure the driver remains out of the truck unless cab protection is provided.

6.16 If necessary to work around power lines, attempt to make arrangements to have power disconnected.

6.17 Use caution when moving a piece of equipment with the truck bed in the raised position.

6.18 If you hit a power line, stay inside the cab until the equipment is clear of the line or power is shut off.

7. MOTOR VEHICLES (AUTOMOBILES AND PICKUPS)

7.1 The operator of a motor vehicle must hold a valid driver's license for the class of vehicle to be operated.

7.2 The operator must inspect the vehicle before placing it in operation. Any deficiencies found that might constitute a safety hazard shall be repaired before operating.

7.3 A competent person shall explain any special controls or operating functions to the operator.

7.4 In case of an incident regarding a third party vehicle on or off the jobsite, the operator should be aware of the following:

- Do not admit fault.
- If in question, call the police. Obtain the number of the police report.
- An incident form should be filled out within 24 hours after the incident. The glove compartment should contain an "In Case of Incident Form," the vehicle registration, and insurance card.
- Report the incident immediately to your supervisor.
- Give a third party the telephone number of the jobsite and the name of the Field Office Manager for contact.
- Follow the incident investigation procedures and notifications presented in the Incident Reporting section of this program.

8. ATTACHMENTS

8.1 General Equipment Daily Inspection Checklist

8.2 Forklift Pre-Operation Inspection Checklist

8.3 Heavy Equipment Maintenance Inventory Log

8.4 Aerial Work Platform Checklist

8.5 Aerial Work Platform Pre-Operation Checklist

8.6 Equipment LOTO Procedure Template

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Attachment 1: General Equipment Daily Inspection Checklist

General Equipment Daily Inspection Checklist								
Supervisor:	Equipment type:		Model #:			Rented from/Owned by:		
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	
Operator instruction manual on board?								
Instruction/Safety decals in place/legible?								
Engine oil level OK, no leaks?								
Hydraulic oil level OK, no leaks?								
Engine coolant level OK, no leaks?								
Battery OK, no fluid leaks?								
Fuel level – Adequate for full shift?								
Tires properly inflated, no lug nuts missing?								
Hydraulic lines/fittings – leaks/worn areas?								
Electrical wiring –not burned, cut, frayed?								
Horn and back-up alarm operational?								
Seat belt in place and operational?								
Fire extinguisher present/charged?								
Rolloverprotection installed/good condition?								
Boom angle indicators operational?								
Mirrors present/not damaged?								
Flaggers Needed? Trained?								
	Inspector:	Inspector:	Inspector:	Inspector:	Inspector:	Inspector:	Inspector:	
	Date:	Date:	Date:	Date:	Date:	Date:	Date:	
Notes								
Use a check mark to indicate OK. Use an “X” to indicate needed repair or unsafe condition. These items shall be checked prior to each shift’s use. Report all items in need of repair or unsafe condition to your supervisor at the time of inspection. Return the completed sheet to your supervisor at the end of each week.								

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Attachment 2: Forklift Pre-Operation Inspection Checklist

FORKLIFT PRE-OPERATION INSPECTION CHECKLIST															
Forklift Information															
Number/Name:							Make/Model:								
Checklist Items - Operator must initial each Item (UNSAT Items require corrective action prior to forklift use)															
Hour meter reading:															
Check the following as appropriate	Operator Name/Date:			Operator Name/Date:			Operator Name/Date:			Operator Name/Date:			Operator Name/Date:		
	SAT	UNSAT	N/A	SAT	UNSAT	N/A	SAT	UNSAT	N/A	SAT	UNSAT	N/A	SAT	UNSAT	N/A
1. Operator Qualified to Operate															
2. Operation and warning signs legible															
3. Panel and foot switches															
4. Turn table															
5. Parking brake															
6. Service brakes															
7. Steering															
8. Oil Level															
9. Radiator fluid level															
10. Platform gate															
11. Boom and limit switches															
12. Movement alarm															
13. Tires (visual)															
14. Restraint protection, including tie off anchor point															
15. Fuel leaks (liquid or propane)															
16. Fire extinguisher full and secure															
17. Fuel lines secure															
18. Electrical lines															
19. Exhaust component															
20. Ground controls															
Comments/Remarks															

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Attachment 3: Heavy Equipment Maintenance Inventory Log

Heavy Equipment Maintenance Inventory Log

[illegible]

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Attachment 4: Aerial Work Platform Checklist

Aerial Work Platform Checklist

EQUIPMENT TYPE:	EQUIPMENT NUMBER:	SERIAL NUMBER:
MAKE:		MODEL NUMBER:
MILEAGE:	DATE CHECKED:	LOCATION:
ENGINE MAKE:	YEAR MANUFACTURED:	HOURS:
	MARK ONE	COMMENTS
	GOOD FAIR REPAIR	(List percentage of wear on all wear items)
1. GENERAL APPEARANCE		
Hood & Panels		
Paint		
Undercarriage		
Tires - Type, Size		
2. ENGINE COMPARTMENT		
Oil Level & Condition		
Filters, Condition		
Fan / Radiator / Belts		
Gov. / Idling Device		
Exhaust System		
Fuel / Water / Oil / Lines		
Coolant — Level - Protect.		
Battery - Test Condition		
3. OPERATOR'S POSITION		
Controls		
Gauges		
Starting / Charging		
Engine Condition		
- Minor Repair		
- Major Repair		
Decals & Labels		
4. AERIAL EQUIPMENT		
Boom Condition		
Wear Pads		
Power Trac		
Hydraulic Fluid & Filter		
Emergency Lowering		
Aux. Power System		
Scissor Arms		
Safety Circuits		
Control Labels		
Basket & Door		
Platform & Railings		
Back-up Alarm		
Stabilizers		
Pads		
5. DOCUMENTATION, ETC		
Operator's Manual		
Service Manual		
Log Book		
Fire Ext.		
6. OTHER COMPONENTS		
Certification By:	Inspected By: Approved	ESTIMATED HOURS TO REPAIR:

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Attachment 5: Aerial Work Platform Pre-Operation Checklist

Aerial Work Platform Pre-Operation Checklist

CHECKLIST MUST BE COMPLETED DAILY PRIOR TO INITIAL OPERATION		
<i>FAULTY EQUIPMENT/ DEVICES WILL BE IDENTIFIED AND REPAIRED BEFORE USE</i>		
A = Acceptable O = Needs Improvement X = Not To Be Operated N/A = Not Applicable		
EQUIPMENT / VEHICLE ID #		LOCATION:
DATE:		OPERATOR NAME:
MARK CONDITION	COMMENTS	
PRE-START UP WALK AROUND		
Manufacturer's Operations Manual is present with equipment		
Battery Tray — opens/closes easily, latches shut		
Turret Turntable — gears/lock pins/stops		
Cover Panels — opens/closes easily, latches/locks shut		
Engine — fluids/filters/belts/hoses		
Batteries — clean/dry/secure/caps-cables/level		
Fuel Tank/Level		
Boom Valley / Under Platform — leaks/debris		
Accessory plugs and cables		
Boom Lift / Arms — general condition / wear		
Articulated Joints — wear/cracks		
Parking Brake		
Brakes		
Tires Pressure & Condition		
Platform — guard rails/toe board		
Seat Belts		
Muffler / Exhaust System		
Rear View Mirrors		
Belts - Fan, Alternator		
Hydraulic System		
Hydraulic Fluid Level		
Hydraulic Cylinders		
Fire Extinguisher		
Fall Protection Anchor Point		
Ladders, Tools, Equipment		
Informational Plates / Markings /Reflectors		
Annual Inspection Certification Sticker — valid/legible		
OPERATING CONTROLS		
Turn Signals		
Brake Lights		

FAULTY EQUIPMENT / DEVICES WILL BE IDENTIFIED AND REPAIRED BEFORE USE

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CHECKLIST MUST BE COMPLETED DAILY PRIOR TO INITIAL OPERATION		
<i>FAULTY EQUIPMENT / DEVICES WILL BE IDENTIFIED AND REPAIRED BEFORE USE</i>		
A= Acceptable O = Needs Improvement X= Not To Be Operated N/A = Not Applicable		
EQUIPMENT / VEHICLE ID #		LOCATION:
DATE:		OPERATOR NAME:
MARK CONDITION		COMMENTS
Emergency Lights		
Head & Tail Lights		
Gauges		
Battery Charge		
Battery Load Test		
Windshield Wipers		
Back-Up/Travel Alarm/ Horn		
Boom / Lift Arms — raise/lower/extend/retract		
Turret Rotate		
Drive — forward and reverse		
Steer — left and right		
Platform — tilt/rotate/extend		
Outriggers / Stabilizers / Pothole Protection		
Extendable Axles		
Function-Enable (dead man) Pedal / Switch		
Manual / Auxiliary Controls		
Wheels Chocked		
WORK ZONE		
Trained Operator		
Required PPE		
Fall Protection Equipment Used		
Wind and Weather Conditions		
Cones, Barricades, Signage		
Established, Clear Work Zone		
Drop-Offs or Holes		
Bumps and Floor / Ground Obstructions		
Debris		
Overhead Obstructions		
Energized Power Lines		
Ground Surface and Support Conditions		

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CHECKLIST MUST BE COMPLETED DAILY PRIOR TO INITIAL OPERATION		
A= Acceptable O = Needs Improvement X = Not To Be Operated N/A = Not Applicable		
EQUIPMENT / VEHICLE ID#	LOCATION:	
DATE:	OPERATOR NAME:	
MARK CONDITION	COMMENTS	
Pedestrian / Vehicle Traffic		
Hazardous Locations		
Review Work Task(s) with Staff		
Traffic Directional Lights		
<i>NOTE: WHEN VEHICLE IS NOT REQUIRED TO HAVE SPECIFIC ITEMS, IT MUST BE SHOWN AS N/A IN BOX.</i>		
THIS FORM MUST BE MAINTAINED IN THE PROJECT FILE.		

Attachment 6: Equipment LOTO Procedure Template

Equipment LOTO Procedure Template

EQUIPMENT:														
EQUIPMENT NO:	LOCATION:													
PURPOSE This seven-step procedure establishes the minimum requirements for the lockout of energy isolating devices whenever servicing or maintenance is done on facility equipment. This procedure will be used to ensure that the equipment is stopped, isolated from all potentially hazardous energy sources and locked out before employees perform any maintenance where the unexpected energization or startup of the equipment or release of energy could cause injury.														
COMPLIANCE WITH THIS PROCEDURE All Employees are required to comply with the restrictions and limitations imposed on them during the use of this procedure. The Qualified Person (if applicable) and Authorized Employees are required to perform the lockout in accordance with this procedure. Other employees, upon observing a piece of equipment which is locked and/or tagged out, will not attempt to start, energize, or use said equipment.														
SEQUENCE OF LOCKOUT/TAGOUT														
1. All affected employees will be notified that the equipment must be shut down and locked out to perform servicing or maintenance. Specific Instructions:														
2. The Qualified Person (if applicable) or Authorized Employee will identify the type and magnitude of the energy that the equipment utilizes, will understand the hazards of the energy, and will know the methods to control the energy. ENERGY <table style="width: 100%; border: none;"> <tr> <td style="width: 33%;">☐ Electrical (44V)</td> <td style="width: 33%;">☐ Natural Gas</td> <td style="width: 33%;">☐ Spring</td> </tr> <tr> <td>☐ Hydraulic</td> <td>☐ Gravity</td> <td>☐ Steam</td> </tr> <tr> <td>☐ Chemical</td> <td>☐ Pneumatic</td> <td>☐ Thermal</td> </tr> <tr> <td>☐ Other</td> <td></td> <td></td> </tr> </table>			☐ Electrical (44V)	☐ Natural Gas	☐ Spring	☐ Hydraulic	☐ Gravity	☐ Steam	☐ Chemical	☐ Pneumatic	☐ Thermal	☐ Other		
☐ Electrical (44V)	☐ Natural Gas	☐ Spring												
☐ Hydraulic	☐ Gravity	☐ Steam												
☐ Chemical	☐ Pneumatic	☐ Thermal												
☐ Other														
3. Shut down operating equipment by the normal stopping procedures (depress stop button, open switch, close valve, etc.). Specific Instructions:														
4. Apply the energy isolating device(s) so that the equipment is isolated from the energy sources(s). Specific Instructions:														
5. Authorized Employee(s) lockout and tagout the energy isolating devices(s) with assigned individual locks and tags. Lockout Equipment Needed:														

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6. Dissipate any stored or residual energy (such as that in capacitors, springs, hydraulic systems, and air, steam, or water pressure, etc.) by methods such as grounding, repositioning, blocking, bleeding down, etc.

Specific Instructions:

7. Ensure that the equipment is disconnected from the energy source(s) by first checking that no personnel are exposed, then verify the isolation of the equipment by operating the normal operating control(s) or by testing to make certain the equipment will not operate. CAUTION: Return controls to "OFF" after verification. THE EQUIPMENT SHOULD NOW BE LOCKED OUT AT ZERO ENERGY STATE.

Specific Instructions:

METHODS OF VERIFICATION

Verification should be determined via start-up attempts, visual observations and testing. For electrical verification, place local on/off switch to ON position and verify equipment will not operate. Return the switch to OFF position and commence work.

DEVELOPED BY: _____ DATE: _____

_____(print) _____(sign)

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SECTION 21 - MATERIAL HANDLING, STORAGE, USE, AND DISPOSAL

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 All materials stored in tiers shall be secured to prevent sliding, falling or collapse.

2.2 Maximum safe load limits of floors shall be conspicuously posted in all storage areas (except for floors on a grade).

2.3 Aisles and passageways shall be kept clear.

2.4 Material stored inside buildings under construction shall not be placed within six feet of any hoist-way, or inside floor openings, or within ten feet of an exterior wall that does not extend above the top of the material stored.

2.5 Materials shall not be stored on scaffolds or runways in excess of supplies needed for immediate use.

2.6 Materials are not permitted to be dropped to any point lying outside the exterior walls of the building unless approved by of the Safety Manager and Superintendent.

2.7 Whenever debris are dropped through holes in the floor without the use of chutes, the area onto which the material is dropped shall be completely enclosed with barricades no less than forty-two inches high and six feet back from the projected edge of the opening above. Warning signs shall be posted. Debris shall not be dropped greater than twenty feet. These operations shall be reviewed by the AECOM TISHMAN CONSTRUCTION project team prior to commencement.

2.8 Materials must be stored in a manner that prevents damage or deterioration to the material and in a configuration that is safe and provides for the optimal and efficient use of storage space.

2.9 Material must not be stacked within 18 inches of the sprinkler heads. Material should be kept clear of light fittings, heating pipes and ceilings.

2.10 Base areas and heights of stacks should be kept as small as circumstances permit. The ratio of height to base dimensions of large stacks should be correctly proportioned so that failure of part or whole of the stack does not occur.

2.11 Keep aisles, passageways, and other access ways clear to provide for the free and safe movement of material-handling equipment or employees. Segregate and label non-compatible materials.

2.12 Individuals should be warned about handling cement or lime about skin burns and ensure that proper PPE is worn.

2.13 Lime must be stored to prevent a premature slaking action that may cause fire.

2.14 Brick stacks must not exceed 7 feet (2.1 meters), and they should be tapered back starting at 4 feet (1.2 meters).

2.15 Always place bricks on planks, asphalt, or concrete. Bricks are never to be placed on uneven or soft surfaces.

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- 2.16 Keep the top of brick stacks level and maintain the taper during un-stacking operations.
- 2.17 Stack blocks in tiers on solid, level surfaces and taper back over the 6-foot (1.8-meter) level.
- 2.18 When stacking lumber, place cross strips on stacks more than 4 feet (1.2 meters) high.
- 2.19 Remove all nails from used lumber before stacking or hauling away.
- 2.20 Stack lumber on level and solidly supported sills to be stable and self-supporting.
- 2.21 Do not stack lumber more than 16 feet (4.9 meters) high.
- 2.22 Store steel rods in separate stacks according to length and size.
- 2.23 Store structural steel to prevent the danger of members sliding off, or the stack toppling.
- 2.24 Use only rated and stamped rebar stands.
- 2.25 Do not stack pipes higher than 5 feet (1.5 meters) unless racked on rated and stamped pipe racks.
- 2.26 While removing sand, gravel, and crushed stone from stockpiles, ensure there are no overhanging or vertical faces at any time.
- 2.27 Do not store material dumped against walls or partitions to a height that will endanger the stability or exceed the resisting strength of such walls and partitions.
- 2.28 All material handling, use, and disposal must meet OSHA 29 CFR 1926.250 Subpart H.
- 2.29 A materials logistic plan shall be developed by each AECOM project team. The plan should limit the amount of materials stored on site by all parties.
- 2.30 All exterior storage around a building should be well organized with proper pathways for egress.
- 2.31 All deliveries of material, products or equipment shall complete a risk assessment and Pre-Task Plan (PTP) that must be performed prior to any loading/unloading activities commencing.
- 2.32 Communication to all affected individuals in the delivery area shall occur prior to offloading.
- 2.33 Each project shall maintain a current site logistics plan clearly identifying Controlled Access Zones (CAZ) for truck loading and unloading, required traffic controls, and best practices for unloading or loading materials. Logistics plans must be updated regularly to reflect evolving conditions. CAZs shall have signage, be flagged or barricaded and kept clean and clear of all non-essential materials and personnel. Plastic, colored chains or 3/8 mylar yellow rope with warning sign should be used in lieu of tape.
- 2.34 Safe Zone for Waiting shall be established for the delivery drivers to wait during loading / unloading operations. If a driver exits the vehicle then all PPE requirements shall apply.
- 2.35 Drivers shall be in full view to personnel loading / unloading the truck at all times. All loading or unloading activities must stop if the driver is not visible to personnel or needs to enter the zone to inspect the load. Alternatively, if safe to do so, the operation can allow the driver to stay in the truck cab during loading and unloading.
- 2.36 A restricted access zone (RAZ) must be established around the truck to prohibit entry into the loading/unloading area. No individual, except those involved in the loading/unloading operations shall be

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permitted to be within the identified restricted access zone. The zone must be equal to the area needed to load / unload plus ten (10) feet around the entire truck area.



2.37 A qualified “spotter”, wearing an ANSI approved high visibility traffic vest, must be utilized during loading / unloading operations. Vehicles must never back up “blind” on a project.

2.38 Individuals on the ground within the zone should never be on the opposite side (blind side) of the truck from the hoisting equipment loading or unloading materials, product or equipment.

2.39 Individuals on top of the truck load while equipment is loading or unloading is prohibited unless the operation requires it.

2.40 All individuals in the loading zones should be trained on these safe work practices and given refresher training periodically. Subcontractors and vendors shall be responsible to train their employees.

3. ATTACHMENTS

None Applicable.

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SECTION 22 - NOISE

1. TERMS and DEFINITIONS

1.1 Ambient Noise (Residual Noise, Background Noise). Noise of a measurable intensity that is normally present in the background in a given environment.

1.2 Audiologist. A professional specializing in the study of rehabilitation of hearing, who is certified by the American Speech, Hearing and Language Association or licensed by a state board of examiners.

1.3 Auditory Trauma. Damage to the hearing mechanism resulting in some degree of permanent or temporary hearing loss. Auditory trauma may be caused by agents other than noise; e.g., head injury; burns; sudden or excessive changes of atmospheric pressure.

1.4 A-Weighted Sound Level. A single number representing the sound level of a noise containing a wide range of frequencies in a manner representative of the ear's response.

1.5 Decibels - A-Weighted (dBA). A unit of measurement of sound level corrected to the A-weighted scale, as defined in ANSI S1.4 Part 3 2014 using a reference level of 0.00002 Newton per square meter.

1.6 Dosimeter. An instrument which registers the occurrence and cumulative duration of noise exceeding a predetermined level at a chosen point in the environment or on a person.

1.7 Earmuff. A hearing protective device that encloses the entire outer ear.

1.8 Earplug. A hearing protective device, having specified or standard acoustic characteristics, which upon insertion occludes the external auditory meatus.

1.9 Frequency. The number of times per second that a sine wave repeats itself. It is expressed in hertz (Hz) or cycles per second (cps).

1.10 Hearing Impairment. Hearing loss exceeding a designated criterion (commonly 25 decibels (dB), re International Safety Operations (ISO) standard, averaged from the threshold levels at 500, 1,000, and 2,000 Hz).

1.11 Hearing Protective Device (HPD). A device inserted into or placed over the ear in order to attenuate air conducted sounds.

1.12 Impulse Noise (Impact Noise). Noise of short duration (typically, less than one second) especially of high intensity, abrupt onset and rapid decay, and often of rapidly changing spectral composition.

1.13 Medical Pathology. A disorder or disease affecting the ear, which should be treated by a physician specialist.

1.14 Noise Reduction Rating (NRR). A rating system for hearing protective devices developed by the Environmental Protection Agency. The NRR is a measure of the ability of a given protective device to attenuate noise, as a function of noise intensity and frequency. The NRR must be shown on the hearing protector package.

1.15 Otolaryngologist. A physician specializing in diagnosis and treatment of disorders of the ear, nose, and throat.

1.16 Representative Exposure. Measurements of an employee's noise dose or 8 hour time weighted average sound level that the employers deem to be representative of exposures of other individuals in the workplace.

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1.17 Standard Threshold Shift. As defined by OSHA, a change in hearing threshold relative to the baseline audiogram of ten (10) dB or more at 2,000, 3,000, and 4,000 Hz in either ear, corrected for aging (presbycusis).

1.18 Sound Level. Ten times the common logarithm of the ratio of the square of the measured A weighted sound pressure to the square of the standard reference pressure of 20 micropascals, given in decibels (dB). For use with the directive, slow time response, in accordance with ANSI S1.4 Part 3- 2014 is required.

2. GENERAL REQUIREMENTS

2.1 A hearing conservation program shall be implemented and protection against the effects of noise exposure shall be provided when the noise exposures equal or exceed an 8-hour time weighted average sound level of 85 decibels measured on the A-weighted scale of a standard sound level meter at slow response. Subcontractors shall implement their own program for their employees and shall meet the minimum requirements found in this program.

For compliance purposes as regulated by OSHA, an effective hearing conservation program shall include as a minimum:

- a. Monitoring of the workplace to determine the representative exposure of individuals to excessive noise levels.
- b. A training program for all individuals exposed to noise levels of this standard will be performed in accordance with OSHA standards.
- c. Workers will be provided personal protective equipment when administrative or engineering controls fail to reduce sound levels to within the levels of the standards.

2.2 Protection against the effect of noise exposure shall be provided when the sound levels exceed:

- 90 decibels for eight hours duration per day.
- 92 decibels for six hours duration per day.
- 95 decibels for four hours duration per day.
- 97 decibels for three hours duration per day.
- 100 decibels for two hours duration per day.
- 102 decibels for one- and one-half hours duration per day.
- 105 decibels for one-hour duration per day.
- 115 decibels for one-fourth hour or less duration per day.

2.3 If the sound levels exceed the levels shown above, feasible administrative or engineering controls shall be utilized. If such controls fail, personal protective equipment shall be provided and used.

2.4 If the work environment exceeds these limits, contact your Regional Safety Director immediately.

2.5 For more information, refer to OSHA 29 CFR 1926.52.

3. FACTORS TO CONSIDER

To assess employee noise exposure, the following factors shall be considered:

3.1 The overall decibel level of the noise exposure. If a noise exposure does not cause auditory fatigue, the exposure is not considered harmful to hearing sensitivity.

3.2 The frequency spectrum of the noise. Noise exposure in the high frequency ranges is generally more harmful than low frequency noise.

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3.3 The daily time distribution of the noise exposure. Noise that is intermittent in character is generally less harmful to hearing than steady state noise exposure. The potential for hearing loss increases as a function of the duration of the exposure over a working lifetime.

3.4 The susceptibility of the worker to hazardous noise. Not every individual will develop identical hearing impairment if exposed to the same noise intensity over the same time period.

4. IMPLEMENTATION

4.1 Monitoring.

- a. Monitoring of employee exposures to noise shall be conducted by the field safety and health staff on an as needed basis.
- b. The monitoring requirement may be met by either area monitoring or personal monitoring that is representative of employee exposures. Personal monitoring is preferred and may be required based on the type(s) of noise sources.

4.2 Employee Training and Information.

- a. Training Program. A training program shall be established as follows:
 - i. All employees who may be exposed to excessive noise levels shall complete a formal training program, which shall include, as a minimum, the following information:
 - The effects of noise on hearing.
 - The purpose of hearing protectors, the advantages, disadvantages, and attenuation of various types, and instructions on selection, fitting, use, and care.
 - The specific nature of the operations which could result in exposure to excessive noise levels.
 - A description of the medical surveillance program, including a description of the purpose of audiometric testing, and an explanation of the test procedures.
 - The engineering controls and safe work practices associated with the employee's work assignment.
 - ii. The training program shall be repeated annually, and each employee receiving such training shall complete a Hearing Protection Training Completion Form.
- b. Safety Meetings. Job site safety meetings and site-specific health and safety plans should include instruction on the need for hearing protective devices in designated areas.

4.3 Control Measures.

- a. Examination of the Problem. To control noise exposure, its three basic elements shall be examined, i.e., source of the sound, travel path, and effect on receiver or listener. Solution of a given noise problem might require alteration or modification of any or all of these three basic elements.
- b. Noise Control at the Source. Controlling noise at the noise source can be achieved by the following:
 - i. Select quiet equipment initially. In selecting quiet equipment the following features shall be considered:

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- Low noise certification.
 - Advertisement of "quiet" operation, evidence of noise control design.
 - Evidence of "lower" and "slower" operating characteristics.
 - Side-by-side noise testing of equipment.
 - "Onsite" or "in operation" inspection of mechanical equipment before purchase.
- ii. Reduce operating noise by considering the following control measures:
- Reduce impact or impulse noise by reducing weight, size, or height of fall of impacting mass.
 - Reduce speed in machines and flow velocities and pressure in fluid systems.
 - Balance rotating parts to control machinery noise and vibration of fans, fly wheels, pulleys, cams, shafts, etc.
 - Reduce frictional resistance between rotating, sliding or moving parts in mechanical systems: frequent lubrication, proper alignment of moving parts; static and dynamic balancing of rotating parts; correction of eccentricity or "out of roundness" of wheels, gears, rollers, pulley, etc.
 - Reduce resistance in air or fluid systems: use of low flow velocities, smooth boundary surfaces of duct or pipe systems, and long radius turns and flared sections in pipes, etc., to reduce turbulence noise.
 - Isolate vibration elements in machinery: install motors, pumps, etc., on most massive part of machine. Use belt or roller drives in place of gear trains. Use flexible hoses and wiring instead of rigid piping and stiff wiring, etc.
 - Apply vibration damping materials such as liquid mastic; pads of rubber, felt, foam or fibrous blankets; or sheet metal viscoelastic laminates or composites to vibrating machine surfaces.
 - Reduce noise leakage from the interior of machines such as compressors by sealing or covering all openings or applying acoustical materials to machine interiors.
- c. Noise Control in the Transmission Path. This can be achieved by the following:
- i. Separate the noise source and receiver as much as possible.
 - ii. Use sound absorbing materials on ceiling, floor, or wall surfaces as close to the machine as possible.
 - iii. Use sound barriers and deflectors in the noise path.
 - iv. Use acoustical lining on inside surfaces of such passageways as ducts, pipe chases, or electrical channels.

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- v. Use mufflers, silencers, or snubbers on all gasoline or diesel engines, regardless of size; and particularly on equipment when large quantities of high-pressure, high velocity gases, liquids, steam or air are discharged into the open air.
 - vi. Use vibration isolators and flexible couplers where the noise transmission path is structure-borne in character.
- d. Protection for the Receiver. When engineering controls fail to reduce the levels to within acceptable levels, the following measures shall be implemented:
- i. Personal protective equipment shall be provided and replaced as necessary at no cost to employees.
 - ii. Supervisors shall ensure that hearing protective devices are worn by all employees who are exposed to a time weighted average of 85 decibels or greater and who have experienced a significant threshold shift.
 - iii. Employees shall be given the opportunity to select their hearing protectors from a variety of suitable protectors.

4.4 Recordkeeping.

- a. Noise exposure measurement records, surveys, and Site-Specific Hearing Conservation Plans will be retained at the project/facility for the duration of the project.
- b. Audiogram records will be retained in the employee's medical records as per AECOM TISHMAN CONSTRUCTION policies and OSHA standards.
- c. Employee training session documentation will be retained in accordance with AECOM TISHMAN CONSTRUCTION policies and OSHA standards.

5. ATTACHMENTS

5.1 Permissible Noise Exposure per OSHA 29 CFR 1910.95

5.2 Example Training Record

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Attachment 1: Permissible Noise Exposure per OSHA 29 CFR 1910.95

A-Weighted Sound Level	Permitted Duration Per Workday	A-Weighted Sound Level	Permitted Duration Per Workday
(dB)	(Hours)	(dB)	(Hours)
80	32	106	0.87
81	27.9	107	0.76
82	24.3	108	0.66
83	21.1	109	0.57
84	18.4	110	0.50
85	16.0	111	0.44
86	13.9	112	0.38
87	12.1	113	0.33
88	10.6	114	0.29
89	9.2	115	0.25
90	8.0	116	0.22
91	7.0	117	0.19
92	6.2	118	0.16
93	5.3	119	0.14
94	4.6	120	0.125
95	4.0	121	0.110
96	3.5	122	0.095
97	3.0	123	0.082
98	2.6	124	0.072
99	2.3	125	0.063
100	2.0	126	0.054
101	1.7	127	0.047
102	1.5	128	0.041
103	1.3	129	0.036
104	1.1	130	0.031
105	1.0		

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Attachment 2: Example Training Record

NAME: _____
(Please Print)

- | | <u>INITIAL</u> |
|---|----------------|
| 1. I have been informed about the health hazards associated with exposure to excessive noise levels and the effect of noise on hearing. | _____ |
| 2. I have been informed about the types of work that may result in exposure to excessive noise levels, and the necessary protective steps to prevent exposure, including engineering controls and required safe work practices. | _____ |
| 3. I understand the purpose for, proper use and limitations of hearing protective devices, and I have received instruction on selection, fitting, use, and care of such devices. | _____ |
| 4. I have been informed about the monitoring and medical surveillance programs, including information on audiometric testing and an explanation of the test procedures. | _____ |
| 5. I have received copies of the applicable regulations governing occupational exposure to excessive noise. | _____ |

Employee Signature: _____

Date: _____

Signature of Trainer: _____

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SECTION 23 - ORIENTATIONS

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 All project employees and subcontractor employees must receive a safety orientation and pass any given orientation quiz prior to starting project work activities. The orientation must be documented with the orientation/training checklist and maintained as part of the project safety file. Orientations shall be made job-specific.

2.2 AECOM TISHMAN CONSTRUCTION safety commitment shall be relayed during this orientation. The depth of training shall be commensurate with the job function(s) to be performed. Videos can only be shown in conjunction with face-to-face instruction.

2.3 Items to be covered in the orientation may include, but are not limited to:

- General project overview.
- Emergency procedures.
- Egress and parking routes.
- Project work rules.
- Incident investigation.
- First Aid/Incident/Near Miss reporting.
- Fitness for duty.
- Lost time injury management policy.
- Fall protection policy (6 foot).
- Discipline policy.
- Actions to be expected for failure to comply with safety requirements.
- Parking and site security procedures and regulations.
- Public protection.
- Scaffold program and standards.
- Hazard communication.
- Hazardous chemical inventory list.
- All SDS's on site and submitted before product is used.
- Project safety recognition program.
- Ladder safety.
- Contractual obligation to comply with our safety program.
- Respiratory protection program or requirements.
- Fire protection.
- Confined spaces/permits.
- Job hazard analysis/Task hazard analysis.
- Safety meetings and documentation.
- Hazard recognition and Hazard abatement.
- Housekeeping.
- Personal protective equipment
- Noise Control / Hearing Protection
- First aid provider identification.
- First aid kit location.
- Clothing/jewelry/hair/radio/headphones/earbuds policy.
- Traffic control.

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- Trenches and excavations.
- Barricade/barricade tape (Yellow tape is cautionary; Red tape is mandatory).
- Lockout/Tagout.
- Hot work permits.
- Rigging requirements and Rigger identification.
- Scaffold use.
- 3rd party crane and hoist inspections.
- Project Emergency Action Plan (EAP)

2.4 Attendance Documentation. Records of attendance for all employee and subcontractor safety orientation and training provided as part of this procedure shall be documented on the training attendance record. Files shall be kept onsite.

2.5 Projects shall create their own site-specific orientation.

2.6 If a worker is found without having gone through orientation, that worker shall be removed from the project until the orientation is completed and the SH&E manager approves their return.

3. ATTACHMENTS

None Applicable.

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SECTION 24 - PHASE I & II ENVIRONMENTAL SURVEYS

1. TERMS and DEFINITIONS

1.1 Phase I Assessment. Phase I Assessments are written environmental reports compiled from using information found when researching property records, property transfer records, library records, property disclosure statements, geo-technical reports and site surveys. Typically asbestos, lead based paint, molds, PCB fluorescent light ballasts, mercury controls and CFC's (Freon) may be encountered during these assessments of older buildings. At times you may also find contaminated/hazardous soils, underground storage tanks (UST), bird dropping residue and other environmental materials that must be removed by licensed contractors.

1.2 Phase II Assessment. Phase II Environmental Assessments are written environmental reports further defining the possible existing environmental materials. This includes taking samples of both physical samples and air monitoring samples, transferring these samples to a certified laboratory to determine the results of hazardous material. Utilizing this formal report may define the scope of work as well as the most efficient means of abatement. Only professional licensed contractors shall retrieve samples for phase II assessments.

2. GENERAL REQUIREMENTS

2.1 A Phase I Survey or equivalent is required for all projects.

2.2 A Phase II Survey is required when a Phase I indicates the potential for environmental contamination in any phase of work.

2.3 A signed copy of the Phase I Survey or its equivalent to be retained and kept on file with the project files.

2.4 In the event that an owner does not have an assessment completed during project negotiations, the owner shall be presented with the attached environmental questionnaire for their completion (Attachment 2).

2.5 If there is resistance from an owner with regard to these procedures, the Safety Department is to be contacted immediately prior to acceptance of a project contract.

2.6 Subcontractors are to be made aware of any areas of their work which may be affected by these surveys.

NOTE: Many owners/clients have these assessments completed by their own consultants, in such cases we need only request copies. If an owner refuses to complete a Phase I Assessment in conjunction with AECOM TISHMAN CONSTRUCTION, please notify the Corporate Safety Department.

2.7 Prior to the acceptance of a project contract, AECOM TISHMAN CONSTRUCTION management is to request a written copy of a Phase I Assessment or equivalent from the owner.

2.8 AECOM TISHMAN CONSTRUCTION management is to review the assessment and confer with the Safety Department as to any potential environmental concerns which may need to be addressed or relayed to the labor force.

2.9 If environmental concerns listed warrant further testing, a Phase II Assessment or equivalent is to be completed which outlines sampling steps that will be taken.

2.10 Only after both assessments have taken place will a definition of abatement work be created and subcontracted out.

2.11 File all copies of the surveys with the permanent project files.

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3. ATTACHMENTS

3.1 Phase I Environmental Assessment – General Format

3.2 Phase I Environmental Site Assessment Questionnaire

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Attachment 1: Phase I Assessment - General Format

Phase I **Environmental Site Assessment** **(General Format)**

1. Executive Summary.
2. Introduction.
 - i. Purpose.
 - ii. Special Terms and Conditions.
 - iii. Limitations and Exceptions.
 - iv. Limiting Conditions.
3. Site Description.
 - i. Location and Legal Description.
 - ii. Site and Vicinity Characteristics.
 - iii. Description of Improvements.
 - iv. Information on Liens.
 - v. Current Site Uses.
 - vi. Past Site Uses.
 - vii. Adjacent Properties, Current and Past Uses.
 - viii. Site Map and Plan.
4. Review of Records.
 - i. Federal and State Sources
 - a. Federal National Priorities List - 1.0 mile
 - b. Federal CERCLIS List - 0.5 miles
 - c. Federal RCRA TSD List - 1.0 mile
 - d. Federal Emergency Response List - site only
 - e. State NPL and CERCLIS List - 1.0 mile
 - f. State Waste Disposal Site Lists - 0.5 miles
 - g. State Registered UST Lists - site and adjoining properties
 - h. State Leaking UST Lists - 0.5 miles
 - i. Citations, Administrative Orders
 - ii. Local Sources.
 - a. Lists of Landfill Sites
 - b. Lists of Hazardous Waste Sites
 - c. Lists of Registered UST
 - d. Record of Emergency Response Reports
 - e. Records of Contaminated Wells
 - f. Health Department
 - g. Fire Department
 - h. Planning Department
 - iii. Potential Sources.
 - a. Building Department

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- b. Regional Pollution Control Agency
- c. Regional Water Quality Agency
- d. Electric Utility Companies

iv. Physical Setting - Geologic, Hydrogeologic, Hydrologic, Topographic.

- a. USGS 7.5 Minute Topographic Map
- b. State Groundwater Maps
- c. State Bedrock Geology Maps
- d. Soil Conservation Survey - Soil Maps

v. Historical Uses - Back to at least 1940. Include environmental uses of site and adjacent areas.

- a. Aerial Photographs
- b. Fire Insurance Maps
- c. Property Tax Maps
- d. Recorded Land Title Records
- e. USGS 7.5 Minute Topological Maps
- f. Building Department Records
- g. Zoning Use Records
- h. Existing Site Assessments

vi. Other Sources.

- a. Environmental Assessments on Adjacent Sites
- b. Environmental Audit Reports
- c. Environmental Permits
- d. Registration of Storage Tanks
- e. Material Safety Data Sheets
- f. Company Right-To-Know Plan
- g. Spill Prevention, Countermeasure and Control Plans
- h. Hydrogeologic Reports
- i. Notices of Violations/Administrative Orders
- j. Hazardous Waste Generator Reports
- k. Geotechnical Studies

5. Site Reconnaissance and Interviews.

i. Interview Sources and Document.

- a. Current Owner/Maintenance Personnel
- b. Major Site Occupant
- c. Any Occupant Handling Hazardous or Petroleum Materials
- d. Adjacent Property Owners
- e. Agency Regulatory Personnel

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Attachment 2: Phase I Site Assessment Questionnaire

Phase I Environmental Site Assessment Questionnaire

RE: _____
Project Name

Description of Site: _____

Address: _____

Question

1a. Is the property used for an industrial use? _____ Yes _____ No _____ Ukn

1b. Is any adjoining property used for an industrial use? _____ Yes _____ No _____ Ukn

2a. Did you observe evidence or do you have any prior knowledge that the property has been used for an industrial use in the past? _____ Yes _____ No _____ Ukn

2b. Did you observe evidence or do you have any prior knowledge that any adjoining property has been used for an industrial use in the past? _____ Yes _____ No _____ Ukn

3a. Is the property used as a gasoline station, motor repair facility, commercial printing facility, dry cleaners, photo developing laboratory, junkyard or landfill, or as a waste treatment, storage, disposal, processing, or recycling facility? _____ Yes _____ No _____ Ukn
(if applicable, identify which) _____

3b. Is any adjoining property used as a gasoline station, motor repair facility, commercial printing facility, dry cleaners, photo developing laboratory, junkyard or landfill, or as a waste treatment, storage, disposal, processing, or recycling facility? _____ Yes _____ No _____ Ukn
(if applicable, identify which) _____

4a. Did you observe evidence or do you have any prior knowledge that the property has been used as a gasoline station, motor repair facility, commercial printing facility, dry cleaners, photo developing laboratory, junkyard or landfill, or as a waste treatment, storage, disposal, processing, or recycling facility? _____ Yes _____ No _____ Ukn

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(if applicable, identify which)

4b. Did you observe evidence or do you have any prior knowledge that any adjoining property has been used as a gasoline station, motor repair facility, commercial printing facility, dry cleaners, photo developing laboratory, junkyard or landfill, or as a waste treatment, storage, disposal, processing, or recycling facility?

_____ Yes _____ No _____ Ukn

(If applicable, identify which)

5a. Are there currently any damaged or discarded automotive or industrial batteries, pesticides, paints, or other chemicals in individual containers of >5 gal (19 L) in volume or 50 gal (190 L) in the aggregate, stored on or used at the property or at the facility?

_____ Yes _____ No _____ Ukn

5b. Did you observe evidence or do you have any prior knowledge that there have been previously any damaged or discarded automotive or industrial batteries, or pesticides, paints, or other chemicals in individual containers of >5 gal (19 L) in volume or 50 gal (190 L) in the aggregate, stored on or used at the property or at the facility?

_____ Yes _____ No _____ Ukn

6a. Are there currently any industrial drums (typically 55 gal (208 L)) or sacks of chemicals located on the property or at the facility?

_____ Yes _____ No _____ Ukn

6b. Did you observe evidence or do you have any prior knowledge that there have been previously any industrial drums (typically 55 gal (208 L)) or sacks of chemicals located on the property or at the facility?

_____ Yes _____ No _____ Ukn

7a. Did you observe evidence or do you have any prior knowledge that fill dirt has been brought onto the property that originated from a contaminated site?

_____ Yes _____ No _____ Ukn

7b. Did you observe evidence or do you have any prior knowledge that fill dirt has been brought onto the property that is of an unknown origin?

_____ Yes _____ No _____ Ukn

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8a. Are there currently any pits, ponds, or lagoons located on the property in connection with waste treatment or waste disposal?

_____ Yes _____ No _____ Unk

8b. Did you observe evidence or do you have any prior knowledge that there have been previously, any pits, ponds, or lagoons located on the property in connection with waste treatment or waste disposal?

_____ Yes _____ No _____ Unk

9a. Is there currently any stained soil, asbestos, or lead on the property?

_____ Yes _____ No _____ Unk

9b. Did you observe evidence or do you have any prior knowledge that there has been previously, any stained soil on the property?

_____ Yes _____ No _____ Unk

10a. Are there currently any registered or unregistered storage tanks (above or underground) located on the property?

_____ Yes _____ No _____ Unk

10b. Did you observe evidence or do you have any prior knowledge that there have been previously, any registered or unregistered storage tanks (above or underground) located on the property?

_____ Yes _____ No _____ Unk

11a. Are there currently any vent pipes, fill pipes, or access ways indicating a fill pipe protruding from the ground on the property or adjacent to any structure located on the property?

_____ Yes _____ No _____ Unk

11b. Did you observe evidence or do you have any prior knowledge that there have been previously, any vent pipes, fill pipes, or access ways indicating a fill pipe protruding from the ground on the property or adjacent to any structure located on the property?

_____ Yes _____ No _____ Unk

12a. Is there currently evidence of leaks, spills or staining by substances other than water, or foul odors, associated with any flooring, drains, walls, ceilings, or exposed grounds on the property?

_____ Yes _____ No _____ Unk

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12b. Did you observe evidence or do you have any prior knowledge that there have been previously any leaks, spills, or staining by substances other than water, or foul odors, associated with any flooring drains, walls, ceilings or exposed grounds on the property? _____ Yes _____ No _____ Ukn

13a. If the property is served by a private well or non-public water system, is there evidence or do you have prior knowledge that contaminants have been identified in the well or system that exceed guidelines applicable to the water system? _____ Yes _____ No _____ Ukn

13b. If the property is served by a private well or non-public water system, is there evidence or do you have prior knowledge that the well has been designated as contaminated by any government environmental/heal agency? _____ Yes _____ No _____ Ukn

14. Does the owner or occupant of the property have any knowledge of environmental liens or governmental notification relating to past or recurrent violations of environmental laws with respect to the property or any facility located on the property? _____ Yes _____ No _____ Ukn

15a. Has the owner or occupant of the property been informed of the past existence of hazardous substances (asbestos, lead, PCBs, etc.) or petroleum products with respect to the property or any facility located on the property? _____ Yes _____ No _____ Ukn

15b. Has the owner or occupant of the property been informed of the current existence of hazardous substances (asbestos, lead, PCBs, etc) or petroleum products with respect to the property or any facility located on the property? _____ Yes _____ No _____ Ukn

15c. Has the owner or occupant of the property been informed of the past existence of environmental violations with respect to the property or any facility located on the property? _____ Yes _____ No _____ Ukn

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15d. Has the owner or occupant of the property been informed of the current existence of environmental violations with respect to the property or any facility located on the property?

_____ Yes _____ No _____ Ukn

16. Does the owner or occupant of the property have any knowledge of any environmental site assessment of the property or facility that indicated the presence of hazardous substances or petroleum products on, or contamination of, the property or recommended further assessment of the property?

_____ Yes _____ No _____ Ukn

17. Does the owner or occupant of the property know of any past, threatened, or pending lawsuits or administrative proceedings concerning a release or threatened release of any hazardous substance or petroleum products involving the property by any owner or occupant of the property?

_____ Yes _____ No _____ Ukn

18a. Does the property discharge waste water (not including sanitary waste or storm water) onto or adjacent to the property and/or into a storm water system?

_____ Yes _____ No _____ Ukn

18b. Does the property discharge waste water (not including sanitary waste or storm water) onto or adjacent to the property and/or into a sanitary sewer system?

_____ Yes _____ No _____ Ukn

19. Did you observe evidence or do you have any prior knowledge that any hazardous substances or petroleum products, unidentified waste materials, tires, automotive or industrial batteries, asbestos containing materials or any other waste materials have been dumped above grade, buried and/or burned on the property?

_____ Yes _____ No _____ Ukn

20. Is there a transformer, capacitor, or any hydraulic equipment for which there are any records indicating the presence of PCBs?

_____ Yes _____ No _____ Ukn

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Government Records/Historical Sources Inquiry

21. Do any of the following Federal government record systems list the property or any property within the search distance noted below:

Approximate Minimum Search Distance, miles (kilometers)

Federal NPL site list 1.0 (1.6) _____ Yes _____ No

Federal CERCLIS list 0.5 (0.8) _____ Yes _____ No

Federal CERCLIS NFRAP site list
property and adjoining properties _____ Yes _____ No

Federal RCRA CORRACTS facilities list
1.0 (1.6) _____ Yes _____ No

Federal RCRA non-CORRACTS TSD
facilities list 0.5 (0.8) _____ Yes _____ No

Federal RCRA generators list property
and adjoining properties _____ Yes _____ No

Federal ERNS list property only _____ Yes _____ No

22. Do any of the following state record systems list the property or any
property within the search distance noted below:

Approximate Minimum Search Distance, miles (kilometers)

State lists of hazardous waste sites identified for investigation or remediation:

State — Equivalent NPL 1.0 (1.6) _____ Yes _____ No

State — Equivalent CERCLIS 0.5 (0.8) _____ Yes _____ No

State landfill and/or solid waste disposal
site lists 0.5 _____ Yes _____ No

State leaking UST lists 0.5 (0.8) _____ Yes _____ No

State registered UST lists property and
adjoining properties _____ Yes _____ No

23. Based upon a review of fire insurance
maps 10.3.1.3 or consultation with the
local fire department serving the property,
all as specified in the guide, are any
buildings or other improvements on the
property or on an adjoining property
identified as having been used for an
industrial use or uses likely to lead to
contamination of the property?

_____ Yes _____ No

24. Please describe any known hazardous materials and their locations on the property:

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Thank you for your time in completing this questionnaire. It will assist AECOM TISHMAN CONSTRUCTION in fulfilling our commitments to you, our employees, and others associated with the project.

This questionnaire was completed by:

Name _____

Title _____

Firm _____

Address _____

Phone number _____

Date _____

Preparer represents that to the best of the preparer's knowledge the above statements and facts are true and correct and to the best of the preparer's actual knowledge no material facts have been suppressed or misstated.

Signature _____ Date _____

SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

SECTION 25 - PERSONAL PROTECTIVE EQUIPMENT

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

The use of PPE (Personal Protective Equipment) is a requirement whenever a hazard(s) cannot engineered out of a work process. AECOM TISHMAN CONSTRUCTION employees shall be trained in the proper use of PPE. Subcontractors shall train their employees in the proper use of PPE as well as any task - specific training that might be required for their task(s). The following requirements must be adhered to on all AECOM TISHMAN CONSTRUCTION projects.

2.1 Head Protection - hard hats with the employer's name and/or logo must be worn at all times while on/in the construction site. Hard hats must be a Class B or better which is described in ANSI Standards Z89.1 and 89.2. Hard hats must be worn per manufacturer's recommendations. By 1/1/27 all AECOM TISHMAN CONSTRUCTION staff shall wear Type II hard hats/ safety helmets. Subcontractors are recommended to do the same.

2.2 Work Attire - individuals shall wear long pants in good repair. Athletic pants such as jogging/warm-ups or pants with holes and loose threads are not acceptable. Shirts must have at least a 4" sleeve and must not contain any vulgar or profane wording or depictions imprinted on it.

2.3 Eye Protection - eye protection is mandatory on all AECOM TISHMAN CONSTRUCTION projects at all times. This protection requires the use of ANSI Z87.1 approved eyewear. Prescription/corrective eyewear must also meet ANSI Z87.1 standards. Approved prescription eyewear must have frames, lenses and side shields compliant with ANSI Z87.2+. When necessary, ANSI-approved prescription safety glasses will be purchased by AECOM TISHMAN CONSTRUCTION employees and each employee shall be reimbursed up to \$200.00 for the glasses as a jobsite expense. These glasses may be purchased every two years. Subcontractors are solely responsible for the purchase and cost of prescription safety glasses for their employees.

2.4 Face Protection - face protection must be utilized anytime there is the potential of injury from splashing liquids, flying debris from chipping/grinding and welding/cutting operations. Refer to ANSI Z87.1 for types of protection to be used.

2.5 Hearing Protection - hearing protection shall be worn when noise levels exceed the Permissible Noise Exposure as established by OSHA guidelines. Noise levels and time duration will dictate the type of hearing protection required to be used. Hearing protection shall be used at all decibel levels above 90.

2.6 Personal Fall Arrest System (PFAS) or Fall Restraint System - consists of a full body harness, lanyard, and lifeline. All individuals required to use a PFAS must be trained in its use. The specific requirements for a PFAS are identified in OSHA 29 CFR 1926.502(d).

2.7 High Visibility Apparel/Material - all individuals shall wear vests or apparel with high visibility and reflective material in compliance with ANSI standards while on the project site. Vests may be color-coded based on worker type and/or to distinguish a particular group.

2.8 Life Vests/Buoyant Work Vests - individuals shall wear life vest/buoyant work vest whenever working over or near water. Ring buoys shall be readily available for emergency rescue operations.

2.9 Hand Protection - a minimum ANSI "level two" or EN "level three" cut resistant or greater work gloves shall be worn by all individuals on all projects where the tasks dictate the need to use hands in the

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specific work and where needed for protection against splinters, sharp edges, jagged surfaces, wire rope, glass, metal splinters, and/or other hazards as determined by the operation that is being completed. All individuals should have gloves on their person while on the project. A higher cut resistance level may be required based on the operation. Special purpose work gloves shall be required when employees are performing certain tasks.

Cut Level	Weight in grams needed to cut with 1" blade travel	
	ANSI/ISEA	EN388 (European)
0	<199	<119
1	200-499	120-249
2	500-999	250-499
3	1000-1499	500-999
4	1500-3499	1000-1999
5	>3500	>2000

2.10 Respiratory Protection - AECOM TISHMAN CONSTRUCTION adopts ANSI Z88.2 as our Respiratory Protection program for our workers who may perform tasks requiring respirators. In addition, all subcontractors under our authority/control will also be required to comply with this program.

2.11 Respirators - where respirators are necessary to protect the health of employees, the employer shall establish and implement a written respiratory protection program with site specific procedures. The program shall identify procedures for selecting respirators, medical evaluations of employees, fit testing procedures and training of employees in the proper use of respirators. More specific information in the use of respirators can be located in OSHA standard 29 CFR 1910.134. See attached 29 CFR 1910.134 Appendix D.

2.12 Foot Protection - leather or equivalent work boots are mandatory for all individuals. Protective toe and metatarsal caps are highly recommended. Each THA shall list any further requirements for the task at hand. Sneakers, sandals, high heels, leather - soled street or dress shoes, and thongs will not be considered approved industrial or construction footwear. Metatarsals shall be utilized for all jack hammering and compactor operations as well as those other operations required by OSHA. In the event an AECOM employee is required to wear non-typical foot protection by regulations or the client, AECOM will provide affected employees with boots that meet the applicable requirements. Otherwise, the purchase of footwear is the responsibility of the employee.

2.13 Other - other types of PPE may be required for a task or operation. These will be determined on a case - by - case basis and be listed on the THA.

3. ATTACHMENTS

3.1 OSHA 29 CFR 1910.134 Appendix D

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Attachment 1: OSHA 29 CFR 1910.134 Appendix D

NOTE: The Appendix D, copied from the OSHA website, is provided as a convenience to the user. This information was current at the time of publication. No responsibility is taken or implied for the accuracy of this information.

Part Number:	1910
Part Title:	Occupational Safety and Health Standards
Subpart:	I
Subpart Title:	Personal Protective Equipment
Standard Number	1910.134 App D
Title	(Mandatory) Information for Employees Using Respirators When Not Required Under Standard.

Appendix D to Sec. 1910.134 (Mandatory) Information for Employees Using Respirators When Not Required Under the Standard.

Respirators are an effective method of protection against designated hazards when properly selected and worn. Respirator use is encouraged, even when exposures are below the exposure limit, to provide an additional level of comfort and protection for workers. However, if a respirator is used improperly or not kept clean, the respirator itself can become a hazard to the worker. Sometimes, workers may wear respirators to avoid exposures to hazards, even if the amount of hazardous substance does not exceed the limits set by OSHA standards. If your employer provides respirators for your voluntary use, or if you provide your own respirator, you need to take certain precautions to be sure that the respirator itself does not present a hazard.

You should do the following:

1. Read and heed all instructions provided by the manufacturer on use, maintenance, cleaning and care, and warnings regarding the respirators limitations.
2. Choose respirators certified for use to protect against the contaminant of concern. NIOSH, the National Institute for Occupational Safety and Health of the U.S. Department of Health and Human Services, certifies respirators. A label or statement of certification should appear on the respirator or respirator packaging. It will tell you what the respirator is designed for and how much it will protect you.
3. Do not wear your respirator into atmospheres containing contaminants for which your respirator is not designed to protect against. For example, a respirator designed to filter dust particles will not protect you against gases, vapors, or very small solid particles of fumes or smoke.
4. Keep track of your respirator so that you do not mistakenly use someone else's respirator.

[63 FR 1152, Jan. 8, 1998; 63 FR 20098, April 23, 1998]

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SECTION 26 - PRECAST, CONCRETE, AND TILT-UP

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 Precast concrete wall units, structural framing, rebar walls and columns, and tilt-up wall panels shall be adequately supported to prevent overturning and to prevent collapse until permanent connections are completed.

2.2 Lifting inserts which are embedded or otherwise attached to tilt-up precast concrete members shall be capable of supporting at least two times the maximum intended load applied or transmitted to them. Manufacturer specifications shall be followed at all times.

Tilt-Up and Precast Rigging

2.3 Lifting inserts which are embedded or otherwise attached to precast concrete members, other than tilt-up members, shall be capable of supporting at least four times the maximum intended load applied or transmitted to them.

2.4 Lifting hardware shall be capable of supporting at least five times the maximum intended load applied/transmitted to the lifting hardware. Any lifting hardware which has been shop fabricated needs to have certification paperwork stating the hardware has been weight/load tested and is capable of picking the maximum intended load including safety ratings.

2.5 No individual shall be permitted under precast concrete members being lifted or tilted into position except those employees required for the erection of those members.

Fall Protection

2.6 Each subcontractor shall submit a site-specific fall protection plan for their workers engaged in the erection of precast concrete members (including, but not limited to the erection of wall panels, columns, beams, and floor and roof tees) and related operations such as grouting of precast concrete members, who are 6' or more above lower levels shall be protected from falling by guardrail systems, safety net systems, or personal fall arrest systems.

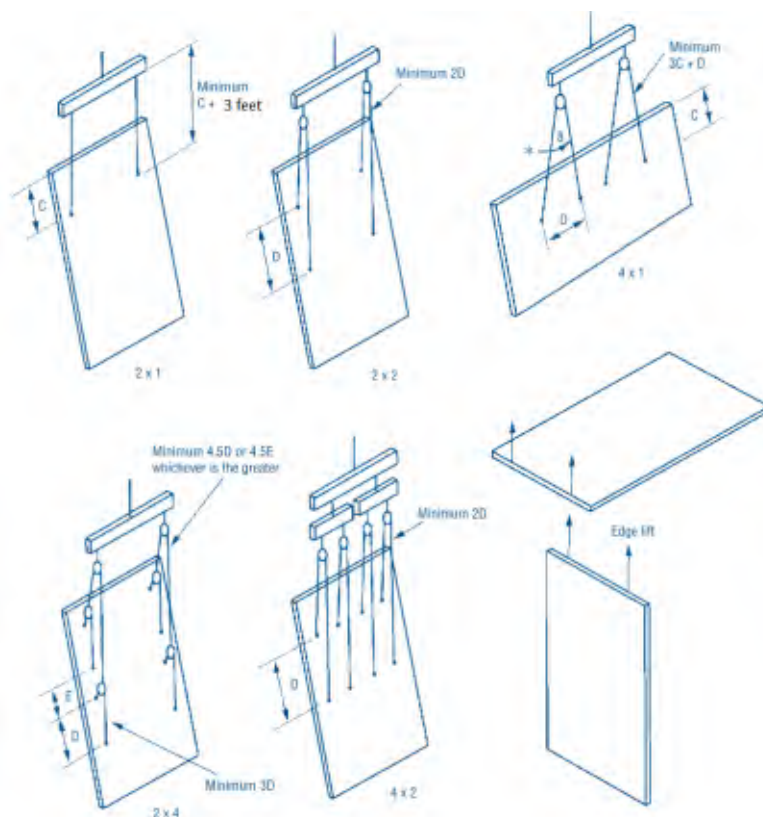
2.7 A JHA shall be provided detailing fall protection methods for installation of precast members. A THA will be completed prior to each crew beginning work each shift, multiple THA's may be required if crew and workplace parameters change during the shift.

2.8 Temporary and/or permanent perimeter safety cables and/or guardrails will be installed at the final interior and exterior perimeters of the precast concrete as soon as installation of precast members, that complete "leading edge" work, is complete.

2.9 In general, the rigging system should be designed to distribute equal loads to all lifting points. In some circumstances the design may require unequal loading on lifting points. Where this is the case, such requirements should be clearly specified in the shop drawings.

2.10 Examples of Rigging Designs.

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2.11 Where fixed length multi-legged slings are to be used to lift an element, the lifting inserts must be designed to carry the full mass of the element on any two lifting inserts.

2.12 To prevent the element slewing sideways during erection, lifting inserts should be located symmetrically about the center of gravity across the width of the element. In determining the center of gravity, the effect of any additional equipment, such as strong backs, needs to be taken into account. When the element is lifted, the bottom edge should be horizontal.

2.13 Lifting inserts may be positioned in the face or edges of a tilt-up or precast concrete element. The actual locations of the lifting inserts are determined according to the:

- Method of lifting (face or edge)
- Mass of the element
- Size of the element
- Shape of the element and presence of openings and cut-outs
- Structural capacity of the element
- Concrete strength at the time of lifting
- Capacity of the lifting inserts, including edge effects and embedment depth

2.14 When it is intended that wall panels be tilted about an edge using anchors placed in the panel face, the geometric center of the face-lift inserts must be above the panel's center of gravity.

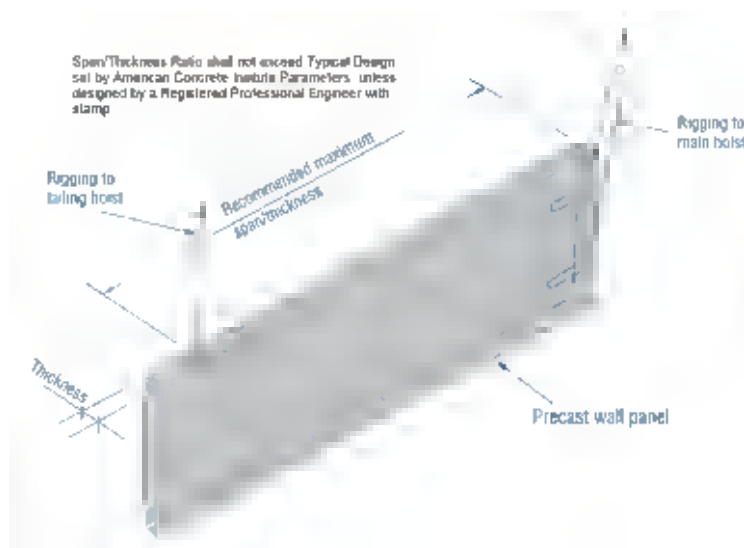
2.15 Face-lifted panels should be designed to hang no more than approximately 10 degrees from the vertical. If this is not possible, consideration should be given to using edge-lifting or a combination of face-lifting and edge-lifting.

2.16 For multi-story construction, precast walls elevated above the ground must be edge-lifted unless there is a working platform on each side of the final location of the element.

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Rotational Lifting

2.17 Buckling and instability can occur during lifting and erection of long, slender elements. Lifting inserts should be located to ensure that compression flange buckling (as in a long slender beam) cannot occur, particularly during rotation of panels with slender side aspects. The span/thickness ratio of the element between lifting points should be limited to typical Span/Thickness ratio in accordance with American Concrete Institute design, unless a Registered Professional Engineer with Stamp has issues specific design along with a detailed buckling analysis is undertaken.



Bracing

2.18 A formal Bracing Plan will be submitted with shop drawings for review. All bracing plans will be reviewed prior to commencement of tilt-up or erection of precast panels by AECOM TISHMAN CONSTRUCTION field management and contracted erector field management and shall be per manufacturer's specifications.

2.19 Panels will be stood up with brace already bolted to vertical portion no higher than $\frac{2}{3}$ of wall height unless designed by a Registered Professional Engineer. Some panels will require multiple bracing struts by design. The JHA and THA process shall coordinate all crew efforts when working around multiple bracing struts per panel.

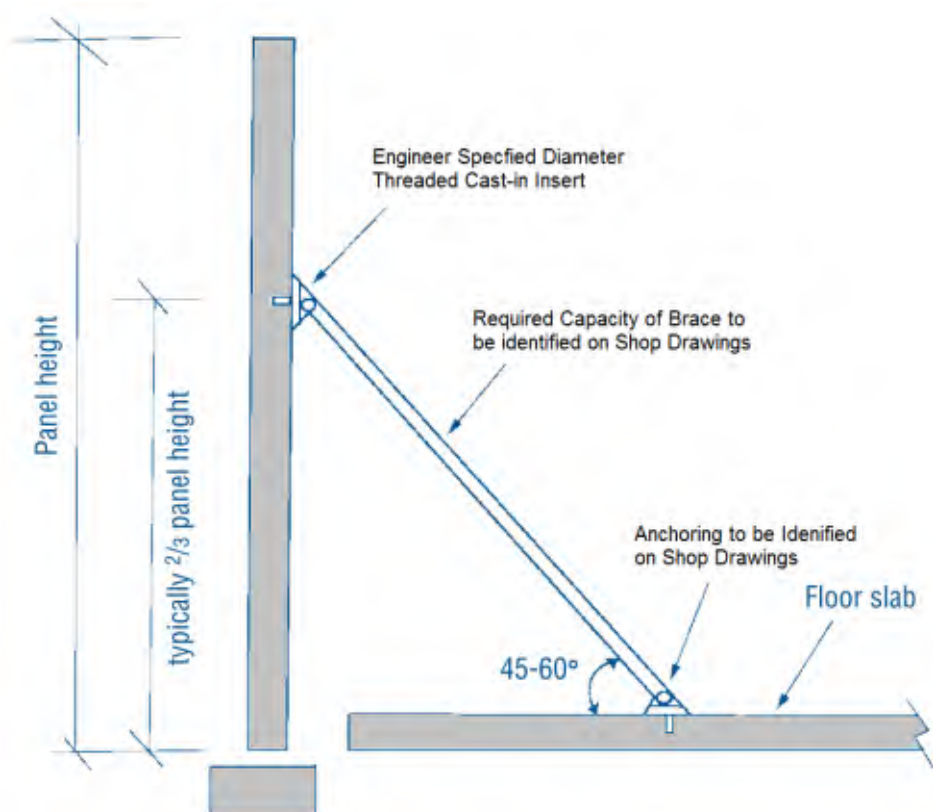
2.20 All bracing angles shall conform to 45 – 60 degrees as noted in diagram below.

2.21 All bracing shall be inspected daily and documented in Project Daily Log or QA/QC log.

2.22 If bracing poses a trip hazard, all efforts will be made by the erecting subcontractor to create, post and maintain a formal barrier keeping non-essential personnel out of the area.

2.23 During erection and bracing a barricade 1 and $\frac{1}{2}$ times the height of the panels will be constructed around the panels to form a drop zone that no employee is permitted until panels are structurally tied in and bracing is removed.

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2.24 All Anchors shall meet the intended manufacturer specification. Double-headed nails shall not be used as an anchor.

Construction Loads and Movement

2.25 Subcontractors must not place loads on any concrete structure until concrete has reached a compressive strength predetermined by the structural engineer for supporting the loads.

2.26 All outrigger platforms on buildings shall be equipped with attached debris nets.

2.27 All loads shall be secured prior to movement.

Reinforcing Steel

2.28 All vertical protrusions at 54" or less including but not limited to reinforcing steel, conduit and stakes must be guarded to eliminate the hazard of impalement.

2.29 Reinforcing steel for walls, piers, columns and other vertical structures must be adequately supported to prevent overturning.

2.30 No employee shall climb the steel or place a ladder to remove the crane hooks until the steel has been adequately supported.

2.31 Reinforcing steel footer mats greater than 4' in height shall have a single walkway provided on top where possible to minimize the chance of an employee falling through for non-rebar workers. (Plywood or equivalent walkway)

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3. FORMWORK/SHORING

3.1 Formwork shall be designed, fabricated, erected, supported, braced and maintained per manufacturer specifications by a licensed professional engineer registered in the project's jurisdiction. Formwork must be capable to safely support all vertical and lateral loads applied to the formwork.

3.2 Concrete subcontractor shall provide drawings stamped by a licensed professional engineer registered in the project's jurisdiction for formwork and falsework for each floor/elevated deck of the project, including typical and non-typical details, and vertical formwork for walls, columns, and stairs. Concrete subcontractor shall provide a location-specific formwork drawing for each floor/elevated deck that supports wet concrete loads and the employees and equipment working above.

3.3 All horizontal formwork shall have a documented inspection prior to the placement of concrete. This inspection will be completed by the concrete subcontractor's superintendent or higher to ensure it has been erected as shown on the drawing. The inspection document shall meet or exceed the requirements of Attachment 2.

3.4 Fabricated formwork (shoring) shall not be loaded beyond the safe working load recommended by the manufacturer.

3.5 All shoring equipment (including reshoring) will be inspected prior to erection to make sure it meets the requirements specified in the formwork drawings.

3.6 Any damaged equipment will be tagged out of service and removed from the area to prevent reuse. Damage includes, but is not limited to, bent components, excessive corrosion, cracked welds and broken locking devices on frames and braces.

3.7 Erected shoring equipment shall be inspected immediately prior to, during and immediately after concrete placement. Any shoring that is found to be damaged or displaced shall be reinforced to prevent failure.

3.8 At least one employee will be responsible to monitor shoring during concrete placement. All other employees will be kept from underneath concrete pour.

3.9 Visual delineation shall be installed to keep people away.

3.10 Base plates, shore heads, extension devices and adjustment screws shall be in firm contact with the footing sill and form material and shall be secured to them when necessary.

3.11 Reshoring shall be provided to safely support slabs and beams after stripping or where such members are subjected to superimposed loads due to construction. Reshoring will be immediately installed after shoring is removed. Daily documented inspections should confirm that no movement or loosening has occurred and that reshores are plumb and true.

3.12 Wherever single post shores are used in more than one tier, the layout shall be designed by a licensed professional engineer registered in the project's jurisdiction and inspected by the concrete subcontractor's superintendent.

3.13 Fabricated single post shores and adjusting devices shall not be used if heavily rusted, bent, broken welds or other defects.

3.14 Post shores shall be positively secured to prevent them from falling over. When used on the outside slab edges of a building, post shores will be tied back to secure them from falling.

3.15 Removal or relocation of reshores shall only be completed by the concrete subcontractor. All attempts shall be made to limit workers in the reshore areas to concrete workers and/or other specially trained individuals.

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4. CONCRETE PLACEMENT

4.1 A visual inspection of concrete buckets must be done prior to each use.

4.2 Buckets shall be routed so that employees, or the fewest number of employees, are exposed to the hazards associated with falling concrete buckets.

4.3 An audible warning signal (e.g. whistle, horn, etc.) loud enough to warn workers shall be provided each time a concrete bucket enters the pour area or staging area.

4.4 An eye wash kit or station shall be provided in the immediate area of all poured or mixing concrete operations.

5. CONCRETE PUMP TRUCKS/PUMPING SYSTEMS

5.1 A copy of the manufacturer's operation manual must be located in the truck or onsite for each pump or placing boom.

5.2 If during an equipment inspection a deficiency is observed, the pump shall not return to service until corrected.

5.3 A JHA is required for all pump truck setups. A concrete pump shall have a clearly labeled emergency stop switch that stops the pumping action and moving parts.

5.4 If quick connect clamps are used, they must be pinned or secured to keep from opening.

5.5 At no time shall any employee stand on the grate or the grate lid of any pump truck.

5.6 Employees responsible for backing up concrete trucks to the pump shall never stand in a caught-between position.

5.7 Concrete pump trucks shall only be cleaned out by the operator of the pump truck. Concrete pump systems shall be cleaned out under the supervision of qualified employees. Each washout station shall be designed to prevent concrete from contaminating soil.

5.8 No pipe bends or flexible delivery hose may be connected to the end of the pipeline during the blowout process. The point of discharge must be controlled and clear of employees. All sections must be secured to prevent whipping.

5.9 When using concrete pump trucks, outriggers shall be fully extended unless otherwise specified by the manufacturer. If the manufacturer does allow pumping without full extension of all outriggers, then the following must be completed:

- Flags must be used on the extended outriggers to designate the safe working area.
- A documented meeting must take place before any short-rigging of pump trucks on site.

5.10 Riding and walk behind trowel machines must be equipped with a control switch that will automatically shut off the power whenever the hands of the operator are removed from the equipment handles or off the seat.

6. REMOVAL OF FORMWORK

6.1 Forms and shores (except those used for slab on grade) shall not be removed until the concrete subcontractor determines that the concrete has gained sufficient strength to support its weight and superimposed loads. Results shall be submitted to AECOM TISHMAN CONSTRUCTION.

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6.2 Concrete subcontractor must not remove any horizontal formwork or shoring until a determination has been made by the testing lab and engineer that the concrete has gained sufficient strength to support its own weight and that of superimposed loads.

6.3 Employees are not permitted on formwork or falsework after the formwork removal begins, such as the removal of horizontal bracing between posts.

6.4 Concrete subcontractor is not permitted to have more floors of form materials available onsite than bracing materials (e.g. can't have 1-1/2 floors of forms but only 1 floor of bracing).

6.5 Reshoring shall not be removed until the concrete being supported has attained adequate strength to support its weight and all loads in place upon it.

6.6 Concrete subcontractor shall provide stripping criteria and reshore calculations and drawings stamped by a licensed professional engineer registered in the project's jurisdiction for each floor/elevated deck of the project including, but not limited to post spacing and reshoring requirements at cantilevered conditions, beams, drops, and other unique conditions.

7. POST-TENSIONING

7.1 No workers, other than those essential to post-tensioning operations, shall be allowed to be behind the jack during post tensioning operations.

7.2 Signs and a warning line/physical barrier shall be posted to limit access to post-tensioning areas during tensioning operations. (Danger/Caution tape shall not be used).

7.3 No employees shall be behind the jack while tensioning cables.

7.4 Prior to drilling and cutting into any concrete slab that contains post-tensioning cables, subcontractors shall request documentation from AECOM TISHMAN CONSTRUCTION on location of cables and do the necessary testing and exploration (using a method approved by the Structural Engineer) to validate the actual location of any post-tensioned elements.

8. PRECAST CONCRETE

8.1 No employee shall be permitted under precast concrete members being lifted or tilted into position except those employees required for the erection of those members.

8.2 A precast erection plan must be submitted to AECOM TISHMAN CONSTRUCTION at least two weeks prior to the start of work. This plan shall include, but not be limited to:

- a) Transportation and storage of precast elements.
- b) Crane requirements (Refer to Crane Operations Section).
- c) Lifting insert/rigging specifications.
- d) Rigging design.
- e) Erection sequence.
- f) Temporary bracing.
- g) Permanent connections.
- h) Safe working distances for other subcontractors.
- i) Fall protection plan.

8.3 Loads may only be unstrapped when the truck is located in the final position.

8.4 Install hole covers on precast floor members prior to lifting into place.

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9. MASONRY CONSTRUCTION

9.1 A limited access zone shall be established whenever a masonry wall is being constructed. This zone shall meet all of the following:

- a) Established for all standing masonry walls.
- b) Be located on the non-scaffold side of the wall and run the entire length of the wall.
- c) Shall be equal to the height of the wall plus 4 feet.
- d) Marked off for access to only those involved in the construction of the wall.
- e) Remain in place until the wall is adequately supported or structurally tied-in to prevent overturn or collapse.
- f) Shall be monitored.

9.2 All masonry walls 8 feet in height and over shall be adequately braced to prevent overturning and collapse. The bracing shall remain in place until the wall is adequately tied into the structure.

10. ATTACHMENTS

10.1 One-Sided Support for a Concrete Pump

10.2 Pre-Pour Formwork Inspection Checklist Example

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Attachment 1: One-Sided Support for a Concrete Pump

One-Sided Support for a Concrete Pump

This practice creates unique safety hazards, and specific safety rules must be followed.

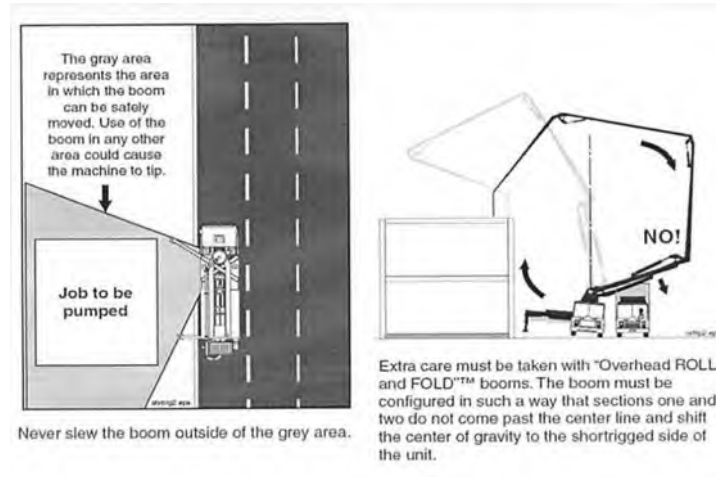


Figure 1

- NOTE: The front outriggers may only be jacked while they are fully extended or fully retracted. Serious outrigger damage will result from jacking in an intermediate position.
- Do *not* position the boom outside the center lines of the fully extended outriggers (Figure 1). When the center of gravity is beyond the outriggers, the unit can tip.
- With an overhead Roll and Fold™ boom, you must be careful to not retract the boom so far that the center of gravity will be on the short rigged side (Figure 1).
- Remember that not all of the outriggers are extended. Forgetting is the most probable cause of an accident.
- Never short rig the machine on the side of the pour.
- Attach a safety flag on each fully extended outrigger that you are to stay between.
- On swing out type outriggers that don't extend, swing out enough to not damage the outrigger lock when you jack up the truck.

Pour Date: _____ Unit#: _____

Sub Operator: _____ Foreman: _____

Safety: _____ Supt: _____

When short rigging, this form must be signed prior to unfolding the boom and placed in the Driver's front window during the pour.

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Attachment 2 : Pre-Pour Formwork Inspection Checklist Example

Pre-Pour Formwork Inspection Checklist Example

Job Name: _____ **Job #** _____ **Date:** _____

Location: _____ **Job Supervisor Signature:** _____

CHECKLIST	Y/N	COMMENTS
1. Are scaffolding frames plumb or as shown on plan?		
2. Is all bracing installed?		
3. Are strings completely seated in J-heads?		
4. Are heads pinned to staffs or screws?		
5. Are strings clipped to heads at joists?		
6. Are screws on bottom of scaffold set in scaffold feet?		
7. Are post shores being substituted for shoring legs?		
8. Are staffs or screws extended beyond plan length?		
9. Are the combinations of accessories correct?		
10. Are scaffold connections correct?		
11. Are stringers to sub-stringer correct?		
12. Are beams in good condition and if not have they been removed and replaced?		
13. Are the cantilevers of joists or stringers correct?		
14. Are speed sets, staffs and brace pins correct and in good condition? Have any cracked speed sets, bent staffs, or missing brace pins been removed and replaced?		
15. Are blocking, bracing, lacing, and cabling per drawings?		
16. Are post shores, Ellis jacks, or single leg posts over extended?		
17. Is the brace size and scaffold spacing per drawing?		
18. Are the reshores the proper size and spaced per drawings?		
19. Are heavily loaded areas reshored per drawings?		
20. Is blocking under stringer or joists secured?		
21. Are stringer and joist spaced per drawings?		

Additional comments:

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SECTION 27 - ROPES, SLINGS, & CHAINS

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 The use of ropes, slings, and chains shall be in accordance with the safe usage recommendations of the manufacturer at all times.

2.2 Hooks used to support loads shall be equipped with safety latches to prevent components from slipping out of the hook.

2.3 Safety latches on hooks shall not be deactivated or made inoperable.

2.4 Qualified riggers shall determine proper placement of materials when moving and lifting the items.

2.5 Hooks, shackles, rings, pad eyes and other fittings that show excessive wear or that have been bent, twisted or otherwise damaged shall be removed from service. Excessive wear is defined as the condition where the original strength has been reduced by twenty (20) percent.

2.6 Lines running horizontally within six feet six inches (6' - 6") of the ground or working level must be boxed-off or guarded or the area must be restricted.

2.7 Slings and their fittings and fastenings, when in use, shall be inspected daily by a qualified craft person for evidence of overloading, excessive wear or damage. Excessive wear is the condition where the original strength has been reduced by twenty (20) percent.

2.8 Defective ropes, slings, and chains shall be removed from service and rendered inoperable.

2.9 Proper storage shall be provided for ropes, slings, and chains while not in use.

2.10 Protection softeners shall be provided between a sling and sharp, unyielding surface of the load to be lifted.

2.11 Wire rope shall be removed from hoisting and load carrying service when one of the following conditions exists:

- In running lines with twelve randomly distributed broken wires in one rope lay, or
- Four broken wires in one strand in one rope lay. A rope lay is the distance in which one strand makes one complete turnaround.

2.12 All ropes, slings and chains shall have a documented inspection as well as permanently- affixed and legible identification markings as prescribed by the manufacturer that indicate the recommended safe working load. Individuals using these shall know the capacities.

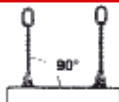
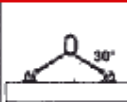
3. ATTACHMENTS

3.1 Capacity Examples

3.2 Identification Tag Examples

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Attachment 1: Capacity Examples

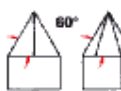
CHAIN SLING CAPACITIES (LBS.) - CHAIN GR-8 - ASME B30.9 DESIGN FACTOR 4/1							
CHAIN SIZE (IN.)						 QT ALLOY	
CHAIN GR - 8 DESIGN FACTOR 4/1	VERTICAL (SINGLE LEG)	TWO LEG OR BASKET HITCH	60 DEGREE SLING ANGLE	45 DEGREE SLING ANGLE	30 DEGREE SLING ANGLE	SINGLE LEG MASTER LINK SIZE (IN.)	DOUBLE LEG MASTER LINK SIZE (IN.)
1/4 - (9/32)	3500	7000	6050	4900	3800	1/2	1/2
3/8	7100	14200	12200	10000	7100	3/4	3/4
1/2	12000	24000	20750	15950	12000	7/8	1
5/8	18100	36200	31350	25500	18100	1	1-1/4
3/4	28300	56600	49000	40000	28300	1-1/4	1-1/2
7/8	34200	68400	59200	48350	34200	1-1/2	1-3/4
1	47700	95400	82600	67450	47700	—	—
1-1/4	72300	144600	125200	102200	72300	—	—



A CHAIN GRAB HOOK APPLICATION WILL RESULT IN A 20% REDUCTION OF CHAIN CAPACITY OF A SINGLE LEG. THE HORIZONTAL ANGLE MUST BE GREATER THAN 30 DEGREES.



HORIZONTAL ANGLE	CAPACITY % OF SINGLE LEG
90	200%
60	170%
45	140%
30	100%





TRIPLE LEG SLINGS HAVE 50% MORE CAPACITY THAN DOUBLE LEG ONLY IF THE CENTER OF GRAVITY IS IN CENTER OF CONNECTION POINT AND LEGS ARE ADJUSTED PROPERLY (EQUAL SHARE OF THE LOAD). QUAD LEG SLINGS OFFER IMPROVED STABILITY BUT DO NOT PROVIDE INCREASED LIFTING CAPACITY.

ROPE Ø	REEVE LOAD			Round Basket Load			Oblong Basket Load			2,3 or 4 Leg Sling		
mm	Straight	Round	Oblong	0°	60°	90°	0°	60°	90°	0°- 60°	90°	120°
8	0.55	0.41	0.27	1.09	0.94	0.77	0.55	0.48	0.39	0.94	0.77	0.55
9	0.69	0.52	0.34	1.38	1.19	0.97	0.69	0.60	0.49	1.19	0.97	0.69
10	0.85	0.64	0.43	1.70	1.47	1.20	0.85	0.74	0.61	1.47	1.20	0.85
11	1.03	0.77	0.52	2.10	1.78	1.45	1.03	0.90	0.73	1.78	1.45	1.03
12	1.23	0.92	0.61	2.50	2.10	1.73	1.23	1.07	0.87	2.10	1.73	1.23
13	1.44	1.08	0.72	2.90	2.50	2.00	1.44	1.25	1.02	2.50	2.00	1.44
14	1.67	1.25	0.83	3.30	2.90	2.40	1.67	1.45	1.19	2.90	2.40	1.67
16	2.20	1.64	1.09	4.40	3.80	3.10	2.20	1.90	1.55	3.80	3.10	2.20
18	2.80	2.10	1.38	5.50	4.80	3.90	2.80	2.40	1.87	4.80	3.90	2.80
20	3.40	2.60	1.70	6.80	5.90	4.80	3.40	3.00	2.40	5.90	4.80	3.40
22	4.10	3.10	2.10	8.30	7.10	5.80	4.10	3.60	2.90	7.10	5.80	4.10
24	4.90	3.70	2.50	9.80	8.50	6.90	4.90	4.30	3.50	8.50	6.90	4.90
26	5.80	4.30	2.90	11.5	10.0	8.10	5.80	5.00	4.10	10.0	8.10	5.80
28	6.70	5.00	3.30	13.4	11.6	9.40	6.70	5.80	4.70	11.6	9.40	6.70
32	8.70	6.50	4.40	17.4	15.1	12.3	8.70	7.60	6.20	15.1	12.3	8.70

SYNTHETIC WEB SLINGS

VERTICAL (90°) LIFTING CAPACITIES

Type 3
Web Sling

Type 4
Web Sling

Type 5
Web Sling

Flat
Eye

Tucked
Eye

Endless

SINGLE PLY (ONE PLY)

SINGLE PLY (ONE PLY)

WIDTH	VERTICAL	CHOKER	BASKET
1"	1,600 lbs	1,200 lbs	3,200 lbs
2"	3,200 lbs	2,400 lbs	6,400 lbs
3"	4,800 lbs	3,600 lbs	9,600 lbs
4"	6,400 lbs	4,800 lbs	12,800 lbs
6"	9,600 lbs	7,200 lbs	19,200 lbs

WIDTH	VERTICAL	CHOKER	BASKET
1"	3,200 lbs	2,550 lbs	6,400 lbs
2"	6,400 lbs	5,100 lbs	12,800 lbs
3"	9,400 lbs	7,050 lbs	18,800 lbs
4"	12,400 lbs	9,300 lbs	24,800 lbs
6"	18,600 lbs	14,000 lbs	37,200 lbs

DOUBLE PLY (TWO PLY)

DOUBLE PLY (TWO PLY)

WIDTH	VERTICAL	CHOKER	BASKET
1"	3,200 lbs	2,400 lbs	6,400 lbs
2"	6,400 lbs	4,800 lbs	12,800 lbs
3"	8,800 lbs	6,600 lbs	17,600 lbs
4"	11,500 lbs	8,600 lbs	23,000 lbs
6"	16,500 lbs	12,400 lbs	33,000 lbs

WIDTH	VERTICAL	CHOKER	BASKET
1"	6,400 lbs	5,100 lbs	12,800 lbs
2"	12,400 lbs	9,900 lbs	24,800 lbs
3"	17,400 lbs	13,200 lbs	35,200 lbs
4"	22,000 lbs	17,600 lbs	44,000 lbs
6"	33,000 lbs	24,800 lbs	66,000 lbs

SLING-CHOKER MANUFACTURING

For non-vertical application, refer to below chart to calculate angle strength loss from the rated capacities

When using these slings, multiply the **RATED CAPACITY** by the **FACTOR** that corresponds with the **ANGLE** of your lift.

ANGLE	FACTOR	ANGLE	FACTOR	ANGLE	FACTOR
85°	0.996	65°	0.906	45°	0.707
80°	0.985	60°	0.866	40°	0.643
75°	0.966	55°	0.819	35°	0.574
70°	0.94	50°	0.766	30°	0.5

SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

Attachment 2: Identification Tag Examples

Chain Sling



Replacement Tag



SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

SECTION 28 - SAFETY PLAN MANAGEMENT RESPONSIBILITIES

1. TERMS and DEFINITIONS

1.1 AECOM TISHMAN CONSTRUCTION. AECOM TISHMAN CONSTRUCTION and all of its subsidiaries and joint ventures in which AECOM TISHMAN CONSTRUCTION has an ownership interest.

1.2 Area/Regional/Divisional Safety Manager (A/RSM). An AECOM TISHMAN CONSTRUCTION employee assigned safety and health oversight responsibilities in a division or area in a region of the company.

1.3 VP of SHE (Safety, Health, and Environment). An AECOM TISHMAN CONSTRUCTION employee assigned responsibilities for overall administration and oversight of the company safety and health program.

1.4 Superintendent/ Senior Superintendent. An AECOM TISHMAN CONSTRUCTION employee assigned specific project oversight responsibilities within a region or division.

1.5 Estimator. An AECOM TISHMAN CONSTRUCTION employee assigned project estimating responsibilities within a region or division.

1.6 General Foreman/ Foreman. An AECOM TISHMAN CONSTRUCTION employee directly responsible for work and control of craft persons in their area of responsibility.

1.7 Officer in Charge (OIC). An AECOM TISHMAN CONSTRUCTION employee assigned project responsibilities for a project(s) and personnel oversight areas of the company.

1.8 Project Engineer/Office Engineer. An AECOM TISHMAN CONSTRUCTION employee assigned project engineering and/or office oversight responsibilities on a specific or multiple projects.

1.9 Project Executive (PX)/Project Director (PD). An AECOM TISHMAN CONSTRUCTION employee assigned responsibilities for overall management operations of a single or multiple projects within a region or division.

1.10 Project Manager (PM)/Senior Project Manager (SPM). An AECOM TISHMAN CONSTRUCTION employee assigned specific administrative and/or management responsibilities of a single or multiple projects within a region or division.

1.11 Purchasing Agent. An AECOM TISHMAN CONSTRUCTION employee assigned purchasing and/or subcontractor relations responsibilities within a region or division.

1.12 Regional Executive Officer (REO), Division President, or Executive Vice President (EVP). An AECOM TISHMAN CONSTRUCTION employee assigned responsibilities for overall operations of a region or an entire division of the company.

1.13 Regional Safety Director (RSD). An AECOM TISHMAN CONSTRUCTION employee assigned safety and health oversight responsibilities over a region of the company.

1.14 Safety Carpenter. An AECOM TISHMAN CONSTRUCTION trade employee or subcontractor assigned safety and health duties on a specific project.

1.15 Senior/Project Safety Manager/Coordinator. An AECOM TISHMAN CONSTRUCTION employee assigned safety and health oversight responsibilities over a specific or multiple projects in a region or division of the company.

2. RESPONSIBILITIES

SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

2.1 Regional Executive Officer (REO), Division President, and Executive Vice President (EVP)

1. Lead by example.
2. Implement a culture for establishing a positive attitude towards safety by all AECOM TISHMAN CONSTRUCTION personnel.
3. Market AECOM TISHMAN CONSTRUCTION's safety program to owners.
4. Ensure region's compliance with AECOM and AECOM TISHMAN CONSTRUCTION's safety program and policies.
5. Ensure adequate funding for the safety program in the annual business plan and establish annual safety budget with the Regional/Area Safety and Health Director for regional safety activities.
6. Participate in safety-related programs, including seminars, incident review meetings, safety award meetings, and other safety awareness-oriented functions.
7. Make safety the first agenda item for discussion at all staff meetings.
8. Include the appropriate safety staff member in project proposals, reviewing potential projects, and pre-planning, as necessary.
9. Require safety inspections and observations by onsite Project Executives, Project Directors, Project Managers, and Superintendents, accompanied by a write-up distributed to the appropriate Safety and Health Director and/or Project Safety Manager. This requirement is part of the monthly safety report.
10. Ensure all client contract safety sections are reviewed with the Regional Safety Director prior to signing the contract to ensure proper understanding of the safety implications in the contract.
11. Meet with the assigned Regional/Area Safety Director on a regular basis (at least monthly) to discuss safety in the region.
12. Include the RSD/ASD in all business planning meetings and activities.
13. Ensure all required personal safety observations and inspections are completed and hold staff accountable.
14. Complete all monthly required senior management activities (SMAs). All SMAs must be documented.

2.2 Officer in Charge (OIC)

1. Lead by example.
2. Maintain an environment where safety is a core value.
3. Ensure that business development and others work with the owner in the acquisition of studies (including asbestos, mold, and lead surveys) on all projects.
4. Provide effective disciplinary support when necessary to reinforce AECOM TISHMAN CONSTRUCTION's commitment to safety when working with non-compliant staff and/or subcontractors.

SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

5. Utilize safety accountability/scoring for staff annual review/bonus purposes. Include the appropriate Safety and Health Director in reviewing the staff safety performance.
6. Ensure cooperation and support of the Regional/Area Safety and Health Director by all staff.
7. Reformat project outcome reports (project profitability/performance reports) to include a reduction in earnings due to losses incurred to any safety related costs that have impacted specific projects.
8. Make safety attitude, management and implementation a primary focus on performance evaluations.
9. Develop a procedure to establish consistency of safety evaluations of key personnel, including management, engineering, and field staff, from project-to-project.
10. Review safety at all Operation Review Meetings.
11. Start all meetings with 5 or more people with a safety moment.
12. Ensure that staff personnel are given clear job descriptions and performance criteria, including adequate and proper training to implement AECOM TISHMAN CONSTRUCTION's safety policies as required by their job responsibilities.
13. Participate in safety-related training programs, incident review meetings, safety award meetings, and other safety-related functions. Keep current on new legislative policies for OSHA, and new AECOM policies and requirements.
14. Determine, in conjunction with the Regional/Area Safety and Health Director, whether a full-time project safety manager is necessary on new projects per AECOM TISHMAN CONSTRUCTION policies.
15. Ensure all required personal safety observations and inspections are completed.
16. Ensure participation by all AECOM TISHMAN CONSTRUCTION employees in all required safety and health training.

2.3 Project Executive (PX) / Project Director (PD) / Project Manager (PM) / Senior Project Manager (SPM)

1. Lead by example.
2. Work with the pre-construction team to ensure an adequate safety budget is included.
3. During project set-up, evaluate specific project exposures and risks. Implement safety, pre-planning programs to properly mitigate risks during subcontract buy-out. Involve Regional/Area Safety Director in project pre-planning activities.
4. Conduct pre-construction staff meetings establishing goals and responsibilities, in addition to reemphasizing AECOM TISHMAN CONSTRUCTION's mission to be the leader in construction safety.
5. Include field staff early in projects to review plans for safety concerns during the pre-planning phase.

SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

6. Conduct a weekly safety inspection of each project under their direction and make a written report to the Regional/Area Safety Director. (Can be a checklist type form). If the PX or PD is not stationed on the project, it may be monthly.
7. Work with Business Development to ensure that a Phase I Environmental Study is conducted on all potential new work where applicable.
8. Ensure project staff is completing required safety responsibilities throughout the duration of the project.
9. Participate in safety-related training programs, incident review meetings, safety award meetings, and other safety-related functions. Keep current on all new legislative policies for OSHA, and new AECOM policies and requirements.
10. Provide support and enforcement of safety policies on each jobsite.
11. Ensure project staff attends safety-related meetings and training functions.
12. Pursue the acquisition of subcontractor safety programs and SDS sheets as necessary.
13. Provide administrative support for the field safety staff.
14. Work with the Safety Department to determine proper distribution of safe-start documents from the project team.
15. Ensure all required personal safety observations and inspections are completed.
16. Communicate and preplan safety into all project phases.
17. Start all meetings with 5 or more people with a safety moment.
18. Ensure a robust project rewards and recognition program for all individuals.

2.4 Purchasing Agent/Team/Assigned Personnel

1. Before a contract is awarded, ensure subcontractors have submitted their current safety qualification information and have an EMR of 1.0 or less. If greater than 1.0, evaluate risks associated with each subcontractor and include money and resources to manage that risk. Project Executive and Regional/Area Safety Director must approve subcontract award to any subcontractor with an EMR greater than 1.0. Contracts may be executed if the EMR is greater than 1.0 only when the RSD has planned a course of action for the subcontractor for the project. No subcontractor shall be considered if they are pending meeting prequalification requirements.
2. On traditional AECOM TISHMAN CONSTRUCTION projects verify subcontractors have all the appropriate insurance.
3. Assist in the proper administration of the AECOM TISHMAN CONSTRUCTION insurance program, if applicable. Provide direction and necessary documentation in the field for the administration of the program.
4. Ensure all appropriate safety exhibits are included in the contract, i.e. specific project hazards and mitigation measures.
5. Ensure this general overview safety program and safety exhibits are included in the subcontractors' bid and contract documents.

SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

6. Implement the AECOM TISHMAN CONSTRUCTION subcontractor prequalification program with safety performance being one of the primary evaluation criteria.
7. Coordinate with the Estimators, the project staff responsible for developing requisitions, and the Regional/Area Safety Director on the scope of safety requirements and how they will be purchased.
8. Ensure that general and critical safety provisions are reviewed as a standard agenda item at every scope review meeting.
9. Start all meetings with 5 or more people with a safety moment.
10. Ensure all current subcontractor safety staffing requirements are included in contracts per AECOM policy. Discuss with the Regional Safety Director.

2.5 Estimator

1. Ensure an adequate safety budget, including cost for Regional/Area Safety and Health Director and specific line items for safety management depending on project exposures. Include a budget for jobsite safety recognition programs.
2. Verify the owner has included an environmental survey in project documents, and obtain copies of all pertinent reports, documents, etc.
3. Discuss and review site-specific safety requirements with the Project Management and the Regional/Area Safety Director via a Pre-Estimate Risk Assessment.
4. Coordinate with the Purchasing Department and the Regional/Area Safety Director on the scope of safety requirements and how they will be purchased.
5. Participate in safety education training seminars as directed by the Safety Department.
6. Identify high-risk activities during budget development. Report findings to Regional/Area Safety Director and procurement staff so appropriate systems may be purchased effectively.

2.6 Project Engineer/Office Engineer

1. When on the jobsite, enforce and uphold all AECOM TISHMAN CONSTRUCTION safety policies.
2. Design safety into all project processes.
3. Assist in overall safety program implementation.
4. Notify the Safety Manager of new subcontractors.
5. Assist the Project Managers in their responsibilities.
6. Ensure all required personal safety observations and inspections are completed.

2.7 Superintendent / Senior Superintendent

1. Lead by example.

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2. Responsible for taking leadership role on their project and for implementing AECOM TISHMAN CONSTRUCTION's safety policies and procedures, including the scheduling and participation in safety pre-planning meetings as established at the initial safety pre-planning meeting. Include a line item for a safety pre-planning meeting in all schedules to be located before each phase of work and/or before any critical job or process.
3. Supervise, manage and require compliance to the project safety program by all personnel working on the project.
4. Support and implement the AECOM TISHMAN CONSTRUCTION Disciplinary Action Policy for project safety program violations, especially fall protection, as a lead person responsible for the project. For more information, refer to the Disciplinary Policies section of this program.
5. Inform risk on all new projects to the Regional/Area Safety Director.
6. Conduct pre-planning meetings with the Regional Safety Director and/or designated Safety Manager prior to the start of new phases of construction.
7. Participate in the development of the Site-Specific Project Safety Program, Fire Prevention and Protection Program, Emergency Action Plan, and additional safety programs, as required.
8. Create a Site Logistics Plan (layout) of the project.
9. Coordinate Safe-Start safety meetings with all subcontractors prior to their start of work.
10. Distribute Site-Specific Project Safety Program, Fire Prevention and Protection Program, Emergency Action Plan, and additional safety programs as required to each subcontractor before they start work and discuss their active participation in the incident prevention effort.
11. Set up the project trailer/office to be compliant with federal, state, and local regulations.
12. Provide the project trailer/office with a first aid kit, fire extinguishers, exit signs, and an evacuation route.
13. Coordinate safety orientation for project staff and all trades persons. The orientation process must include documentation with a hardhat sticker, and/or other identification system. Safety staff will assist in the orientations whenever possible and participate in the orientations regularly.
14. Walk the job on a daily basis and pay special attention to any hazardous conditions. Conduct a formal weekly documented inspection.
15. In the event a safety staff member is not assigned to a project, the Project Superintendent shall act as the safety coordinator for the duration of the project if none is assigned as a permanent project position by the Safety Department. The Superintendent is still responsible for job site safety requirements.
16. Implement a Task Hazard Analysis (THA) system for all jobs and/or processes to be completed by all crews.
17. Must complete a minimum of the OSHA 30 Hour Outreach in Construction course as part of Superintendent job qualifications. The Superintendent must complete all required AECOM safety training and regionally required governmental training such as NYC-DOB requirements.
18. Conduct a monthly project safety meeting and weekly coordination staff meeting with safety as the first topic of discussion. Assign safety to an operations staff member on a weekly rotation. The

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Operations staff member will discuss safety at the weekly meeting. Subjects for discussion should cover but are not limited to:

- Superintendents' observations regarding safety.
 - Reports of the Project Safety Manager and actions taken on any recommendations.
 - Incidents which have occurred during the past month and methods of eliminating or protecting against them.
 - Conditions and/or actions that may affect the public and methods for correcting them.
 - Communication from the Safety Department.
 - Reports of foreman's Tool Box Safety Meetings.
 - Adherence to Hazard Communication Policy.
19. Issue safety information to job staff, foremen and subcontractors once a month concerning safety subjects pertinent to the job.
20. Require that each AECOM TISHMAN CONSTRUCTION and subcontractor's foreman hold Tool Box Safety Meetings with their crew at least once each week to discuss the following:
- Minutes of staff safety meeting as they affect the work.
 - Instruct the employees in safe and efficient planning of their work.
 - The safety subject assigned at the staff safety meeting. Safety subjects shall be pertinent to the current work activity.
 - Injuries or near misses that have occurred to their employees.
 - Solicit comments and suggestions relating to safety.
 - Issued Safety Department communications (as directed by the RSD).
 - Minutes shall include dates and signatures.
21. Require all persons employed on the job, AECOM TISHMAN CONSTRUCTION employees, and subcontractors to wear proper hard hats, safety glasses, reflective vests, footwear, gloves and clothing as a condition of employment (also includes visitors and employees of owners who are in AECOM TISHMAN CONSTRUCTION work areas per our Personal Protective Equipment (PPE) policy).
22. Provide, require (and properly train where necessary) the use of personal protective equipment by all AECOM TISHMAN CONSTRUCTION employees.
23. Require that all subcontractors provide their employees with the proper safety equipment required by the project safety program, and federal, state, and local requirements.
24. Review and sign off on (for review purposes only) all major JHAs, safety required permits, and/or crane documentation as determined by the Safety Department.

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25. Require all subcontractor crew foreman / leaders / superintendents to comply with our safety documentation requirements.
26. Start all meetings with 5 or more people with a safety moment.
27. Ensure all required personal safety observations and inspections are completed.

2.8 Vice President of Safety, Health, and Environmental (SHE)

1. Lead by example.
2. Maintain an overview of performance of the entire AECOM TISHMAN CONSTRUCTION group on a summary basis and communicate the results to the Regional/Area Safety and Health Directors and Upper Management.
3. Develop and implement, in cooperation with Upper Management, true safety accountability for all employees. Safety performance will be a major determinant in success and progression of each employee's career.
4. Provide all Regional/Area Safety and Health Directors/Managers any OSHA/EPA/AECOM changes or updates to the Safety, Health and Environmental Regulations as issued.
5. Provide liaison with such organizations as ANSI, NSC, NCSE, and AGC to ensure AECOM TISHMAN CONSTRUCTION is current on new developments.
6. Review and evaluate methods, products, and devices to make recommendations regarding their use.
7. Provide advice on special problems and be alerted regarding serious incidents, OSHA citations, etc.
8. Follow the guidelines of subcontractor pre-qualification procedures.
9. Develop and maintain relations with Federal OSHA and local agencies as necessary.
10. Issue safety communication and other informative documents.
11. Assist all regions and divisions on business development as needed.
12. Approve or disapprove payment of all OSHA or other regulatory citations.
13. Coordinate, oversee, and implement all AECOM TISHMAN CONSTRUCTION programs.
14. Conduct periodic Project Safety Reviews.

2.9 Regional/Area Safety Director (RSD/ASD)

1. Lead by example.
2. Work with Business Development personnel and the Project Executive/Director to ensure that an Environmental Survey be completed on all new projects.
3. Conduct an initial pre-planning meeting with job staff responsible for creating trade requisitions. Develop safety scope for purchasing and develop safety goals/targets for life of the project.

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4. Require and approve safety and environmental job-specific safety programs for all projects prior to commencement of work.
5. Establish and implement emergency action plans for the region and project sites.
6. Monitor project site programs and ensure a robust rewards and recognition program is in place and being used.
7. Monitor overall construction safety performance through project management, managers and staff.
8. Require that monthly safety reports are prepared by each job and are submitted to the Safety Department Analyst no later than the 5th of each month. The Regional/Area Safety and Health Director is to ensure that each project has received safety reporting guidelines, which contain all the necessary documents for the development of the monthly safety report.
9. Monitor and ensure that each project is inspected using the Project Safety Evaluation (PSE) process at least one time per quarter. This is a minimum mandatory requirement. Audits are being appraised in accordance with the safety rating standards and will be reviewed with the superintendent/safety manager/coordinator at the conclusion of the process.
10. Inspect each project as necessary before/while environmental remediation is taking place.
11. Follow up with projects for correcting hazards noted by inspection of insurance carriers, AECOM TISHMAN CONSTRUCTION staff, local, state and federal agencies or others.
12. Attend senior staff meetings to report on/review safety and environmental issues.
13. Keep current on all new legislative policies, AECOM TISHMAN CONSTRUCTION and AECOM policies and requirements, and current industry trends. Keep personnel informed of any new or changed local, state or federal environmental guidelines or laws. Participate with local associations.
14. Attend progress meetings or safety meetings as deemed necessary.
15. Review incident, safety, and environmental reports. Investigate all serious/fatal and/or catastrophic accidents, notify appropriate parties, and maintain permanent file.
16. Establish and make available safety and environmental training. Develop annual safety training planner and submit to VP of SHE.
17. Maintain safety and environmental reference materials.
18. Provide projects with necessary reporting forms and posters as needed.
19. Establish and maintain liaison with insurance carriers for loss prevention and claims service, as necessary.
20. Establish and maintain liaison with appropriate local, state or federal agencies.
21. Maintain region-wide incident log that tracks the project incident history of all subcontractors. Update incident log monthly and distribute to the Safety Department. Meetings shall be held with substandard subcontractors to discuss deficiencies and inquire as to a recommended course of action to improve their safety performance. Subcontractors shall be advised that poor safety

SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

performance will have an adverse effect on their ability to conduct future business with AECOM TISHMAN CONSTRUCTION and AECOM. Conduct follow up meetings as necessary.

22. Oversee the interview and hiring process for full-time safety professionals when required on our projects.
23. Work with purchasing to meet the subcontractor prequalification requirements.
24. Conduct safety observations and inspections.
25. Ensure all safety policies and procedures are implemented in your region/area.

2.10 Project Safety Manager/Coordinator

1. Lead by example.
2. Assist in the development of the Project Safety Program, which must receive approval from the Regional/Area Safety and Health Director.
3. Establish and conduct jobsite orientation, in conjunction with the Superintendent, for every new individual on the project. Administer and record their participation in the orientation program and issue identification to those individuals completing the orientation.
4. Attend all initial meetings with project staff and subcontractor representatives to clearly define their role within the confines of the Project Safety Program.
5. Conduct and document Safe-Start meetings with each subcontractor, safety representative and/or foremen to establish safety procedures prior to subcontractors' activity on site.
6. Establish and conduct regular (weekly) safety meetings with subcontractor representatives and issue minutes of meeting. Interface with project staff and each subcontractor safety representative relating to safety regulations to ensure proper compliance.
7. Ensure that subcontractors are conducting the proper training requirements as per the OSHA standards. Subcontractors are solely responsible for this training.
8. Ensure and maintain a log of each subcontractor's toolbox safety meetings held with their employees.
9. Review each subcontractor's safety program and ensure that it meets or exceeds the Project Safety Program requirements as applicable. Ensure the THA process is implemented.
10. Ensure that each subcontractor designates a safety representative that is trained per OSHA standards, competent for the subcontractor's scope of work, follows AECOM TISHMAN CONSTRUCTION guidelines, and has the proper authority to correct safety issues and hazards relating to their safety compliance. Receive the names of their competent person(s) for their specific work in writing and file.
11. Conduct regular (daily) jobsite and work area inspections. Conduct formal weekly jobsite inspections and complete the safety checklist noting safety violations and corrective actions.
12. Conduct safety observations and near miss reports.

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13. Record, notify, and prepare written reports of any violations or unsafe practices to subcontractors for immediate correction actions. Give copies to Project Manager, Project Superintendent, Subcontractor, Project Director and Regional Safety and Health Director, as appropriate per project.
14. Stop any violation or unsafe practice where there is imminent danger to life or property at once. Report immediately to the Project Superintendent and Regional/Area Safety and Health Director.
15. Monitor the subcontractor's compliance with testing and recording all potential unsafe environmental conditions such as air quality, noise levels and chemical exposures.
16. Assist Project Superintendent in establishing and implementing proper fire prevention, evacuation, and fire control procedures.
17. Investigate all serious and near miss incidents and generate proper reports to be distributed to Project Superintendent, Project Manager and Regional/Area Safety and Health Director.
18. Establish and maintain all job safety records, including accident and incident reports, investigations and required OSHA reports.
19. In conjunction with the Superintendent and Project Manager, establish and maintain a safety recognition program for the project that will award employees and/or subcontractors that excel in safety excellence. Inform the RSD/ASD of the program.
20. Organize and contact the proper safety representatives from each subcontractor if necessary for inspections by governmental agencies.
21. Conduct a monthly project safety meeting in conjunction with the Superintendent to present the state of the project safety programs and overall project safety compliance to all relevant project management.
22. Compile a written monthly project report by the 5th of each month, which addresses the previous month's recordable and lost time injuries. Provide the same information for project history from the beginning to current date. Also, categorize the types of injuries consistent with OSHA reporting requirements. Maintain a running total of safe hours worked by all jobsite personnel compared to total man hours worked on site from the beginning of the project to the current date.
23. Assist in the management of claims as necessary.
24. Attend the Project Staff Meeting to brief the staff on safety issues on the project and to keep informed of the progress of the job.
25. In addition to the above requirements, the Owner Controlled Insurance Program (OCIP), the Owner, and/or Regional/Area Safety and Health Director may add other duties.

2.11 Safety Carpenter

1. May be defined as a Project Safety Coordinator when assigned.
2. Work with the Superintendent in maintaining project safety measures.
3. Conduct safety inspections and meetings as directed.
4. Correct hazards on the project.
5. Conduct safety observations.

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6. Follow the direction of the Superintendent and Safety Manager.
7. Meet with the Safety Manager and Superintendent on a regular basis.
8. Participate in daily toolbox talks and THA processes.

2.12 General Foreman/ Foreman

1. Lead by example.
2. Utilize the method of Task Hazard Analysis (THA) to include Safety Task Assignments as necessary for all members of their crew. Demonstrate and explain to workers the safety and health precautions and actions that must be taken before proceeding with tasks.
3. Have all tools, materials, equipment, and personal protective equipment needed to perform the job safely on hand prior to the start of work.
4. Ensure that individuals have been trained in the proper use, care, and inspection of all safety equipment.
5. Screen individuals under their supervision for craft knowledge, current job knowledge, safety and health knowledge, and physical limitations to better assign the right person to the tasks at hand.
6. Check individuals at the start of each shift to be sure they are ready to perform their assigned tasks. Be on the lookout for unusual symptoms of a physical, mental, or emotional nature that could interfere with employee performance.
7. Follow-up with employees to be sure they are inspecting their tools/equipment as well as following the established safety requirements and safe work practices. Correct any non compliance.
8. Coordinate planning and work activities with others performing work within or adjacent to the assigned work areas and cooperate with all on-site safety personnel.
9. Hold weekly Toolbox Talks with crew using and completing the Toolbox Talk forms provided by the company. Return the completed form to the Superintendent or Project Safety Manager.
10. Ensure that prompt first aid is provided for each injured employee and promptly report all incidents to the Superintendent or Project Safety Manager.
11. Cooperate with the Project Safety Manager in completing the Incident Investigation Report.
12. Correct all hazardous conditions as identified.

2.13 All AECOM TISHMAN CONSTRUCTION Employees

1. Lead by example.
2. All AECOM TISHMAN CONSTRUCTION personnel must adapt, enforce, and follow AECOM and AECOM TISHMAN CONSTRUCTION's safety policies.
3. Use all required safety equipment.
4. Speak up when a safety hazard is recognized.

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5. Seek out and obtain all required and necessary safety training.
6. Be held responsible and accountable for their own action and/or inactions.
7. Support the disciplinary actions for project safety program violations as established by the lead person on the project.
8. Conduct safety observations.

2.14 Subcontractor Employees

1. All subcontracting personnel are required to meet or exceed all of AECOM and AECOM TISHMAN CONSTRUCTION's safety and health policies, as well as all federal, state, and local policies in addition to their own company program.
2. Speak up when a safety hazard is recognized.
3. Are solely responsible for their own actions and/or inactions.

3. ATTACHMENTS

None Applicable

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SECTION 29 - SCAFFOLDING

1. TERMS and DEFINITIONS

1.1 Competent Person. One who is capable of identifying existing and predictable hazards and authorized to take prompt corrective measures to eliminate them.

1.2 Qualified Person. One who is recognized by a degree, certificate or professional standing, or possesses extensive knowledge, training and experience.

2. GENERAL REQUIREMENTS

2.1 In light of the numerous types of scaffolds available for use, the information contained below is the general requirements for all scaffolds. All scaffolding must be inspected prior to each shift by a competent person. For more in-depth information pertaining to a specific scaffold, please review OSHA 29 CFR 1926.451.

2.2 Footings or anchorage shall be sound, rigid, and capable of carrying the maximum intended load without settling or displacement. Unstable objects shall not be used to support scaffold or planks.

2.3 No scaffold shall be moved, erected, dismantled, or altered except under the supervision of competent person. The scaffold must be designed by a qualified person and constructed and loaded in accordance with that design. Scaffold manufacturer's recommendations must be reviewed and followed.

2.4 Guardrails and toe-boards shall be installed on all open sides and ends of platforms more than six feet (6') above the ground or floor.

2.5 Guardrails shall be two inches by four inches (2" x 4") or equal, approximately forty-two inches (42") high with a one-inch by six inches (1" x 6") mid rail. Supports shall not be spaced more than eight feet (8') apart. Toe-boards shall be a minimum of four inches (4") high.

2.6 Where persons are required to work or pass under a scaffold, a screen shall be installed between the toe-board and guardrail, extending the entire opening, of Number 18 U.S. Standard Wire, one half inch (1/2") mesh or solid plywood, debris netting or equal.

2.7 Each scaffold and scaffold component shall be capable of supporting, without failure, its own weight and at least four times the maximum intended load applied or transmitted to it.

2.8 Any scaffold component or accessory damaged or weakened shall be immediately repaired or replaced.

2.9 All scaffold planking shall be "Scaffold Grade." All scaffold planks shall be maintained in good condition. Planks shall not be used if:

- Cracks exceed 1.5 times the width of the board.
- Notches are deeper than 1/3 the width of the plank.
- Saw kerfs are present.

2.10 All planking shall overlap a minimum of twelve inches (12") or be secured from movement.

2.11 Swing gates, diversions, or similar means shall be installed on fixed ladders on supported scaffolds at each working level to allow for a safe transition from the ladder to the working platform. (This avoids having to climb over or through guardrails/cross braces to access each level) A landing platform shall be provided at intervals not to exceed thirty-five feet (35').

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2.12 Scaffold planks shall extend over their end supports not less than six inches (6") or more than twelve inches (12") unless they are secured to preclude any movement.

2.13 Poles, legs, or uprights shall be plumb and securely and rigidly braced to prevent swaying and displacement.

2.14 Overhead protection shall be provided for workers on a scaffold exposed to overhead hazards.

2.15 Slippery conditions shall be eliminated as soon as possible after they occur.

2.16 Climbing of braces is prohibited.

2.17 Scaffold systems shall not be used for purposes other than as intended and designed for by the manufacturer unless subcontractor provides drawings stamped by a licensed Professional Engineer registered in the project's jurisdiction.

2.18 A scaffold tagging system shall be used in which all scaffolds are tagged by the competent person. Tags shall be secured at each access point and shall be color-coded accordingly:

- Green: Indicates the scaffold has been inspected and is safe to use.
- Yellow: Indicates the scaffold has restrictions for use (e.g. areas that require use of a personal fall arrest system).
- Red: Indicates the scaffold is unsafe to use.

2.19 Suspension scaffolds shall have inspection tags placed on each suspension scaffold support device (i.e. outrigger beams) in addition to the inspection tag placed on the platform of the suspended scaffold.

2.20 Tags shall be readily visible, made of materials that will withstand the environment in which they are used, be legible and shall include:

- The competent person's name and signature.
- Dates of initial and last inspections.

2.21 Direct connections to roofs and floors, and counterweights used to balance adjustable suspension scaffolds, shall be capable of resisting at least 4 times the tipping moment imposed by the scaffold operating at the rated load of the hoist, or 1.5 (minimum) times the tipping moment imposed by the scaffold operating at the stall load of the hoist, whichever is greater.

2.22 All scaffold heights over one hundred twenty-five feet (125') and non-typical scaffolds shall be designed by a registered Professional Engineer (PE).

2.23 Scaffolds in use by any persons shall rest upon a suitable footing and shall stand plumb. The casters or wheels shall be locked to prevent any movement.

2.24 Each platform on all working levels shall be fully planked between the uprights and the guardrail supports.

2.25 The use of shore or lean-to scaffold is strictly forbidden.

2.26 Scaffolds and scaffold components shall be inspected for visible defects by a competent person before each shift and after any occurrence that could affect a scaffold's structural integrity.

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2.27 Debris shall not be allowed to accumulate on scaffold platforms. Individuals shall be prohibited from working on scaffolds covered with snow, ice, or other slippery material except as necessary for removal of such materials.

2.28 Legs shall be set on adjustable bases or plain bases placed on mudsills or other foundations adequate to support the maximum rated load.

2.29 Only material that can be used in one (1) work shift shall be stored on any scaffold provided it does not exceed the acceptable weight for the type of scaffold in use.

2.30 A competent person shall be in charge of all scaffold operations at all times.

2.31 Scaffold components manufactured by different manufacturers shall not be intermixed unless components fit together, and a competent person approves the procedure.

2.32 Fall protection must be provided for employees erecting or dismantling scaffolds where the installation and use of protection is feasible and does not create a greater hazard. A competent person shall determine the feasibility and safety of providing fall protection. If it is determined that the work cannot be done by using a conventional fall protection system, a written fall protection plan must be developed and submitted to the AECOM TISHMAN CONSTRUCTION Regional Safety Director for approval, prior to performing the work. See the Fall Protection section of this program for further requirements on general fall protection.

2.33 Scaffolds shall not be erected or used in the immediate vicinity of power lines or electrical conductors until such are insulated, de-energized, or otherwise rendered safe against accidental contact. The clearance between scaffolds and power lines shall be as follows:

Insulated lines, Voltage	Minimum Distance	Alternatives
Less than 300 volts 300 - 50 kv More than 50 kv	3 feet 10 feet 10 feet plus 4 inches for each 1 kv > 50 kv.	2 times the length of the line insulator, but never less than 10 feet.
Uninsulated lines, Voltage	Minimum Distance	Alternatives
Less than 50 kv More than 50 kv	10 feet 10 feet plus 4 inches for each 1 kv > 50 kv.	2 times the length of the line insulator, but never less than 10 feet.

EXCEPTIONS: Scaffolds and material may be closer to power lines than specified above where such clearance is necessary for performance of work, but only after the utility company, or electrical system operator has been notified of the need to work closer and the utility company or electrical system operator has de-energized the lines, relocated the lines, or installed protective coverings to prevent accidental contact with the lines.

2.34 A competent person must be onsite at all times when scaffolds are in use.

2.35 Atypical scaffolds (e.g. unique scaffold systems for atriums, pools, gyms, concourses, etc.) shall be designed by a registered Professional Engineer and shall be constructed and loaded in accordance with such design.

2.36 Each scaffold platform and walkway shall be at least 18 inches (20 inches in California) wide.

2.37 There shall be more than 14 inches from the face of the building to the edge of the edge plate.

2.38 Makeshift devices, such as but not limited to boxes and barrels, shall not be used on top of scaffold platforms to increase the working level height of employees.

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2.39 Ladders shall not be used on scaffolds to increase the working level height of employees, except on large area scaffolds where the following criteria are satisfied:

- When the ladder is placed against a structure that is not part of the scaffold, the scaffold shall be secured against the sideways thrust exerted by the ladder.
- The platform units shall be secured to the scaffold to prevent movement.
- The ladder legs shall be on the same platform or other means shall be provided to stabilize the ladder against unequal platform deflection, and the ladder legs shall be secured to prevent slipping or being pushed off the platform.

2.40 Wind/debris screens shall consist of flame - resistant material. Scaffolds shall also be designed to support the forces imposed by the wind/debris screens.

2.41 Guys, ties, and braces shall be installed according to the scaffold manufacturer's recommendations or at the closest horizontal member to the 4:1 height and be repeated vertically at locations of horizontal members every 20 feet or less thereafter for scaffolds 3 feet wide or less, and every 26 feet or less thereafter for scaffolds greater than 3 feet wide. The top guy, tie, or brace of completed scaffolds shall be placed no further than 4:1 height from the top (3:1 in California). Such guys, ties and braces shall be installed at each end of the scaffold and at the horizontal intervals not to exceed 30 feet (measured from one end [not both] towards the other). No wood means shall be used as a tieback.

2.42 Adequate mud sills or other rigid footing, capable of withstanding the maximum intended load, must be used. All stationary metal scaffold legs, including those of outriggers, shall rest upon base plates available from the manufacturer for this service. When the scaffold or outrigger is resting on earth or soft material, the base plates shall rest on and be secured to the equivalent of a 2-inch by 10-inch by 10-inch wooden base.

2.43 Engineered plans shall be required prior to erecting or altering suspended scaffolds. Pre-engineered plans from the manufacturer are acceptable only if the setup is done exactly per the manufacturer's design, specifications and procedures. Engineered plans are required when making any alteration to the manufacturer's pre-engineered plan.

2.44 All parts of all suspended scaffolds shall have a minimum safety factor of four (4). A minimum safety factor of six (6) is required for support ropes.

2.45 A 14" distance shall be maintained from the face of the building to the edge of the plate.

3. FALL PROTECTION AND FALLING OBJECT PROTECTION

3.1 All provisions for access and fall protection shall be delineated in the JHA/THA prior to commencing any activity that requires work in elevated areas 6 ft or greater in height (including scaffolds). All plans shall be submitted to AECOM TISHMAN CONSTRUCTION for review prior to commencing work.

3.2 Fall protection shall be provided for individuals erecting or dismantling scaffolds where the installation and use of protection is feasible and does not create a greater hazard as mentioned above.

3.3 Personal fall arrest systems used on scaffolds shall be attached by lanyard to a vertical lifeline, horizontal lifeline, or scaffold structure member as permitted by the manufacturer.

3.4 When vertical lifelines are used, they shall be fastened to a fixed safe point of anchorage, shall be independent of the scaffold, and shall be protected from sharp edges and abrasions. Safe points of anchorage include structural members of buildings, but do not include standpipes, vents, other piping systems, electrical conduit, outrigger beams, or counterweights.

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3.5 When working from single-point or two-point adjustable suspension scaffolds, employees shall be protected by both a vertical lifeline (independent from the scaffold and its anchorage point) and a guardrail system.

3.6 Guardrail systems shall be installed along all open sides and ends of platforms. Guardrail systems shall be installed before the scaffold is released for use by employees.

3.7 In addition to wearing hard hats, each employee on a scaffold shall be provided with additional protection from falling hand tools, debris, and other small objects through the installation of toe-board, screens or through the erection of debris nets, catch platforms, or canopy structures that contain or deflect the falling objects. When falling objects are too large, heavy, or massive to be contained or deflected by any of the above listed measures, such potential falling objects shall be placed away from the edge of the surface from which they could fall and secured as necessary to prevent their falling.

3.8 Where there is danger of tools, materials, or equipment falling from a scaffold and striking employees below, the following provisions shall apply:

- The area below the scaffold to which objects can fall shall be barricaded, and employees shall not be permitted to enter the hazard area, or
- A toe-board and screen (mesh or equivalent) shall be provided and secured between the toe-board and the guardrail extending from the toe-board or platform to the top of the guardrail.
- A guardrail system shall be installed with openings small enough to prevent passage of potential falling objects.
- Tool tethers shall be used wherever work must be completed at height and there is potential for a tool to fall to a lower level.

3.9 A canopy structure, debris net, or catch platform strong enough to withstand the impact forces of the potential falling objects shall be erected over the employees below. Canopies when used for falling object protection, shall comply with the following criteria:

- Canopies shall be installed between the falling object hazard and the employees.
- When canopies are used on suspension scaffolds for falling object protection, the scaffold shall be equipped with additional suspension ropes.

3.10 Baker scaffolds shall have guardrails attached at heights of four (4) feet or more.

4. TRAINING REQUIREMENTS

4.1 Scaffolds shall be erected, moved, dismantled, or altered only under the supervision and direction of a competent person qualified in scaffold erection, moving, dismantling or alteration. Such activities shall be performed only by experienced and trained employees selected for such work by the competent person.

4.2 Each individual involved in erecting, disassembling, moving, operating, using, repairing, maintaining, or inspecting a scaffold shall be trained by a competent person to recognize any hazards associated with the work in question. The training shall include the following topics, as applicable:

- The nature of scaffold hazards.

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- The correct procedures for erecting, disassembling, moving, operating, repairing, inspecting, and maintaining the type of scaffold in question.
- The design criteria, maximum intended load-carrying capacity, and intended use of the scaffold.
- Any other pertinent requirements of the scaffold.

4.3 Proof of training for any individual or competent person shall be made available upon request.

4.4 When there is reason to believe that an individual or competent person lacks the skill or understanding needed for safe work involving the erection, use or dismantling of scaffolds, each such individual shall be retrained as required per OSHA regulations.

5. ATTACHMENTS

5.1 Scaffold Tag Examples

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Attachment 1: Scaffold Tag Examples

Front

SCAFFOLD # _____
SCAFFOLDING IDENTIFICATION TAG
Date Erected: _____
Expected Removal Date: _____
Job Number: _____
I have inspected and approved the Scaffold built and consider it to be safe and adequate for completion of the work specified.
Inspected By: _____
Date: _____

REINSPECTED	
NAME	DATE

MODIFICATION DATE	
NAME	DATE

Back

SCAFFOLDING IDENTIFICATION TAG
Erected by: _____
SAFE FOR USE
DO NOT ALTER DO NOT OVERLOAD

Front

SCAFFOLD # _____
SCAFFOLDING IDENTIFICATION TAG
Date Erected: _____
Expected Removal Date: _____
Job Number: _____
I have inspected and approved the Scaffold built and consider it to be safe and adequate for completion of the work specified.
Inspected By: _____
Date: _____

REINSPECTED	
NAME	DATE

MODIFICATION DATE	
NAME	DATE

Back

SCAFFOLDING IDENTIFICATION TAG
The following client company representative authorizes the use of this scaffold subject to fulfillment of the conditions listed under the preventive measures section of this tag.
NAME: _____
CAUTION
POTENTIAL OR UNUSUAL HAZARD
WHAT IS THE POTENTIAL HAZARD: _____
PREVENTIVE MEASURES TO BE TAKEN: _____

Front

SCAFFOLD # _____
SCAFFOLDING IDENTIFICATION TAG
Date Erected: _____
Expected Removal Date: _____
Job Number: _____
I have inspected and approved the Scaffold built and consider it to be safe and adequate for completion of the work specified.
Inspected By: _____
Date: _____

REINSPECTED	
NAME	DATE

MODIFICATION DATE	
NAME	DATE

Back

SCAFFOLDING IDENTIFICATION TAG
Erected by: _____
STOP
UNSAFE FOR USE
UNDER ERECTION ☐
BEING DISMANTLED ☐
REPAIRS REQUIRED ☐
OVERHEAD PROTECTION ONLY ☐

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SECTION 30 - SITE/WORKPLACE ACCESS AND SECURITY

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 A site logistics plan shall be developed for all projects and updated regularly to reflect evolving conditions. Controlled access to the project and to personnel hoists shall be established. Overhead protection shall be provided in access areas where necessary. Site security shall be reviewed and documented. Example site security plan templates are included in the attachments.

2.2 All access to the project site, both vehicular and pedestrian shall be kept clear at all times. Superintendents, subcontractors, supervisor and management staff shall continually monitor the site for compliance.

2.3 Vertical access shall be by stair tower or approved vertical hoisting device. Job - made ladders are not permitted over heights of four feet.

2.4 Project Superintendent and Project Safety Manager shall identify and post fire and emergency access and egress routes as construction proceeds.

2.5 The Project Superintendent shall establish an Emergency Response Plan emphasizing how to escort/direct emergency vehicles utilizing AECOM TISHMAN CONSTRUCTION employees equipped with two-way communications. This plan shall include start of escort/direction beginning at the main entrance to the project. This plan shall be updated as project conditions change.

2.6 Proper signage shall be posted on the access routes.

2.7 Access routes into buildings with potential overhead falling hazards shall be properly covered or have overhead protection. Local standards and regulations may apply.

2.8 All individuals working on the project shall have verified project orientation prior to gaining access. Visitors shall be escorted and sign a visitor's release form.

2.9 If security guards are used, the project superintendent or designee should request a copy of the post orders for the security guards from the security company. The superintendent or designee should review the post orders to ensure they are appropriate. AECOM Global Security & Resilience is available to assist in the review.

2.10 Controlled Access Zones (CAZs) must be included in the site logistics plan. CAZs shall have signage, be flagged, or barricaded and kept clean and clear of all non-essential materials and personnel. Plastic, colored chains or 3/8 mylar yellow rope with warning sign should be used in lieu of tape.

2.11 For additional information and resources, refer to the [Global Security and Resilience](#) page on Ecosystem.

3. ATTACHMENTS

3.1 Example Site Security Plan Template

3.2 Example Site Security Plan Template Instructions

3.3 Example of Completed Site Security Plan Template

Construction Management Site Security Plan

Project Information

1. AECOM Operating Unit:
2. Project name:
3. Client:
4. Location of Project:
5. Approximate number of people working on the site:
6. Approximate size of the project site:

Site Security Plan

Perimeter security

1. What Fencing is in place:
2. What gates are present onsite (Type and Number):
3. How is access to the site controlled:
4. Information on security guards:
5. Information on security lighting on site:
6. Information on cleared area outside of project site perimeter:
7. Information on CCTV/intrusion detection measures:

Security incident reporting

8. Include those in project management to be notified:
9. Include others in AECOM to be notified:
 - All security incidents to be entered into IndustrySafe as security incident.

Security incident response

10. How are people on site notified of a security incident:
11. Procedure for how to deal with an unauthorized person on site:
12. Procedure for how to deal with an unruly/belligerent/argumentative person on site:
13. Under what conditions would you want the personnel on site to take a specific action:

Resilience/Crisis Management Plan

1. Attach a copy of the project's site resilience/crisis management plan.
 - Listing of the project emergency response team including positions and identifying individuals in those positions
 - Incident Management Team Lead
 - Incident Management Alternate Team Lead
 - Safety
 - Security
 - HR
 - Comms
 - Legal

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Attachments

1. Expanded Emergency notification call list
 - Include AECOM Construction Management emergency notification phone number.
2. Emergency information distribution process
3. Unauthorized persons on the site response process
4. Belligerent or argumentative persons on site response process
5. Site layout drawings, maps (both street and satellite)

Instructions for: Construction Management Site Security Plan

Project Information

1. AECOM Operating Unit: *List the AECOM Operating Unit.*
2. Project name: *List the name of the project.*
3. Client: *List the client's name.*
4. Location of Project: *List the location of the project. Be as descriptive as possible.*
5. Approximate number of people working on the site: *List the number of people who work on the site.*
6. Approximate size of the project site: *List the approximate size of the project.*

Site Security Plan

Perimeter security

1. What Fencing is in place: *Provide the specifics on any fencing that is in place (size, construction, is the fence topped? If yes with what, how is the fence secured?)*
2. What gates are present onsite (Type and Number): *How many person gates, vehicle gates and where are they located on site?*
3. How is access to the site controlled: *Provide information on how access to the site is controlled.*
4. Information on security guards: *Number of guards, armed vs unarmed, guard posts, guard escalation process.*
5. Information on security lighting on site: *Does the site have security lighting (lighting designed to light the exterior perimeter of the project preferably pointing out from the site)?*
6. Information on cleared area outside of project site perimeter: *Is there an area outside of the perimeter that has been cleared of brush, obstacles, etc. that allow for clear site lines?*
7. Information on CCTV/intrusion detection measures: *Does the project have CCTV cameras and/or intrusion detection measures in place? If yes please provide information on any CCTV cameras or intrusion detection measures present.*

Security incident reporting

8. Include those in project management to be notified: *List the names and cell phone numbers of the individuals in project management who should be notified of security incidents.*
9. Include others in AECOM to be notified: *List the names and cell phone numbers of the individuals in AECOM outside of the project who need to be notified of security incidents.*
 - All security incidents to be entered into IndustrySafe as security incident.

Security incident response

10. How are people on site notified of a security incident: *List how security incidents on site or in the vicinity are communicated to personnel on site. This could be by radio/siren/whistle/phone/text/WhatsApp.*
11. Procedure for how to deal with an unauthorized person on site: *Who will confront the unauthorized person? Will it be by a lone person or two people? Will it be by security or project management? How will the fact there is an unauthorized person on site be communicated to personnel? Will the police be notified? Will this result in upward communications to management, client, GSR?*
12. Procedure for how to deal with an unruly/belligerent/argumentative person on site: *What conditions would require the police to be notified? Would security handle this? If the person is a sub would the subcontractor*

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be expected to handle this or project management? What would constitute the person's removal from the site?

13. Under what conditions would you want the personnel on site to take a specific action: *Are there situations that could occur where you would want personnel to take a specific action? ie: shelter in place, seek shelter, leave the site, etc.*

Resilience/Crisis Management Plan

1. Attach a copy of the project's site resilience/crisis management plan.
 - Listing of the project emergency response team including positions and identifying individuals in those positions
 - Incident Management Team Lead
 - Incident Management Alternate Team Lead
 - Safety
 - Security
 - HR
 - Comms
 - Legal

Attachments

1. Expanded Emergency notification call list: *Include those in project management to be notified, and others in AECOM to be notified.*
 - Include AECOM Construction Management emergency notification phone number.
2. Emergency information distribution process: *How will emergency information be distributed on site?*
3. Unauthorized persons on the site response process: *How will unauthorized persons on the site be handled?*
4. Belligerent or argumentative persons on site response process: *How will belligerent or argumentative persons be handled? Include any action that needs to be taken by those on site in response to the situation.*
5. Site layout drawings, maps (both street and satellite). *Include site layout drawings, maps.*

Construction Management Site Security Plan

Project Information

1. AECOM Operating Unit: Tishman Construction
2. Project name: XYZ Building
3. Client: Jones Reality
4. Location of Project: Los Angeles, CA
5. Approximate number of people working on the site: 250
6. Approximate size of the project site: 10 acres

Site Security Plan

Perimeter security

1. What Fencing is in place: 8-foot chain link fence. Anchored into the ground and topped with 1.5 foot of barbed wire
2. What gates are present onsite (Type and Number): 2 - person gates and 1 - vehicle gate
3. How is access to the site controlled: Access through each entry gate is controlled by a gate and a security guard.
4. Information on security guards: All security guards onsite are unarmed. Guards are assigned to each of the gates through which personnel enter the site and gate through which vehicles enter the site. Post orders are in place for each guard post and have been provided to Global Security & Resilience for review.
5. Information on security lighting on site: In addition to interior site lighting, exterior site lighting is installed so the area outside the perimeter fence is lit up so anyone walking along the perimeter is easily visible to those inside and outside of the project site.
6. Information on cleared area outside of project site perimeter: The area on the outside of the perimeter fence is clear of visual obstructions allowing anyone on the outside of the fence to be seen.
7. Information on CCTV/intrusion detection measures: CCTV cameras are located on the perimeter of the project site to allow for a complete view of the exterior of the project site. The cameras are connected to a digital recording system which allows for footage to be retained for 30 days.

Security incident reporting

8. Include those in project management to be notified: Project Superintendent Jones, Deputy Project Superintendent Smith
9. Include others in AECOM to be notified: Global Security & Resilience (GSR@aecom.com), Jack Mattera GSR Corp VP.

All security incidents to be entered into IndustrySafe as security incident.

Security incident response

10. How are people on site notified of a security incident: Text, WhatsApp
11. Procedure for how to deal with an unauthorized person on site: AECOM GSR and police to be called
12. Procedure for how to deal with an unruly/belligerent/argumentative person on site: AECOM GSR and police to be called.
13. Under what conditions would you want the personnel on site to take a specific action: EAP to be followed as applicable.

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Resilience/Crisis Management Plan

Attach a copy of the project's site resilience/crisis management plan.

1. Listing of the project emergency response team including positions and identifying individuals in those positions

- Incident Management Team Lead
- Incident Management Alternate Team Lead
- Safety
- Security
- HR
- Comms
- Legal

Attachments

1. Expanded Emergency notification call list
 - Include AECOM Construction Management emergency notification phone number.
2. Emergency information distribution process
3. Unauthorized persons on the site response process
4. Belligerent or argumentative persons on site response process
5. Site layout drawings, maps (both street and satellite)

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SECTION 31 - STEEL ERECTION

1. TERMS and DEFINITIONS

1.1 Christmas–Treeing. The act of lifting multiple like size pieces during one lifting operation.

2. GENERAL REQUIREMENTS OSHA 29 CFR 1926 Subpart R

2.1 Steel Erection Regulations

- a) Develop a written steel erection plan that includes construction specifications and safety provisions before the actual steel erection of the structure. Submit to AECOM TISHMAN CONSTRUCTION a minimum of 60 days prior to commencement of Steel Erection, for site management team to review.
- b) At a minimum of 14 business days prior to beginning steel erecting activities. The erector and contract holder will participate in a Steel Erection Planning Meeting. This meeting shall review but not limited to the erecting sequences, crane placement, coordination with other trades, laydown area, emergency procedures, fall protection plan, hot work, and site logistics.
- c) Protect the safety of others, whether public or other workers at all times. Confirm that appropriate safety signage (e.g., “Personnel Working Overhead”) and/or barricades are posted/placed where necessary to keep people out of dangerous areas.
- d) Individuals who may indicate a fear of heights or are subject to dizziness shall be kept on groundwork.
- e) Riding loads or riding on the headache ball of the crane is not permitted.
- f) Use personal lifts, ladders, and/or scaffolds instead of climbing steel.
- g) Employees working above any adjacent working surfaces shall review and conform to the applicable fall protection requirements.
- h) Bend or cover projecting or protruding reinforcing rods that create tripping or falling hazards. Use rebar caps significant enough to provide impalement protection (minimum ability to withstand a 250-pound [113-kilogram] weight drop at 7 feet, 6 inches [2.5 meters] without the rebar breaking through).
- i) Do not hoist excess material to a structure until it is ready to be put into position and fastened before cutting any large or heavy structural steel member, secure member, or support by ropes, cables, or other means to prevent dropping, uncontrolled swinging, and uncontrolled rolling.
- j) Keep the working area in orderly condition with necessary equipment and materials safely arranged.
- k) In setting steel, securely bolt each piece before the load line is unhooked at a minimum of two bolts at each connection point.
- l) When setting steel trusses, securely tie or cross-brace until permanent braces are in place.
- m) Check loads to make sure there are no sharp edges that will cut into lifting slings. Use softeners where applicable, lifting lugs and shackles.
- n) Attach a tag line or lines to loads where their use will not create a greater hazard. Tag lines are to be a minimum of 10 feet long. Additional Tag lines may be required based on the size of the piece or complexity of the pick.

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- o) Securely plank or sheet over with metal decking or guard rail all unused openings in floors, whether temporary or permanent.

2.2 Steel Assembly

- a) Use extreme care when storing, handling, and placing steel members. Long steel members or loads will be controlled by taglines to prevent swinging hazards.
- b) Use pneumatic hand tools with extreme caution, with special attention to the following:
 - Secure power sources and bleed off hose lines before disconnecting tools or hose sections.
 - Wire or tie together airline hose connections to prevent accidental separation.
 - Appropriate safety goggles or other eye protection will be provided and worn by workers using pneumatic hand tools, or when performing other work that is hazardous to the eyes.
 - Riveting will not be done in areas where combustible materials are present unless precautions are taken to prevent fires.
 - Connections of the equipment used in plumbing-up shall be properly secured.
 - Turn buckles shall be secured to prevent unwinding under stress.
 - Plumbing-up guys and related equipment shall be placed so that employees can get at connection points.
 - During the final placing of solid web structural members, the load shall not be released from the hoisting line until the members are secured with not less than two bolts, or the equivalent, at each connection, drawn up wrench tight.
 - When bolts or drift pins are knocked out, means will be provided to keep them from falling.
 - Impact wrenches will have the means to secure the socket with a locking device for retaining the socket.
 - Complete bolt-up/plumb-up will be kept within 48 feet (14.6 meters) or 4 floors; whichever is less, of the uppermost steel framework.

2.3 Christmas - Treeing

- a) A maximum of three (3) like size pieces will be permitted to be lifted.
- b) Only single piece lift methods will be utilized for steel erection when winds are sustained above 20 mph, and/ or employees show inability to adequately control all pieces while Christmas - Treeing.
- c) Pieces shall be kept a minimum of 7 feet apart vertically, while suspended in Christmas - Tree formation.
- d) Verify certification of rigging assembly whether supplied by a manufacturer or shop fabricated by the erector.
- e) Documented Daily Rigging Inspections of all Rigging for Christmas – Treeing shall be kept on file.

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- f) Review cranes load chart and radius to ensure Christmas - Treeing does not cause a critical lift condition with total weight.
- g) The erector shall document their plan for multiple lifts (Christmas-Treeing), in their site-specific erection plan.

2.4 Decking / Flooring Requirements

- a) Where erection is being done by means of a crane operating on the ground, a tight and substantial floor will be maintained within two stories or 25 feet (7.6 meters), whichever is less, below and directly under that portion of each tier of beams on which bolting, riveting, or welding is being done.
- b) In the erection of a building having double-wood floor construction, the rough flooring will be completed as the building progresses, including the tier below the one on which floor joists are being installed.
- c) Permanent stairways will be installed as soon as working conditions permit.
- d) When persons are working below riveting, bolting, welding, or painting operations, a temporary floor or protective covering will be erected above them to prevent accidents caused by falling objects.
- e) Deck over all openings, (except very large openings), pour stop / flash around and pour concrete. Steel deck will not be cut until material (ductwork, etc) is installed. Use offset lug connections or shear tabs in lieu of standard double connections to reduce falls and partial collapses.

2.5 Site Layout and Construction Sequence

- a) AECOM TISHMAN CONSTRUCTION shall notify the steel erector and fabricator of the proper curing of concrete in footings, piers, etc. for steel columns in writing, prior to the commencement of steel erection (see Attachment 1: Notification to Begin Steel Erection Form).
- b) AECOM TISHMAN CONSTRUCTION is to provide the steel erector with a safe site layout (see Attachment 2: Steel Erection Site Layout). The erector shall notify the general contractor with any requests for access road maintenance in writing.
- c) Material hoisting routes shall be determined by the steel erector, as part of their steel erection plan.
- d) The steel erector shall notify AECOM TISHMAN CONSTRUCTION in writing of any damages to anchor bolts prior to erection. (See Attachment 3: Anchor Rods/Bolts Inspection and Acceptance Letter).

2.6 Site Specific Erection Plan

- a) The erector shall pre-plan all lifts and outline key elements, including coordination with AECOM TISHMAN CONSTRUCTION before erection begins for the following circumstances:
 - Safety latches on hooks have been deactivated or made inoperable.
 - When joists at or near columns that span more than 60' are not being set in tandem with all bridging/bracing installed.
 - When bundles of decking are being placed on steel joists before all bridging has been installed and anchored and all joist bearing ends attached.

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- Critical lifts (Exceeds 75% of the load chart, multiple crane lift, or as designated by AECOM TISHMAN CONSTRUCTION SH&E dept.) shall be reviewed in a critical lift meeting.

2.7 Hoisting and Rigging

- a) Document the Competent and Qualified Person(s) for the erector who will conduct rigging inspections prior to each work shift.
- b) The erector shall assign a Competent and Qualified Rigger(s) to do all rigging if individuals are working under loads.
- c) The following crane utilization procedures must be provided by the erector:
 - A Competent and Qualified Person(s) must inspect the crane prior to each work shift (documentation required).
 - A Qualified Operator(s) must operate the crane (documentation required).
 - The Qualified Operator(s) or Competent person(s) shall determine if the crane manufacturer allows multiple lift rigging (Christmas - Treeing).
- d) The erector shall minimize employee exposure to overhead loads through pre-planning.

2.8 Structural Steel Assembly

- a) Erector must maintain structural stability at all times during the erection process.
- b) Erector must safely land deck bundles and protect against fall hazards from interior openings.
- c) Erector must protect deck holes and openings.

2.9 Column Anchorage

- a) Erector is required to install a minimum of four anchor bolts per column.
- b) The structural engineer of record must approve and document all repairs to anchor bolts/rods that have been damaged, replaced or field modified.

2.10 Beams and Columns

- a) Erector must ensure connectors are protected from collapse hazards associated with making double bolt connections.

2.11 Open Web Style Joists

- a) Erector must install all bridging and attachments required to minimize risk of collapse of lightweight steel joist by ensuring the following:
 - Erector must stabilize steel joists and girders before releasing hoisting cables.
 - Erector must comply with requirements for attaching steel joists and steel joist girders (includes requirement for “K”, “LH” and “DLH” series steel joists).
 - Erector must comply with requirements for the erection of steel joists (short span and long span).

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- Erector must comply with requirements for the erection of erection bridging (short span and long span).
- Erector must comply with requirements for landing and placing loads on joists.

Note: Installation of open web steel joists, bridging and attachments to be accomplished per fabricator instructions. Instruction for installation must be included in steel erection safety plan.

2.12 Systems - Engineered Metal Buildings

- a) Erector must comply with requirements to minimize collapse in the erection of these specialized structures.

2.13 Falling Object Protection

- a) Erector must assure that all materials, equipment, and tools that are not being used are secured against accidental displacement.
- b) AECOM TISHMAN CONSTRUCTION will bar other construction processes below steel erection unless adequate overhead protection is provided for the employees working below. Construction processes below steel erection will be barred until adequate overhead protection is installed per design specification.

2.14 Fall Protection

- a) All steel erection activities will be required to comply with the AECOM TISHMAN CONSTRUCTION 6' Fall Protection requirement.
- b) The erector must submit a Site-Specific Fall Protection Plan for the scope of work to be performed on the project.
- c) The erector's fall protection plan must comply with the following:
 - Where applicable and feasible, all attempts will be made by the erecting company to limit the exposure to personal fall hazards by means of aerial work platform equipment, scissor lifts, swing staging, spider baskets, scaffolding, personal fall restraint and by use of personal fall protection equipment.
 - Each employee engaged in a steel erection activity that is on a walking/working surface with an unprotected side or edge 6' or greater above a lower level shall be protected from fall hazards by guardrail systems, safety net systems, and personal fall arrest systems, positioning device systems or fall restraint systems.
 - Each connector shall have completed connector training (OSHA 29 CFR 1926.761). Documentation of training is required. The erector must ensure that the training being performed is understood.
 - Each connector must wear and utilize fall restraint/arrest equipment.
 - The use of controlled access zones are required below all steel erecting and steel decking operations related to operations of steel erection. Controlled access zones for decking will be submitted and reviewed by AECOM TISHMAN CONSTRUCTION.

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- d) Erector must install perimeter cables as soon as the metal decking is complete and signed off.
 - The installation of the guardrail shall meet the requirements OSHA 29 CFR 1926.502(b), 1926 Subpart M Appendix B and CFR 1926.251(f)(2) Table H-1, H-2. Also, refer to AECOM TISHMAN CONSTRUCTION Policy, Fall Protection.
 - AECOM TISHMAN CONSTRUCTION may accept responsibility for the fall protection equipment when the Erector is complete. Acceptance must be documented only after a complete thorough inspection of the fall protection systems is complete (See Attachment 4: Perimeter Cable System Inspection and Acceptance Letter). The responsibility of maintaining the cable system will be assigned by AECOM TISHMAN CONSTRUCTION by contract provision.
 - Perimeter Cable Guard Rail shall be constructed with cabling of 3/8" minimum diameter.

2.15 Training

- a) Erector must ensure that a qualified person has completed and documented site-specific training for all their exposed workers in fall protection and any worker exposure for special, high risk activities (i.e. as prescribed in OSHA 29 CFR 1926 Subpart R).
- b) Erector must ensure that training is understood.
- c) Erector shall designate their assigned "Competent Person." Competent Person is defined as one who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them.
- d) Erector shall also designate their assigned "Qualified Persons." Qualified Person means one who, by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training, and experience, has successfully demonstrated the ability to solve or resolve problems relating to the subject matter, the work, or the project.

2.16 Records

- a) The following documentation will be kept on file:
 - Completed Steel Erection Plan.
 - Anchor bolt Survey/Modification/Design change
 - FAA Approval
 - Concrete Strength Break test data

3. ATTACHMENTS

3.1 Notification to Begin Steel Erection Form

3.2 Steel Erection Site Layout

3.3 Anchor Rods/ Bolts Inspection and Acceptance Letter

3.4 Perimeter Cable System Inspection and Acceptance Letter

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Attachment 1: Notification to Begin Steel Erection Form

NOTIFICATION TO BEGIN STEEL ERECTION

TO:

RE: Concrete and Masonry Compressive Design Strength - LOCATION

In accordance with the steel erection standards, the concrete or masonry in the above-mentioned location(s) have attained the required minimum compression design strength to begin steel erection.

Repairs, replacements and modifications to the anchor bolts are as noted below:

- _____ No repairs, replacements or modifications have been made to date.
- _____ Project Engineer of Record has approved repairs, replacement or modifications. Subsequent to this notification, Steel Fabricator/Erector must report all repair and/or modification to AECOM in writing.

29 CFR 1926.752 Site layout, site-specific erection plan, and construction sequence

(a) Approval to begin steel erection. Before authorizing the commencement of steel erection, the controlling contractor shall ensure that the steel erector is provided with the following written notifications:

1. The concrete in the footings, piers and walls and the mortar in the masonry piers and walls has attained, on the basis of an appropriate ASTM standard test method of field-cured samples, either 75 percent of the intended minimum compressive design strength or sufficient strength to support the loads imposed during steel erection.
2. Any repairs, replacements and modification to the anchor bolts were conducted in accordance with § 1926.755(b).

Reference or attach the RFI, if applicable.	RFI Attached	N/A
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Signatures below signify that steel erection may commence in the area identified above, and proper notification has been given and received.

AECOM		
Project Manager (Print)	(Initial)	Date
Project Superintendent (Print)	(Initial)	Date

Steel Fabricator/Erector		
Project Manager (Print)	(Initial)	Date
Project Superintendent (Print)	(Initial)	Date

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Attachment 2: Steel Erection Site Layout

STEEL ERECTION SITE LAYOUT

DATE:

RE: Steel Erection Site Layout

Location: _____

In accordance with the steel erection standards, the site layout in the above-mentioned location meets the requirements to begin steel erection.

Steel Fabricator/Erector acknowledges adequate site layout and assumes responsibility for the maintenance of it for the duration of steel erection. Contractor will provide and maintain access road into and through the site.

Steel Fabricator/Erector will notify Contractor in writing should the conditions of the site layout and access roads require maintenance.

29 CFR 1926.752 Site layout, site-specific erection plan, and construction sequence

(c) Site layout. The controlling contractor shall ensure that the following is provided and maintained:

1. Adequate access roads into and through the site for the safe delivery and movement of derricks, cranes, trucks, and other necessary equipment, and the materials to be erected and means and methods for pedestrian and vehicular control. Exception: this requirement does not apply to roads outside of the construction site.
2. A firm, properly graded, drained area, readily accessible to the work with adequate safe storage of materials and the safe operation of the erection equipment.

Signatures below signify that steel erection may commence in the area identified above, and proper notification has been given and received.

AECOM		
_____ (Print) Project Manager (Signature)	_____ (Initial)	_____ Date
_____ (Print) Project Superintendent (Signature)	_____ (Initial)	_____ Date

Steel Fabricator/Erector		
_____ Project Manager (Print)	_____ (Initial)	_____ Date
_____ Project Superintendent (Signature)	_____ (Initial)	_____ Date

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Attachment 3: Anchor Rods/Bolts Inspection and Acceptance Letter

ANCHOR RODS/BOLTS INSPECTION AND ACCEPTANCE LETTER

Date:

To:

From:

RE: **Acceptance of Anchor Rods/Bolts** - (Location) _____

In accordance with the steel erection standards, the Structural Engineer of Record has inspected and approved the repair, replacement, or modification of the anchor rods/bolts in the above-mentioned location.

The steel erection standard is referenced below:

29 CFR 1926.755 Column Anchorage

(b) Repair, Replacement or field modification of anchor rods (anchor bolts).

1. Anchor rods (anchor bolts) shall not be repaired, replaced or field modified without the approval of the project structural engineer of record.
2. Prior to erection of the column, the controlling contractor shall provide written notification to the steel erector if there has been any repair, replacement or modification of the anchor rods (anchor bolts) of that column.

Attach applicable RFI from the Project Engineer of Record.

Signatures below signify that the Anchor Rods/Bolts described above have been inspected and accepted, and that proper notification has been given and received.

AECOM		
_____ (Print) Project Manager (Signature)	_____ (Initial)	_____ Date
_____ (Print) Project Superintendent (Signature)	_____ (Initial)	_____ Date

Steel Fabricator/Erector		
_____ (Print) Project Manager (Signature)	_____ (Initial)	_____ Date
_____ (Print) Project Superintendent (Signature)	_____ (Initial)	_____ Date

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Attachment 4: Perimeter Cable System Inspection and Acceptance Letter

PERIMETER CABLE SYSTEM INSPECTION AND ACCEPTANCE LETTER

To:

From:

Date:

Re: Acceptance of Perimeter Cable System - _____
LOCATION

In accordance with the new steel erection standards, AECOM has inspected & accepted the steel cable perimeter gable system installed on the second floor and roof of the public works facility, and on the roof of the

LOCATION

The new steel erection standard is referenced below:

Custody of Perimeter Cable System (Fall Protection) — provided by the steel erectors shall remain in the area where steel erection activity has been completed, to be used by other trades, only if the controlling contractor or its authorized representative:

1. Has directed the steel erector to leave the fall protection in place; and;
 2. Has inspected and accepted control and responsibility for the fall protection prior to authorizing persons other than steel electors to work in the area.
- 29 C.F.R. 1926.760(e), (e)(l), and (e)(2) (2001)

AECOM requires all subcontractors working in the _____, to conduct their own inspection of the perimeter safety cable system and provide AECOM. with written inspection and acceptance of the system. Each subcontractor's competent person must sign their acceptance of the system prior to employees being allowed to work in the areas referenced above.

After inspection and acceptance, the responsible subcontractors working in the area shall maintain the perimeter cable system. Should AECOM identify discrepancies in the system the responsible subcontractor(s) for the discrepancy will be required to correct the problem immediately. Should a subcontractor fail to correct the deficiency a Stop Work order may be issued until corrective measures are taken.

Signature below signifies that the Perimeter Cable System described above has been inspected and accepted by:

Erecting Company: _____

Competent Person: _____
Print Name Signature Date
Title: _____

AECOM Accepting Company

Competent Person: _____
Print Name Signature Date
Title: _____

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SECTION 32 - SUBCONTRACTOR SAFETY RESPONSIBILITIES

1. TERMS and DEFINITIONS

1.1 Pre-Construction. Prior to mobilization or commencement of any work beginning on the project.

1.2 Safe-Start Meeting. A pre-construction meeting held between the AECOM TISHMAN CONSTRUCTION Project Manager, Project Superintendent, Project Safety Manager, and/or other applicable AECOM employees, and the subcontractor Project Superintendent, Project Manager, Project Foreman, Project Safety Manager and/or other applicable subcontractor employees.

2. GENERAL REQUIREMENTS

2.1 Subcontractors shall participate in a Safe-Start Meeting prior to work on the project.

2.2 Subcontractors shall provide the following documentation to the AECOM TISHMAN CONSTRUCTION Project Manager at least two weeks prior to this Safe-Start Meeting:

- Subcontractor company safety program.
- Hazard-specific work plan.
- Safety Data Sheets (SDS) and Globally Harmonized System (GHS) compliant program.
- Designated project safety coordinator meeting AECOM TISHMAN CONSTRUCTION requirements.
- Project persons trained in First Aid/CPR.
- Job Hazard Analysis (JHAs) as required.
- Company disciplinary program.
- Documentation of training records for designated Competent Persons.
- Additional safety submittals as requested.

2.3 AECOM TISHMAN CONSTRUCTION should cover the following topics and expectations at Safe-Start meeting:

- Phases of construction.
- Project Orientation process.
- Emergency procedures.
- Project work rules.
- Incident investigation and reporting procedures- including submitting monthly work hours.
- Return to work policy and submitting applicable medical paperwork.
- Fall protection policy (6 foot) and subcontractor's fall rescue plan.

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- Discipline policy.
- Fit for Duty policy.
- Site security procedures and regulations.
- Hazard communication policy and SDS procedures.
- Project incentive program and committees.
- Specific licensing requirements (as applicable).
- AECOM's written safety program.
- Contractual obligation to comply with our safety program.
- Fire protection policy and hot work permit program.
- Confined spaces policy.
- Job hazard analysis.
- Safety meetings and documentation.
- Housekeeping policy.
- Designated safety representative and safety audit requirements.
- Competent Person for applicable tasks.
- Designated First Aid Provider for the subcontractor.
- Personal protective equipment policy and proper work attire.
- Specific owner/project-imposed safety requirements.
- Rigger qualification policy.
- Compliance with federal, state, and local safety standards.
- Daily THA requirements.
- Traffic control responsibilities.
- Equipment inspections.
- Subcontractor work plan.
- Crane packet requirements.
- Incident reporting as applicable.

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- Lockout Tagout
- Other items as deemed necessary for a specific project or by a specific region.

2.4 Subcontractors shall actively participate in project safety program and attend all required safety meetings.

2.5 Subcontractors shall conduct and document weekly site safety inspections of their own work activities, tools and equipment. Also provide and document corrective action for all items noted on inspection report.

2.6 Subcontractors shall maintain a first-aid kit on site and eye wash stations as applicable.

2.7 Subcontractors shall maintain and replace safety protection systems damaged and/or removed by their operations.

2.8 Subcontractors shall submit accident, injury and incident reports, including near misses within 24 hours of incident occurrence.

2.9 Subcontractors shall properly install contractually required safety systems as outlined in the general conditions or the scope of work (i.e., guardrail, fence, fall protection systems, floor opening covers, etc.).

2.10 Subcontractors shall conduct weekly employee safety toolbox meetings and copy AECOM TISHMAN CONSTRUCTION project management personnel. This includes subcontractors who work a partial week.

2.11 Subcontractors shall conduct new employee orientation for their employees per federal OSHA and make those employees available for project orientations and all other safety related activities.

2.12 Subcontractors shall conduct task hazard analysis (THAs) for their operations by crew and have all employees involved in the process. Copies shall be kept on the Foreman and made available to AECOM TISHMAN CONSTRUCTION upon request. Copies shall also be filed as applicable.

2.13 Subcontractors shall be solely responsible for the safety and security of their employees and areas of work. All subcontractors shall submit a copy of their company safety program and certificate of insurance to the AECOM TISHMAN CONSTRUCTION Project Manager prior to beginning work on the project as mentioned above.

2.14 Subcontractors shall at all times conduct work in a safe and practical manner in conformance with the Federal Occupational Safety and Health Act of 1970 as well as state and local safety laws and regulations. Subcontractors will abide by safety and health standards for all their work including but not limited to: pollution control, water supply, fire protection, sanitation facilities, waste disposal and other related items.

2.15 Each subcontractor shall maintain and provide written safety records as requested to the AECOM TISHMAN CONSTRUCTION Project Management.

2.16 Subcontractors shall furnish all information concerning safety of their operations on the project as required by AECOM TISHMAN CONSTRUCTION.

2.17 Subcontractors shall notify the AECOM TISHMAN CONSTRUCTION Project Management Team immediately of any near miss, injury, or property damage.

2.18 Subcontractors shall indoctrinate their employees and their sub-tier subcontractors related to the safety and health requirements of their operations and to enforce compliance with safe work procedures.

2.19 Subcontractors shall provide an individual as their designated competent person who will have the responsibility to resolve matters concerning safety and to act as liaison among the subcontractors, their tier subcontractors, and AECOM TISHMAN CONSTRUCTION. This individual must be onsite at all times work

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is being performed and have the authority necessary to immediately correct unsafe practices or hazardous conditions. This designee must meet all AECOM TISHMAN CONSTRUCTION requirements. See Attachment 1: "Designated Competent Person Form."

2.20 Subcontractors shall give full cooperation to the owner's and AECOM's representatives who periodically visit the project.

2.21 Subcontractors shall inform AECOM TISHMAN CONSTRUCTION of any impending federal or state inspection prior to a site inspection. AECOM TISHMAN CONSTRUCTION will receive copies of all federal and state inspection reports, citations, penalties, abatement dates, etc. Subcontractors are solely responsible for compliance.

2.22 All tiered subcontractors and employees acting unsafe or exposed to an unsafe condition shall be directed to cease activity until the condition is brought into compliance with site safety procedures. The cost thereof will be borne by the non-complying subcontractor and no costs will be charged to the owner or AECOM TISHMAN CONSTRUCTION. In addition, if a subcontractor refuses to correct unsafe conditions, AECOM TISHMAN CONSTRUCTION shall cease the work activity until corrective action has been taken, or correct the situation using other means and implement a Deductive Change Order for expenses incurred.

2.23 Failure to comply with the contract safety requirements shall be considered as non-compliance with the contract requirements and may result in remedial action including but not limited to withholding of payment, cancellation of contract and/or removal from the job site.

2.24 Subcontractors shall comply with the AECOM TISHMAN CONSTRUCTION personal protective equipment policy including but not limited to: requiring all persons to wear ANSI approved hard hats, ANSI approved safety glasses, proper work boot footwear made of all leather uppers and thick soled bottoms with anti-skid materials, reflective vests or highly visible shirts, hearing protection, glove policy, and proper work attire with all shirts covering shoulders. Under no conditions are dress shoes, athletic-type footwear (tennis, running, basketball, cross-country, etc.) or sandals allowed. Subcontractors will wear mandated full hand protection applicable to task per subcontractor's task hazard analysis and AECOM TISHMAN CONSTRUCTION glove policy.

2.25 Subcontractors shall provide their employees with Class II or greater ANSI approved attire when working in or around traffic.

2.26 Subcontractors shall observe good housekeeping at all times. Waste, debris, and garbage shall be removed daily or placed in appropriate waste containers. All materials, tools, and equipment shall be stored in a safe and orderly fashion. If any subcontractor fails to maintain a satisfactory clean-up program, AECOM TISHMAN CONSTRUCTION will give 24 hours notice, perform the cleanup, and implement a Deductive Change Order to the responsible subcontractor for the costs.

2.27 Subcontractors shall supply proper equipment, take necessary precautions to maintain the equipment according to current regulations and specifications, and accept responsibility to assure that necessary safety equipment is supplied and used when required.

2.28 Subcontractors shall be responsible for supplying, maintaining and inspecting all fire extinguishers in their office, work areas, and storage and refueling areas. In the event a fire extinguisher is discharged or damaged, it shall be removed from service, immediately tagged inoperable, and replaced with a properly - working and compliant unit.

2.29 Subcontractors shall report spills of hazardous materials immediately (including cutting oil, fuel, solvents, antifreeze, etc.). Each subcontractor is responsible for the associated cleanup costs.

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2.30 Subcontractor pipe cutting equipment must have secondary containment (drip pans, sandboxes). Cardboard and other permeable products are not to be used as containment. All non-contaminated rainwater must be removed to a chemical sewer within 24 hours.

2.31 Subcontractors shall maintain all containers in good condition, and the containers must be appropriate for the materials contained. All containers must be labeled with their contents and precautions for use per Hazard Communication standards. All containers must be stored properly.

2.32 Subcontractors are responsible for proper disposal of its own hazardous wastes. A copy of the completed Uniform Hazardous Waste Manifest shall be provided to the AECOM TISHMAN CONSTRUCTION Project Manager.

2.33 Subcontractors shall provide at least a minimum of one OSHA 30 hour trained supervisor when craft levels meet or exceed 5 individuals for the subcontractor.

2.34 A full-time designated safety person, with a minimum training of the OSHA 500 certification, minimum CHST or equivalent certification, and minimum 5 years relevant construction safety experience, shall be required when subcontractor levels meet or exceed 35 individuals and/or the value of their work in their contract is \$10mm or greater. This person shall have no craft responsibilities other than safety. Multiple persons will be assigned as the subcontractor levels compound.

2.35 Subcontractors must have Fitness for Duty (Drug Testing) Program that meets or exceeds the AECOM TISHMAN CONSTRUCTION Program. Pre-employment, post-incident, and reasonable suspicion testing are required. The most recent policy can be requested from AECOM TISHMAN CONSTRUCTION to ensure compliance.

2.36 Subcontractors shall follow any and all local laws and regulations with regards to their staff safety oversight with the aforementioned requirements being the minimum.

3. ATTACHMENTS

3.1 Designated Competent Person Form

3.2 Safety/Competent Person Summary Contact List

3.3 Safe - Start Submittal Checklist

3.4 Subcontractor Contact List

3.5 Fitness for Duty Certification

3.6 Chemical Inventory List (compliant with GHS)

3.7 Safety and Health Training Certification

3.8 Rigging/Signaler Training & Third Party Crane Inspection Acknowledgement

3.9 Task Hazard Analysis Form Example

3.10 Safe-Start Form/Checklist

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Attachment 1: Designated Competent Person Form

Re. Designated Competent Person

As required by Subcontract Agreement and the Occupational Safety and Health Standards for the Construction Industry (OSHA 29 CFR 1926.20 (b) (2)), employer is required to provide for frequent and regular inspections of the jobsites, materials and equipment, to be made by a competent person designated by the employer.

Definition of Competent Person per OSHA 29 CFR 1926.32 (f) "Competent Person" means one who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous or dangerous to employees, and who has authorization to take prompt corrective measures to eliminate them.

Please find information below designating the competent person for the project located at:

Subcontractor

Scope of Work

Designated Competent Person

Title of Position

Signature of Officer

Title of Officer

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Attachment 2: Safety/Competent Person Summary Contact List

SAFETY/COMPETENT PERSON SUMMARY CONTACT LIST

Safety Contact

NAME	TITLE	COMPANY NAME	E-MAIL ADDRESS	CONTACT NUMBER

Competent Person

NAME	COMPETENT AREAS	COMPANY NAME	SUPERVISOR (SIGNATURE)	CONTACT NUMBER

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Attachment 3: Safe - Start Submittal Checklist

SUBCONTRACTOR NAME *Also List Lower Tier Subcontractors	Competent Persons Contact List	Subcontractor List	Fitness For Duty Certification	Chemical Inventory List (Site Specific)	Safety & Health Training Certification(s)	Equipment Inspection Forms	Rigging Training & Third-Party Crane Inspection Acknowledgement	AECOM TISHMAN CONSTRUCTION Safety Program Acknowledgement	Hazcom Program (On Site*)	Job Hazard Analysis	Demo/Steel Plans	Special CCIP/OCIP Enrollment	Safety Program (On Site*)	Notes
1														
2														
3														
4														
5														
6														
7														
9														

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Attachment 4: Subcontractor Contact List

Subcontractor Contact List

Begin with your company and list all lower tier subcontractors. NOTE: All lower tier subcontractors are required to attend the Safe - Start Meeting.

Subcontractor Name	# of employees or other persons performing work directly or indirectly on behalf of subcontractor	Date of Work From – To	Key Contacts		Work to be Performed
			Name	Telephone	
		From: _____ To: _____	Office: _____ Field: _____	Office: _____ Field: _____	
		From: _____ To: _____	Office: _____ Field: _____	Office: _____ Field: _____	
		From: _____ To: _____	Office: _____ Field: _____	Office: _____ Field: _____	
		From: _____ To: _____	Office: _____ Field: _____	Office: _____ Field: _____	
		From: _____ To: _____	Office: _____ Field: _____	Office: _____ Field: _____	

Attachment 5: Fitness for Duty Certification

Fitness for Duty Certification

This will certify that all _____
(subcontractor)

employees, or other persons working directly or indirectly on its behalf, who will work on the

(project name)

have had pre-employment drug screens and that the results of the drug screens for each employee assigned to this project were negative.

We use _____ to conduct drug and alcohol testing.
(name of drug testing laboratory)

We further certify that our company Fitness for Duty program meets the requirements of the AECOM TISHMAN CONSTRUCTION Fitness for Duty program.

(Signed by subcontractor Company Officer)

(Print Name)

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Attachment 6: Chemical Inventory List

COMPANY NAME: _____ PAGE NO. ____ OF ____

PROJECT NAME: _____ PROJECT NO. _____

CHEMICAL INVENTORY LIST

DATE ON SITE	COMMON NAME & CHEMICAL NAME	MANUFACTURER/ SUPPLIER	SDS ON FILE	SDS ORDERED	LOCATION ON SITE

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Attachment 7: Safety and Health Training Certification

Safety & Health Training Certification

This will certify that all _____
(subcontractor) employees, or other persons working directly or indirectly on its behalf, who will work on the _____
(project name) have been trained and
educated in the following areas of Construction Safety and Health. **(Check all applicable boxes)**

- | | | |
|--|---|---|
| ☐ Fall Protection | ☐ Hazard Communication | ☐ Aerial Lifts |
| ☐ Scaffolding | ☐ Electrical – LO/TO | ☐ Silica |
| ☐ Rigging Certification | ☐ PPE | ☐ Mobile Equipment |
| ☐ Master Rigger | ☐ NFPA 70E | ☐ Hand and Power Tools |
| ☐ Rigger 1 | ☐ Forklift | ☐ General Construction |
| ☐ Rigger 2 | ☐ Respiratory Protection | ☐ Other: _____ |
| ☐ Fire Extinguishers | ☐ Traffic Flagger | |
| ☐ Ladders | ☐ Trenching/Excavation | |

***NOTE: Documentation of training must be made available upon request.**

(Signed by subcontractor Company Officer)

(Print Name)

Rigging/Signaler Training & Third Party Crane Inspection Acknowledgement

This will certify that _____ employees, or other persons working directly
(subcontractor)

or indirectly on its behalf, who will be identified as Riggers/Signalers on the

_____ project
(project name)

have been trained on Rigger/Signaler Qualifications and Training as required by AECOM TISHMAN CONSTRUCTION's SH&E Program.

This will also certify that **all mobile cranes** will receive an inspection from a Third Party Crane Inspection Company **PRIOR** to operating on site. All deficiencies according to manufacturer specifications that would affect the safe operation of the crane shall be corrected **PRIOR** to operating on site.

***All Riggers/Signalers will be identified by acceptable means of identification and shall have appropriate communications.**

(Signed by subcontractor Company Officer)

(Print Name)

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Attachment 9: Task Hazard Analysis Form Example

Task Hazard Analysis Form

SUBCONTRACTOR: _____

JOB/TASK: _____ **WORKAREA:** _____

DATE: _____

STEPS OF TASK	POTENTIAL HAZARDS	CONTROL/ELIMINATION OF HAZARDS
END OF SHIFT COMMENTS:		

WORKER ACKNOWLEDGMENT

<i>I have been trained to do the tasks required. I acknowledge receiving and understanding these instructions. I will fully comply with the assigned task instructions.</i>		<i>I have completed the tasks in a safe manner without damage to property or injury to myself.</i>
WORKER SIGNATURE AT START OF SHIFT	EMPLOYEE NAME PRINTED	EMPLOYEE SIGNATURE AT END OF SHIFT
EMERGENCY EVACUATION ROUTE: EMERGENCY PHONE #:		
CREW SUPERVISOR REVIEWED AND APPROVED THA:		DATE:
Additional Information :		

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Attachment 10: Safe-Start Form/Checklist

Pre-Construction Subcontractor Safe - Start Orientation Checklist

Review with subcontractor safety/operations representative before work starts.

Subcontractor: _____
Project Name: _____ Project No. _____
Subcontractor Safety Representative: _____ Mobile No. _____
Subcontractor Superintendent: _____ Mobile No. _____

Place a checkmark in each block after it has been explained.

1. AECOM TISHMAN CONSTRUCTION Safety and Health Program Overview

- ☐ Statement of Policy.
- ☐ Purpose.

2. Project Safety Requirements for Subcontractors

- ☐ Introduction – Responsibilities.
- ☐ Subcontractor Safety Representative Responsibilities –
**Read this section of AECOM TISHMAN CONSTRUCTION SH&E Program*
- ☐ Job Hazard Analysis (JHA).
- ☐ Task Hazard Analysis (THA).
- ☐ Safe Work Plan folders & content.
- ☐ AECOM TISHMAN CONSTRUCTION Programs.

3. New Hire Safety Orientation

- ☐ A Project - Specific Orientation is required for each person working directly or indirectly on Subcontractor's behalf prior to starting work.
- ☐ Upon completion of the orientation and passing any given orientation competency quiz, each worker will be given a sticker or another identification system to indicate the worker has received initial safety orientation and is authorized to be on site.
- ☐ The sticker must be worn on the hardhat as applicable as a condition of employment.

4. Project Safety Meetings

- ☐ Subcontractors must attend all AECOM TISHMAN CONSTRUCTION Project Safety Meetings.
- ☐ Monthly Stand – Down.
- ☐ Monthly Safety Committee Meetings.
- ☐ Principals Meeting.
- ☐ Weekly Toolbox Talks – Document in Safe Work Plan folder -- Submit documentation weekly to AECOM TISHMAN CONSTRUCTION.
 - ☐ Must include name of company, date, signatures of attendees, topics, and comments.
 - ☐ A pre-shift meeting shall be held by each subcontractor to review the Task Hazard Analysis (THA) with their employees. The THA shall be documented in each subcontractor's Safe Work Plan folder. Submit documentation weekly to AECOM TISHMAN CONSTRUCTION.

5. General Safety Procedures

- ☐ All Subcontractors must ensure that their employees and any other person working directly or indirectly on behalf of the subcontractor, regardless of tier, understand and agree to comply with their company's Safety and Health and HAZCOM Program, the contract documents, the AECOM

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TISHMAN CONSTRUCTION SH&E Program, OSHA Standards and all other federal, state, and local codes, laws, and regulations.

- ☐ Subcontractors shall review all information, policies, and procedures found here within, with all lower tier subs under their contract during the Safe - Start Meeting.
- ☐ Prior to a vendor or supplier performing work under the subcontractor's direction, a detailed plan shall be submitted to AECOM TISHMAN CONSTRUCTION for review and approval.
- ☐ Subcontractors shall require that each of their employees and any other person working directly or indirectly on behalf of the subcontractor wear personal protective equipment required for the performance of their work and enforces its use as a condition of employment.

6. Applicable Safety and Health Requirements

- ☐ Personal protection equipment:
 - ☐ ANSI approved eye protection and head protection shall be worn onsite at all times.
 - ☐ Safety vests will be required at all times while working on the project.
 - ☐ Class two safety vests must be worn by all persons working on the roadway.
 - ☐ Minimum ANSI level two or EN level three cut-resistant gloves shall be worn by all individuals.
 - ☐ Proper leather work boots and work clothing.
- ☐ Scaffolding – Discuss competent person, inspection, proper erection/ dismantling, and fall protection.
- ☐ Excavation – Discuss competent person, daily documented inspection, and proper sloping, etc. For further detail, please refer to the Excavation and Trenching section of this program.
- ☐ Fire Prevention – Discuss fire extinguishers, use of pressurized gas cylinders, temporary heat, flammable storage, and hot work permits.
- ☐ Powder-Actuated Tools – must have certificate for particular tool.
- ☐ Electrical – GFCI's are required for all phases of the project. All temporary permits shall be marked per OSHA regulations.
- ☐ Confined Space / Respirator Use and Requirements / Air Monitoring/Silica.

7. Subcontractor Safety and Health Enforcement Policy

- ☐ Safety Violation examples: Violations of fall protection, eye protection, fighting, drugs and alcohol use, unsafe behavior, etc.
- ☐ Project Safety Representative will be held accountable for the immediate correction of hazards and unsafe acts by their employees, subcontractors, and vendors.
- ☐ AECOM TISHMAN CONSTRUCTION reserves the right to stop work, correct hazard and back charge, send notice to the company management and insurance company, terminate/ remove employee(s), and/or withhold payments to correct the hazard.

8. Fitness For Duty Program

- ☐ Subcontractors must have Fitness for Duty (Drug Testing) Program that meets or exceeds the AECOM TISHMAN CONSTRUCTION Program.
- ☐ Pre-employment, post incident, and reasonable suspicion testing are required.

9. Job Hazard Analysis & Task Hazard Analysis (THA)

- ☐ Submit copy to AECOM TISHMAN CONSTRUCTION prior to work for approval.
- ☐ Review phases/types of work and hazards to address.
- ☐ Review daily requirement for THA.
- ☐ Review Safe Work Plan folder and contents.
- ☐ Review sample and instructions for JHA Form.
- ☐ Must include fall protection (if applicable) and clean-up activities.

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- ☐ AECOM TISHMAN CONSTRUCTION can provide sample JHA & THA Forms.
- ☐ Required for work performed onsite by vendors and suppliers (pump trucks, cranes, boom truck deliveries, scaffold erection and hoist installation).

10. Work at Elevated Locations (Fall Protection Program)

- ☐ Review of the 6-foot 100% fall protection policy. Please refer to the Fall Protection section of this program.
- ☐ Review of the Zero Tolerance Policy for fall protection violations.
- ☐ Discuss Guardrail Systems, Safety Net Systems, Personal Fall Arrest Systems, Fall Protection Plans, Controlled Access Zones, Warning Line Systems, Holes and Falling Objects, Training, and Steel Erection.

11. Subcontractor Safety Documentation Checklist and Forms

- ☐ Emergency/Safety Contacts List (List First Aid/CPR certified personnel, Competent Persons, and Emergency Contacts with phone numbers).
- ☐ Subcontractor List.
- ☐ Fitness for Duty Certification.
- ☐ Hazardous Material Inventory List with corresponding SDS's (site specific).
- ☐ Safety & Health Training Certifications.
- ☐ Training documentation for designated competent persons.
- ☐ Rigging/Signaler Training & Third Party Crane Inspection Acknowledgement.
- ☐ Hazard Communication Program.
- ☐ JHA & Daily THA – Site Specific.
- ☐ Demolition / Steel Erection Plans, Pre-Cast, Crane Plans, etc.
- ☐ Safety & Health Program.
- ☐ First tier subcontractors shall attend a Safe - Start Meeting with each lower tier subcontractor with AECOM TISHMAN CONSTRUCTION's project team and obtain the above listed documentation for each lower tier subcontractor under their contract. This documentation must then be submitted to and approved by AECOM TISHMAN CONSTRUCTION prior to their start of work. Each lower tier subcontractor is also required to attend the Safe - Start Meeting with a representative from the first tier subcontractor.

12. Incident Reporting

- ☐ All Incidents/Injuries/Property Damage – notify AECOM TISHMAN CONSTRUCTION immediately.
- ☐ Complete an Incident Investigation Report - submit within 24 hours.
- ☐ Incident review meetings.

Subcontractor's Company Officer, Project Manager, Superintendent, Safety Manager and other employees shall attend incident review meetings as required.

- ☐ Emergency Action Plan (Fire/Medical Procedures).
- ☐ Post Emergency Phone Numbers and Clinic Directions at gang box/trailer.
- ☐ One person onsite with First Aid and CPR training.
- ☐ First Aid Kits and Fire Extinguishers must be located in conspicuous location.

13. Emergency Action Plan

- ☐ Reviewed with subcontractor.
- ☐ Informed of location of EAP in office.

14. Special Insurance Program: ___ Yes ___ No

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- ☐ Subcontractor required to be enrolled: ____ Yes ____ No
 - ☐ Subcontractor enrolled and Enrollment Forms turned in: ____ Yes ____ No
 - ☐ All claims must be reported to AECOM TISHMAN CONSTRUCTION Safety and Site Superintendent.
 - ☐ All lower tier subcontractors are enrolled: ____ Yes ____ No
 - ☐ Light Duty Program will be provided by subcontractor: ____ Yes ____ No
 - ☐ Site Specific Orientation required by all employees and lower tier subcontractors.
 - ☐ All injured persons will go to designated clinics.
-

As the Subcontractor Safety Representative for my company, I understand that I am required to abide by the rules and procedures noted above, as well as to communicate these rules and procedures to the employees of my company and any subcontractor brought onsite.

Subcontractor Safety Representative: _____ Date: _____
Orientation Conducted By: _____ Date: _____

Copy # 1 – Project File

Copy # 2 – Subcontractor Safety Representative

SAFETY, HEALTH, AND ENVIRONMENTAL PROGRAM

SECTION 33 - TRAINING AND PROJECT MEETINGS

1. TERMS and DEFINITIONS

None Applicable.

2. GENERAL REQUIREMENTS

2.1 All project employees and subcontractor employees must receive proper safety training prior to starting project work activities and as needed throughout their operations. Subcontractors are solely responsible to properly train their employees.

2.2 Project Safety Orientation. All employees and subcontractor personnel onsite shall attend a project safety orientation prior to any work commencing.

2.3 Task Hazard Analysis (THA). Daily safety planning meetings are to be conducted by individual crew Foremen and/or Superintendents to review the work anticipated for that day. This meeting, called, "Task Hazard Analysis (THA)" is conducted to review foreseeable hazards and the actions required in recognizing, avoiding, reducing, and/or controlling those hazards. This process is an excellent tool when used by the entire crew to plan safety into their work. The THA form examples or other AECOM TISHMAN CONSTRUCTION approved forms found in this program or provided by subcontractors are to be used to document this daily process. If a subcontractor has a form that meets the minimum intent found in these examples, it may be used instead as long as the AECOM TISHMAN CONSTRUCTION Safety Manager approves the use.

2.4 Weekly Tool Box Meetings. Tool Box Training Meeting topics will be prepared and will generally be germane to the work that is currently being performed on the project. Meeting subject matter and attendance will be documented in the permanent project files. The Project Safety Manager will review meeting attendance sheets. All suggestions, concerns or other comments are to be followed up as soon as possible with resultant actions reported to the writer of the meeting report. All project AECOM TISHMAN CONSTRUCTION and subcontractor employees are required to participate in a weekly tool box talk. Subcontractors can pick their own topics and/or may be periodically given a topic as situations dictate.

2.5 Monthly Project Stand-Downs. At least once a month, AECOM TISHMAN CONSTRUCTION project management shall conduct a project-wide safety stand-down. This stand-down shall cover safety topics relevant to the project, items required by AECOM, and/or items requested by the owner or subcontractors. All project individuals must participate with a record of the stand-down being kept in the project files.

2.6 Personal Protective Equipment (PPE) and Specific Tasks. AECOM TISHMAN CONSTRUCTION employees shall be trained in the proper use of PPE. Subcontractors shall train their employees in the proper use of PPE as well as any task - specific training that might be required for their task(s).

2.7 AECOM University. Every employee of AECOM shall attend and successfully complete courses required for the position they hold. Speak to your Safety Director to find out what the requirements are. These may include the OSHA 10 hour, OSHA 30 hour, Reasonable Suspicion Training for Supervisors, START, SULU, Safety 4 Life, or other assigned coursework.

2.8 Outside Training. Outside training programs will be conducted as needed for the project. Some examples of outside training where an instructor may be brought in include:

- First Aid/CPR.
- Excavation and Trenching "Competent Person."
- Crane Inspection.
- Safe Rigging.
- Special Security Training.

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- Marine Operations.
- Hazardous Substance Training.
- Emergency Procedures.
- Emergency Drills.
- Underground Requirements.
- MSHA Certification.
- Blasting.

2.9 Project Safety Committees. Projects may be required to establish a project safety committee at the onset of the project. The committee is to be made up of a representative from each prime subcontractor at a minimum and is to be initially co-chaired by the Lead Superintendent and the assigned Project Safety Manager through the first meeting. The committee shall meet at least once a month and review all aspects of safety on the project, including but not limited to incidents that have occurred, safety alerts recently issued, safety concerns, and implementation of the project's incentive program. After the committee has their sit-down meeting, all members shall jointly complete a safety walkthrough of the project with recommendations submitted to the Lead Superintendent. Meeting minutes shall be taken and filed on the project for review. All action items recommended by the committee shall be responded to in an appropriate time frame with a follow-up in the committee. Each project shall have the flexibility to design these committees to meet the needs of their individual project.

2.10 Job-Specific. The AECOM TISHMAN CONSTRUCTION Safety Department and project team will determine the need for job-specific training and will make it available for AECOM employees as necessary.

2.11 Online Training. AECOM provides numerous online safety training courses that are available through AECOM University. Employees are encouraged to participate in these courses and complete the required coursework if applicable.

2.12 AECOM TISHMAN CONSTRUCTION employees shall receive direction from management on required annual training. All employees shall take the training.

2.13 Subcontractors are solely responsible to train their own employees on all applicable AECOM, AECOM TISHMAN CONSTRUCTION, federal, state, and local standards, policies, and procedures.

2.14 Principals Meeting. Once a quarter each project shall conduct a subcontractor principals meeting. Principals include company owners and/ or officers. Each active subcontractor on the project shall meet on the project and conduct a safety walkthrough with the AECOM project team. After each walkthrough, the principals group shall meet and discuss the walkthrough findings and any other relevant project safety information. Each project shall create a specific format for their principals meeting.

3. ATTACHMENTS

3.1 Tool Box Talk Meeting Example Form

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Attachment 1: Tool Box Talk Example Form

AECOM TISHMAN CONSTRUCTION

TOOL BOX SAFETY MEETING

Date: _____

Subjects Discussed:

NAMES/ SIGNATURES OF EMPLOYEES ATTENDING MEETING

Name (Print)

Signature

_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____